spintent

SPANISH PARSE INTENTS

v0.99 2026-09-05*

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CTAN: `https://www.ctan.org/pkg/spintent`

`https://github.com/pablgonz/spintent`

Resumen

El paquete spintent proporciona una serie de utilidades para docentes de educación primaria y secundaria que necesiten crean documentos PDF *accesibles* (*tagged* PDF) en español usando LuaLaTeX.

Índice

Motivación y agradecimientos

Desde que desarrollé `scontents`[7] en 2019 y, más tarde, `enumext`[8] en 2024, me he ido interesando cada vez más por la accesibilidad en los documentos PDF. Al principio era un tema prácticamente desconocido para mí, pero a medida que fui aprendiendo sobre él comprendí la importancia que tiene, especialmente en el ámbito educativo.

El equipo del proyecto de LaTeX[19] ha realizado un trabajo extraordinario en este campo. Hoy es posible generar documentos etiquetados, validarlos con herramientas como `veraPDF`[20] o inspeccionar su estructura con `ngpdf`[21]. Sin embargo, siempre me quedó una duda: cuando un estudiante utiliza `NVDA`[11] junto con `MathCAT`[12], ¿realmente escucha lo que el o la docente quiso expresar?

Basta con generar un documento accesible siguiendo las recomendaciones de `The LaTeX Tagged PDF Project` y escucharlo con `NVDA/MathCAT` para darse cuenta de que, en muchos casos, la *verbalización* no coincide con lo que uno espera. Esto no resulta extraño: las reglas del español, y en particular el lenguaje matemático, dependen en gran medida del contexto en que se utilizan.

El objetivo de spintent nace precisamente de esta necesidad: proporcionar herramientas para que el o la docente pueda indicar explícitamente el contexto de determinadas expresiones matemáticas desde el propio código fuente de LaTeX, de modo que el documento PDF resultante no solo sea *accesible*, sino que también produzca una *verbalización más natural* y adecuada para el estudiante al utilizar `NVDA/MathCAT`

El nuevo estándar PDF `2.0`[14, 18, 16], las especificaciones de `WTPDF`[15] y la guía más reciente de la PDF Association, «Best Practice Guide: Math in PDF»[17], coinciden en que el camino hacia una accesibilidad matemática de calidad pasa por el uso de `MathML`. En ese contexto, LuaLaTeX ofrece una plataforma especialmente adecuada para incorporar esta tecnología.

*Este archivo describe la documentación para la versión v0.99 revisada por última vez el 2026-09-05.

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Licencia y requerimientos

El paquete spintent carga y requiere el paquete unicode-math[1] y que se ejecute bajo LuaLaTeX por lo que es necesario contar con una distribución moderna de TeX como TeX Live 2026. Ha sido probado con las clases estándar proporcionadas por LaTeX: book, report, article y letter en tamaños 10pt, 11pt y 12pt.

Se otorga permiso para copiar, distribuir y/o modificar este paquete bajo los términos de LaTeX Project Public License (LPPL), versión 1.3 o posterior (https://www.latex-project.org/lppl.txt). El paquete tiene la condición de mantenido al estilo «*A Work in Progress*[1] ».

El requerimiento mínimo es LaTeX release 2026-11-01.

Este paquete solo se puede utilizar solo con LuaLaTeX y *tagged* PDF activo. Está presente en TeX Live y MiKTeX, para instalarlo, utilice el gestor de paquetes de su distribución. Si desea una instalación manual, descargue spintent.zip y descomprímalo, luego ejecute `luatex spintent.ins`, mueva todos los archivos a las ubicaciones correspondientes y luego ejecute `mktexlsr`. Para generar la documentación, ejecute `arara spintent.dtx`.

```
spintent.sty  ⇒  TDS:tex/lualatex/spintent/
spintent.lua  ⇒  TDS:tex/lualatex/spintent/
spintent.pdf  ⇒  TDS:doc/lualatex/spintent/
README.md     ⇒  TDS:doc/lualatex/spintent/
spintent.dtx  ⇒  TDS:source/lualatex/spintent/
spintent.ins  ⇒  TDS:source/lualatex/spintent/
```

1 Descripción

El nombre spintent puede entenderse como «*Spanish Parse Intent*» o también como «*School Parse Intent*». Ambas interpretaciones reflejan el objetivo del paquete y el público para el que fue diseñado: docentes de educación primaria y secundaria.
Está pensado para crear «*apuntes*», «*hojas de ejercicios*», «*clases escritas*» enfocadas en el estudiante. Solo un docente sabe cual es el nivel de aprendizaje del estudiante. El paquete intenta sobrescribir muchas de las reglas dictadas por la heurística de NVDA/MathCAT teniendo esto presente.

El paquete spintent usa y abusa del atributo `:intent` de MathML por medio del comando `\MathMLintent` definido en latex-lab-mathintent[30], un encapsulado de `\luamml_attribute:een` proveída por luamml[2]. !Gracias Marcel¡

1.1 Carga del paquete

El paquete spintent trabaja solo con LuaLaTeX y *tagged* PDF activo usando `tagging=on` dentro del comando `\DocumentMetadata` al inicio del archivo. Las llaves (⟨*keys*⟩) aceptadas por `\DocumentMetadata` y en general para el código de *tagged* PDF están documentadas en los paquetes y tagpdf[29] documentmetadata-support[30], es deber del usuario revisar la documentación asociada a ellos.

```
\DocumentMetadata
  {
    lang=es-CL,
    pdfversion=2.0,
    pdfstandard=ua-2,
    tagging=on,
    tagging-setup={math/setup={mathml-SE}},
  }
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage{spintent}
```

1.2 El paquete babel

El paquete babel-spanish[10] está marcado como «parcialmente compatible» para *tagged* PDF, pero, es absolutamente necesario para la creación de documentos en español. En caso de encontrar problemas de compatibilidad con otros paquetes se puede desactivar usando:

```
\usepackage[spanish,shorthands=off]{babel}
```

Muchas de las «shorthands» proveidas por babel-spanish hoy no son necesarias, más allá de `\sptext` para las abreviaturas y ordinales el resto son solo atajos de teclas para escritura que hoy en día no se ocupan. El paquete spintent provee el comando `\spshort` (§3.4) que cubre estos casos, de esta manera podemos cargar babel[9] con todas sus funcionalidades sin problemas.

[1] *A Work in Progress* es un álbum en vivo de Neil Peart (∗ 1952–† 2020), baterista y principal letrista de Rush. El nombre no pretende formar parte de la terminología de la LPPL; simplemente describe el estado permanente de este paquete mientras sigo aprendiendo sobre *accesibilidad* en PDF.

1.3 Fuentes tipográficas

Como spintent es un paquete LuaLaTeX debemos distinguir entre las fuentes tipográficas `OpenType/TrueType` versus las fuentes de `8bit` estándar usadas por LaTeX para el *modo texto* y el *modo matemático*. Para evitar problemas con *tagged* `PDF` preferiremos no combinar fuentes `OpenType/TrueType` con fuentes `8bit` y trabajaremos solo con fuentes `OpenType/TrueType` que tengan soporte para matemáticas.

La carga de las fuentes tipográficas para LuaLaTeX se maneja por medio del paquete `fontspec`[3], si no se espcifica ninguna se usarán las fuentes `Latin Modern` para el *modo texto* y `Latin Modern Math` para el *modo matemático*.

El tipo de fuente tipográfica escogida es preferencia de cada usuario, en lo personal utilizo las fuente `Libertinus` para el *modo texto* por medio del paquete `libertinus-otf`[4] y la fuente `Erewhon-Math` para el modo matemático cargada por el paquete `fourier-otf`[5]. Adicionalmente cargo la fuente `Source Code Pro` para el texto tipo «`máquina de escribir`» por medio del paquete `sourcecodepro`[6].

```
% \DocumentMetadata{...}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage[osf,nomath,mono=false]{libertinus-otf}
\usepackage[osf,semibold]{sourcecodepro}
\usepackage[no-text]{fourier-otf}
\usepackage{spintent}
```

1.4 El paquete `hyperref`

El paquete `hyperref`[23] es *«casi obligatorio»* cuando se crean documentos *tagged* `PDF`, se encarga de las referencias cruzadas `\label` y `\ref`, las notas al pie `\footnote`, los hipervínculos `\href` y muchos otros elementos como marcadores, índices y metadatos.

```
\DocumentMetadata
  {
    lang=es-CL,
    pdfversion=2.0,
    pdfstandard=ua-2,
    tagging=on,
    tagging-setup={math/setup={mathml-SE}},
  }
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage[osf,nomath,mono=false]{libertinus-otf}
\usepackage[osf,semibold]{sourcecodepro}
\usepackage[no-text]{fourier-otf}
\setmathfont{Erewhon-Math}[range={cal,bfcal},StylisticSet=1]
\usepackage{spintent, hyperref}
\hypersetup
  {
    colorlinks         = true,
    pdfstartview       = FitH,
    pdfdisplaydoctitle = true,
    pdftitle           = {Test para el paquete spintent},
    pdfinfo            =
      {
        Title   = {Test},
        Subject = {spintent},
      },
  }
```

2 El comando \setspintent

\setspintent `\setspintent{⟨comando⟩}{⟨key-uno = valor, key-dos = valor, key-tres = valor, ...⟩}`

El comando `\setspintent` establece las llaves (⟨*keys*⟩) de forma global para los comandos provistos por spintent. Puede utilizarse tanto en el preámbulo como en el cuerpo del documento tantas veces como se desee.

Las llaves (⟨*keys*⟩) definidas en el *argumento opcional* [⟨*key = val*⟩] de los comandos tienen prioridad sobre las establecidas por este, anulando las configuraciones globales de `\setspintent`.

Muchas de las utilidades que provee el paquete spintent implementan el sistema [⟨*key = val*⟩] y están pensadas para ser utilizadas en la generación de *«hojas de ejercicios»* o *«clases escritas»*. Se recomienda usar `\setspintent` y establecerlas de manera global para cada documento, para mantener la coherencia en la verbalización de NVDA/MathCAT a lo largo del mismo.

El sistema [⟨*key = val*⟩] utilizado por spintent se implementa usando el módulo `l3keys` de `expl3`[24]. Por diseño, las llaves que reciben un *valor* después del signo '=' NO aceptan *valores vacíos* y es obligatorio pasar uno valido; las llaves documentadas como ⟨*valor prohibido*⟩ no reciben *ningun valor*. En caso de omitir un *valor* valido o pasar un *valor vacío* a una llave que acepte *valor* o pasar un *valor* a una llave del tipo ⟨*valor prohibido*⟩ se devolverá un "error" y se detendrá la compilación del documento.

3 The big four

El paquete spintent provee cuatro grandes comandos (*«the big four*[2]*»*): `\spnum` para el manejo de números, `\spunit` para el manejo de unidades, `\spmoney` para el manejo de dinero y `\spshort` para el manejo de abreviaturas, acrónimos y números ordinales.

Tambien se incluyen las utilidades `\spnewunit` y `\spunitalias` asociadas a al comando `\spunit`.

3.1 El comando \spnum

\spnum

`\spnum{⟨número⟩}`
`\spnum[⟨key = val⟩]{⟨+número⟩}`
`\spnum[⟨key = val⟩]{⟨-número⟩}`
`\spnum[⟨key = val⟩]{⟨número,número⟩}`
`\spnum[⟨key = val⟩]{⟨número.número⟩}`
`\spnum[⟨key = val⟩]{⟨número,número;número⟩}`
`\spnum[⟨key = val⟩]{⟨número.número;número⟩}`
`\spnum[⟨key = val⟩]{⟨número;número⟩}`
`\spnum[⟨key = val⟩]{⟨número e exponente⟩}`
`\spnum[⟨key = val⟩]{⟨número E exponente⟩}`

El comando `\spnum` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*número*⟩}, donde la *«parte entera»* puede iniciar con los signos '+' o '−', la *«parte decimal finita»* **debe** iniciar con una *coma* '`,`' o con un *punto* '`.`', y la *«parte decimal infinita (periódica)»* **debe** iniciar con un *punto y coma* '`;`'.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩] y se ejecuta `$\spnum{+23}$` se imprimirá '+23' en la salida PDF y NVDA/MathCAT verbalizará *«más veintitrés»*, si se ejecuta `$\spnum{3,14}$` se imprimirá '3,14' en la salida PDF y NVDA/MathCAT verbalizará *«tres coma catorce»*, si se ejecuta `$\spnum{3,1;4}$` se imprimirá '$3{,}1\overline{4}$' en la salida PDF y NVDA/MathCAT verbalizará *«tres coma uno con cuatro periódico»*.

Además, el *argumento obligatorio* {⟨*número*⟩} admite al final de este un sufijo de *«notación científica»*. Este se indica mediante las letras '`e`' o '`E`', acompañadas de los signos opcionales '+' o '−' seguido por los dígitos del exponente. Por ejemplo `$\spnum{3,14e3}$` imprimirá '$3{,}14 \cdot 10^3$' en la salida PDF y NVDA/MathCAT verbalizará *«tres coma catorce por diez al cubo»*.

La *coma* '`,`' o el *punto* '`.`' son *separadores decimales* y NO de millares, por lo tanto solo pueden aparecer una «única vez» dentro del {⟨*argumento*⟩}. El {⟨*argumento*⟩} puede contener espacios matemáticos validos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo. Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

3.1.1 Llaves para \spnum

`cifras-sep` = {⟨*true* | *false*⟩} default: *true*

Establece el *espaciado* utilizado para agrupar los dígitos en bloques de tres, tanto en la parte entera como en la parte decimal finita. Si se establece en `true` se siguen las normas de la RAE y se utiliza un *espacio fino* '`\,`' entre las cifras; si se establece en `false`, se respetan los espacios matemáticos introducidos en el argumento. Por ejemplo `$\spnum{100000}$` imprimirá '100 000' en la salida PDF y `$\spnum[cifras-sep=false]{10\,00\,00}$` imprimirá '10 00 00' en la salida PDF.

`read-sign` = {⟨*terse* | *clear*⟩} default: *terse*

Establece la *forma* en la que NVDA/MathCAT verbalizará los signos '+' o '−'. Si se establece en `terse`, NVDA/MathCAT verbalizará *«más»* o *«menos»* **antes** de leer el número; si se establece en `clear`, NVDA/MathCAT verbalizará *«positivo»* o *«negativo»* **después** de leer el número. Por ejemplo `$\spnum{-23}$` imprimirá '−23' en la salida PDF y NVDA/MathCAT verbalizará *«menos veintitrés»* y `$\spnum[read-sign=clear]{-23}$` imprimirá '−23' en la salida PDF y NVDA/MathCAT verbalizará *«veintitrés negativo»*.

`read-period-sep` = {⟨*coma* | *punto*⟩} default: *coma*

[2]Los *«cuatro grandes»* del Thrash Metal: Metallica, Megadeth, Slayer y Anthrax, que me acompañan mientras desarrollo spintent.

Establece la *forma* en la que NVDA/MathCAT verbalizará e imprimirá el signo de separación '`;`' de la *«parte decimal infinita (periódica)»* cuando no se tiene la *«parte decimal finita»*. Si se establece en `coma`, se imprimirá '`,`' en la salida PDF y NVDA/MathCAT verbalizará *«coma»*; si se establece en `punto`, se imprimirá '`.`' en la salida PDF y NVDA/MathCAT verbalizará *«punto»*. Por ejemplo `$\spnum{3;4}$` imprimirá '$3{,}\overline{4}$' en la salida PDF y NVDA/MathCAT verbalizará *«tres coma cuatro periódico»* y `$\spnum[read-period-sep=punto]{3;4}$` imprimirá '$3.\overline{4}$' en la salida PDF y NVDA/MathCAT verbalizará *«tres punto cuatro periódico»*

Esta llave es necesaria SOLO cuando se utilizan *«decimales periódicos puros»*, es decir, cuando no se ha incluido en el {⟨*argumento*⟩} una *coma* '`,`' o un *punto* '`.`' antes del signo de separación '`;`' periódica.

`notation-sym` = {⟨*times* | *cdot*⟩} default: *cdot*

Establece el *símbolo* que se utilizará para imprimir la *«notación científica»* al usar '`e`' o '`E`'. Si se establece en `times` imprimirá '$\times$' en la salida PDF y NVDA/MathCAT verbalizará *«por»*; si se establece en `cdot` imprimirá '$\cdot$' en la salida PDF y NVDA/MathCAT verbalizará *«por»*. Por ejemplo `$\spnum{3,14e3}$` imprimirá '$3{,}14 \cdot 10^{3}$' en la salida PDF y NVDA/MathCAT verbalizará *«tres coma catorce por diez al cubo»* y `$\spnum[notation-sym=times]{3,14e3}$` imprimirá '$3{,}14 \times 10^{3}$' en la salida PDF y NVDA/MathCAT verbalizará *«tres coma catorce por diez al cubo»*.

Decir *«notación científica»* aquí es un poco exagerado, en realidad es *«multiplicación por potenicas de* 10*»*, no se verfica si la entrada cumple o no con la condición de estár escrita en *«notación científica»*.

3.2 El comando `\spunit`

`\spunit`

`\spunit{⟨unidad⟩}`
`\spunit[⟨key = val⟩]{⟨unidad⟩}`
`\spunit[⟨key = val⟩]{⟨número unidad⟩}`
`\spunit[⟨key = val⟩]{⟨número unidad*unidad⟩}`
`\spunit[⟨key = val⟩]{⟨número unidad*unidad/unidad⟩}`
`\spunit[⟨key = val⟩]{⟨número unidad*unidad/unidad*unidad⟩}`

El comando `\spunit` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*unidad*⟩}, el cual puede contener una *«parte numérica»* opcional la cual se procesa internamente por el comando `\spnum` al inicio del {⟨*argumento*⟩} seguido de una {⟨*unidad*⟩} o un {⟨*alias*⟩} valido descritos en §3.2.2 o una {⟨*unidad*⟩} creada por el comando `\spnewunit` (§3.2.3) o un {⟨*alias*⟩} creado por el comando `\spunitalias` (§3.2.4).

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩] y se ejecuta `$\spunit{23 m}$` se imprimirá '23 m' en la salida PDF y NVDA/MathCAT verbalizará *«veintitrés metros»*.

La multiplicación entre las {⟨*unidades*⟩} se realiza mediante un *asterisco* '`*`' y las potencias de estas se indican con un *circunflejo seguido del exponente entre llaves* '`^{⟨exponente⟩}`'. Por ejemplo si se ejecuta `$\spunit{23 m*s^{2}}$` se imprimirá '$23\,\mathrm{m}\,\mathrm{s}^{2}$' en la salida PDF y NVDA/MathCAT verbalizará *«veintitrés metros segundos al cuadrado»*.

Se pueden usar *fracciones lineales* con las {⟨*unidades*⟩} utilizando una *«única barra»* '`/`' para la separación entre las {⟨*unidades*⟩} del numerador y el denominador, teniendo presente que el denominador NO puede contener números de ningún tipo. Por ejemplo si se ejecuta `$\spunit{23 m/s^{2}}$` se imprimirá '$23\,\mathrm{m}/\mathrm{s}^{2}$' en la salida PDF y NVDA/MathCAT verbalizará *«veintitrés metros por segundos al cuadrado»*.

Solo imprimiremos *fracciones lineales*, si se requiere otra forma fraccionaria para explicaciones y otros usos se debe usar el comando por separado. El {⟨*argumento*⟩} puede contener espacios matemáticos validos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo, NO puede contener más de una *«única barra»* '`/`' y NO puede contener números en el denominador. Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

3.2.1 Llaves para `\spunit`

El comando `\spunit` posee las ⟨*meta-keys*⟩ `cifras-sep`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1).

`print-cdot` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprime* `\cdot` '$\cdot$' o un *espacio fino* '`\,`' para la multiplicación de {⟨*unidades*⟩}. Si se establece en `true`, se imprimirá '$\cdot$' entre las {⟨*unidades*⟩} en la salida PDF; si se establece en `false` se imprimirá un *espacio fino* '`\,`' entre las {⟨*unidades*⟩} en la salida PDF.

`read-cdot` = {⟨*true* | *false*⟩} default: *false*

Establece la *forma* en la que NVDA/MathCAT verbalizará la multiplicación de {⟨*unidades*⟩}. Si se establece en `true`, NVDA/MathCAT verbalizará «por» entre las {⟨*unidades*⟩}; si se establece en `false`, NVDA/MathCAT no verbalizará nada entre las {⟨*unidades*⟩}.

`read-slash` = {⟨*por* | *partido-por*⟩} default: *por*

Establece la *forma* en la que NVDA/MathCAT verbalizará la *«barra»* '`/`' de separación entre el numerador y el denominador. Si se establece en `por`, NVDA/MathCAT verbalizará *«por»*; si se establece en `partido-por`, NVDA/MathCAT verbalizará *«partido por»*.

3.2.2 Argumentos para `\spunit`

Resumen de {⟨*unidades*⟩} y {⟨*alias*⟩} soportados por `\spunit`.

Entrada	Alias soportados	Nombre unidad	Salida	NVDA/MathCAT
m	metro, metros	Metro	m	*«metro»*
g	gramo, gramos	Gramo	g	*«gramo»*
s	segundo, segundos	Segundo (Tiempo)	s	*«segundo (Tiempo)»*
l	litro, l-litro	Litro (minúscula)	l	*«litro (minúscula)»*
h	hora, horas	Hora	h	*«hora»*
d	dia, dias	Día	d	*«día»*
A	amperio, ampere	Amperio	A	*«amperio»*
V	voltio, volt	Voltio	V	*«voltio»*
W	vatio, watt	Vatio	W	*«vatio»*
J	julio, joule	Julio	J	*«julio»*
N	newton	Newton	N	*«newton»*
Pa	pascal	Pascal	Pa	*«pascal»*
Hz	hercio, hertz	Hercio	Hz	*«hercio»*
Ω	ohmio, ohm, Omega	Ohmio	Ω	*«Ohmio»*
F	faradio	Faradio	F	*«Faradio»*
C	culombio	Culombio	C	*«Culombio»*
K	kelvin, Kelvin	Kelvin	K	*«Kelvin»*
℃	celsius, degC, ℃	Grados Celsius	°C	*«Grados Celsius»*
℉	fahrenheit, degF, ℉	Grados Fahrenheit	°F	*«Grados Fahrenheit»*
cal	caloria, calorias	Caloría	cal	*«Caloría»*
b	bit	Bit	b	*«Bit»*
B	byte, bytes	Byte	B	*«Byte»*
in	pulgada, pulgadas	Pulgada	in	*«Pulgada»*
ft	pie, pies	Pie	ft	*«Pie»*
yd	yarda, yardas	Yarda	yd	*«Yarda»*
mi	milla, millas	Milla	mi	*«Milla»*
lb	libra, libras	Libra	lb	*«Libra»*
oz	onza, onzas	Onza	oz	*«Onza»*
gal	galon, galones	Galón	gal	*«Galón»*
Å	angstrom	Angstrom	Å	*«Angstrom»*
µ	micro	Micro (Prefijo)	µ	*«Micro (Prefijo)»*
℧	mho	Mho / Siemens	℧	*«Mho / Siemens»*
′	minmin	Minuto (Arco)	′	*«Minuto (Arco)»*
″	segseg	Segundo (Arco)	″	*«Segundo (Arco)»*
u	u-masa	Unidad masa atómica	u	*«Unidad masa atómica»*
		Unidades directas (sin alias):		
L	-	Litro (Mayúscula)	L	*«Litro (Mayúscula)»*
ℓ	-	Litro (Cursiva)	ℓ	*«Litro (Cursiva)»*
w	-	Vatio (Minúscula)	w	*«Vatio (Minúscula)»*
°K	-	Grado Kelvin	°K	*«Grado Kelvin»*
Bq	-	Becquerel	Bq	*«Becquerel»*
Da	-	Dalton	Da	*«Dalton»*
Gy	-	Gray	Gy	*«Gray»*
S	-	Siemens	S	*«Siemens»*
Sv	-	Sievert	Sv	*«Sievert»*
T	-	Tesla	T	*«Tesla»*
Wb	-	Weber	Wb	*«Weber»*
atm	-	Atmósfera	atm	*«Atmósfera»*
au	-	Unidad astronómica	au	*«Unidad astronómica»*
bar	-	Bar	bar	*«Bar»*
cd	-	Candela	cd	*«Candela»*
dB	-	Decibelio	dB	*«Decibelio»*
dyn	-	Dina	dyn	*«Dina»*
eV	-	Electronvoltio	eV	*«Electronvoltio»*
erg	-	Ergio	erg	*«Ergio»*
ha	-	Hectárea	ha	*«Hectárea»*
kat	-	Katal	kat	*«Katal»*
kg	-	Kilogramo	kg	*«Kilogramo»*
km	-	Kilómetro	km	*«Kilómetro»*
lm	-	Lumen	lm	*«Lumen»*

Continúa en la siguiente página…

Continuación de la tabla 1

Entrada	Alias soportados	Nombre unidad	Salida	MathCAT
lx	-	Lux	lx	*«Lux»*
cm	-	Centímetro	cm	*«Centímetro»*
min	-	Minuto (Tiempo)	min	*«Minuto (Tiempo)»*
mm	-	Milímetro	mm	*«Milímetro»*
mol	-	Mol	mol	*«Mol»*
pc	-	Pársec	pc	*«Pársec»*
rad	-	Radián	rad	*«Radián»*
rpm	-	Rev. por minuto	rpm	*«Rev. por minuto»*
sr	-	Estereorradián	sr	*«Estereorradián»*
t	-	Tonelada	t	*«Tonelada»*
a	-	Año / Área (are)	a	*«Año / Área (are)»*
y	-	Año (Year)	y	*«Año (Year)»*
°	-	Grado (Arco)	°	*«Grado (Arco)»*

3.2.3 El comando `\spnewunit`

`\spnewunit` `\spnewunit{⟨nueva unidad⟩}{⟨verbalización NVDA⟩}`

El comando `\spnewunit` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: {⟨*nueva unidad*⟩} que establece lo que se imprimirá en la salida PDF y {⟨*verbalización* NVDA⟩} que establece como NVDA/MathCAT *verbalizará* la {⟨*nueva unidad*⟩}.

La {⟨*nueva unidad*⟩} no debe estar registrada como {⟨*unidad*⟩} o {⟨*alias*⟩} para `\spunit` descritos en §3.2.2. Si la {⟨*nueva unidad*⟩} ya está registrada se devolverá un "error" y se detendrá la compilación del documento.

La declaración de la {⟨*nueva unidad*⟩} es «global» y se almacena en el módulo `spintent.lua` como una {⟨*unidad*⟩} valida para pasar como {⟨*argumento*⟩} a `\spunit`. Por ejemplo: `\spnewunit{px}{píxeles}` creará la {⟨*nueva unidad*⟩} 'px' y cuando se utilice `\spunit{100 px}` se imprimirá '100 px' en la salida PDF y NVDA/MathCAT verbalizará *«cien píxeles»*.

3.2.4 El comando `\spunitalias`

`\spunitalias` `\spunitalias{⟨nuevo alias⟩}{⟨expresión de unidades⟩}`

El comando `\spunitalias` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: {⟨*nuevo alias*⟩} que establece es el nuevo {⟨*alias*⟩} de {⟨*argumento*⟩} para `\spunit` y {⟨*expresión de unidades*⟩} que puede contener multiplicación o división de {⟨*unidades*⟩} ya registradas que se imprimirán en la salida PDF.

El {⟨*nuevo alias*⟩} no debe estar registrado como {⟨*unidad*⟩} o {⟨*alias*⟩} para `\spunit` descritos en §3.2.2 o estar definido como {⟨*unidad*⟩} por medio del comando `\spnewunit`. Si el {⟨*nuevo alias*⟩} ya está registrado se devolverá un "error" y se detendrá la compilación del documento.

La declaración de un {⟨*nuevo alias*⟩} es «global» y se almacena en el módulo `spintent.lua` como una {⟨*alias*⟩} valido para pasar como {⟨*argumento*⟩} a `\spunit`. Por ejemplo: `\spunitalias{caudal}{m^{3}/s}` creará el {⟨*alias*⟩} 'caudal' y cuando se utilice `$\spunit{50 caudal}$` se imprimirá '50 m^3/s' en la salida PDF y NVDA/MathCAT verbalizará *«cincuenta metros al cubo por segundos»*.

3.3 El comando `\spmoney`

`\spmoney`
`\spmoney{⟨cifra⟩}`
`\spmoney{⟨cifra moneda⟩}`
`\spmoney[⟨key = val⟩]{⟨cifra moneda⟩}`

El comando `\spmoney` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*cifra*⟩} que contiene la *«cifra numérica»* positiva o negativa la se cual procesa internamente por el comando `\spnum` al inicio del {⟨*argumento*⟩} acompañada de un *«símbolo de moneda»* o *«alias de moneda»* opcional descritos en §3.3.2.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩] y se ejecuta `$\spmoney{500}$` se imprimirá '$500' en la salida PDF y NVDA/MathCAT verbalizará *«quinientos pesos»*.

El comportamiento de `\spmoney` depende de como se pase el *argumento obligatorio* que tiene prioridad por sobre los valores por defecto. Por ejemplo: `$\spmoney{500 dolares}$` imprimirá '$500' en la salida PDF y NVDA/MathCAT verbalizará *«quinientos dolares»* aunque la moneda por defecto sean pesos.

Si se pasa un *«alias de moneda»* del tipo ISO (código de divisas) como CLP o USD (mayúsculas) se modifica la verbalización en NVDA/MathCAT agregando el país y se ajusta la salida PDF al estandar ISO. Por ejemplo: `$\spmoney{500 CLP}$` imprimirá 'CLP500' en la salida PDF y NVDA/MathCAT verbalizará *«quinientos pesos chilenos»*.

Si se pasa un *«alias de moneda»* del tipo ISO (código de divisas) como clp o usd (minúsculas) se modifica la verbalización en NVDA/MathCAT agregando el país, pero, NO se modifica la salida PDF imprimiendo el *«símbolo de moneda»* correspondiente. Por ejemplo: `$\spmoney{500 clp}$` imprimirá '\$500' en la salida PDF y NVDA/MathCAT verbalizará *«quinientos pesos chilenos»*

Pasar de manera directa en el *argumento obligatorio* un *«símbolo de moneda»* como '\$', '€', '£' o un *«alias de moneda»* como `peso`, `euro` o `libra` produce la salida PDF y verbalización estándar esperada para NVDA/MathCAT.

- La posición del *«símbolo de moneda»* a la izquierda o a la derecha de la *«cifra numérica»* y la verbalización para NVDA/MathCAT se manejan desde el módulo Lua `spintent.lua` y no se proveen llaves para modificar este comportamiento.

- El *«símbolo de moneda»* es tomado de la fuente tipográfica definida para el *modo matemático* al momento de ejecutar el comando, se debe tener presente en caso de querer usar un *«símbolo de moneda»* poco habitual que esté soportada por spintent. El {⟨*argumento*⟩} puede contener espacios matemáticos validos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo. La *coma* '`,`' o el *punto* '`.`' como *separador decimal* solo pueden aparecer una «única vez». Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

3.3.1 Llaves para `\spmoney`

El comando `\spmoney` posee la ⟨*meta-key*⟩ `cifras-sep` pasada al comando `\spnum` (§3.1.1).

`currency` = {⟨*moneda* | *alias*⟩} default: *peso*

Establece el *símbolo de moneda* por defecto que se imprimirá en la salida PDF y cómo lo verbalizará NVDA/MathCAT cuando SOLO se pasa una *«cifra numérica»* como {⟨*argumento*⟩} al comando `\spmoney`. El *«símbolo de moneda»* o el *«alias de moneda»* pasado a esta llave debe ser alguno de los descritos en §3.3.2.

`sub-units` = {⟨*true* | *false*⟩} default: *false*

Establece si se usarán *subunidades* de la moneda en curso para la verbalización en NVDA/MathCAT. La moneda en curso debe aceptar *subunidades*. Por ejemplo: `$\spmoney[sub-units=true]{100,5 dolares}$` imprimirá '\$100,5' en la salida PDF y NVDA/MathCAT verbalizará *«cien dólares con cincuenta centavos»*.

`dolar-math` = {⟨*true* | *false*⟩} default: *true*

Establece si el *símbolo* '\$' se imprimirá en *modo texto* o en *modo matemático*. Si se establece en `true`, se usará la fuente tipográfica definida para el *modo matemático* en curso para imprimir '\$' en la salida PDF; si se establece en `false`, se usará la la fuente tipográfica definida para el *modo texto* para imprimir '\$' en la salida PDF.

3.3.2 Monedas y alias para `\spmoney`

Resumen de los {⟨*símbolos de moneda*⟩} y {⟨*alias de moneda*⟩} soportados como {⟨*argumento*⟩} para `\spmoney`.

3.4 El comando `\spshort`

`\spshort`

`\spshort{`⟨*abreviaturas* | *acrónimos* | *números ordinales*⟩`}`
`\spshort[`⟨*key = val*⟩`]{`⟨*abreviaturas* | *acrónimos* | *números ordinales*⟩`}`
`\spshort[`⟨*read-arg*={⟨*verbalización del argumento para* NVDA⟩}⟩`]{`⟨*comandos LaTeX*⟩`}`

El comando `\spshort` trabaja *solo en modo texto* y recibe *un argumento obligatorio* del tipo {⟨*abreviaturas*⟩}, {⟨*acrónimos*⟩} o {⟨*números ordinales*⟩} descritos en §3.4.2.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩] y se ejecuta `\spshort{dr.a}` se imprimirá 'Dr.ª' en la salida PDF y NVDA/MathCAT verbalizará *«doctora»*; si se ejecuta `\spshort{onu}` se imprimirá 'ONU' en la salida PDF y NVDA/MathCAT verbalizará *«organización de las naciones unidas»* y si se ejecuta `\spshort{mineduc}` se imprimirá 'Mineduc' en la salida PDF y NVDA/MathCAT verbalizará *«ministerio de educación»*.

Para la escritura de {⟨*números ordinales*⟩} se puede ejecutar `\spshort{1.a}` que imprimirá '1.ª' en la salida PDF y NVDA/MathCAT verbalizará *«primera»*; si se ejecuta `\spshort{1.o}` imprimirá '1.º' en la salida PDF y NVDA/MathCAT verbalizará *«primero»*.

Para crear un nueva {⟨*abreviatura*⟩}, {⟨*acrónimo*⟩} o {⟨*número ordinal*⟩} que no se encuentre registrada por spintent dentro del módulo Lua `spintent.lua` descritas en §3.4.2 se debe utilizar la llave `read-arg` y definir esta dentro del *argumento obligatorio* que puede contener {⟨*comandos LaTeX*⟩} *en modo texto* los cuales se imprimirán en salida PDF.

- Para la escritura de {⟨*números ordinales*⟩} se debe tener cuidado y no confundir la *«letra o voladita»* 'º' con el símbolo de grado '°'. Si la fuente tipográfica del *modo texto* está utilizando `Numbers=OldStyle` los números pueden quedar muy separados de la *«letra voladita»*, la solución para esto es agregar `\addfontfeatures{Numbers=Lining}` dentro de un grupo junto al comando: `{\addfontfeatures{Numbers=Lining}\spshort{1.o}}`.

- El comando `\spshort` está pensado como un reemplazo para *shorthands* y el comando `\sptext` provistos por el paquete `babel` en español compatible con *tagged* PDF. Es de las pocas utilidades que provee el paquete spintent que trabaja *solo en modo texto*. Internamente utiliza la llave `E` *(expansion text)* proveída por el paquete `tagpdf`[29].

Entrada	Alias	Nombre	Salida	NVDA/MathCAT
$	`peso, pesos`	peso	$	*«peso / pesos»*
€	`euro, euros`	euro	€	*«euro / euros»*
£	`libra, libras`	libra	£	*«libra / libras»*
¥	`yen, yenes`	yen	¥	*«yen / yenes»*
¥	`yuan, yuanes`	yuan	¥	*«yuan / yuanes»*
₩	`won, wons`	won	₩	*«won / wons»*
₪	`shekel, shekels, sequel, sequeles`	séquel	₪	*«séquel / séqueles»*
₹	`rupia, rupias`	rupia	₹	*«rupia / rupias»*
₽	`rublo, rublos`	rublo	₽	*«rublo / rublos»*
₺	`lira, liras`	lira	₺	*«lira / liras»*
₴	`grivna, grivnas`	grivna	₴	*«grivna / grivnas»*
¢	`centavo, centavos`	centavo	¢	*«centavo / centavos»*
		Divisas en formato ISO (Mayúsculas)		
`CLP`	—	peso chileno	`CLP`	*«peso chileno / pesos chilenos»*
`MXN`	—	peso mexicano	`MXN`	*«peso mexicano / pesos mexicanos»*
`USD`	—	dólar	`USD`	*«dólar estadounidense / dólares estadounidenses»*
`EUR`	—	euro	`EUR`	*«euro / euros»*
`GBP`	—	libra	`GBP`	*«libra / libras»*
`JPY`	—	yen	`JPY`	*«yen / yenes»*
`CNY`	—	yuan	`CNY`	*«yuan / yuanes»*
`KRW`	—	won	`KRW`	*«won / wons»*
`ILS`	—	séquel	`ILS`	*«séquel / séqueles»*
`INR`	—	rupia	`INR`	*«rupia / rupias»*
`RUB`	—	rublo	`RUB`	*«rublo / rublos»*
`TRY`	—	lira	`TRY`	*«lira / liras»*
`UAH`	—	grivna	`UAH`	*«grivna / grivnas»*
		Códigos ISO en minúsculas		
`clp`	—	peso chileno	$	*«peso chileno / pesos chilenos»*
`mxn`	—	peso mexicano	$	*«peso mexicano / pesos mexicanos»*
`usd`	`dolar, dolares`	dólar	$	*«dólar estadounidense / dólares estadounidenses»*
`eur`	—	euro	€	*«euro / euros»*
`gbp`	—	libra	£	*«libra / libras»*
`jpy`	—	yen	¥	*«yen / yenes»*
`cny`	—	yuan	¥	*«yuan / yuanes»*
`krw`	—	won	₩	*«won / wons»*
`ils`	—	séquel	₪	*«séquel / séqueles»*
`inr`	—	rupia	₹	*«rupia / rupias»*
`rub`	—	rublo	₽	*«rublo / rublos»*
`try`	—	lira	₺	*«lira / liras»*
`uah`	—	grivna	₴	*«grivna / grivnas»*

3.4.1 Llaves para \spshort

`read-arg` = {⟨*verbalización* NVDA⟩} default: *no usado*

Establece la *verbalización del argumento* {⟨*comandos* LaTeX⟩} de la nueva {⟨*abreviatura*⟩}, {⟨*acrónimo*⟩} o {⟨*número ordinal*⟩} para NVDA. El valor de esta llave *siempre* debe pasarse entre '{ }'. Por ejemplo: `\spshort[read-arg={octavo}]{8.\textsuperscript{\emph{avo}}}` creará una nueva {⟨*abreviatura*⟩} que imprimirá '8.*avo*' en la salida PDF y NVDA verbalizará *«octavo»*.

`small-caps` = {⟨*true* | *false*⟩} default: *false*

Establece si se usarán *versalitas* para imprimir {⟨*acrónimos*⟩} y para la *«letra volada»* en {⟨*abreviaturas*⟩} y {⟨*números ordinales*⟩} respetando las reglas de la RAE para el uso de estas. Si se establece en `true`, las abreviaturas con mayúscula como `EE. UU.` y los acrónimos de cinco o más letras como `Unicef` no se verán afectados; si se establece en `false`, no se aplicarán *versalitas*. Por ejemplo: `\spshort[small-caps=true]{dr.a}` imprimirá 'Dr.ᴬ' en la salida PDF y NVDA verbalizará *«doctora»*.

3.4.2 Argumentos para \spshort

El cuadro 2 resume las abreviaturas, expresiones latinas y títulos registrados en el módulo Lua `spintent.lua`.

Cuadro 2: Abreviaturas soportadas

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
	Abreviaturas con letras voladas		
`dr.a`	`dr.ª`	Dr.ª	*«doctora»*
`dr.as`	—	Dr.ªˢ	*«doctoras»*
`prof.a`	—	Prof.ª	*«profesora»*
`d.a`	`d.ª`	D.ª	*«doña»*
`m.a`	—	M.ª	*«maría»*
`n.o`	`n.º`	N.º	*«número»*
`n.os`	—	N.ºˢ	*«números»*
`c.ia`	—	C.ⁱᵃ	*«compañía»*
	Abreviaturas comunes y expresiones latinas		
`pág.`	`pag.`	pág.	*«página»*
`vol.`	—	vol.	*«volumen»*
`etc.`	—	etc.	*«etcétera»*
`et al.`	`et. al.`	et al.	*«y otros»*
`ibíd.`	`ibid.`	ibíd.	*«ibídem»*
`op. cit.`	`op.cit.`	op. cit.	*«obra citada»*
`loc. cit.`	`loc.cit.`	loc. cit.	*«lugar citado»*
`v. gr.`	`v.gr.`	v. gr.	*«verbigracia»*
`i. e.`	`i.e.`	i. e.	*«esto es»*
`e. g.`	`e.g.`	e. g.	*«por ejemplo»*
`p. ej.`	`p.ej.`	p. ej.	*«por ejemplo»*
	Títulos, profesiones y cargos		
`sr.`	—	Sr.	*«señor»*
`sra.`	—	Sra.	*«señora»*
`srta.`	—	Srta.	*«señorita»*
`dr.`	—	Dr.	*«doctor»*
`dra.`	—	Dra.	*«doctora»*
`dres.`	—	Dres.	*«doctores»*
`dras.`	—	Dras.	*«doctoras»*
`prof.`	—	Prof.	*«profesor»*
`profa.`	—	Profa.	*«profesora»*
`profs.`	—	Profs.	*«profesores»*
`ing.`	—	Ing.	*«ingeniero»*
`ings.`	—	Ings.	*«ingenieros»*
`lic.`	—	Lic.	*«licenciado»*
`v. b.`	`v.b.`	V. B.	*«visto bueno»*
	Abreviaturas compuestas e institucionales		
`s. a.`	`s.a.`	S. A.	*«sociedad anónima»*
`ee. uu.`	`ee.uu.`	EE. UU.	*«estados unidos»*
`dd. hh.`	`dd.hh.`	DD. HH.	*«derechos humanos»*
`jj. oo.`	`jj.oo.`	JJ. OO.	*«juegos olímpicos»*

Continúa en la siguiente página...

Cuadro 2: *(Continuación - Abreviaturas soportadas)*

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
`ff. aa.`	`ff.aa.`	FF. AA.	*«fuerzas armadas»*
`rr. ee.`	`rr.ee.`	RR. EE.	*«relaciones exteriores»*
`cc. aa.`	`cc.aa.`	CC. AA.	*«comunidades autónomas»*
`p. d.`	`p.d.`	P. D.	*«posdata»*
`a. c.`	`a.c.`	a. C.	*«antes de Cristo»*
`d. c.`	`d.c.`	d. C.	*«después de Cristo»*
`d. n. i.`	`d.n.i.`	D. N. I.	*«documento nacional de identidad»*
`r. u. t.`	`r.u.t.`	R. U. T.	*«rol único tributario»*
`r. u. n.`	`r.u.n.`	R. U. N.	*«rol único nacional»*

El cuadro 3 detalla las siglas y acrónimos registrados en el módulo Lua `spintent.lua`.

Cuadro 3: Siglas y acrónimos soportados

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
	Siglas (Deletreo o lectura expandida)		
`dni`	—	DNI	*«documento nacional de identidad»*
`rut`	—	RUT	*«rol único tributario»*
`run`	—	RUN	*«rol único nacional»*
`onu`	—	ONU	*«organización de las naciones unidas»*
`rae`	—	RAE	*«real academia española»*
`ong`	`ongs`	ONG	*«organización no gubernamental»*
`urss`	—	URSS	*«unión de repúblicas socialistas soviéticas»*
`oea`	—	OEA	*«organización de los estados americanos»*
`oms`	—	OMS	*«organización mundial de la salud»*
`fmi`	—	FMI	*«fondo monetario internacional»*
`bid`	—	BID	*«banco interamericano de desarrollo»*
`otan`	—	OTAN	*«organización del tratado del atlántico norte»*
`ue`	—	UE	*«unión europea»*
`ru`	—	RU	*«reino unido»*
	Acrónimos silábicos		
`unicef`	—	Unicef	*«fondo de las naciones unidas para la infancia»*
`unesco`	—	Unesco	*«organización de las naciones unidas para la educación, la ciencia y la cultura»*
`mercosur`	—	Mercosur	*«mercado común del sur»*
`cepal`	—	Cepal	*«comisión económica para américa latina»*
`mineduc`	—	Mineduc	*«ministerio de educación»*
`conicet`	—	Conicet	*«consejo nacional de investigaciones»*

El cuadro 4 muestra ejemplos de números ordinales registrados en el módulo Lua `spintent.lua`..

Cuadro 4: Números ordinales soportados

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
	Sufijos para ordinales (Ejemplos)		
`1.a`	`1.ª`	1.ª	*«primera»*
`2.o`	`2.º`	2.º	*«segundo»*
`3.er`	—	$3.^{er}$	*«tercer»*
`4.os`	—	$4.^{os}$	*«cuartos»*
`5.as`	—	$5.^{as}$	*«quintas»*

4 Utilidades generales

Además de los *«cuatro grandes»* y sus asociados, el paquete spintent provee tres utilidades de uso general: `\spsiglo` para la escritura y lectura de siglos, `\sptime` para la escritura y lectura de tiempo y `\spdate` para la escritura y lectura de fechas.

El comando `\spsiglo`

`\spsiglo` `\spsiglo{⟨siglo⟩}`
`\spsiglo[⟨key = val⟩]{⟨siglo⟩}`

El comando `\spsiglo` trabaja en *modo texto* y en *modo matemático*, recibe el *argumento obligatorio* {⟨*siglo*⟩}, el cual puede ser un *número arábigo* '`21`' o un *número romano* '`XXI`' mayor que '`0`' y menor que '`4000`'.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩] y se ejecuta `\spsiglo{21}` o `\spsiglo{XXI}` se imprimirá 'xxi' en la salida PDF y NVDA/MathCAT verbalizará *«veintiuno»*.

El *argumento obligatorio* {⟨*siglo*⟩} *«siempre»* es convertido a *número romano en versalitas* y es verbalizado por NVDA/MathCAT como *número arábigo*. Esta es la única utilidad que provee el paquete spintent que trabaja en *modo texto* y en *modo matemático*. Internamente utiliza la llave `alt` proveída por el paquete `tagpdf` en *modo texto* y usa `\MathMLintent` para el *modo matemático*.

Llaves para `\spsiglo`

`print-era` = {⟨*ac* | *dc* | *none*⟩} default: *none*

Establece si se *imprime* o no la abreviatura después de imprimir el argumento {⟨*siglo*⟩}. Si se establece en `ac`, `\spsiglo[print-era=ac]{XXI}` imprimirá 'xxi a. C.' en la salida PDF y NVDA/MathCAT verbalizará *«veintiuno antes de cristo»*; si se establece en `dc`, `\spsiglo[print-era=dc]{XXI}` imprimirá 'xxi d. C.' en la salida PDF y NVDA/MathCAT verbalizará *«veintiuno después de cristo»*; si se establece en `none`, `\spsiglo[print-era=none]{XXI}` imprimirá 'xxi' en la salida PDF y NVDA/MathCAT verbalizará *«veintiuno»*.

El comando `\sptime`

`\sptime` `\sptime{⟨horas : minutos⟩}`
`\sptime{⟨horas : minutos : segundos⟩}`

El comando `\sptime` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*horas : minutos*⟩} o {⟨*horas : minutos : segundos*⟩}. Por ejemplo: `$\sptime{23:45}$` imprimirá '23:45' en *modo texto* la salida PDF y NVDA/MathCAT verbalizará *«las veintitrés y cuarenta y cinco minutos»*.

El comando `\spdate`

`\spdate` `\spdate{⟨DD/ MM/YYYY⟩}` `\spdate{⟨DD- MM-YYYY⟩}`
`\spdate{⟨YYYY/ MM/DD⟩}` `\spdate{⟨YYYY- MM-DD⟩}`

El comando `\spdate` trabaja en *modo matemático* y recibe el *argumento* obligatorio {⟨DD/MM/YYYY⟩}, {⟨DD-MM-YYYY⟩} o {⟨YYYY-MM-DD⟩}. Por ejemplo: `$\spdate{1981-01-02}$` y `$\spdate{02/01/1981}$` imprimirán en *modo texto* la salida PDF '1981-01-02' y '02/01/1981' respectivamente y en ambos casos, NVDA/MathCAT verbalizará *«dos de enero de mil novecientos ochenta y uno»*.

Si se pasa el argumento {⟨YYYY/MM/DD⟩}, internamente se convertirá en {⟨DD/MM/YYYY⟩} y se imprimirá de esa forma por compatibilidad con NVDA/MathCAT.

5 Utilidades para símbolos

El paquete spintent provee tres comandos `\spread`, `\spdsym` y `\spmsym` para trabajar con *«símbolos matemáticos»* descritos en esta sección. Están pensandos desde el punto de vista docente, para situaciones donde la lectura de un *«símbolo»* (acorde al contexto) es relevante.

A diferencia de las otras utilidades implementados por spintent (que analizan y dan formato a la salida), estos no procesan los *«símbolos matemáticos»*, solo afectan la *forma* (semántica) en que serán verbalizados por NVDA/MathCAT.

El comando `\spread`

`\spread` `\spread{⟨verbalización NVDA⟩}{⟨símbolo matemático⟩}`

El comando `\spread` trabaja en *modo matemático* y recibe dos *argumentos obligatorios*: El primer *argumento* {⟨*verbalización* NVDA⟩} establece la *«forma»* en la cual NVDA/MathCAT verbalizará al segundo *argumento* {⟨*símbolo matemático*⟩}. Por ejemplo: `$\spread{es-paralela}{\parallel}$` imprimirá '∥' en la salida PDF y NVDA/MathCAT verbalizará *«es paralela»*; `$\spread{es-paralelo}{\parallel}$` imprimirá '∥' en la salida PDF y NVDA/MathCAT verbalizará *«es paralelo»*.

El comando \spusep

\spusep \spusep{⟨separador⟩}
\spusep[⟨key = val⟩]{⟨separador⟩}

El comando \spusep trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨separador⟩}, el cual debe ser uno de los *«delimitadores»* soportados: '(', ')', '[', ']', '\{' o '\}'. Imprime el *delimitador* indicado con el tamaño establecido por la llave size y controla si NVDA/MathCAT lo verbaliza o no mediante la llave silent.

Si no se proporciona el *argumento opcional* [⟨key = val⟩], $\spusep{(}$ imprimirá '(' en la salida PDF y NVDA/MathCAT verbalizará *«paréntesis izquierdo»*.

El motivo de existencia de \spusep es que los comandos \left, \right, \Uleft y \Uright provistos por LuaTEX crean un *grupo* implícito que dificulta el código para *tagged* PDF cuando se necesita interrumpir algo entre ellos, por ejemplo al construir expresiones complejas con \spnZ, \spmQ o \spdiv. El comando \spusep imprime el *delimitador* sin crear ningún grupo.

Llaves para \spusep

size = {⟨*none* | *big* | *Big* | *bigg* | *Bigg*⟩} default: *none*

Establece el *tamaño* del {⟨delimitador⟩} que se imprimirá en la salida PDF. Los valores corresponden a los cuatro tamaños fijos provistos por LaTeX: si se establece en none, el {⟨delimitador⟩} se imprime en el tamaño por defecto del modo matemático en curso; si se establece en big, se usa \bigl/\bigr; si se establece en Big, se usa \Bigl/\Bigr; si se establece en bigg, se usa \biggl/\biggr; y si se establece en Bigg, se usa \Biggl/\Biggr. El tamaño se aplica automáticamente en el lado correcto (izquierdo o derecho) según el {⟨delimitador⟩} pasado.

silent = {⟨*true* | *false*⟩} default: *false*

Establece si NVDA/MathCAT *verbalizará* el {⟨delimitador⟩} impreso por \spusep. Si se establece en true, el {⟨delimitador⟩} se marcará con el *intent* :silent y NVDA/MathCAT *no* lo verbalizará; si se establece en false, NVDA/MathCAT verbalizará el {⟨delimitador⟩} de forma normal.

El comando \spdsym

\spdsym \spdsym
\spdsym[⟨key = val⟩]

El comando \spdsym trabaja en *modo matemático* y recibe el *argumento opcional* [⟨key = val⟩] mediante el cual se establece la *«forma»* en la que se imprimirá el *símbolo de división* y como se verbalizará en NVDA/MathCAT.

Si no se proporciona el *argumento opcional* [⟨key = val⟩], $\spdsym$ imprimirá ':' en la salida PDF y NVDA/MathCAT verbalizará *«dividido»*.

Llaves para \spdsym

prefix = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será *verbalizado* por NVDA/MathCAT *«antes»* de *verbalizar* el {⟨valor⟩} establecido por la llave read-sym. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

suffix = {⟨*sufijo*⟩} default: *no usado*

Establece el *sufijo* que será *verbalizado* por NVDA/MathCAT *«después»* de *verbalizar* el {⟨valor⟩} establecido por la llave read-sym. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

set-sym = {⟨*old-style* | *two-dots* | *slash*⟩} default: *two-dots*

Establece el *símbolo de división* que se imprimirá en la salida PDF. Si se establece en old-style, imprimirá '÷'; si se establece en two-dots, imprimirá ':' y si se establece slash, imprimirá '/'.

read-sym = {⟨*dividido* | *dividido-por* | *dividido-entre*⟩} default: *dividido*

Establece la *forma* con la cual NVDA/MathCAT verbalizará el *símbolo de división*. Si se establece en dividido, NVDA/MathCAT verbalizará *«dividido»*; si se establece en dividido-por, NVDA/MathCAT verbalizará *«dividido por»* y si se establece en dividido-entre, NVDA/MathCAT verbalizará *«dividido entre»*.

El comando \spmsym

\spmsym \spmsym
\spmsym[⟨key = val⟩]

El comando \spmsym trabaja en *modo matemático* y recibe el *argumento opcional* [⟨key = val⟩] mediante el cual se establece la *«forma»* en la que se imprimirá el *símbolo de multiplicación* y como se verbalizará en NVDA/MathCAT.

Si no se proporciona el *argumento opcional* [⟨key = val⟩], $\spmsym$ imprimirá '·' en la salida PDF y NVDA/MathCAT verbalizará *«por»*.

Llaves para `\spmsym`

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será *verbalizado* por NVDA/MathCAT *«antes»* de *verbalizar* el {⟨*valor*⟩} establecido por la llave `read-sym`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`suffix` = {⟨*sufijo*⟩} default: *no usado*

Establece el *sufijo* que será *verbalizado* por NVDA/MathCAT *«después»* de *verbalizar* el {⟨*valor*⟩} establecido por la llave `read-sym`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`set-sym` = {⟨*cross* | *dot* | *asterisk*⟩} default: *dot*

Establece el *símbolo de multiplicación* que se imprimirá en la salida PDF. Si se establece en `cross`, imprimirá '×'; si se establece en `dot`, imprimirá '·' y si se establece en `asterisk`, imprimirá '*'.

`read-sym` = {⟨*por* | *multiplicado-por* | *veces*⟩} default: *por*

Establece la *forma* con la cual NVDA/MathCAT verbalizará el *símbolo de multiplicación*. Si se establece en `por`, NVDA/MathCAT verbalizará *«por»*; si se establece en `multiplicado-por`, NVDA/MathCAT verbalizará *«multiplicado por»* y si se establece en `veces`, NVDA/MathCAT verbalizará *«veces»*.

`not-print` = {⟨*true* | *false*⟩} default: *false*

Deshabilita la *impresión* del *símbolo de multiplicación*. Cuando se establece esta llave, NVDA/MathCAT verbalizará el *símbolo de multiplicación* acorde al valor establecido por la llave `read-sym`, pero, no se imprimirá este en la salida PDF.

6 Utilidades para números

El paquete spintent provee utilidades para trabajar con números descritas en esta sección.

El comando `\spdiv`

`\spdiv` `\spdiv{⟨dividendo⟩}{⟨divisor⟩}`
`\spdiv[⟨key = val⟩]{⟨dividendo⟩}{⟨divisor⟩}`

El comando `\spdiv` trabaja en *modo matemático* y recibe dos *argumentos obligatorios* {⟨*dividendo*⟩} y {⟨*divisor*⟩}.

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩], `$\spdiv{20}{26}$` imprimirá '20 : 26' en la salida PDF y NVDA/MathCAT verbalizará *«veinte dividido veinte y seis»*

Llaves para `\spdiv`

El comando `\spdiv` posee las ⟨*meta-keys*⟩ `cifras-sep`, `read-sign`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1) junto a las ⟨*meta-keys*⟩ `set-sym` y `read-sym` pasadas al comando `\spdsym` (pág. 12).

`wrap-parens` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprimen* paréntesis redondos '(...)' alrededor de los *argumentos* {⟨*dividendo*⟩} y {⟨*divisor*⟩} pasados a `\spdiv`. Si se establece en `true`, se imprimirán paréntesis redondos alrededor de ambos *argumentos*; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens` = {⟨*true* | *false*⟩} default: *true*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos '(...)' alrededor de los *argumentos* {⟨*dividendo*⟩} y {⟨*divisor*⟩} pasados a `\spdiv`. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en `false`, NVDA/MathCAT no los verbalizará.

El comando `\spmul`

`\spmul` `\spmul{⟨factor⟩}{⟨factor⟩}`
`\spmul[⟨key = val⟩]{⟨factor⟩}{⟨factor⟩}`

El comando `\spmul` trabaja en *modo matemático* y recibe dos *argumentos obligatorios* {⟨*factor*⟩} y {⟨*factor*⟩}.

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩], `$\spmul{20}{26}$` imprimirá '20 · 26' en la salida PDF y NVDA/MathCAT verbalizará *«veinte por veinte y seis»*.

Llaves para `\spmul`

El comando `\spmul` posee las ⟨*meta-keys*⟩ `cifras-sep`, `read-sign`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1) junto a las ⟨*meta-keys*⟩ `set-sym` y `read-sym` pasadas al comando `\spmsym` (pág. 13).

`wrap-parens` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprimen* paréntesis redondos '(...)' alrededor de los *argumentos* {⟨*factor*⟩} y {⟨*factor*⟩} pasados a `\spmul`. Si se establece en `true`, se imprimirán paréntesis redondos alrededor de ambos *argumentos*; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens` = {⟨*true* | *false*⟩} default: *true*

Establece si `NVDA/MathCAT` *verbalizará* los paréntesis redondos '`(...)`' alrededor de los *argumentos* `{⟨factor⟩}` y `{⟨factor⟩}` pasados a `\spmul`. Si se establece en `true`, `NVDA/MathCAT` verbalizará la lectura de los paréntesis; si se establece en `false`, `NVDA/MathCAT` no los verbalizará.

El comando `\spnD`

`\spnD` `\spnD`
`\spnD(⟨número natural⟩)`
`\spnD[⟨key = val⟩](⟨número natural⟩)`

El comando `\spnD` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* `(⟨número natural⟩)`, por ejemplo: '`(6)`'. Si se usa sin `(⟨argumento⟩)`, `$\spnD$` imprimirá 'D(n)' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«divisores de n»*.

Si se usa con `(⟨argumento⟩)`, por ejemplo, `$\spnD(6)$`, imprimirá 'D(6)' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«divisores de seis»*.

Llaves para `\spnD`

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA/MathCAT` *«antes»* de verbalizar `\spnD(⟨argumento⟩)`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`result = {⟨true | false⟩}` default: *false*

Establece si se *imprime* el resultado de calcular los *divisores* del `(⟨número natural⟩)` pasado a `\spnD`. Si se establece en `true`, `$\spnD[result=true](6)$` imprimirá '$D(6) = \{1, 2, 3, 6\}$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«los divisores de seis son uno, dos, tres y seis»*; si se establece en `false`, no se imprimirá en la salida `PDF` el resultado de calcular los *divisores*.

`result-num = {⟨número natural⟩}` default: *5*

Establece la *cantidad* de números que se mostrarán en la salida `PDF` al activar `result=true`. Si se establece en `2`, solo se imprimirán dos valores en la salida `PDF`; si no se establece, se imprimirán como máximo `5` valores y luego se añadirá '...' en la salida `PDF`.

`sample-arg = {⟨true | false⟩}` default: *false*

Desactiva la *validación* interna del `(⟨argumento⟩)` pasado a `\spnD`, permitiendo usar ejemplos *«no numéricos»* como entrada. Si se establece en `true`, `$\spnD[sample-arg=true](m)$` imprimirá 'D(m)' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«divisores de m»*; si se establece en `false`, el `(⟨argumento⟩)` debe ser un número natural, en caso contrario se devolverá un "error".

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización para `NVDA/MathCAT` de `\spnD(⟨argumento⟩)` cuando se establece `sample-arg=true` o cuando `\spnD` se usa sin `(⟨argumento⟩)`. Si se establece en `true` `NVDA/MathCAT` NO lo verbalizará; si se establece en `false`, `NVDA/MathCAT` SÍ lo verbalizará.

El comando `\spnM`

`\spnM` `\spnM`
`\spnM(⟨número natural⟩)`
`\spnM[⟨key = val⟩](⟨número natural⟩)`

El comando `\spnM` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* `(⟨número natural⟩)`, por ejemplo: '`(6)`'. Si se usa sin `(⟨argumento⟩)`, `$\spnM$` imprimirá 'M(n)' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«múltiplos de n»*.

Si se usa con `(⟨argumento⟩)`, por ejemplo, `$\spnM(6)$`, imprimirá 'M(6)' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«múltiplos de seis»*.

Llaves para `\spnM`

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA/MathCAT` *«antes»* de verbalizar `\spnM(⟨argumento⟩)`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`result = {⟨true | false⟩}` default: *false*

Establece si se *imprime* el resultado de calcular los *múltiplos* del `(⟨número natural⟩)` pasado a `\spnM`. Si se establece en `true`, `$\spnM[result=true](6)$` imprimirá '$M(6) = \{6, 12, 18, 24, 30, \dots\}$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«los múltiplos de seis son seis, doce, dieciocho, veinticuatro, treinta puntos suspencivos»*; si se establece en `false`, no se imprimirá en la salida `PDF` el resultado de calcular los *múltiplos*.

`result-num = {⟨número natural⟩}` default: *5*

Establece la *cantidad* de números que se mostrarán en la salida `PDF` al activar `result=true`. Si se establece en `2`, solo se imprimirán dos valores en la salida `PDF` seguidos de '...'; si no se establece, se imprimirán por defecto los `5` primeros valores y luego se añadirá '...' en la salida `PDF`.

`sample-arg = {⟨true | false⟩}` default: *false*

Desactiva la *validación* interna del (⟨*argumento*⟩) pasado a `\spnM`, permitiendo usar ejemplos *«no numéricos»* como entrada. Si se establece en `true`, `$\spnM[sample-arg=true](m)$` imprimirá 'M(m)' en la salida PDF y NVDA/MathCAT verbalizará *«múltiplos de m»*; si se establece en `false`, el (⟨*argumento*⟩) debe ser un número natural, en caso contrario se devolverá un "error".

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spnM(`⟨*argumento*⟩`)` cuando se establece `sample-arg=true` o cuando `\spnM` se usa sin (⟨*argumento*⟩). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false` NVDA/MathCAT lo verbalizará.

El comando `\spmcd`

`\spmcd` `\spmcd`
`\spmcd(`⟨*números separados por coma*⟩`)`
`\spmcd[`⟨*key* = *val*⟩`](`⟨*números separados por coma*⟩`)`

El comando `\spmcd` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (⟨*números separados por coma*⟩), el cual debe ser una lista de *números naturales*, por ejemplo: '`(2, 3, 6)`'. Si se usa sin (⟨*argumento*⟩) `$\spmcd$` imprimirá 'mcd' en la salida PDF, y NVDA/MathCAT verbalizará *«máximo común divisor»* o *«m c d»*, acorde al {⟨*valor*⟩} establecido por la llave `read-cmd`.

Si se usa con (⟨*argumento*⟩) `$\spmcd(2, 3, 6)$` imprimirá 'mcd(2, 3, 6)' en la salida PDF, y NVDA/MathCAT verbalizará *«máximo común divisor entre dos, tres y seis»* o *«m c d entre dos, tres y seis»* acorde al {⟨*valor*⟩} establecido por la llave `read-cmd`.

Llaves para `\spmcd`

`upper-case` = {⟨*true* | *false*⟩} default: *false*

Establece si se usarán *mayúsculas* o *minúsculas* en la salida PDF para el comando `\spmcd`. Si se establece en `true`, se imprimirá 'MCD' en la salida PDF; si se establece en `false`, se imprimirá 'mcd' en la salida PDF.

`read-cmd` = {⟨*terse* | *clear*⟩} default: *clear*

Establece la *forma* en la que NVDA/MathCAT verbalizará el comando `\spmcd`. Si se establece en `terse`, NVDA/MathCAT verbalizará *«m c d»*; si se establece en `clear`, NVDA/MathCAT verbalizará *«máximo común divisor»*.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *«antes»* de verbalizar el valor establecido por la llave `read-cmd`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`result` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprime* el resultado de calcular el *«máximo común divisor»* del (⟨*argumento*⟩) pasado a `\spmcd`. Si se establece en `true`: `$\spmcd[result=true](2, 3, 6)$` imprimirá 'mcd(2, 3, 6) = 1' en la salida PDF y NVDA/MathCAT verbalizará *«máximo común divisor entre dos, tres y seis es igual a uno»*; si se establece en `false` no se imprimirá en la salida PDF el resultado de calcular el *«máximo común divisor»*.

`sample-arg` = {⟨*true* | *false*⟩} default: *false*

Desactiva la *validación* interna del (⟨*argumento*⟩) pasado a `\spmcd`, permitiendo usar ejemplos *«no numéricos»* como entrada. Si se establece en `true`: `$\spmcd[sample-arg=true](m, n)$`, imprimirá 'mcd(m, n)' en la salida PDF y NVDA/MathCAT verbalizará *«máximo común divisor entre m y n»*; si se establece en `false` el (⟨*argumento*⟩) debe ser una lista de *números naturales*, en caso contrario se devolverá un "error".

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spmcd(`⟨*argumento*⟩`)` cuando se establece `sample-arg=true` o cuando `\spmcd` se usa sin (⟨*argumento*⟩). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\spmcm`

`\spmcm` `\spmcm`
`\spmcm(`⟨*números separados por coma*⟩`)`
`\spmcm[`⟨*key* = *val*⟩`](`⟨*números separados por coma*⟩`)`

El comando `\spmcm` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (⟨*números separados por coma*⟩), el cual debe ser una lista de *números naturales*, por ejemplo: '`(2, 3, 6)`'. Si se usa sin (⟨*argumento*⟩) `$\spmcm$` imprimirá 'mcm' en la salida PDF, y NVDA/MathCAT verbalizará *«mínimo común múltiplo»* o *«m c m»*, acorde al {⟨*valor*⟩} establecido por la llave `read-cmd`.

Si se usa con (⟨*argumento*⟩) `$\spmcm(2, 3, 6)$` imprimirá 'mcm(2, 3, 6)' en la salida PDF, y NVDA/MathCAT verbalizará *«mínimo común múltiplo entre dos, tres y seis»* o *«m c m entre dos, tres y seis»* acorde al {⟨*valor*⟩} establecido por la llave `read-cmd`.

Llaves para `\spmcm`

`upper-case` = {⟨*true* | *false*⟩} default: *false*

Establece si se usarán *mayúsculas* o *minúsculas* en la salida PDF para el comando `\spmcm`. Si se establece en `true` se imprimirá 'MCM' en la salida PDF; si se establece en `false` se imprimirá 'mcm' en la salida PDF.

`read-cmd` = {⟨*terse* | *clear*⟩} default: *clear*

Establece la *forma* en la que NVDA/MathCAT verbalizará el comando `\spmcm`. Si se establece en `terse`, NVDA/MathCAT verbalizará *«m c m»*; si se establece en `clear`, NVDA/MathCAT verbalizará *«mínimo común múltiplo»*.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *«antes»* de verbalizar el valor establecido por la llave `read-cmd`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`result` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprime* el resultado de calcular el *«mínimo común múltiplo»* del (⟨*argumento*⟩) pasado a `\spmcm`. Si se establece en `true`: `$\spmcm[result=true](2, 3, 6)$` imprimirá '$\mathrm{mcm}(2,3,6)=6$' en la salida PDF y NVDA/MathCAT verbalizará *«mínimo común múltiplo entre dos, tres y seis es igual a seis»*; si se establece en `false` no se imprimirá en la salida PDF el resultado de calcular el *«mínimo común múltiplo»*.

`sample-arg` = {⟨*true* | *false*⟩} default: *false*

Desactiva la *validación* interna del (⟨*argumento*⟩) pasado a `\spmcm`, permitiendo usar ejemplos *«no numéricos»* como entrada. Si se establece en `true`: `$\spmcm[sample-arg=true](m, n)$`, imprimirá '$\mathrm{mcm}(m,n)$' en la salida PDF y NVDA/MathCAT verbalizará *«mínimo común múltiplo entre m y n»*; si se establece en `false` el (⟨*argumento*⟩) debe ser una lista de *números naturales*, en caso contrario se devolverá un "error".

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spmcm(`⟨*argumento*⟩`)` cuando se establece `sample-arg=true` o cuando `\spmcm` se usa sin (⟨*argumento*⟩). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false` NVDA/MathCAT SÍ lo verbalizará.

El comando `\spnZ`

`\spnZ`

`\spnZ{`⟨*número entero*⟩`}`
`\spnZ[`⟨*key = val*⟩`]{`⟨*número entero*⟩`}`

El comando `\spnZ` trabaja en *modo matemático* y recibe el *argumento* obligatorio {⟨*número entero*⟩} el cual se procesa internamente bajo el comando `\spnum`. Este comando está dedicado exclusivamente al manejo de *«números enteros»*, y añade algunas utilidades prácticas para el demarcado e interacción con NVDA/MathCAT.

Llaves para `\spnZ`

El comando `\spnZ` posee las ⟨*meta-keys*⟩ `cifras-sep` y `read-sign` pasadas al comando `\spnum` (§3.1.1).

`wrap-parens` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprimen* paréntesis redondos '(...)' alrededor del argumento {⟨*número entero*⟩} pasado a `\spnZ`. Si se establece en `true`, se imprimirán paréntesis redondos alrededor del {⟨*número entero*⟩}; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens` = {⟨*true* | *false*⟩} default: *false*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos '(...)' alrededor del *argumento* {⟨*número entero*⟩} pasado a `\spnZ`. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura del paréntesis '(...)'; si se establece en `false`, NVDA/MathCAT no los verbalizará.

El comando `\spmQ`

`\spmQ`

`\spmQ{`⟨*número entero*⟩`}{`⟨*numerador*⟩`}{`⟨*denominador*⟩`}`
`\spmQ[`⟨*key = val*⟩`]{`⟨*número entero*⟩`}{`⟨*numerador*⟩`}{`⟨*denominador*⟩`}`

El comando `\spmQ` trabaja en *modo matemático* y recibe *tres argumentos obligatorios*: {⟨*número entero*⟩}, {⟨*numerador*⟩} y {⟨*denominador*⟩}, los procesa internamente y devuelve un *«número mixto»*.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `$\spmQ{2}{5}{7}$` imprimirá '$2\frac{5}{7}$' en la salida PDF y NVDA/MathCAT verbalizará *«dos y cinco séptimos»*.

En el caso de las fracciones, y en especial con los *«números mixtos»*, no se pueden manejar muchas cosas desde el lado de spintent. Por ejemplo, la entrada `$3\frac{5}{7}$` con *tagged* PDF se verbaliza por NVDA/MathCAT como *«tres más cinco séptimos»*, lo cual no está mal, pero pedagógicamente hablando no es del todo correcto.

Llaves para `\spmQ`

`join-word` = {⟨*entero* | *enteros* | *y*⟩} default: *y*

Establece la *forma* en la que NVDA/MathCAT verbaliza la conexión entre el entero y la fracción que forman el *número mixto*. Si se establece en `entero`, al usar `$\spmQ[join-word=entero]{1}{5}{7}$` imprimirá '$1\frac{5}{7}$' en la salida PDF, y NVDA/MathCAT verbalizará *«un entero cinco séptimos»*; si se establece en `enteros`, al usar `$\spmQ[join-word=enteros]{2}{5}{7}$` imprimirá '$2\frac{5}{7}$' en la salida PDF, y NVDA/MathCAT verbalizará *«dos enteros cinco séptimos»*; si se establece en `y`, al usar `$\spmQ[join-word=y]{2}{5}{7}$` imprimirá '$2\frac{5}{7}$' en la salida PDF, y NVDA/MathCAT verbalizará *«dos y cinco séptimos»*.

`use-spnZ` = {⟨*true* | *false*⟩} default: *true*

Establece si el {⟨*numerador*⟩} y el {⟨*denominador*⟩} que forman la fracción del *número mixto* se procesan internamente bajo el comando `\spnZ`, añadiendo así *semántica de entero* en la estructura MathML de la salida PDF. Si se establece en `true`, {⟨*numerador*⟩} y {⟨*denominador*⟩} se pasarán a `\spnZ` y NVDA/MathCAT los verbalizará correctamente como *números ordinales*; si se establece en `false`, los argumentos se tipografían directamente sin procesamiento semántico adicional.

`print-dfrac` = {⟨*true* | *false*⟩} default: *false*

Establece si la fracción del *número mixto* se imprime usando `\dfrac` en lugar de `\frac`. Si se establece en `true`, al usar `$\spmQ[print-dfrac=true]{2}{5}{7}$` imprimirá la fracción en tamaño *display* '$2\dfrac{5}{7}$' en la salida PDF; si se establece en `false`, la fracción se imprime con `\frac` siguiendo el tamaño matemático del contexto en curso.

`wrap-parens` = {⟨*true* | *false*⟩} default: *false*

Establece si se *imprimen* paréntesis redondos '(...)' alrededor del *número mixto* completo devuelto por `\spmQ`. Si se establece en `true`, se imprimirán paréntesis redondos cuyo tamaño se ajusta automáticamente al estilo matemático en curso; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens` = {⟨*true* | *false*⟩} default: *true*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos '(...)' alrededor del *número mixto* cuando la llave `wrap-parens` está activa. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en `false`, NVDA/MathCAT no los verbalizará.

7 Utilidades para geometría plana

El paquete spintent provee comandos para trabajar con los objetos y medidas de la *geometría plana* descritos en esta sección. Todos los comandos trabajan en *modo matemático* y comparten el mismo módulo `spintent.lua` para construir la verbalización correcta.

Todos los comandos de esta sección aceptan como puntos solo *letras mayúsculas* como base, con modificadores opcionales de subíndice '`_{...}`' o prima '`'`'. La única excepción es `\sprecta` cuando el *argumento* es de la forma {⟨*letra*⟩}, que acepta *letras minúsculas*. Los subíndices siempre deben escribirse de la forma correcta usando `_{...}` como `A_{...}` y nunca como `A_n` sin llaves que funciona solo por coincidencia.

El comando `\spPoint`

`\spPoint` `\spPoint{⟨punto⟩}`
`\spPoint[⟨key = val⟩]{⟨punto⟩}`

El comando `\spPoint` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto*⟩} el cual es una *única letra mayúscula* con un modificador opcional de subíndice '`_{...}`' o prima '`'`'.

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩], `$\spPoint{A}$` imprimirá 'A' en la salida PDF y NVDA/MathCAT verbalizará *«punto A»*; `$\spPoint{A_{1}}$` imprimirá 'A_1' y verbalizará *«punto A sub uno»*.

El comando `\spPoint` solo acepta *un punto*. Para pares o listas de puntos (rectas, rayos, lados, ángulos, arcos, polígonos), véanse los comandos correspondientes de esta sección.

Llaves para `\spPoint`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento {⟨*punto*⟩}. Si se establece en `school`, NVDA/MathCAT verbalizará el argumento {⟨*punto*⟩} sin el prefijo «mayúscula», `$\spPoint{A}$` se verbaliza *«punto A»*; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término *«punto»*. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término *«punto»*. Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\sprecta`

`\sprecta` `\sprecta{⟨punto, punto⟩}` `\sprecta{⟨letra⟩}`
`\sprecta[⟨key = val⟩]{⟨punto, punto⟩}` `\sprecta[⟨key = val⟩]{⟨letra⟩}`

El comando `\sprecta` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma {⟨*letra*⟩} o {⟨*punto, punto*⟩}. Si el *argumento* es de la forma {⟨*punto, punto*⟩} se aplica `\overleftrightarrow` en la impresión; si el *argumento* no contiene coma y es de la forma {⟨*letra*⟩} (mayúscula o minúscula), se imprimirá solo la {⟨*letra*⟩} sin aplicar `\overleftrightarrow`.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `$\sprecta{A, B}$` imprimirá '$\overleftrightarrow{AB}$' en la salida PDF y NVDA/MathCAT verbalizará *«recta A B»*; `$\sprecta{m}$` imprimirá 'm' en la salida PDF y NVDA/MathCAT verbalizará *«recta m»*.

Llaves para `\sprecta`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento {⟨*punto, punto*⟩} y el argumento {⟨*letra*⟩} cuando está en mayúscula. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\sprecta{A, B}$` se verbaliza *«recta A B»*; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término *«recta»*. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término *«recta»*. Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\sprayo`

`\sprayo` `\sprayo{⟨punto, punto⟩}`
`\sprayo[⟨key = val⟩]{⟨punto, punto⟩}`

El comando `\sprayo` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto, punto*⟩}. Representa un *rayo* o *semirrecta* con origen en el primer punto y dirección hacia el segundo, aplicando `\overrightarrow` en la impresión.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `$\sprayo{A, B}$` imprimirá '$\overrightarrow{AB}$' en la salida PDF y NVDA/MathCAT verbalizará *«rayo A B»*.

Llaves para `\sprayo`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento {⟨*punto, punto*⟩}. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\sprayo{A, B}$` se verbaliza *«rayo A B»*; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término establecido por la llave `read-sty`. Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

`read-sty` = {⟨*rayo* | *semirrecta*⟩} default: *rayo*

Establece el *término* que verbalizará NVDA/MathCAT el ejecutar el comando sin modificar la impresión de este. Si se utiliza `$\sprayo[read-sty=rayo]{A, B}$`, NVDA/MathCAT verbalizará *«rayo A B»*; si se utiliza `$\sprayo[read-sty=semirrecta]{A, B}$`, NVDA/MathCAT verbalizará *«semirrecta A B»*.

La impresión '$\overrightarrow{AB}$' en la salida PDF es idéntica en ambos casos. La elección depende de la terminología del libro de texto o del currículo nacional: en Chile y España se prefiere «semirrecta»; en México y otros países de América Latina se usa «rayo».

El comando `\spside`

`\spside` `\spside{⟨punto, punto⟩}`
`\spside[⟨key = val⟩]{⟨punto, punto⟩}`

El comando `\spside` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto, punto*⟩}. Representa un *lado*, *segmento* o *trazo* determinado por dos puntos, aplicando `\overline` en la impresión.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `$\spside{A, B}$` imprimirá '$\overline{AB}$' en la salida PDF y NVDA/MathCAT verbalizará *«lado A B»*.

Llaves para `\spside`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que `NVDA/MathCAT` verbalizará al argumento {⟨*punto*, *punto*⟩}. Si se establece en `school`, `NVDA/MathCAT` verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spside{A, B}$` se verbaliza *«lado A B»*; si se establece en `mathcat`, `NVDA/MathCAT` aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA/MathCAT` *antes* de verbalizar el término establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en `NVDA/MathCAT` del término establecido por la llave `read-sty`. Si se establece en `true`, `NVDA/MathCAT` NO verbalizará el término; si se establece en `false`, `NVDA/MathCAT` SÍ lo verbalizará.

`read-sty` = {⟨*lado* | *segmento* | *trazo*⟩} default: *lado*

Establece el *término* que verbalizará `NVDA/MathCAT` el ejecutar el comando sin modificar la impresión de este. Si se utiliza `$\spside[read-sty=lado]{A, B}$`, `NVDA/MathCAT` verbalizará *«lado A B»*; si se utiliza `$\spside[read-sty=segmento]{A, B}$`, `NVDA/MathCAT` verbalizará *«segmento A B»* y si se utiliza `$\spside[read-sty=trazo]{A, B}$`, `NVDA/MathCAT` verbalizará *«trazo A B»*

`read-arg` = {⟨*verbalización* NVDA⟩} default: *no usado*

Sobrescribe el *término* establecido por la llave `read-sty` que será verbalizado por `NVDA/MathCAT` al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'. Por ejemplo: `$\spside[read-arg={hipotenusa}]{A, B}$` imprimirá '$\overline{AB}$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«hipotenusa A B»*.

El comando `\spmside`

`\spmside`

`\spmside{⟨punto, punto⟩}`
`\spmside[⟨key = val⟩]{⟨punto, punto⟩}`

El comando `\spmside` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto*, *punto*⟩}. Representa la *medida* del *lado*, *segmento* o *trazo* determinado por dos puntos. La distinción entre `\spside` y `\spmside` es semántica: `\spside` representa el *objeto geométrico* '$\overline{AB}$'; `\spmside` representa su *medida* (un número real positivo).

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩], `$\spmside{A, B}$` imprimirá 'AB' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«medida del lado A B»*.

Llaves para `\spmside`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que `NVDA/MathCAT` verbalizará al argumento {⟨*punto*, *punto*⟩}. Si se establece en `school`, `NVDA/MathCAT` verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spmside{A, B}$` se verbaliza *«medida del lado A B»*; si se establece en `mathcat`, `NVDA/MathCAT` aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA/MathCAT` *antes* de verbalizar *«medida del»* seguido por el término establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en `NVDA/MathCAT` de *«medida del»* seguido por el término establecido por la llave `read-sty`. Si se establece en `true`, `NVDA/MathCAT` NO verbalizará *«medida del»* seguido por el término; si se establece en `false`, `NVDA/MathCAT` SÍ lo verbalizará.

`read-sty` = {⟨*lado* | *segmento* | *trazo*⟩} default: *lado*

Establece el *término* que verbalizará `NVDA/MathCAT` el ejecutar el comando sin modificar la impresión de este. Si se utiliza `$\spmside[read-sty=lado]{A, B}$`, `NVDA/MathCAT` verbalizará *«medida del lado A B»*; si se utiliza `$\spmside[read-sty=segmento]{A, B}$`, `NVDA/MathCAT` verbalizará *«medida del segmento A B»* y si se utiliza `$\spmside[read-sty=trazo]{A, B}$`, `NVDA/MathCAT` verbalizará *«medida del trazo A B»*

`read-arg` = {⟨*verbalización* NVDA⟩} default: *no usado*

Sobrescribe el *término* establecido por la llave `read-sty` que será verbalizado por `NVDA/MathCAT` al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'. Por ejemplo: `$\spmside[read-arg={medida cateto}]{A, B}$` imprimirá '$\overline{AB}$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«medida cateto A B»*.

`print-m` = {⟨*true* | *false*⟩} default: *false*

Establece *si se imprime* la notación visual 'm' antes del argumento. Si se establece en `true`, `$\spmside[print-m=true]{A,` imprimirá 'm $\overline{AB}$' en la salida `PDF`, si se establece en `false` imprimirá 'AB' en la salida `PDF`. En ambos casos `NVDA/MathCAT` verbalizará *«medida del lado A B»*.

El comando `\spangle`

`\spangle`

`\spangle{`⟨*punto* | *letra griega* | *número*⟩`}` `\spangle{`⟨*punto, punto, punto*⟩`}`
`\spangle[`⟨*key* = *val*⟩`]{`⟨*punto* | *letra griega* | *número*⟩`}` `\spangle[`⟨*key* = *val*⟩`]{`⟨*punto, punto, punto*⟩`}`

El comando `\spangle` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{`⟨*punto* | *letra griega* | *número*⟩`}` o `{`⟨*punto, punto, punto*⟩`}` aplicando `\angle` o `\rightangle` en la impresión.

Si no se proporciona el *argumento opcional* `[`⟨*key* = *val*⟩`]`, `$\spangle{A}$` imprimirá '$\angle A$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«ángulo en A»*; `$\spangle{\alpha}$` imprimirá '$\angle\alpha$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«ángulo alfa»* y `$\spangle{A, O, B}$` imprimirá '$\angle AOB$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«ángulo A O B»*.

Llaves para `\spangle`

`read-cmd` = `{`⟨*school* | *mathcat*⟩`}` default: *school*

Establece la *forma* en la que `NVDA/MathCAT` verbalizará al argumento `{`⟨*punto*⟩`}` o `{`⟨*punto, punto, punto*⟩`}`. Si se establece en `school`, `NVDA/MathCAT` verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spangle{A, O, B}$` se verbaliza *«ángulo A O B»*; si se establece en `mathcat`, `NVDA/MathCAT` aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = `{`⟨*prefijo*⟩`}` default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA/MathCAT` *antes* de verbalizar el término *«ángulo»*. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = `{`⟨*true* | *false*⟩`}` default: *false*

Establece si se *silencia* la verbalización en `NVDA/MathCAT` del término *«ángulo»*. Si se establece en `true`, `NVDA/MathCAT` NO verbalizará el término; si se establece en `false`, `NVDA/MathCAT` SÍ lo verbalizará.

`recto` = `{`⟨*true* | *false*⟩`}` default: *false*

Establece si se usará `\rightangle` en la impresión añadiendo el término *«recto»* a la verbalización en `NVDA/MathCAT`. Si se establece en `true`, `$\spangle[recto=true]{A}$` imprimirá '$\llcorner A$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«ángulo recto en A»*, `$\spangle[recto=true]{A, O, B}$` imprimirá '$\llcorner AOB$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«ángulo recto A O B»*; si se establece en `false`, imprimirá usando `\angle` en la salida `PDF` y `NVDA/MathCAT` omitirá el término *«recto»*.

El comando `\spmangle`

`\spmangle`

`\spmangle{`⟨*punto* | *letra griega* | *número*⟩`}` `\spmangle{`⟨*punto, punto, punto*⟩`}`
`\spmangle[`⟨*key* = *val*⟩`]{`⟨*punto* | *letra griega* | *número*⟩`}` `\spmangle[`⟨*key* = *val*⟩`]{`⟨*punto, punto, punto*⟩`}`

El comando `\spmangle` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{`⟨*punto* | *letra griega* | *número*⟩`}` o `{`⟨*punto, punto, punto*⟩`}` aplicando `\measuredangle` o `\measuredrightangle` en la impresión. La distinción entre `\spangle` y `\spmangle` es semántica: `\spangle` representa el *objeto geométrico* '$\angle AOB$'; `\spmangle` representa su *medida* (un valor en grados).

Si no se proporciona el *argumento opcional* `[`⟨*key* = *val*⟩`]`, `$\spmangle{A}$` imprimirá '$\measuredangle A$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«medida del ángulo en A»*; `$\spmangle{\alpha}$` imprimirá '$\measuredangle\alpha$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«medida del ángulo alfa»* y `$\spmangle{A, O, B}$` imprimirá '$\measuredangle AOB$' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«medida del ángulo A O B»*.

Llaves para `\spmangle`

`read-cmd` = `{`⟨*school* | *mathcat*⟩`}` default: *school*

Establece la *forma* en la que `NVDA/MathCAT` verbalizará al argumento `{`⟨*punto*⟩`}` o `{`⟨*punto, punto, punto*⟩`}`. Si se establece en `school`, `NVDA/MathCAT` verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spmangle{A, O, B}$` se verbaliza *«medida del ángulo A O B»*; si se establece en `mathcat`, `NVDA/MathCAT` aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = `{`⟨*prefijo*⟩`}` default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA/MathCAT` *antes* de verbalizar el término *«medida del ángulo»*. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`silent-cmd` = `{`⟨*true* | *false*⟩`}` default: *false*

Establece si se *silencia* la verbalización en `NVDA/MathCAT` del término *«medida del ángulo»*. Si se establece en `true`, `NVDA/MathCAT` NO verbalizará el término; si se establece en `false`, `NVDA/MathCAT` SÍ lo verbalizará.

`recto` = `{`⟨*true* | *false*⟩`}` default: *false*

Establece si se usará `\measuredrightangle` en la impresión añadiendo el término *«recto»* a la verbalización en `NVDA/MathCAT`. Si se establece en `true`, `$\spmangle[recto=true]{A}$` imprimirá '⊾A' en la salida `PDF` y `NVDA/MathCAT` verbalizará *«medida del ángulo recto en A»*, `$\spmangle[recto=true]{A, O, B}$`

imprimirá '$\measuredangle AOB$' en la salida PDF y NVDA/MathCAT verbalizará *«medida del ángulo recto A O B»*; si se establece en false, imprimirá usando \measuredangle en la salida PDF y NVDA/MathCAT omitirá el término *«recto»*.

print-m = {⟨*true* | *false*⟩} default: *false*

Establece *si se imprime* la notación visual 'm' antes del argumento. Si se establece en true, $\spmangle[print-m=true]{A, imprimirá 'm $\angle AOB$' en la salida PDF, si se establece en false imprimirá '$\measuredangle AOB$' en la salida PDF. En ambos casos NVDA/MathCAT verbalizará *«medida del ángulo A O B»*.

El comando \spang

\spang \spang{⟨*grados*⟩}
\spang{⟨*grados* : *minutos*⟩}
\spang{⟨*grados* : *minutos* : *segundos*⟩}

El comando \spang trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*grados*⟩}, {⟨*grados* : *minutos*⟩} o {⟨*grados* : *minutos* : *segundos*⟩}. Los procesa internamente bajo los comandos \spnum y \spunit para imprimir y verbalizar el *«ángulo sexagesimal»*. Por ejemplo, $\spang{45:30:15}$ imprimirá '45°30′15″' en la salida PDF y NVDA/MathCAT verbalizará *«cuarenta y cinco grados treinta minutos quince segundos»*.

El comando \sparc

\sparc \sparc{⟨*punto*, *punto*⟩} \sparc{⟨*punto*, *punto*, *punto*⟩}
\sparc[⟨*key* = *val*⟩]{⟨*punto*, *punto*⟩} \sparc[⟨*key* = *val*⟩]{⟨*punto*, *punto*, *punto*⟩}

El comando \sparc trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma {⟨*punto*, *punto*⟩} o {⟨*punto*, *punto*, *punto*⟩}. Representa un arco determinado por dos o tres puntos aplicando \widearc en la impresión.

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩], $\sparc{A, B}$ imprimirá '$\widehat{AB}$' en la salida PDF y NVDA/MathCAT verbalizará *«arco A B»*.

Llaves para \sparc

read-cmd = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento {⟨*punto*, *punto*⟩} o {⟨*punto*, *punto*, *punto*⟩}. Si se establece en school, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: $\sparc{A, B}$ se verbaliza *«arco A B»*; si se establece en mathcat, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

prefix = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término *«arco»*. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

silent-cmd = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término *«arco»*. Si se establece en true, NVDA/MathCAT NO verbalizará el término; si se establece en false, NVDA/MathCAT SÍ lo verbalizará.

El comando \spmarc

\spmarc \spmarc{⟨*punto*, *punto*⟩} \spmarc{⟨*punto*, *punto*, *punto*⟩}
\spmarc[⟨*key* = *val*⟩]{⟨*punto*, *punto*⟩} \spmarc[⟨*key* = *val*⟩]{⟨*punto*, *punto*, *punto*⟩}

El comando \spmarc trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma {⟨*punto*, *punto*⟩} o {⟨*punto*, *punto*, *punto*⟩}. Representa la *medida* del *arco* determinado por dos o tres puntos. La distinción entre \sparc y \spmarc es semántica: \sparc representa el *objeto geométrico* '$\widehat{AB}$'; \spmarc representa su *medida* (un valor en grados).

Si no se proporciona el *argumento opcional* [⟨*key* = *val*⟩], $\spmarc{A, B}$ imprimirá 'm $\widehat{AB}$' en la salida PDF y NVDA/MathCAT verbalizará *«medida del arco A B»*.

Llaves para \spmarc

read-cmd = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento {⟨*punto*, *punto*⟩} o {⟨*punto*, *punto*, *punto*⟩}. Si se establece en school, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: $\spmarc{A, B}$ se verbaliza *«medida del arco A B»*; si se establece en mathcat, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

prefix = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término *«medida del arco»*. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

silent-cmd = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término *«medida del arco»*. Si se establece en true, NVDA/MathCAT NO verbalizará el término; si se establece en false, NVDA/MathCAT SÍ lo verbalizará.

El comando `\spPoly`

`\spPoly` `\spPoly{⟨punto, punto, punto, ...⟩}`
`\spPoly[⟨key = val⟩]{⟨punto, punto, punto, ...⟩}`

El comando `\spPoly` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto, punto, punto, ...*⟩}, una lista de al menos tres {⟨*puntos*⟩} separados por comas. El tipo de *símbolo* en la impresión y la verbalización se determinan acorde al número de puntos en el argumento: Si son tres puntos se aplica `\triangle` y añade el término *«triángulo»* a la verbalización; si son cuatro puntos se aplica `\mdlgwhtsquare` y añade el término *«cuadrado»* a la verbalización; si son cinco o más puntos no se aplica ningún símbolo ni se añade verbalización.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `$\spPoly{A, B, C}$` imprimirá '$\triangle ABC$' en la salida PDF y NVDA/MathCAT verbalizará *«triángulo A B C»*; `$\spPoly{A, B, C, D}$` imprimirá '$\square ABCD$' y verbalizará *«cuadrado A B C D»*; `$\spPoly{A, B, C, D, E}$` imprimirá '$ABCDE$' y verbalizará *«A B C D E»*.

Llaves para `\spPoly`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento {⟨*punto, punto, punto, ...*⟩}. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spPoly{A, B, C}$` se verbaliza *«triángulo A B C»*; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el argumento o los términos *«tríangulo»* y *«cuadrado»* establecido por la llave `print-sym`. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`print-sym` = {⟨*true* | *false*⟩} default: *true*

Establece si se usará `\triangle` o `\mdlgwhtsquare` en la impresión añadiendo el término correspondiente a la verbalización en NVDA/MathCAT. Si se establece en `true`, `$\spPoly[print-sym=true]{A, B, C}$` imprimirá '$\triangle ABC$' en la salida PDF y NVDA/MathCAT verbalizará *«triángulo A B C»*; si se establece en `false`, imprimirá 'ABC' en la salida PDF y NVDA/MathCAT omitirá el término *«triángulo»*, verbalizando solo *«A B C»*.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Alias de `print-sym=false`, incluido por consistencia con el resto de los comandos de geometría plana.

El comando `\spAfig`

`\spAfig` `\spAfig`
`\spAfig(⟨punto, punto, punto, ...⟩)`
`\spAfig(⟨número | letra | letra_{número}⟩)`
`\spAfig[⟨key = val⟩](⟨punto, punto, punto, ...⟩)`

El comando `\spAfig` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis*, el cual es *completamente opcional*, representando el *área* de una figura poligonal en la impresión aplicando `\mathcal{A}`.

El argumento admite tres formas: una *lista de vértices* (⟨*punto, punto, punto, ...*⟩) (al menos tres); un *número o letra sueltos* (⟨*1*⟩), (⟨*n*⟩); o una *letra de una tabla fija* (F, P, B, L) con subíndice numérico opcional, (⟨*F*⟩), (⟨*F_{1}*⟩).

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩] ni el argumento delimitado por paréntesis, `$\spAfig$` imprimirá '$\mathcal{A}$' en la salida PDF y NVDA/MathCAT verbalizará *«área»*. Con vértices, `$\spAfig(A, B, C)$` imprimirá '$\mathcal{A}_{\triangle ABC}$' y verbalizará *«área del triángulo A B C»*. Con un número o letra sueltos, `$\spAfig(1)$` imprimirá '$\mathcal{A}_1$' y verbalizará *«área uno»*; `$\spAfig(n)$` imprimirá '$\mathcal{A}_n$' y verbalizará *«área n»*. Con una letra de la tabla, `$\spAfig(F)$` imprimirá '$\mathcal{A}_F$' y verbalizará *«área de la figura»*; `$\spAfig(F_{1})$` imprimirá '$\mathcal{A}_{F_1}$' y verbalizará *«área de la figura uno»*.

Llaves para `\spAfig`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spAfig(A, B, C)$` se verbaliza *«área del triángulo A B C»*; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término *«área»*. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`print-sym` = {⟨*auto* | *triangle* | *square* | *none*⟩} default: *auto*

Establece si se usará `\triangle` o `\mdlgwhtsquare` en la impresión. Con vértices, `auto` infiere el símbolo según el número de puntos (tres o cuatro); `none` lo omite y NVDA/MathCAT verbaliza solo los vértices sin nombrar la figura: `$\spAfig[print-sym=none](A, B, C)$` imprimirá '$\mathcal{A}_{ABC}$' y verbalizará *«área A B C»*.

Forzar `triangle`/`square` exige que coincida con el número real de vértices; en caso contrario se produce un error. Con un número o letra sueltos, `auto` y `none` se comportan igual (sin símbolo); `triangle`/`square` anida el argumento bajo el símbolo: `$\spAfig[print-sym=triangle](1)$` imprimirá '$\mathcal{A}_{\triangle_1}$' y verbalizará *«área del triángulo uno»*. Esta llave no puede combinarse con una letra de la tabla (`F`, `P`, `B`, `L`).

`read-sub` = {⟨*true* | *false*⟩} default: *false*

Establece si `NVDA`/`MathCAT` antepone la palabra «sub» al verbalizar el número o letra del argumento único. Si se establece en `true`, `$\spAfig[read-sub=true](1)$` verbaliza *«área sub uno»*; si se establece en `false`, verbaliza *«área uno»*. No tiene efecto en la notación de vértices ni cuando el argumento es una letra de la tabla sin subíndice.

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia por completo* la verbalización en `NVDA`/`MathCAT`, sin afectar la impresión. Si se establece en `true`, `NVDA`/`MathCAT` NO verbalizará nada; si se establece en `false`, `NVDA`/`MathCAT` verbalizará normalmente.

`read-arg` = {⟨*nombre*⟩} default: *no usado*

Sobrescribe el *nombre de la figura* determinado automáticamente por el número de puntos en el argumento, que será verbalizado por `NVDA`/`MathCAT` al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'. Por ejemplo: `$\spAfig[read-arg={trapecio}](A, B, C, D)$` imprimirá '$\mathcal{A}_{\square ABCD}$' en la salida `PDF` y `NVDA`/`MathCAT` verbalizará *«área del trapecio A B C D»*.

El comando `\spPfig`

`\spPfig`

`\spPfig`
`\spPfig(`⟨*punto, punto, punto, ...*⟩`)`
`\spPfig(`⟨*número* | *letra* | *letra_{número}*⟩`)`
`\spPfig[`⟨*key = val*⟩`](`⟨*punto, punto, punto, ...*⟩`)`

El comando `\spPfig` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis*, el cual es *completamente opcional*, representando el *perímetro* de una figura poligonal en la impresión aplicando `\mathcal{P}`.

El argumento admite las mismas tres formas que `\spAfig` (§7): una *lista de vértices*, un *número o letra sueltos*, o una *letra de la tabla fija* (`F`, `P`, `B`, `L`) con subíndice numérico opcional.

Si no se proporciona el *argumento opcional* `[`⟨*key = val*⟩`]` ni el argumento delimitado por paréntesis, `$\spPfig$` imprimirá '$\mathcal{P}$' en la salida `PDF` y `NVDA`/`MathCAT` verbalizará *«perímetro»*. Con vértices, `$\spPfig(A, B, C)$` imprimirá '$\mathcal{P}_{\triangle ABC}$' y verbalizará *«perímetro del triángulo A B C»*. Con un número o letra sueltos, `$\spPfig(2)$` imprimirá '$\mathcal{P}_2$' y verbalizará *«perímetro dos»*. Con una letra de la tabla, `$\spPfig(P_{2})$` imprimirá '$\mathcal{P}_{P_2}$' y verbalizará *«perímetro del polígono dos»*.

Llaves para `\spPfig`

`read-cmd` = {⟨*school* | *mathcat*⟩} default: *school*

Establece la *forma* en la que `NVDA`/`MathCAT` verbalizará al argumento. Si se establece en `school`, `NVDA`/`MathCAT` verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spPfig(A, B, C)$` se verbaliza *«perímetro del triángulo A B C»*; si se establece en `mathcat`, `NVDA`/`MathCAT` aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = {⟨*prefijo*⟩} default: *no usado*

Establece el *prefijo* que será verbalizado por `NVDA`/`MathCAT` *antes* de verbalizar el término *«perímetro»*. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'.

`print-sym` = {⟨*auto* | *triangle* | *square* | *none*⟩} default: *auto*

Establece si se usará `\triangle` o `\mdlgwhtsquare` en la impresión, con la misma semántica que en `\spAfig` (§7).

`read-sub` = {⟨*true* | *false*⟩} default: *false*

Establece si `NVDA`/`MathCAT` antepone la palabra «sub» al verbalizar el número o letra del argumento único, con la misma semántica que en `\spAfig` (§7).

`silent-cmd` = {⟨*true* | *false*⟩} default: *false*

Establece si se *silencia por completo* la verbalización en `NVDA`/`MathCAT`, sin afectar la impresión. Si se establece en `true`, `NVDA`/`MathCAT` NO verbalizará nada; si se establece en `false`, `NVDA`/`MathCAT` verbalizará normalmente.

`read-arg` = {⟨*nombre*⟩} default: *no usado*

Sobrescribe el *nombre de la figura* determinado automáticamente por el número de puntos en el argumento, que será verbalizado por `NVDA`/`MathCAT` al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '`{ }`'. Por ejemplo: `$\spPfig[read-arg={trapecio}](A, B, C, D)$` imprimirá '$\mathcal{P}_{\square ABCD}$' en la salida `PDF` y `NVDA`/`MathCAT` verbalizará *«perímetro del trapecio A B C D»*.

8 Referencias

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9 Change history

v0.99 (ctan), 2026-09-05 – First public release.

10 Índice de la Documentación

Los números en cursiva indican las páginas donde se describe la entrada correspondiente.

11 Implementation

The most recent publicly released version of spintent is available at CTAN: `https://www.ctan.org/pkg/spintent`. While general feedback via email is welcomed, specific bugs or feature requests should be reported through the issue tracker: `https://github.com/pablgonz/spintent/issues`.

The documentation presented here is far from professional, it contains a lot of obvious information that to the eye of a TEXpert are superfluous, but, after so many years developing this project is the only way to remember what does what.

11.1 General conventions

Functions of the form `\__spintent_luafun_some:n` or `\__spintent_luafun_some:nn` are defined (created) in the Lua module `spintent.lua` and only their variants are declared in `expl3`.

Similarly, variables of the form `\l__spintent_luaset_some_tl` or `\l__spintent_luaset_some_str` are defined in the Lua module `spintent.lua` and only declared in `expl3`.

All variables and functions defined in this package are private and are NOT intended to work or be used by another package or module.

11.2 Initial set up

Start the DocStrip guards.

```
⟨*package⟩
```

Identify the internal prefix (LaTeX3 DocStrip convention) for `l3doc` class.

```
⟨@@=spintent⟩
```

11.3 Declaration of the package

First we will make sure we have a minimum (super updated) release version of LaTeX to work correctly, otherwise, we will return an “error”.

```
\NeedsTeXFormat{LaTeX2e}[2026-11-01]
\IfFormatAtLeastF { 2026-11-01 }
  {
    \PackageError { spintent }
      { LaTeX~kernel~too~old~need~2026-11-01~or~later }
      {
        spintent~requires~a~LaTeX2e~release~2026-11-01~or~
        later.~Update~your~TeX~distribution.
      }
  }
```

Now declare the spintent package and we will add the only key `mathcal` that can be passed as a package option.

```
\ProvidesExplPackage {spintent} {2026-09-05} {0.99} {Spanish parse intents}
\keys_define:nn { spintent }
  {
    mathcal .usage:n     = load,
    mathcal .bool_set:N = \l__spintent_mathcal_bool,
    mathcal .initial:n  = true,
    mathcal .value_required:n = true,
  }
\ProcessKeyOptions [ spintent ]
```

We check if the requirements for running the spintent package are met; that is: `\DocumentMetadata` must be active, LuaTEX must be running, and the `unicode-math` or `lua-unicode-math` package must be loaded. If neither is present, we will load `unicode-math`. If the conditions cannot be met, we will return a “fatal error”.

```
\IfDocumentMetadataTF
  {
    \sys_if_engine_luatex:TF
      {
        \msg_new:nnn { spintent } { package-not-load }
          {
            The~'#1'~package~will~be~loaded~as~a~dependency.
          }
        \IfPackageLoadedF { unicode-math }
          {
            \IfPackageLoadedF { lua-unicode-math }
              {
                \msg_info:nnn { spintent } { package-not-load } { unicode-math }
                \RequirePackage{unicode-math}
              }
          }
        \msg_new:nnn { spintent } { env-success }
          { Everything~looks~correct,~spintent~will~make~the~intent! }
```

```
        \msg_info:nn { spintent } { env-success }
      }
      {
        \msg_new:nnnn { spintent } { fatal-engine-requires-luatex }
          { The~spintent~package~requires~LuaLaTeX. }
          { This~package~relies~on~Lua~module.~Compile~with~lualatex. }
        \msg_fatal:nn { spintent } { fatal-engine-requires-luatex }
      }
  }
  {
    \msg_new:nnnn { spintent } { fatal-metadata-missing }
      { \token_to_str:N \DocumentMetadata \c_space_tl is~missing. }
      {
        The~spintent~package~requires~tagged~PDF~active.\\
        You~must~declare~\token_to_str:N \DocumentMetadata { tagging=on } ~
      }
    \msg_fatal:nn { spintent } { fatal-metadata-missing }
  }
```

11.4 Some kernel variants

\str_if_eq:nVTF
\str_if_eq:nVF
\str_uppercase:V
\str_lowercase:V
\seq_set_from_clist:Ne

Non-standard kernel variants.

```
\cs_generate_variant:Nn \str_if_eq:nnTF { V }
\cs_generate_variant:Nn \str_if_eq:nnF { V }
\cs_generate_variant:Nn \str_uppercase:n { V }
\cs_generate_variant:Nn \str_lowercase:n { V }
\cs_generate_variant:Nn \seq_set_from_clist:Nn { Ne }
```

(End of definition for \str_if_eq:nVTF and others.)

11.5 Variables for the key mathcal and \rightanglesqr

\l__spintent_luaset_cal_fam_ok_str
\l__spintent_luaset_sym_fam_ok_str

First, we declare the variables \l__spintent_luaset_cal_fam_ok_str and \l__spintent_luaset_sym_fam_ok_ before loading the module spintent.lua. This is necessary because of the implementation of the functions __spintent_luafun_classic_mathcal:nn and __spintent_luafun_right_angle_sqr: within the Lua module; otherwise, expl3 will complain about duplicate variable declarations.

```
\str_new:N \l__spintent_luaset_cal_fam_ok_str
\str_new:N \l__spintent_luaset_sym_fam_ok_str
```

(End of definition for \l__spintent_luaset_cal_fam_ok_str and \l__spintent_luaset_sym_fam_ok_str.)

11.6 Loading the Lua module and functions

__spintent_luafun_spunit_lookup_alias:V
__spintent_luafun_spmoney_lookup_data:V
__spintent_luafun_spmQ_scale_int:Vn
__spintent_luafun_clean_split_and_set:V

Now we load the Lua module spintent.lua and generate the variants for the functions __spintent_luafun_spunit used by \spunit (§11.15), __spintent_luafun_spmoney_lookup_data:V used by \spmoney (§11.17), __spintent_luafun_spmQ_scale_int:Vn used by \spmQ (§11.30) and __spintent_luafun_clean_split_and used by \spnum (§11.14) defined in it.

```
\lua_load_module:n { spintent.lua }
\cs_generate_variant:Nn \__spintent_luafun_spunit_lookup_alias:n { V }
\cs_generate_variant:Nn \__spintent_luafun_spmoney_lookup_data:n { V }
\cs_generate_variant:Nn \__spintent_luafun_spmQ_scale_int:nn { VV, Vn }
\cs_generate_variant:Nn \__spintent_luafun_clean_split_and_set:n { V }
```

(End of definition for __spintent_luafun_spunit_lookup_alias:V and others.)

11.7 Internal implementation for \mathcal and \rightanglesqr

To simplify user configuration and obtain the expected output style pdfTEX for \mathcal, we will provide a fallback to the font NewCMMath-Regular.otf, which contains the glyphs for these characters. Using this same font, we will obtain the symbol \rightanglesqr for the key recto.

mathcal-fallback
right-angle-sqr-fallback

First, we'll define two error messages in case our fallback fails.

```
\hook_gput_code:nnn { begindocument } { spintent }
  {
    \__spintent_luafun_init_math_fallbacks:
  }
\msg_new:nnn { spintent } { mathcal-fallback }
  {
    Using~classic~\token_to_str:N \mathcal \c_space_tl as~fallback~(font~not~found~or~
    no~free~math~family~available).~\msg_line_context:
  }
\msg_new:nnn { spintent } { right-angle-sqr-fallback }
  {
    Using~\token_to_str:N \rightangle \c_space_tl as~fallback~for~the~right-angle~
```

```
    glyph.~\msg_line_context:
  }
```

(*End of definition for* `mathcal-fallback` *and* `right-angle-sqr-fallback`.)

`\__spintent_mathcal:n` Now we define the internal function `\__spintent_mathcal:n` for `\mathcal` which checks the state of the variable `\l__spintent_mathcal_bool` set by the package key `mathcal`; if its state is true, we check the string variable `\l__spintent_luaset_cal_fam_ok_str` and if the latter has the value true, we call the function `\__spintent_luafun_classic_mathcal:n` and otherwise (we cannot find the source in the system) we return a warning message and directly use `\mathcal`. Otherwise, that is, `mathcal=false` we simply use `\mathcal`.

```
\cs_new_protected:Nn \__spintent_mathcal:n
  {
    \bool_if:NTF \l__spintent_mathcal_bool
      {
        \str_if_eq:VnTF \l__spintent_luaset_cal_fam_ok_str { true }
          { \__spintent_luafun_classic_mathcal:n {#1} }
          {
            \msg_warning:nn { spintent } { mathcal-fallback }
            \mathcal {#1}
          }
      }
      { \mathcal {#1} }
  }
```

(*End of definition for* `\__spintent_mathcal:n`.)

`\__spintent_right_angle_sqr:` The internal function `\__spintent_right_angle_sqr:` establishes the internal copy of the command `\rightanglesqr`. First, we check the string variable `\l__spintent_luaset_sym_fam_ok_str`. If it is true, we call the function `\__spintent__luafun_right_angle_sqr:`; otherwise (we do not find the font in the system), we return a warning message and use `\rightangle` instead.

```
\cs_new_protected:Nn \__spintent_right_angle_sqr:
  {
    \str_if_eq:VnTF \l__spintent_luaset_sym_fam_ok_str { true }
      { \__spintent_luafun_right_angle_sqr: }
      {
        \msg_warning:nn { spintent } { right-angle-sqr-fallback }
        \rightangle
      }
  }
```

(*End of definition for* `\__spintent_right_angle_sqr:`.)

11.8 Internal copy of the unicode invisible separator

`\__spintent_invisible_sep:`
`\c__spintent_invisible_sep_tl`

The function `\__spintent_invisible_sep:` is an internal copy of the Unicode invisible separator `U-2063` that uses `\luamml_annotate:en` from the package `luamml`[2] and let `<mo>...</mo>` in the `MathML` structure embedded in the `PDF` output.

```
\cs_new_protected:Nn \__spintent_invisible_sep:
  {
    \__spintent_luafun_invisible_sep:
  }
```

The token list constant `\c__spintent_invisible_sep_tl` is an internal copy of the Unicode invisible separator `U-2063` using the primitive `\Umathchardef` from LuaTeX which leaves `<mi>...</mi>` in the structure `MathML` embedded in the output `PDF`.

```
\hook_gput_code:nnn { begindocument } { spintent }
  {
    \Umathchardef \c__spintent_invisible_sep_tl = "0 "0 "2063
  }
```

Both copies are used for different purposes, the function `\__spintent_invisible_sep:` will be used to force the start of a `<mrow>` in the structure `MathML`, the token list constant `\c__spintent_invisible_sep_tl` will be used to inject text into the structure `MathML` when we use `\MathMLintent{⟨text⟩}`.

(*End of definition for* `\__spintent_invisible_sep:` *and* `\c__spintent_invisible_sep_tl`.)

11.9 Normalize argument and affixes

We normalize the {⟨*argument*⟩} to support affixes (prefixes/suffixes) by trimming leading and trailing spaces, converting internal spaces to hyphens, collapsing consecutive hyphens, and removing leading or trailing hyphens.

__spintent_collapse_hyphens:N

A recursive helper to collapse multiple hyphens into a single one.

```
\cs_new_protected:Nn \__spintent_collapse_hyphens:N
  {
    \tl_if_in:NnT #1 { -- }
      {
        \tl_replace_all:Nnn #1 { -- } { - }
        \__spintent_collapse_hyphens:N #1
      }
  }
```

(*End of definition for* `\__spintent_collapse_hyphens:N`.)

__spintent_sanitize_affix:Nn
__spintent_sanitize_affix:cn

The function `\__spintent_sanitize_affix:Nn` checks if the {⟨*argument*⟩} passed in '`#2`' is *blank*. If so, it clears the target token list variable passed in '`#1`'. Otherwise, trimming bounds, converting spaces '␣' to hyphens '-', collapsing multiple hyphens, and safely stripping leading/trailing hyphens. For example, if you pass a target variable and raw text:

```
\__spintent_sanitize_affix:Nn \l__spintent_spsomesym_prefix_tl{ foo bar baz }
```

The function will *"store"* the normalized string `foo-bar-baz` inside `\l__spintent_spsomesym_prefix_tl`, ready to be conditionally appended.

```
\cs_new_protected:Nn \__spintent_sanitize_affix:Nn
  {
    \tl_if_blank:nTF {#2}
      { \tl_clear:N #1 }
      {
        \tl_set:Ne #1 {#2}
        \tl_trim_spaces:N #1
        \tl_replace_all:Nnn #1 { ~ } { - }
        \__spintent_collapse_hyphens:N #1
        \tl_put_left:Nn  #1 { \q_mark }
        \tl_put_right:Nn #1 { \q_stop }
        \tl_replace_all:Nnn #1 { \q_mark - } { \q_mark }
        \tl_replace_all:Nnn #1 { - \q_stop } { \q_stop }
        \tl_remove_all:Nn #1 { \q_mark }
        \tl_remove_all:Nn #1 { \q_stop }
      }
  }
\cs_generate_variant:Nn \__spintent_sanitize_affix:Nn { cn }
```

(*End of definition for* `\__spintent_sanitize_affix:Nn`.)

11.10 The command \setspintent

The `\setspintent` command is used to configure ⟨*keys*⟩ *"globally"* for the utilities provided by spintent.

11.10.1 Internal variables and error messages for \setspintent

\l__spintent_setspintent_input_arg_cmd_tl
unknown-command-setup

The variable `\l__spintent_setspintent_input_arg_cmd_tl` will *store* the command or command name passed to the *"first"* {⟨*argument*⟩} '`#1`' of the command `\setspintent`.

```
\tl_new:N \l__spintent_setspintent_input_arg_cmd_tl
```

Error message if the {⟨*argument*⟩} '`#1`' passed to `\setspintent` is a *"non-existent"* command.

```
\msg_new:nnnn { spintent } { unknown-command-setup }
  {
    Cannot~configure~unknown~command~'\exp_not:c{#1}'.
  }
  {
    You~attempted~to~use~\token_to_str:N \setspintent{#1}{...},~but~the~
    command~\exp_not:c{#1}~is~not~defined.~Please~check~for~typos.
  }
```

(*End of definition for* `\l__spintent_setspintent_input_arg_cmd_tl` *and* `unknown-command-setup`.)

__spintent_setspintent_set_keys:nn
__spintent_setspintent_set_keys:en

The function __spintent_setspintent_set_keys:nn will check if the command passed as {⟨*argument*⟩} exists in '#1' and set the ⟨*keys*⟩ passed as {⟨*argument*⟩} in '#2' to the correct path; otherwise, it will return an error.

```
\cs_new_protected:Nn \__spintent_setspintent_set_keys:nn
  {
    \cs_if_exist:cTF { #1 }
      {
        \keys_set:nn { spintent / #1 } { #2 }
      }
      {
        \msg_error:nnn { spintent } { unknown-command-setup } { #1 }
      }
  }
\cs_generate_variant:Nn \__spintent_setspintent_set_keys:nn { en }
```

(*End of definition for* __spintent_setspintent_set_keys:nn.)

__spintent_setspintent_parse_args:nn
__spintent_setspintent_parse_args:Vn

The function __spintent_setspintent_parse_args:nn will check if the argument passed in #1 begins with '\' or is just a *command name*. If it begins with '\', it will remove it using \cs_to_str:N and then execute __spintent_setspintent_set_keys:en. Otherwise, i.e., if '\' was not passed, it will directly execute __spintent_setspintent_set_keys:nn.

```
\cs_new_protected:Nn \__spintent_setspintent_parse_args:nn
  {
    \tl_if_single:nTF { #1 }
      {
        \token_if_cs:NTF #1
          {
            \__spintent_setspintent_set_keys:en { \cs_to_str:N #1 } { #2 }
          }
          {
            \__spintent_setspintent_set_keys:nn { #1 } { #2 }
          }
      }
      {
        \__spintent_setspintent_set_keys:nn { #1 } { #2 }
      }
  }
\cs_generate_variant:Nn \__spintent_setspintent_parse_args:nn { Vn }
```

(*End of definition for* __spintent_setspintent_parse_args:nn.)

\setspintent

Now we define the user command \setspintent.

```
\NewDocumentCommand \setspintent { m m }
  {
    \tl_if_blank:nTF { #1 }
      {
        \msg_error:nnn { spintent } { unknown-command-setup } { <empty> }
      }
      {
        \tl_set:Nn \l__spintent_setspintent_input_arg_cmd_tl { #1 }
        \tl_trim_spaces:N \l__spintent_setspintent_input_arg_cmd_tl
        \__spintent_setspintent_parse_args:Vn \l__spintent_setspintent_input_arg_cmd_tl { #2 }
      }
  }
```

(*End of definition for* \setspintent. *This function is documented on page* 3.)

11.11 Handling unknown keys

\l__spintent_command_name_str
unknown-choice
cmd-key-unknown
cmd-key-value-unknown

We define the variable \l__spintent_command_name_str which will *store* the *"name"* of each command within the implementation along with the messages unknown-choice used by ⟨*keys*⟩ that use .choice:n in their implementation and the messages cmd-key-unknown, cmd-key-value-unknown used by the ⟨*key*⟩ unknown in commands that use [⟨*key* = *val*⟩].

```
\str_new:N  \l__spintent_command_name_str
\msg_new:nnn { spintent } { unknown-choice }
  {
    The~value~'#3'~for~'#1'~key~is~invalid~use~('#2').
  }
\msg_new:nnnn { spintent } { cmd-key-unknown }
  {
    The~key~'#1'~is~unknown~by~command~
```

```
    \c_backslash_str \l__spintent_command_name_str \c_space_tl
    and~is~being~ignored~\msg_line_context:.
  }
  {
    The~command~\c_backslash_str \l__spintent_command_name_str \c_space_tl
    does~not~have~a~key~called~'#1'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
\msg_new:nnnn { spintent } { cmd-key-value-unknown }
  {
    The~key~'#1=#2'~is~unknown~by~command~
    \c_backslash_str \l__spintent_command_name_str \c_space_tl
    and~is~being~ignored~\msg_line_context:.
  }
  {
    The~command~\c_backslash_str \l__spintent_command_name_str \c_space_tl
    does~not~have~a~key~called~'#1'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
```

(*End of definition for* `\l__spintent_command_name_str` *and others.*)

`\__spintent_unknown_keys:n`
`\__spintent_unknown_keys:nn`

The internal functions `\__spintent_unknown_keys:n` and `\__spintent_unknown_keys:nn` used by the ⟨*key*⟩ unknown.

```
\cs_new_protected:Nn \__spintent_unknown_keys_cmd:n
  {
    \exp_args:NV \__spintent_unknown_keys_cmd:nn \l_keys_key_str {#1}
  }
\cs_new_protected:Nn \__spintent_unknown_keys_cmd:nn
  {
    \tl_if_blank:nTF {#2}
      { \msg_error:nnn { spintent } { cmd-key-unknown } {#1} }
      { \msg_error:nnnn { spintent } { cmd-key-value-unknown } {#1} {#2} }
  }
```

(*End of definition for* `\__spintent_unknown_keys:n` *and* `\__spintent_unknown_keys:nn`*.*)

11.12 Definition of core base variables

`\__spintent_luafun_clean_split_and_set:n`
`\__spintent_luafun_clean_split_and_set:V`

The function `\__spintent_luafun_clean_split_and_set:n` defined in the Lua module `spintent.lua`, is common to several commands in this implementation. It analyzes, processes, cleans and splits the {⟨*argument*⟩} passed to commands using internal functions defined in the Lua module `spintent.lua`. It isolates signs, integer components, decimal separators, decimal parte, decimal periodic part, exponents and final physical *"units"*, assigning them directly to `expl3` variables.

This function assigns the states of the variables '`_str`' or '`_tl`' described below to `true`, `false`, `error` or *stores* information according to the role it fulfills within the implementation.

(*End of definition for* `\__spintent_luafun_clean_split_and_set:n`*.*)

`\l__spintent_luaset_sign_tl`
`\l__spintent_luaset_part_int_tl`
`\l__spintent_luaset_dec_sep_tl`
`\l__spintent_luaset_part_dec_tl`
`\l__spintent_luaset_part_period_tl`
`\l__spintent_luaset_has_integer_str`
`\l__spintent_luaset_has_natural_str`

The variable `\l__spintent_luaset_sign_tl` *stores* the explicit '`+`' or '`-`' sign if present. The variable `\l__spintent_luaset_part_int_tl` *stores* the raw digit sequence of the *"integer part"*. The variable `\l__spintent_luaset_dec_sep_tl` *stores* the matched '`,`' or '`.`' used as the decimal separator. The variable `\l__spintent_luaset_part_dec_tl` *stores* the raw digits of the *"finite decimal part"* that follow the separator, and the variable `\l__spintent_luaset_part_period_tl` *stores* the digits of the *"infinite periodic decimal part"* that follow a semicolon '`;`'.

The variable `\l__spintent_luaset_has_integer_str` *store* the state `true` when it is *only* an *"integer"*. The variable `\l__spintent_luaset_has_natural_str` *store* the state `true` when it is *only* an *"natural number"*.

```
\tl_new:N  \l__spintent_luaset_sign_tl
\tl_new:N  \l__spintent_luaset_part_int_tl
\tl_new:N  \l__spintent_luaset_dec_sep_tl
\tl_new:N  \l__spintent_luaset_part_dec_tl
\tl_new:N  \l__spintent_luaset_part_period_tl
\str_new:N \l__spintent_luaset_has_integer_str
\str_new:N \l__spintent_luaset_has_natural_str
```

(*End of definition for* `\l__spintent_luaset_sign_tl` *and others.*)

\l__spintent_luaset_only_part_int_str
\l__spintent_luaset_only_part_dec_str
\l__spintent_luaset_only_part_period_str
\l__spintent_luaset_format_part_int_str
\l__spintent_luaset_format_part_dec_str

The literal entries of the numbers that we will pass to the *intent function* :number of \MathMLintent are *stored* in \l__spintent_luaset_only_part_int_str for the *"integer part"*, in \l__spintent_luaset_only_part_dec_ for the *"finite decimal part"*, and in \l__spintent_luaset_only_part_period_str for the *"infinite periodic decimal part"*.

Formatted versions (cifras-sep=true) are *stored* in \l__spintent_luaset_format_part_int_str for the *"integer part"* and \l__spintent_luaset_format_part_dec_str for the *"finite decimal part"*.

```
\str_new:N \l__spintent_luaset_only_part_int_str
\str_new:N \l__spintent_luaset_only_part_dec_str
\str_new:N \l__spintent_luaset_only_part_period_str
\str_new:N \l__spintent_luaset_format_part_int_str
\str_new:N \l__spintent_luaset_format_part_dec_str
```

(*End of definition for* \l__spintent_luaset_only_part_int_str *and others.*)

\l__spintent_luaset_millons_str
\l__spintent_luaset_decimal_period_str
\l__spintent_luaset_not_decimal_period_str
\l__spintent_luaset_not_decimal_not_period_str
\l__spintent_luaset_dec_is_all_zeros_str
\l__spintent_luaset_has_exponent_str
\l__spintent_luaset_exponent_tl

The flag variable \l__spintent_luaset_millons_str indicates whether the integer part translates exactly to a multiple of one million when passed to \MathMLintent. To properly structure the reading and writing of numbers in MathML, we use the flag variables \l__spintent_luaset_decimal_period_str, \l__spintent_luaset_not_decimal_period_str, and \l__spintent_luaset_not_decimal_not_period_st The flag variable \l__spintent_luaset_dec_is_all_zeros_str monitors whether the *"finite decimal part"* is completely zero (useful for *"pure periodic decimals"*).

The flag variable \l__spintent_luaset_has_exponent_str indicates whether active *"scientific notation"* was detected at the end of the entered number, passed with 'e' or 'E'. If true, it will store the digits of the exponent in \l__spintent_luaset_exponent_tl.

```
\str_new:N \l__spintent_luaset_millons_str
\str_new:N \l__spintent_luaset_decimal_period_str
\str_new:N \l__spintent_luaset_not_decimal_period_str
\str_new:N \l__spintent_luaset_not_decimal_not_period_str
\str_new:N \l__spintent_luaset_dec_is_all_zeros_str
\str_new:N \l__spintent_luaset_has_exponent_str
\tl_new:N  \l__spintent_luaset_exponent_tl
```

(*End of definition for* \l__spintent_luaset_millons_str *and others.*)

\l__spintent_luaset_has_units_str
\l__spintent_luaset_status_str
\l__spintent_luaset_arg_above_tl
\l__spintent_luaset_arg_below_tl
\l__spintent_luaset_denom_has_number_str

Since the function __spintent_luafun_clean_split_and_set:n is common to \spnum (§11.14) and \spunit (§11.15) (as well as others), it determines the following unit-related variables after processing the {⟨*arguments*⟩} via the Lua module spintent.lua.

First, it checks if there is any residual text *after* the processed number that is a candidate for physical *"units"*. It sets the flag variable \l__spintent_luaset_has_units_str accordingly. Following this, it verifies if this text contains more than one "/". It then sets the variable \l__spintent_luaset_status_str to valid or multislash. If the structure is valid, it splits the unit into \l__spintent_luaset_arg_above_tl and \l__spintent_luaset_arg_below_tl. Finally, it analyzes \l__spintent_luaset_arg_below_tl; if the denominator starts with a number, it sets \l__spintent_luaset_denom_has_number_str to true, otherwise to false.

```
\str_new:N \l__spintent_luaset_has_units_str
\str_new:N \l__spintent_luaset_status_str
\tl_new:N  \l__spintent_luaset_arg_above_tl
\tl_new:N  \l__spintent_luaset_arg_below_tl
\str_new:N \l__spintent_luaset_denom_has_number_str
```

(*End of definition for* \l__spintent_luaset_has_units_str *and others.*)

11.13 Common messages for text mode and math mode

Messages shared by the different commands for text mode or math mode.

```
\msg_new:nnnn { spintent } { need-math-mode }
  {
    Math~mode~required~for~'#1'~\msg_line_context:.
  }
  {
    Use~inside~$...$~or~in~math~environment.
  }
\msg_new:nnnn { spintent } { need-text-mode }
  {
    Text~mode~required~for~'#1'~\msg_line_context:.
  }
  {
    Use~outside~of~$...$~or~math~environment.
  }
```

11.14 The command `\spnum`

The user command `\spnum` is *first* of *"the big four"* provided by the spintent package and is fundamental to the implementation of other commands. It handles the processing of numerical input, scientific notation, and units of measurement, using the Lua module `spintent.lua`, which is responsible for separating and assigning variables defined in `expl3`.

From the `expl3` side, we handle the printing logic and the `intent` *functions* passed to `\MathMLintent` to achieve a valid `MathML` structure that is accepted by `MathCAT`.

11.14.1 Definition of keys and variables for `\spnum`

`\l__spintent_spnum_read_terse_bool`
`\l__spintent_spnum_read_period_sep_str`
`\l__spintent_spnum_notation_sym_tl`
`\l__spintent_spnum_intent_read_sign_str`
`l__spintent_spnum_intent_read_dec_sep_str`

The variable `\l__spintent_spnum_read_terse_bool` handled by the key `read-sign`, controls the `intent` passed to `\MathMLintent` to spoken the '+' or '−'. If set to `clear`, it will say *"positivo"* or *"negativo" after* reading the number; if set to `terse`, it will say *"más"* or *"menos" before* reading the number.

The variable `\l__spintent_spnum_read_period_sep_str` handled by the key `read-period-sep`, controls how the decimal separator ';' (periodic part) is printed and spoken when the {⟨*argument*⟩} is a *pure decimal periodic* and is passed the `intent` to `\MathMLintent`. If set to `coma`, it will say *"coma"* and print ',', if set to `punto` it will say *"punto"* and print '.'.

The variable `\l__spintent_spnum__notation_sym_tl`, handled by the key `notation-sym`, determines whether '×' will be printed when set to `times` or '·' when set to `cdot`.

The variable `\l__spintent_spnum_intent_read_sign_str` will *store* the spoken translation used in the `intent` function passed to `\MathMLintent` for the '+' or '−' according to the value passed to the key `read-sign`.

The variable `\l__spintent_spnum_intent_read_dec_sep_str` *stores* the spoken translation that will be used in the `intent` function passed to `\MathMLintent` for reading the decimal separation as a *"coma"* or a *"punto"* according to the value found in the variable `\l__spintent_luaset_dec_sep_tl`.

```
\bool_new:N \l__spintent_spnum_read_terse_bool
\str_new:N  \l__spintent_spnum_read_period_sep_str
\tl_new:N   \l__spintent_spnum_notation_sym_tl
\str_new:N  \l__spintent_spnum_intent_read_sign_str
\str_new:N  \l__spintent_spnum_intent_read_dec_sep_str
```

(*End of definition for* `\l__spintent_spnum_read_terse_bool` *and others.*)

`cifras-sep`
`read-sign`
`read-period-sep`
`notation-sym`

Now we add the keys `cifras-sep`, `read-sign`, `read-period-sep` and `notation-sym`.

```
\keys_define:nn { spintent/spnum }
  {
    cifras-sep                .bool_set:N = \l__spintent_spnum_cifras_sep_bool,
    cifras-sep                .initial:n = true,
    cifras-sep                .value_required:n = true,
    read-sign                 .choices:nn =
      { terse , clear }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \bool_set_true:N  \l__spintent_spnum_read_terse_bool }
            {2} { \bool_set_false:N \l__spintent_spnum_read_terse_bool }
          }
      },
    read-sign                 .initial:n  = terse,
    read-sign                 .value_required:n = true,
    read-sign / unknown       .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-sign } { terse , ~clear } { \exp_not:n {#1} },
    read-period-sep           .choices:nn =
      { coma , punto }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spnum_read_period_sep_str { coma } }
            {2} { \str_set:Nn \l__spintent_spnum_read_period_sep_str { punto } }
          }
      },
    read-period-sep           .initial:n  = coma,
    read-period-sep           .value_required:n = true,
    read-period-sep / unknown .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-period-sep } { coma, ~punto } { \exp_not:n {#1} },
    notation-sym              .choices:nn =
      { times , cdot }
```

```
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \tl_set:Nn \l__spintent_spnum_notation_sym_tl { \times } }
            {2} { \tl_set:Nn \l__spintent_spnum_notation_sym_tl { \cdot } }
          }
      },
    notation-sym              .initial:n  = cdot,
    notation-sym              .value_required:n = true,
    notation-sym / unknown    .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { notation-sym } { times , ~ cdot } { \exp_not:n {#1} },
    unknown                   .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spnum }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `cifras-sep` *and others.*)

11.14.2 Internal functions for \spnum

`\__spintent_spnum_render_part:n`
`\__spintent_spnum_render_int:`
`\__spintent_spnum_render_dec:`

The function `\__spintent_spnum_render_part:n` processes the {⟨*argument*⟩} passed in '`#1`' used by the functions `\__spintent_spnum_render_int:` and `\__spintent_spnum_render_dec:` to define the *intent function* `:number` for the integer and decimal part passed to `\MathMLintent`. It evaluates whether the key `cifras-sep` is active or not.

```
\cs_new_protected:Nn \__spintent_spnum_render_part:n
  {
    \MathMLintent { \str_use:c { l__spintent_luaset_only_part_#1_str } :number }
      {
        \bool_if:NTF \l__spintent_spnum_cifras_sep_bool
          { \str_use:c { l__spintent_luaset_format_part_#1_str } }
          { \tl_use:c { l__spintent_luaset_part_#1_tl } }
      }
  }
\cs_new_protected:Nn \__spintent_spnum_render_int:
  {
    \__spintent_spnum_render_part:n { int }
  }
\cs_new_protected:Nn \__spintent_spnum_render_dec:
  {
    \__spintent_spnum_render_part:n { dec }
  }
```

(*End of definition for* `\__spintent_spnum_render_part:n`, `\__spintent_spnum_render_int:`, *and* `\__spintent_spnum_render_dec:`.)

`\__spintent_spnum_render_only_period:`
`\__spintent_spnum_print_period_bar:`
`\__spintent_spnum_render_period_sep:`

The function `\__spintent_spnum_render_only_period:` isolates the *"infinite periodic decimal part"* from the *"finite decimal part"*, assigning the *intent function* `:number` passed to `\MathMLintent` which will be used by `\MathMLarg` in the function `\__spintent_spnum_print_period_bar:`.

```
\cs_new_protected:Nn \__spintent_spnum_render_only_period:
  {
    \MathMLintent { \l__spintent_luaset_only_part_period_str :number }
      {
        \l__spintent_luaset_part_period_tl
      }
  }
```

The function `\__spintent_spnum_print_period_bar:` inserts the function `\__spintent_spnum_render_only_p` as an argument within the *intent function* `_periódico:postfix($dp)` passed to `\MathMLintent`, generating the correct spoken for the *"infinite periodic decimal part"*.

```
\cs_new_protected:Nn \__spintent_spnum_print_period_bar:
  {
    \MathMLintent { _periódico:postfix($dp) }
      {
        {
          \overline
            {
              \MathMLarg { dp } { \__spintent_spnum_render_only_period: }
            }
        }
      }
```

```
  }
```

The function `\__spintent_spnum_render_period_sep:` sets how the decimal separator handled by the key `read-period-sep` will be printed and spoken when a *"pure periodic decimal"* is passed.

```
\cs_new_protected:Nn \__spintent_spnum_render_period_sep:
  {
    \str_if_eq:VnTF \l__spintent_spnum_read_period_sep_str { coma }
      {
        \MathMLintent { _coma:prefix($pp) }
          {
            \mathord{,}
            \MathMLarg { pp } { \__spintent_spnum_print_period_bar: }
          }
       }
       {
         \MathMLintent { _punto:prefix($pp) }
           {
             \mathord{.}
             \MathMLarg { pp } { \__spintent_spnum_print_period_bar: }
           }
       }
  }
```

(*End of definition for* `\__spintent_spnum_render_only_period:`, `\__spintent_spnum_print_period_bar:`, *and* `\__spintent_spnum_render_period_sep:`.)

`\__spintent_spnum_render_decimal_sep:`

The function `\__spintent_spnum_render_decimal_sep:` parses the value of the variable `\l__spintent_luaset_` and sets the variable `\l__spintent_spnum_intent_read_dec_sep_str` with the function `_coma:prefix($dp)` for reading a *"coma"* or `_punto:prefix($dp)` for reading a *"punto"* to `\MathMLintent`. We print the decimal separator stored in `\l__spintent_luaset_dec_sep_tl` using `\mathord` to remove spaces around it and then use the command `\MathMLarg` where we insert `\__spintent_invisible_sep:` for compatibility with `MathML` and finally call the function `\__spintent_spnum_render_dec:` to print the finite decimal part.

```
\cs_new_protected:Nn \__spintent_spnum_render_decimal_sep:
  {
    \str_if_eq:VnTF \l__spintent_luaset_dec_sep_tl { , }
      { \str_set:Nn \l__spintent_spnum_intent_read_dec_sep_str { _coma } }
      { \str_set:Nn \l__spintent_spnum_intent_read_dec_sep_str { _punto } }
    \MathMLintent { \l__spintent_spnum_intent_read_dec_sep_str :prefix($dp) }
      {
        \mathord { \l__spintent_luaset_dec_sep_tl }
        \MathMLarg { dp }
          {
            \__spintent_invisible_sep:
            \__spintent_spnum_render_dec:
          }
      }
    % period part
    \str_if_eq:VnT \l__spintent_luaset_decimal_period_str { true }
      {
        \str_if_eq:VnTF \l__spintent_luaset_dec_is_all_zeros_str { true }
          {
            \__spintent_invisible_sep:
            \__spintent_spnum_print_period_bar:
          }
          {
            \MathMLintent { _con:prefix($pnum) }
              {
                \__spintent_invisible_sep:
                \MathMLarg { pnum }
                  {
                    \__spintent_spnum_print_period_bar:
                  }
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_spnum_render_decimal_sep:`.)

`\__spintent_spnum_print_part_dec:`
`\__spintent_spnum_print_part_period:`

The function `\__spintent_spnum_print_part_dec:` checks if a decimal part exists. If it does, it determines whether the decimal separator is a comma or a period to set the correct speech intent (*"coma"* or *"punto"*).

It then prints the separator, renders the decimal digits, and if a periodic sequence follows, it decides how to attach the periodic bar.

The function `\__spintent_spnum_print_part_period:` handles cases where a periodic part exists. It evaluates if it should render the periodic separator directly, particularly when the number has a periodic part but lacks a standard decimal part.

```
\cs_new_protected:Nn \__spintent_spnum_print_part_dec:
  {
    \tl_if_empty:NF \l__spintent_luaset_part_dec_tl
      {
        \__spintent_spnum_render_decimal_sep:
      }
  }
\cs_new_protected:Nn \__spintent_spnum_print_part_period:
  {
    \tl_if_empty:NF \l__spintent_luaset_part_period_tl
      {
        \str_if_eq:VnTF \l__spintent_luaset_not_decimal_period_str { true }
          { \__spintent_spnum_render_period_sep: }
          {
            \tl_if_empty:NT \l__spintent_luaset_part_dec_tl
              {
                \__spintent_spnum_render_period_sep:
              }
          }
      }
  }
```

(End of definition for `\__spintent_spnum_print_part_dec:` *and* `\__spintent_spnum_print_part_period:`*.)*

`\__spintent_spnum_print_integer:`
`\__spintent_spnum_print_num_start:`
`\__spintent_spnum_print_number_core:`

The function `\__spintent_spnum_print_integer:` checks if the number consists only of an integer. If so, it simply renders it. If a decimal or periodic part exists, it sets up a `MathML` intent to link the integer part (respecting digit separators if enabled) to the subsequent decimal or periodic functions.

The function `\__spintent_spnum_print_num_start:` checks if an explicit sign is present. If it finds one, it evaluates the terse reading flag to decide whether the sign should be read as a prefix (*"más"* or *"menos"*) or as a postfix (*"positivo"* or *"negativo"*), and wraps the intent around the rest of the number.

The function `\__spintent_spnum_print_number_core:` acts as the main wrapper. It triggers the sign and base number evaluation, and then checks if a scientific exponent is present. If it is, it appends the correct base-10 notation symbol and exponent to complete the sequence.

Note for `\__spintent_invisible_sep:`.

```
\cs_new_protected:Nn \__spintent_spnum_print_integer:
  {
    \str_if_eq:VnTF \l__spintent_luaset_not_decimal_not_period_str { true }
      { \__spintent_spnum_render_int: }
      {
        \MathMLintent { \l__spintent_luaset_only_part_int_str :number:prefix($next) }
          {
            \bool_if:NTF \l__spintent_spnum_cifras_sep_bool
              { \l__spintent_luaset_format_part_int_str }
              { \l__spintent_luaset_part_int_tl }
            \MathMLarg { next }
              {
                \tl_if_empty:NTF \l__spintent_luaset_part_dec_tl
                  { \__spintent_spnum_print_part_period: }
                  { \__spintent_spnum_print_part_dec: }
              }
          }
      }
  }
\cs_new_protected:Nn \__spintent_spnum_print_num_start:
  {
    \tl_if_empty:NTF \l__spintent_luaset_sign_tl
      {
        \__spintent_invisible_sep: % need
        \__spintent_spnum_print_integer:
      }
      {
        \bool_if:NTF \l__spintent_spnum_read_terse_bool
          {
            \str_if_eq:VnTF \l__spintent_luaset_sign_tl { + }
```

```
                { \str_set:Nn \l__spintent_spnum_intent_read_sign_str { _más:prefix($n)} }
                { \str_set:Nn \l__spintent_spnum_intent_read_sign_str { _menos:prefix($n)} }
            }
            {
              \str_if_eq:VnTF \l__spintent_luaset_sign_tl { + }
                { \str_set:Nn \l__spintent_spnum_intent_read_sign_str { _positivo:postfix($n) } }
                { \str_set:Nn \l__spintent_spnum_intent_read_sign_str { _negativo:postfix($n) } }
            }
          \__spintent_invisible_sep: % need
          \MathMLintent { \l__spintent_spnum_intent_read_sign_str }
            {
              \l__spintent_luaset_sign_tl
              \MathMLarg { n } { \__spintent_spnum_print_integer: }
            }
        }
    }
\cs_new_protected:Nn \__spintent_spnum_print_number_core:
  {
    \__spintent_spnum_print_num_start:
    \str_if_eq:VnT \l__spintent_luaset_has_exponent_str { true }
      {
        \MathMLintent { _por } { \l__spintent_spnum_notation_sym_tl }
        10^{ \l__spintent_luaset_exponent_tl }
      }
  }
```

(*End of definition for* `\__spintent_spnum_print_integer:`*,* `\__spintent_spnum_print_num_start:`*, and* `\__spintent_spnum_print_number_core:`*.*)

11.14.3 Definition of error messages

spnum-has-units
spnum-no-digits

We define two error messages for the `\spnum` command. The first "error" triggers if the user includes *"units"* in the {⟨*argument*⟩} passed in '#1', advising them to use the `\spunit` command instead. The second "error" is raised if the provided {⟨*argument*⟩} contains no numerical digits.

```
\msg_new:nnnn { spintent } { spnum-has-units }
  {
    The~\token_to_str:N \spnum \c_space_tl command~does~not~accept~units~('#1').
  }
  {
    Use~\token_to_str:N \spunit \c_space_tl if~you~need~to~print~numbers~with~units.
  }
\msg_new:nnnn { spintent } { spnum-no-digits }
  {
    The~\token_to_str:N \spnum \c_space_tl command~requires~a~valid~numerical~value.
  }
  {
    You~provided~'#1',~which~contains~no~digits.
  }
```

(*End of definition for* `spnum-has-units` *and* `spnum-no-digits`*.*)

11.14.4 Implementation of the \spum command

\spnum

The public user command `\spnum` accepts an optional [⟨*key* = *val*⟩] argument in '#1' and a mandatory {⟨*rational number*⟩} in '#2'. It first checks if it is executed inside math mode, raising an error if it is not. If correct, it opens a local group to apply the options and passes the input to Lua for parsing via `\__spintent_luafun_clean_split_and_set:n`.

Before printing, it validates the extracted data. It throws an error if it finds trailing units in the input or if the integer part is completely empty (no valid digits). If all checks pass, it calls `\__spintent_spnum_print_number_core:` to render the final number, and finally closes the group.

```
\NewDocumentCommand \spnum { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \keys_set:nn { spintent/spnum } { #1 }
          \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #2 } }
          \str_if_eq:VnTF \l__spintent_luaset_has_units_str { true }
            { \msg_error:nnn { spintent } { spnum-has-units } { #2 } }
            {
              \tl_if_empty:NTF \l__spintent_luaset_part_int_tl
                { \msg_error:nnn { spintent } { spnum-no-digits } { #2 } }
                { \__spintent_spnum_print_number_core: }
```

```
        }
      \group_end:
    }
    { \msg_error:nnn { spintent } { need-math-mode } { \spnum } }
}
```

(*End of definition for* \spnum. *This function is documented on page* 3.)

11.15 The command \spunit

The user command \spunit is the *second* of *"the big four"* provided by the spintent package and is used to parse and print physical units and angular systems. It sends the {⟨*argument*⟩} to Lua module spintent.lua via __spintent_luafun_clean_split_and_set:n to separate and clean the {⟨*argument*⟩}.

This ensures that all *"units"* (such as SÍ fractions, products, or sexagesimal angles) are processed consistently by the spintent.lua.

11.15.1 Definition of variables

\l__spintent_spunit_luaset_canonical_str
_spintent_spunit_luaset_lookup_status_str
\l__spintent_spunit_luaset_read_str
\l__spintent_spunit_luaset_is_compact_str
l__spintent_spunit_luaset_compact_exp_str
spintent_spunit_luaset_is_sexagesimal_str
\l__spintent_spunit_luaset_status_str

These string variables store the data returned by the Lua module spintent.lua. They hold the canonical *"unit symbol"*, the dictionary lookup status, the correct speech text for NVDA/MathCAT, and specific flags that indicate if the *"unit"* is compact or represents a sexagesimal angle.

```
\str_new:N  \l__spintent_spunit_luaset_canonical_str
\str_new:N  \l__spintent_spunit_luaset_lookup_status_str
\str_new:N  \l__spintent_spunit_luaset_read_str
\str_new:N  \l__spintent_spunit_luaset_is_compact_str
\str_new:N  \l__spintent_spunit_luaset_compact_exp_str
\str_new:N  \l__spintent_spunit_luaset_is_sexagesimal_str
\str_new:N  \l__spintent_spunit_luaset_status_str
```

(*End of definition for* \l__spintent_spunit_luaset_canonical_str *and others.*)

\l__spintent_spunit_exponent_tl
\l__spintent_spunit_unit_name_tl
\l__spintent_spunit_read_slash_str
\l__spintent_spunit_exp_seq
\l__spintent_spunit_mult_seq
\l__spintent_spunit_is_mult_bool
\l__spintent_spunit_read_cdot_bool

These variables are used to process and format the *"unit"*. The sequences \l__spintent_spunit_mult_seq and \l__spintent_spunit_exp_seq split compound *"units"* at multiplication '*' or exponent '^' boundaries. The boolean variables control specific rendering options, such as whether the multiplication dot '\cdot' is explicitly read aloud or silently skipped.

```
\tl_new:N   \l__spintent_spunit_exponent_tl
\tl_new:N   \l__spintent_spunit_unit_name_tl
\str_new:N  \l__spintent_spunit_read_slash_str
\seq_new:N  \l__spintent_spunit_exp_seq
\seq_new:N  \l__spintent_spunit_mult_seq
\bool_new:N \l__spintent_spunit_is_mult_bool
\bool_new:N \l__spintent_spunit_read_cdot_bool
```

(*End of definition for* \l__spintent_spunit_exponent_tl *and others.*)

11.15.2 Definition of error messages

spunit-invalid-unit
spunit-no-units
inline-numeric-denominator

We define three error messages for the \spunit command. The first "error" triggers if the *"unit expression"* provided in '#1' is unknown or malformed. The second "error" occurs if no *"units"* are found in the {⟨*argument*⟩}, advising the user to use \spnum for pure numbers. The third "error" is raised if a numeric coefficient is illegally placed in the denominator of an inline fraction (after a slash).

```
\msg_new:nnnn { spintent } { spunit-invalid-unit }
  {
    The~expression~'#1'~contains~an~unknown~or~malformed~unit~structure.
  }
  {
    Check~spelling~or~verify~the~unit~is~defined~in~the~dictionary.
  }
\msg_new:nnnn { spintent } { spunit-no-units }
  {
    The~\token_to_str:N \spunit \c_space_tl command~requires~at~least~one~unit~('#1').
  }
  {
    Use~\token_to_str:N \spnum \c_space_tl if~you~only~need~to~format~a~pure~number.
  }
\msg_new:nnnn { spintent } { inline-numeric-denominator }
  {
    Illegal~numeric~coefficient~in~inline~denominator~'#1'.
  }
  {
    The~inline~mode~of~\token_to_str:N \spunit \c_space_tl
    does~not~allow~numbers~after~the~slash~(/).
  }
```

(*End of definition for* spunit-invalid-unit, spunit-no-units, *and* inline-numeric-denominator.)

11.15.3 Definition of keys for \spunit

read-cdot
read-slash
cifras-sep
read-period-sep
notation-sym

Now we add the keys print-cdot, read-cdot, read-slash, cifras-sep, read-period-sep and notation-sym.

```
\keys_define:nn { spintent/spunit }
  {
    print-cdot           .bool_set:N = \l__spintent_spunit_print_cdot_bool,
    print-cdot           .initial:n  = false,
    print-cdot           .value_required:n = true,
    read-cdot            .bool_set:N = \l__spintent_spunit_read_cdot_bool,
    read-cdot            .initial:n  = false,
    read-cdot            .value_required:n = true,
    read-slash           .choices:nn =
      { por , partido-por }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spunit_read_slash_str { por } }
            {2} { \str_set:Nn \l__spintent_spunit_read_slash_str { partido-por } }
          }
      },
    read-slash           .initial:n  = por,
    read-slash           .value_required:n = true,
    read-slash / unknown .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-slash } { por, ~partido-por } { \exp_not:n {#1} },
    cifras-sep           .meta:nn = { spintent/spnum } { cifras-sep = {#1} },
    cifras-sep           .value_required:n = true,
    read-period-sep      .meta:nn = { spintent/spnum } { read-period-sep = {#1} },
    read-period-sep      .value_required:n = true,
    notation-sym         .meta:nn = { spintent/spnum } { notation-sym = {#1} },
    notation-sym         .value_required:n = true,
    unknown              .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spunit }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* read-cdot *and others.*)

11.15.4 Internal functions for \spunit

__spintent_spunit_format_canonical:
__spintent_spunit_process_base_exp:n

The function __spintent_spunit_format_canonical: handles the visual formatting of the *"unit symbol"*. It specifically checks for the *liter symbol* 'ℓ' to print it as \ell, and formats all other valid *"units"* in standard upright text using \mathrm.

The function __spintent_spunit_process_base_exp:n evaluates a *"single unit"* along with its exponent. It first looks up the unit '#1' in the dictionary. If the *"unit"* is not found, it flags it as invalid. If found, it handles the spacing and multiplication dot '\cdot' intents, and then renders the base unit and its exponent (either using a pre-defined compact exponent from Lua or by manually splitting the string at the '^' symbol).

```
\cs_new_protected:Nn \__spintent_spunit_format_canonical:
  {
    \str_case:VnF \l__spintent_spunit_luaset_canonical_str
      {
        { ℓ } { \ell }
      }
      { \mathrm { \l__spintent_spunit_luaset_canonical_str } }
  }
\cs_new_protected:Nn \__spintent_spunit_process_base_exp:n
  {
    \tl_clear:N \l__spintent_spunit_exponent_tl
    \__spintent_luafun_spunit_lookup_alias:n { #1 }
    \str_if_eq:VnTF \l__spintent_spunit_luaset_lookup_status_str { notfound }
      {
        \str_set:Nn \l__spintent_luaset_status_str { invalid }
      }
      {
        \bool_if:NF \l__spintent_spunit_is_mult_bool
          {
            \bool_if:NTF \l__spintent_spunit_print_cdot_bool
              {
                \bool_if:NTF \l__spintent_spunit_read_cdot_bool
```

```
                  { \MathMLintent { por } { \cdot } }
                  { \MathMLintent { :silent } { \cdot } }
              }
              {
                \bool_if:NTF \l__spintent_spunit_read_cdot_bool
                  { \MathMLintent { por } { \, } }
                  { \, }
              }
          }
        \str_if_eq:VnTF \l__spintent_spunit_luaset_is_compact_str { true }
          {
            {
              \MathMLintent { \l__spintent_spunit_luaset_read_str }
                {
                  \__spintent_spunit_format_canonical:
                }
            }
            ^ { \l__spintent_spunit_luaset_compact_exp_str }
          }
          {
            \seq_set_split:Nnn \l__spintent_spunit_exp_seq { ^ } { #1 }
            \seq_pop_left:NN \l__spintent_spunit_exp_seq \l__spintent_spunit_unit_name_tl
            \seq_if_empty:NF \l__spintent_spunit_exp_seq
              {
                \seq_pop_left:NN \l__spintent_spunit_exp_seq \l__spintent_spunit_exponent_tl
              }
            \MathMLintent { \l__spintent_spunit_luaset_read_str }
              {
                \__spintent_spunit_format_canonical:
              }
            \tl_if_empty:NF \l__spintent_spunit_exponent_tl
              {
                ^ { \l__spintent_spunit_exponent_tl }
              }
          }
        \bool_set_false:N \l__spintent_spunit_is_mult_bool
      }
  }
```

(*End of definition for* `\__spintent_spunit_format_canonical:` *and* `\__spintent_spunit_process_base_exp:n`.)

`\__spintent_spunit_process_mult:n`
`\__spintent_spunit_process_mult:V`

The function `\__spintent_spunit_process_mult:n` handles compound *"units"* joined by an explicit multiplication star '*'. It splits the {⟨*argument*⟩} at each *"star symbol"* and applies `\__spintent_spunit_process_base_ex` to evaluate and render each resulting unit individually.

```
\cs_new_protected:Nn \__spintent_spunit_process_mult:n
  {
    \bool_set_true:N \l__spintent_spunit_is_mult_bool
    \seq_set_split:Nnn \l__spintent_spunit_mult_seq { * } { #1 }
    \seq_map_function:NN \l__spintent_spunit_mult_seq \__spintent_spunit_process_base_exp:n
  }
\cs_generate_variant:Nn \__spintent_spunit_process_mult:n { V }
```

(*End of definition for* `\__spintent_spunit_process_mult:n`.)

`\__spintent_print_spunit_hierarchy:`

The function `\__spintent_print_spunit_hierarchy:` handles the layout of inline *"unit"* fractions. It first processes the *"units"* before the slash (the numerator). If a denominator exists, it prints the slash / '/' with its corresponding speech intent, and then processes the remaining *"units"*.

```
\cs_new_protected:Nn \__spintent_print_spunit_hierarchy:
  {
    \tl_if_empty:NF \l__spintent_luaset_arg_above_tl
      {
        \__spintent_spunit_process_mult:V \l__spintent_luaset_arg_above_tl
        \tl_if_empty:NF \l__spintent_luaset_arg_below_tl
          {
            \MathMLintent { \l__spintent_spunit_read_slash_str } { / }
            \__spintent_spunit_process_mult:V \l__spintent_luaset_arg_below_tl
          }
      }
  }
```

(*End of definition for* `\__spintent_print_spunit_hierarchy:`.)

__spintent_spunit_safe_exec:n

The function `\__spintent_spunit_safe_exec:n` performs the main validation and layout for *"units"*. It first checks if the {⟨*argument*⟩} has units, is a valid expression, and does not contain illegal numbers in the denominator, throwing the appropriate "errors" if any check fails.

If the unit is valid, it checks if it is a sexagesimal angle (degrees, minutes, or seconds). For sexagesimal *"units"*, it prints the number and the symbol *"without any space in between"*, applying a slight negative space '`\!`' for `arcseconds`. For standard units, it prints the number followed by a thin space '`\,`' and then processes the unit sequence.

```
\cs_new_protected:Nn \__spintent_spunit_safe_exec:n
  {
    \str_if_eq:VnTF \l__spintent_luaset_has_units_str { false }
      { \msg_error:nnn { spintent } { spunit-no-units } { #1 } }
      {
        \str_if_eq:VnTF \l__spintent_luaset_status_str { valid }
          {
            \str_if_eq:VnTF
              \l__spintent_luaset_denom_has_number_str { true }
              {
                \msg_error:nnn { spintent }
                  { inline-numeric-denominator } { #1 }
              }
              {
                \__spintent_luafun_spunit_lookup_alias:V \l__spintent_luaset_arg_above_tl
                \str_if_eq:VnTF \l__spintent_spunit_luaset_is_sexagesimal_str { true }
                  {
                    \tl_if_empty:NF \l__spintent_luaset_part_int_tl
                      {
                        \__spintent_spnum_print_number_core:
                      }
                    \MathMLintent { \l__spintent_spunit_luaset_read_str }
                      {
                        \mathord
                          {
                            \str_case:Vn \l__spintent_spunit_luaset_canonical_str
                              {
                                { " } { \prime \! \prime }
                                { ' } { \prime          }
                                { ° } { °               }
                              }
                          }
                      }
                  }
                  {
                    \tl_if_empty:NF \l__spintent_luaset_part_int_tl
                      {
                        \__spintent_spnum_print_number_core: \,
                      }
                    \__spintent_print_spunit_hierarchy:
                  }
              }
          }
          { \msg_error:nnn { spintent } { spunit-invalid-unit } { #1 } }
      }
  }
```

(*End of definition for* `\__spintent_spunit_safe_exec:n`.)

11.15.5 Implementation of the `\spunit` command

\spunit

The public user command `\spunit` accepts an optional [⟨*key* = *val*⟩] argument in '`#1`' and a mandatory {⟨*unit expression*⟩} in '`#2`'. It first checks if it is executed inside *math mode*, raising an "error" if it is not. If correct, it opens a local group to apply the options and passes the {⟨*argument*⟩} to Lua module for parsing via `\__spintent_luafun_clean_split_and_set:n` function.

Finally, it delegates the execution to the function `\__spintent_spunit_safe_exec:n` to perform the safety checks and render the parsed data, before closing the group.

```
\NewDocumentCommand \spunit { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \keys_set:nn { spintent/spunit } { #1 }
          \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #2 } }
```

```
          \__spintent_spunit_safe_exec:n { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spunit } }
  }
```

(*End of definition for* `\spunit`*. This function is documented on page 4.*)

11.16 The commands `\spnewunit` and `\spunitalias`

The user commands `\spnewunit` and `\spunitalias` allow users to add new custom *"units"* or *"aliases"* to the package's dictionary. Similar to `\spunit`, these commands delegate the processing and validation to the `spintent.lua` via `\__spintent_luafun_define_custom_unit:nn` and `\__spintent_luafun_define_spunit`

During the definition, the Lua module checks if the new *"unit"* or *"alias"* is valid and ensures it does not conflict with existing *"units"*. It returns the result of this operation to TEX using the `\l__spintent_spunit_luaset_status_s` variable. If a collision is detected or the `{⟨argument⟩}` is invalid, the package throws the corresponding "error" message.

11.16.1 Definition of error messages

spnewunit-duplicate
spunitalias-duplicate
spunitalias-invalid-expr

We define three error messages for the creation of *"units"* and *"aliases"*. The first two "errors" trigger if the user attempts to define a *"new unit"* or alias passed in '`#1`' that already exists in the dictionary. The third "error" is raised by `\spunitalias` if the target expression provided in '`#2`' is invalid (for example, if it contains unknown *"units"* or incorrect syntax).

```
\msg_new:nnnn { spintent } { spnewunit-duplicate }
  {
    The~unit~symbol~or~alias~'#1'~is~already~defined!
  }
  {
    You~cannot~redefine~with~\token_to_str:N \spnewunit.
  }
\msg_new:nnnn { spintent } { spunitalias-duplicate }
  {
    The~alias~'#1'~is~already~defined~or~reserved!
  }
  {
    You~cannot~redefine~an~existing~unit~or~another~alias.
  }
\msg_new:nnnn { spintent } { spunitalias-invalid-expr }
  {
    The~expression~'#2'~for~alias~'#1'~is~invalid.
  }
  {
    It~must~contain~only~defined~units,~exponents~and~one~'/'.
  }
```

(*End of definition for* `spnewunit-duplicate`*,* `spunitalias-duplicate`*, and* `spunitalias-invalid-expr`*.*)

`\spnewunit` The public command `\spnewunit` defines a new custom unit. It passes the new unit symbol '`#1`' and its speech text '`#2`' to Lua via `\__spintent_luafun_define_custom_unit:nn`. If the module reports that the unit already exists, it throws the corresponding duplicate error.

```
\NewDocumentCommand \spnewunit { m m }
  {
    \mode_if_math:TF
      {
        \msg_error:nnn { spintent } { need-text-mode } { \spnewunit }
      }
      {
        \__spintent_luafun_define_custom_unit:nn { #1 } { #2 }
        \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { duplicate }
          {
            \msg_error:nnn { spintent } { spnewunit-duplicate } { #1 }
          }
      }
  }
```

(*End of definition for* `\spnewunit`*. This function is documented on page 6.*)

`\spunitalias` The public command `\spunitalias` creates a shortcut for a complex unit expression. It passes the new alias name '`#1`' and the target expression '`#2`' to Lua via `\__spintent_luafun_define_spunit_alias:nn`. It throws an error if the alias already exists, or if the provided target expression contains invalid syntax or unknown units.

```
\NewDocumentCommand \spunitalias { m m }
```

```
  {
    \mode_if_math:TF
      {
        \msg_error:nnn { spintent } { need-text-mode } { \spunitalias }
      }
      {
        \__spintent_luafun_define_spunit_alias:nn { #1 } { #2 }
        \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { duplicate }
          {
            \msg_error:nnn { spintent } { spunitalias-duplicate } { #1 }
          }
        \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { invalid-expr }
          {
            \msg_error:nnnn { spintent } { spunitalias-invalid-expr } { #1 } { #2 }
          }
      }
  }
```

(*End of definition for* `\spunitalias`. *This function is documented on page* 6.)

11.17 The command `\spmoney`

The user command `\spmoney` is the *third* of *"the big four"* provided by the spintent package, designed to format monetary values and currencies. It delegates {⟨*argument*⟩} parsing to the Lua module `spintent.lua` via `\__spintent_luafun_clean_split_and_set:n`. The module then retrieves the correct *"currency symbol"*, pluralization rules, and layout structures from the database.

The command accepts *optional arguments* to override the default *"currency symbol"* or enable *"subunit"* rendering. Finally, it constructs the `MathML` intent for `MathCAT`.

\l__spintent_spmoney_luaset_symbol_str
\l__spintent_spmoney_luaset_position_str
\l__spintent_spmoney_luaset_sing_str
\l__spintent_spmoney_luaset_plur_str
\l__spintent_spmoney_luaset_conde_str
\l__spintent_spmoney_luaset_subsing_str
\l__spintent_spmoney_luaset_subplur_str
\l__spintent_spmoney_luaset_print_iso_str
\l__spintent_spmoney_luaset_currency_arg_str
\l__spintent_spmoney_luaset_status_str

These string variables store the currency data returned by the Lua module `spintent.lua`. Upon a successful database lookup, the module populates these variables with the *"currency symbol"*, its layout position (`pre` or `post`), the appropriate singular and plural names for both the main unit and subunits, and the execution status flags. TeX then uses this information to render the visual layout and construct the correct `MathML` speech intents.

```
\str_new:N  \l__spintent_spmoney_luaset_symbol_str
\str_new:N  \l__spintent_spmoney_luaset_position_str
\str_new:N  \l__spintent_spmoney_luaset_sing_str
\str_new:N  \l__spintent_spmoney_luaset_plur_str
\str_new:N  \l__spintent_spmoney_luaset_conde_str
\str_new:N  \l__spintent_spmoney_luaset_subsing_str
\str_new:N  \l__spintent_spmoney_luaset_subplur_str
\str_new:N  \l__spintent_spmoney_luaset_print_iso_str
\str_new:N  \l__spintent_spmoney_luaset_currency_arg_str
\str_new:N  \l__spintent_spmoney_luaset_status_str
```

(*End of definition for* `\l__spintent_spmoney_luaset_symbol_str` *and others.*)

\l__spintent_spmoney_sub_units_bool
\l__spintent_spmoney_intent_money_str
\l__spintent_spmoney_intent_sign_str
\l__spintent_spmoney_intent_sub_unit_str
\l__spintent_spmoney_res_main_voice_str
\l__spintent_spmoney_subnum_int
\l__spintent_spmoney_extracted_curr_str
\l__spintent_spmoney_extracted_num_tl

The boolean variable tracks whether subunit rendering is enabled. The string variables assemble the specific speech text (intents) for `NVDA/MathCAT`, including the sign, the main currency, and the subunits. Finally, the remaining variables temporarily store the numeric value and currency code extracted from the user's input.

```
\bool_new:N \l__spintent_spmoney_sub_units_bool
\str_new:N  \l__spintent_spmoney_intent_money_str
\str_new:N  \l__spintent_spmoney_intent_sign_str
\str_new:N  \l__spintent_spmoney_intent_sub_unit_str
\str_new:N  \l__spintent_spmoney_res_main_voice_str
\int_new:N  \l__spintent_spmoney_subnum_int
\str_new:N  \l__spintent_spmoney_extracted_curr_str
\tl_new:N   \l__spintent_spmoney_extracted_num_tl
```

(*End of definition for* `\l__spintent_spmoney_sub_units_bool` *and others.*)

11.17.1 Definition of error messages

currency-no-subunits
invalid-currency

We define two messages to handle invalid currency configurations. The first "error" warns the user if subunit rendering is requested for a currency that does not support fractional divisions (such as JPY or CLP), safely ignoring the flag. The second "error" is triggered if the provided currency identifier is not registered in the Lua database.

```
\msg_new:nnnn { spintent } { currency-no-subunits }
  {
    The~currency~'#1'~does~not~support~subunits.~
    The~option~'sub-units=true'~is~ignored.
  }
```

```
  {
    Monies~like~the~Japanese~Yen~(JPY)~or~Chilean~Peso~(CLP)~
    do~not~have~fractional~subunits~registered~in~Lua~module.~
  }
\msg_new:nnnn { spintent } { invalid-currency }
  {
    Invalid~currency~identifier~'#1'~in~\token_to_str:N \spmoney .
  }
  {
    The~provided~symbol,~alias,~or~ISO~code~does~not~exist~in~
    the~Lua~module.
  }
```

(*End of definition for* `currency-no-subunits` *and* `invalid-currency`.)

11.17.2 Auxiliary functions for \spmoney

`\__spintent_spmoney_set_currency:n`
`\__spintent_spmoney_normalize_key:n`
`\__spintent_spmoney_normalize_key:V`

The function `\__spintent_spmoney_set_currency:n` validates the currency identifier provided in '#1' against the database. It delegates the lookup to the helper function `\__spintent_spmoney_normalize_key:n`, which interfaces with the Lua module `spintent.lua`. If the module returns a `notfound` status, an "error" is triggered. Otherwise, it stores the valid currency identifier in `\l__spintent_spmoney_luaset_currency_arg_str` for subsequent processing.

```
\cs_new_protected:Nn \__spintent_spmoney_set_currency:n
  {
    \__spintent_spmoney_normalize_key:n {#1}
    \str_if_eq:VnTF \l__spintent_spmoney_luaset_status_str { notfound }
      {
        \msg_error:nnn { spintent } { invalid-currency } {#1}
      }
      { \str_set:Nn \l__spintent_spmoney_luaset_currency_arg_str {#1} }
  }
\cs_new_protected:Nn \__spintent_spmoney_normalize_key:n
  {
    \__spintent_luafun_spmoney_normalize_key:n { #1 }
  }
\cs_generate_variant:Nn \__spintent_spmoney_normalize_key:n { V }
```

(*End of definition for* `\__spintent_spmoney_set_currency:n` *and* `\__spintent_spmoney_normalize_key:n`.)

11.17.3 Definition of keys for \spmoney

`currency`
`sub-units`
`dolar-math`
`cifras-sep`

Now we add the keys `currency`, `sub-units`, `dolar-math` and `cifras-sep`.

```
\keys_define:nn { spintent / spmoney }
  {
    currency     .code:n       = { \__spintent_spmoney_set_currency:n {#1} },
    currency     .initial:n    = { pesos },
    currency     .value_required:n = true,
    sub-units    .bool_set:N   = \l__spintent_spmoney_sub_units_bool,
    sub-units    .value_required:n = true,
    dolar-math   .bool_set:N   = \l__spintent_spmoney_dolar_math_bool,
    dolar-math   .initial:n    = { true },
    dolar-math   .value_required:n = true,
    cifras-sep   .meta:nn      = { spintent/spnum } { cifras-sep = {#1} },
    cifras-sep   .value_required:n = true,
    unknown      .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spmoney }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `currency` *and others.*)

11.17.4 Internal functions for \spmoney

`\__spintent_spmoney_validate_money:`

The function `\__spintent_spmoney_validate_money:` verifies if the parsed input contains an embedded currency identifier by checking `\l__spintent_luaset_has_units_str`. If a currency is detected, it extracts and normalizes the key from `\l__spintent_luaset_arg_above_tl`. It throws an "error" if the currency is not found in the database. If valid, it updates `\l__spintent_spmoney_luaset_currency_arg_str` and retrieves the full currency metadata from the `spintent.lua` via `\__spintent_luafun_spmoney_lookup_data:V`.

```
\cs_new_protected:Nn \__spintent_spmoney_validate_money:
  {
    \str_if_eq:VnT \l__spintent_luaset_has_units_str { true }
      {
```

```
        \__spintent_spmoney_normalize_key:V \l__spintent_luaset_arg_above_tl
        \str_if_eq:VnTF \l__spintent_spmoney_luaset_status_str { notfound }
          { \msg_error:nne { spintent } { invalid-currency } { \l__spintent_luaset_arg_above_tl }
          {
            \str_set:NV
              \l__spintent_spmoney_luaset_currency_arg_str
              \l__spintent_luaset_arg_above_tl
          }
      }
    \str_set:Ne \l__spintent_spmoney_luaset_status_str { found }
    \__spintent_luafun_spmoney_lookup_data:V \l__spintent_spmoney_luaset_currency_arg_str
  }
```

(*End of definition for* `\__spintent_spmoney_validate_money:`.)

`\__spintent_spmoney_print_symbol:`

The function `\__spintent_spmoney_print_symbol:` renders the *"currency symbol"*. If `\l__spintent_spmoney_lu` is `true`, it outputs the ISO code using `\mathrm`. Otherwise, it prints the *"standard symbol"*. For dollar signs '$', it checks `\l__spintent_spmoney_dolar_math_bool` to determine whether to render it directly as a math character '$' or wrap it inside `\text`.

```
\cs_new_protected:Nn \__spintent_spmoney_print_symbol:
  {
    \str_if_eq:VnTF \l__spintent_spmoney_luaset_print_iso_str { true }
      { \mathrm { \l__spintent_spmoney_luaset_currency_arg_str } }
      {
        \str_if_eq:VnTF \l__spintent_spmoney_luaset_symbol_str { $ }
          {
            \bool_if:NTF \l__spintent_spmoney_dolar_math_bool
              { \$ } { \text { \$ } }
          }
          { \l__spintent_spmoney_luaset_symbol_str }
      }
  }
```

(*End of definition for* `\__spintent_spmoney_print_symbol:`.)

`\__spintent_spmoney_render_layout:`

The function `\__spintent_spmoney_render_layout:` determines the layout order of the *"currency symbol"* relative to the numeric value. It evaluates `\l__spintent_spmoney_luaset_position_str`; if set to `pre`, it renders the *"currency symbol" before* the number node. Otherwise, it places the number first, followed by the *"currency symbol"*.

```
\cs_new_protected:Nn \__spintent_spmoney_render_layout:
  {
    \str_if_eq:VnTF \l__spintent_spmoney_luaset_position_str { pre }
      {
        \__spintent_spmoney_print_symbol:
        \MathMLarg { n } { \__spintent_spmoney_render_number_node: }
      }
      {
        \MathMLarg { n } { \__spintent_spmoney_render_number_node: }
        \__spintent_spmoney_print_symbol:
      }
  }
```

(*End of definition for* `\__spintent_spmoney_render_layout:`.)

`\__spintent_spmoney_render_number_node:`
`pintent_spmoney_render_only_integer_node:`

The function `\__spintent_spmoney_render_number_node:` renders the full numeric value, encompassing both the integer and decimal components. Conversely, the function `\__spintent_spmoney_render_only_integer_` outputs exclusively the integer portion, omitting any decimal data.

```
\cs_new_protected:Nn \__spintent_spmoney_render_number_node:
  {
    \__spintent_spnum_render_int:
    \__spintent_spnum_print_part_dec:
  }
\cs_new_protected:Nn \__spintent_spmoney_render_only_integer_node:
  {
    \__spintent_spnum_render_int:
  }
```

(*End of definition for* `\__spintent_spmoney_render_number_node:` *and* `\__spintent_spmoney_render_only_integer_node:`.)

__spintent_spmoney_print_int:
__spintent_spmoney_intent_int:

The function `\__spintent_spmoney_print_int:` evaluates the presence of an explicit sign in `\l__spintent_luase` If a sign is detected, it assigns a prefix speech intent ‘más’ or ‘menos’ and embeds the sign and subsequent layout inside a `\MathMLintent`. If no sign is present, it routes execution based on whether subunits are enabled.

The function `\__spintent_spmoney_intent_int:` determines the correct grammatical inflection for the currency string (singular, plural, or the prepositional ‘de’ variant for exact millions). It evaluates the integer component, decimal boundaries, and zero-fills to select the appropriate string register, applying it as a postfix MathML intent over the layout rendered by `\__spintent_spmoney_render_layout:`.

```
\cs_new_protected:Nn \__spintent_spmoney_print_int:
  {
    \tl_if_empty:NTF \l__spintent_luaset_sign_tl
      {
        \bool_if:NTF \l__spintent_spmoney_sub_units_bool
          { \__spintent_spmoney_set_and_print_sub_units: }
          { \__spintent_spmoney_intent_int: }
      }
      {
        \str_if_eq:VnTF \l__spintent_luaset_sign_tl { + }
          {
            \str_set:Nn \l__spintent_spmoney_intent_sign_str { _más:prefix ($n) }
          }
          {
            \str_set:Nn \l__spintent_spmoney_intent_sign_str { _menos:prefix ($n) }
          }
        \__spintent_invisible_sep:
        \MathMLintent { \l__spintent_spmoney_intent_sign_str }
          {
            \l__spintent_luaset_sign_tl
            \MathMLarg { n }
              {
                \bool_if:NTF \l__spintent_spmoney_sub_units_bool
                  { \__spintent_spmoney_set_and_print_sub_units: }
                  { \__spintent_spmoney_intent_int: }
              }
          }
      }
  }
\cs_new_protected:Nn \__spintent_spmoney_intent_int:
  {
    \tl_if_empty:NTF \l__spintent_luaset_dec_sep_tl
      {
        \str_if_eq:VnTF \l__spintent_luaset_only_part_int_str { 1 }
          {
            \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_sing_st
          }
          {
            \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur_st
          }
        \str_if_eq:VnT \l__spintent_luaset_millons_str { true }
          {
            \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_conde_s
          }
      }
      {
        \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur_str
        \str_if_eq:VnT \l__spintent_luaset_millons_str { true }
          {
            \str_if_eq:VnTF \l__spintent_luaset_dec_is_all_zeros_str { true }
              {
                \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_con
              }
              {
                \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plu
              }
          }
      }
    \__spintent_invisible_sep: % need for MathCAT
    \MathMLintent { \l__spintent_spmoney_intent_money_str :postfix($n) }
      {
        \__spintent_spmoney_render_layout:
```

```
        }
    }
```

(*End of definition for* `\__spintent_spmoney_print_int:` *and* `\__spintent_spmoney_intent_int:`.)

`\__spintent_spmoney_check_sub_units:`
`spintent_spmoney_set_and_print_sub_units:`

The function `\__spintent_spmoney_check_sub_units:` verifies whether the selected currency supports *"subunits"*. If the database returns empty subunit data, it triggers a warning and automatically disables the *"subunit"* rendering flag.

The function `\__spintent_spmoney_set_and_print_sub_units:` processes and formats the speech intents when *"subunits"* are active. It normalizes the decimal component (for instance, padding a single-digit decimal like '.5' to '50'), evaluates the correct singular or plural inflections for both the main *"currency"* and the *"subunits"*, and delegates the final MathML tree assembly to `\__spintent_spmoney_build_tree:`.

```
\cs_new_protected:Nn \__spintent_spmoney_check_sub_units:
  {
    \str_if_empty:NT \l__spintent_spmoney_luaset_subsing_str
      {
        \msg_warning:nne { spintent } { currency-no-subunits }
          { \l__spintent_spmoney_luaset_currency_arg_str }
        \bool_set_false:N \l__spintent_spmoney_sub_units_bool
      }
  }
\cs_new_protected:Nn \__spintent_spmoney_set_and_print_sub_units:
  {
    \int_compare:nTF { \str_count:N \l__spintent_luaset_only_part_dec_str == 1 }
      {
        \int_set:Nn \l__spintent_spmoney_subnum_int
          { \l__spintent_luaset_only_part_dec_str * 10 }
        \str_set:Ne \l__spintent_luaset_only_part_dec_str
          { \l__spintent_luaset_only_part_dec_str 0 }
      }
      {
        \int_set:Nn \l__spintent_spmoney_subnum_int
          { \l__spintent_luaset_only_part_dec_str }
      }
    \int_compare:nTF { \l__spintent_spmoney_subnum_int == 1 }
      {
        \str_set:Ne \l__spintent_spmoney_intent_sub_unit_str
          { \l__spintent_spmoney_luaset_subsing_str :postfix($sub) }
      }
      {
        \str_set:Ne \l__spintent_spmoney_intent_sub_unit_str
          { \l__spintent_spmoney_luaset_subplur_str :postfix($sub) }
      }
    \str_if_eq:VnTF \l__spintent_luaset_only_part_int_str { 1 }
      {
        \str_set:NV \l__spintent_spmoney_res_main_voice_str
          \l__spintent_spmoney_luaset_sing_str
      }
      {
        \str_set:NV \l__spintent_spmoney_res_main_voice_str
          \l__spintent_spmoney_luaset_plur_str
      }
    \str_if_eq:VnT \l__spintent_luaset_millons_str { true }
      {
        \str_set:NV \l__spintent_spmoney_res_main_voice_str
          \l__spintent_spmoney_luaset_conde_str
      }
    \__spintent_spmoney_build_tree:
  }
```

(*End of definition for* `\__spintent_spmoney_check_sub_units:` *and* `\__spintent_spmoney_set_and_print_sub_units:`.)

`\__spintent_spmoney_build_tree:`
`_spintent_spmoney_print_integer_continue:`
`\__spintent_spmoney_print_sep_con:`
`\__spintent_spmoney_print_dec_sub_unit:`

The function `\__spintent_spmoney_build_tree:` structures the final MathML tree assembly for fractional *"currencies"*. It arranges the integer component, the decimal separator, and the *"subunit"* component based on the layout position (`pre` or `post`) of the *"currency symbol"*.

The function `\__spintent_spmoney_print_integer_continue:` isolates the rendering of the integer node with its associated main *"currency"* intent. Furthermore, `\__spintent_spmoney_print_sep_con:` applies the specific speech intent 'con' to the decimal separator, while `\__spintent_spmoney_print_dec_sub_unit:` binds the computed *"subunit"* intent to the decimal digits, ensuring accurate accessibility output.

```
\cs_new_protected:Nn \__spintent_spmoney_build_tree:
```

```
  {
    \str_if_eq:VnTF \l__spintent_spmoney_luaset_position_str { pre }
      {
        \MathMLintent { \l__spintent_spmoney_res_main_voice_str :postfix($n) }
          {
            \__spintent_spmoney_print_symbol:
            \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
          }
        \__spintent_spmoney_print_sep_con:
        \__spintent_spmoney_print_dec_sub_unit:
      }
      {
        \MathMLintent { \l__spintent_spmoney_res_main_voice_str :postfix($n) }
          {
            \__spintent_invisible_sep:
            \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
          }
        \__spintent_spmoney_print_sep_con:
        \__spintent_spmoney_print_dec_sub_unit:
        \MathMLintent {:silent } { \__spintent_spmoney_print_symbol: }
      }
  }
\cs_new_protected:Nn \__spintent_spmoney_print_integer_continue:
  {
    \MathMLintent { \l__spintent_spmoney_res_main_voice_str :postfix($n) }
      {
        \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
      }
  }
\cs_new_protected:Nn \__spintent_spmoney_print_sep_con:
  {
    \MathMLintent { con }
      {
        \mathord { \l__spintent_luaset_dec_sep_tl }
      }
  }
\cs_new_protected:Nn \__spintent_spmoney_print_dec_sub_unit:
  {
    \MathMLintent { \l__spintent_spmoney_intent_sub_unit_str }
      {
        \__spintent_invisible_sep:
        \MathMLarg { sub }
          {
            \__spintent_spnum_render_dec:
          }
      }
  }
```

(*End of definition for* `\__spintent_spmoney_build_tree:` *and others.*)

`\__spintent_spmoney_parse_and_route:n`

The internal function `\__spintent_spmoney_parse_and_route:n` processes the {⟨*argument*⟩} passed in '`#1`' using Lua module to extract the numeric value and any *"currency"* identifier. If a *"currency"* is found (`\l__spintent_spmoney_extracted_curr_str`), it sets the corresponding variables. It then calls `\__spintent_spmoney_validate_money:` to verify the *"currency"*. If *"subunit"* rendering is enabled (`\l__spintent_spmoney_sub_units_bool`), it calls `\__spintent_spmoney_check_sub_units:`. Finally, it calls `\__spintent_spmoney_print_int:` to format the output.

```
\cs_new_protected:Nn \__spintent_spmoney_parse_and_route:n
  {
    \__spintent_luafun_spmoney_prepare_input:n { #1 }
    \__spintent_luafun_clean_split_and_set:V \l__spintent_spmoney_extracted_num_tl
    \str_if_empty:NF \l__spintent_spmoney_extracted_curr_str
      {
        \str_set:Ne \l__spintent_luaset_has_units_str { true }
        \str_set:NV \l__spintent_luaset_arg_above_tl \l__spintent_spmoney_extracted_curr_str
      }
    \__spintent_spmoney_validate_money:
    \bool_if:NT \l__spintent_spmoney_sub_units_bool
      { \__spintent_spmoney_check_sub_units: }
    \__spintent_spmoney_print_int:
  }
```

(*End of definition for* `\__spintent_spmoney_parse_and_route:n`.)

\spmoney

The public document command \spmoney formats monetary values with accessible speech intents. It accepts an optional argument '#1' for local key-value configurations and a mandatory argument '#2' containing the numeric value and optional embedded currency identifier. It first evaluates the typesetting context, triggering a fatal error if executed outside of math mode. Upon validation, it establishes a local TeX group to safely isolate the key assignments, delegates the raw input string to __spintent_spmoney_parse_and_route:n for parsing and rendering, and subsequently closes the group to prevent scope leakage.

```
\NewDocumentCommand \spmoney { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \keys_set:nn { spintent / spmoney } { #1 }
          \__spintent_spmoney_parse_and_route:n { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmoney } }
  }
```

(*End of definition for* \spmoney. *This function is documented on page 6.*)

11.18 The command \spshort

The user command \spshort is the *fourth* and *final* of *"the big four"*. It manages the semantic injection and typographical formatting of *text mode* abbreviations, ordinals, and acronyms. Unlike *math mode* commands, which rely on MathML attributes, this command uses the PDF *tagging tools* provided by the tagpdf[29] package. Specifically, it uses the E (expanded) key to ensure that NVDA interpret the full form of the abbreviation, ordinal, or acronym in *text mode*.

\l__spintent_spshort_luaset_status_str
\l__spintent_spshort_luaset_layout_str
\l__spintent_spshort_luaset_expanded_str
\l__spintent_spshort_luaset_output_str
\l__spintent_spshort_luaset_base_str
\l__spintent_spshort_luaset_suffix_str

These internal string variables serve as the primary data interface with the Lua module spintent.lua. During processing, the Lua module *"stores"* within these registers the execution status, the targeted typographical layout parameters, the phonetic replacement text for the E key intended for NVDA, the final sanitized output string, and the isolated structural components for ordinals (base and suffix).

```
\str_new:N  \l__spintent_spshort_luaset_status_str
\str_new:N  \l__spintent_spshort_luaset_layout_str
\str_new:N  \l__spintent_spshort_luaset_expanded_str
\str_new:N  \l__spintent_spshort_luaset_output_str
\str_new:N  \l__spintent_spshort_luaset_base_str
\str_new:N  \l__spintent_spshort_luaset_suffix_str
```

(*End of definition for* \l__spintent_spshort_luaset_status_str *and others.*)

11.18.1 Definition of error messages for \spshort

unknown-abbreviation

The internal "error" message unknown-abbreviation is triggered when the command \spshort encounters an invalid lookup state from the Lua module. It acts as a strict fallback exception when the provided {⟨*argument*⟩} cannot be mapped to any registered database entry in spintent.lua nor successfully resolved through the grammatical ordinal expansion rules.

```
\msg_new:nnnn { spintent } { unknown-abbreviation }
  { The~abbreviation~or~ordinal~'#1'~is~invalid~or~not~recognized. }
  { Verify~the~spelling~or~check~the~grammar~rules~for~ordinals. }
```

(*End of definition for* unknown-abbreviation.)

11.18.2 Accessibility interface

__spintent_spshort_tag_span:nn
__spintent_spshort_tag_span:en

The internal function __spintent_spshort_tag_span:nn processes the {⟨*argument*⟩} passed in '#2' using the tools provided by tagpdf. By utilizing the tag and E keys, it creates a native PDF /E (#1) structure and expands the {⟨*argument*⟩} passed in '#1' for proper reading with NVDA.

```
\cs_new_protected:Nn \__spintent_spshort_tag_span:nn
  {
    \tag_mc_end_push:
      \tag_struct_begin:n { tag=Span, E={ #1 } }
        \tag_mc_begin:n { tag=Span }
          \tag_suspend:n { \spshort }
            #2
          \tag_resume:n { \spshort }
        \tag_mc_end:
      \tag_struct_end:
    \tag_mc_begin_pop:n {}
  }
\cs_generate_variant:Nn \__spintent_spshort_tag_span:nn { en }
```

(*End of definition for* __spintent_spshort_tag_span:nn.)

11.18.3 Definition of keys for \spshort

small-caps read-arg

Now we add the keys small-caps and read-arg.

```
\keys_define:nn { spintent / spshort }
  {
    small-caps .bool_set:N = \l__spintent_spshort_smallcaps_bool,
    small-caps .initial:n  = false,
    small-caps .value_required:n = true,
    read-arg   .str_set:N  = \l__spintent_spshort_read_arg_str,
    read-arg   .initial:n  = {},
    read-arg   .value_required:n = true,
    unknown    .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spshort }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for small-caps and read-arg.*)

11.18.4 Internal functions for \spshort

__spintent_spshort_print:nn

The internal function `\__spintent_spshort_print:nn` acts as the primary formatter for text abbreviations. It clears the string variable `\l__spintent_spshort_read_arg_str` and evaluates the optional ⟨*keys*⟩ passed in '#1'. It then checks if a manual reading override was provided via those keys. If that string variable remains empty, it calls `\__spintent_spshort_process_and_render:n` to process the {⟨*argument*⟩} passed in '#2'. Otherwise, if an override is detected, it directly delegates the execution to `\__spintent_spshort_render_bypass:n`.

```
\cs_new_protected:Nn \__spintent_spshort_print:nn
  {
    \str_clear:N \l__spintent_spshort_read_arg_str
    \keys_set:nn { spintent / spshort } {#1}
    \str_if_empty:NTF \l__spintent_spshort_read_arg_str
      {
        \__spintent_spshort_process_and_render:n {#2}
      }
      {
        \__spintent_spshort_render_bypass:n {#2}
      }
  }
```

(*End of definition for __spintent_spshort_print:nn.*)

__spintent_spshort_process_and_render:n

The internal function `\__spintent_spshort_process_and_render:n` interfaces with the Lua module spintent.lua function `\__spintent_luafun_spshort_process:n` to resolve the abbreviation or ordinal {⟨*argument*⟩} passed in '#1'.

It subsequently evaluates the execution state stored within the string variable `\l__spintent_spshort_luaset_statu` and dynamically routes execution to the corresponding rendering subroutine `success`, `fallback` or `error`. Any unrecognized status automatically defaults to the "error" handler.

```
\cs_new_protected:Nn \__spintent_spshort_process_and_render:n
  {
    \__spintent_luafun_spshort_process:n {#1}
    \str_case:VnF \l__spintent_spshort_luaset_status_str
      {
        { success }  { \__spintent_spshort_render_success: }
        { fallback } { \__spintent_spshort_render_fallback: }
        { error }    { \__spintent_spshort_render_error:n {#1} }
      }
      { \__spintent_spshort_render_error:n {#1} }
  }
```

(*End of definition for __spintent_spshort_process_and_render:n.*)

__spintent_spshort_render_success:
__spintent_spshort_render_fallback:
__spintent_spshort_render_bypass:n
__spintent_spshort_render_error:n

The function `\__spintent_spshort_render_success:` injects the key `E` and executes the targeted typographical layout. The `\__spintent_spshort_render_fallback:` function handles default cases, applying optional formatting such as SMALL CAPS to unrecognized or partially resolved terms.

Conversely, the function `\__spintent_spshort_render_bypass:n` directly maps the user-defined *"speech"* to the {⟨*argument*⟩}, bypassing the automated lookup Lua module. Finally, the function `\__spintent_spshort_rende` triggers a structural exception while yielding the unformatted {⟨*argument*⟩} to prevent fatal compilation.

```
\cs_new_protected:Nn \__spintent_spshort_render_success:
  {
```

```
    \__spintent_spshort_tag_span:en
      { \l__spintent_spshort_luaset_expanded_str }
      { \__spintent_spshort_execute_layout: }
  }
\cs_new_protected:Nn \__spintent_spshort_render_fallback:
  {
    \__spintent_spshort_tag_span:en
      { \l__spintent_spshort_luaset_expanded_str }
      {
        \bool_if:NTF \l__spintent_spshort_smallcaps_bool
          { \textsc{ \l__spintent_spshort_luaset_output_str } }
          { \l__spintent_spshort_luaset_output_str }
      }
  }
\cs_new_protected:Nn \__spintent_spshort_render_bypass:n
  {
    \__spintent_spshort_tag_span:en { \l__spintent_spshort_read_arg_str } { #1 }
  }
\cs_new_protected:Nn \__spintent_spshort_render_error:n
  {
    \msg_error:nnn { spintent } { unknown-abbreviation } {#1}
  }
```

(*End of definition for* `\__spintent_spshort_render_success:` *and others.*)

`\__spintent_spshort_execute_layout:`

The internal function `\__spintent_spshort_execute_layout:` executes the typographical layout. It evaluates the descriptor *"stored"* within the string variable `\l__spintent_spshort_luaset_layout_str` (such as `linear_regular`, `superscript`, or `small_caps_pure`) and maps it to the corresponding LaTeX formatting. Furthermore, it manages complex nested configurations, dynamically applying combinations such as superscripts integrated with conditional SMALL CAPS based on the active boolean variables.

```
\cs_new_protected:Nn \__spintent_spshort_execute_layout:
  {
    \str_case:VnF \l__spintent_spshort_luaset_layout_str
      {
        { linear_regular }
          {
            \bool_if:NTF \l__spintent_spshort_smallcaps_bool
              { \textsc{ \l__spintent_spshort_luaset_output_str } }
              { \l__spintent_spshort_luaset_output_str }
          }
        { linear_caps }
          { \l__spintent_spshort_luaset_output_str }
        { linear_mixed }
          { \l__spintent_spshort_luaset_output_str }
        { superscript }
          {
            \l__spintent_spshort_luaset_base_str
            \bool_if:NTF \l__spintent_spshort_smallcaps_bool
              {
                \textsuperscript{
                  \textsc{
                    \str_lowercase:V \l__spintent_spshort_luaset_suffix_str
                  }
                }
              }
              {
                \textsuperscript{ \l__spintent_spshort_luaset_suffix_str }
              }
          }
        { small_caps_pure }
          {
            \bool_if:NTF \l__spintent_spshort_smallcaps_bool
              {
                \textsc{
                  \str_lowercase:V \l__spintent_spshort_luaset_output_str
                }
              }
              { \l__spintent_spshort_luaset_output_str }
          }
        { acronym_long }
          { \l__spintent_spshort_luaset_output_str }
```

```
      }
      { \l__spintent_spshort_luaset_output_str }
  }
```

(*End of definition for* `\__spintent_spshort_execute_layout:`.)

`\spshort`

The public document command `\spshort` manages the accessible formatting of *text mode* abbreviations, acronyms, and ordinals. It first triggers an error if invoked within a *math mode*, upon validation, puts `\mode_leave_vertical:` and establishes a local group to call the function `\__spintent_spshort_print:nn` for processing and printing.

```
\NewDocumentCommand \spshort { O{} m }
  {
    \mode_if_math:TF
      {
        \msg_error:nnn { spintent } { need-text-mode } { \spshort }
      }
      {
        \mode_leave_vertical:
        \group_begin:
          \__spintent_spshort_print:nn {#1} {#2}
        \group_end:
      }
  }
```

(*End of definition for* `\spshort`. *This function is documented on page 7*.)

11.19 The command `\spsiglo`

The public document command `\spsiglo` provides a semantic interface for typesetting century designations (e.g., siglo XXI). It accepts both Arabic and Roman numerals as {⟨*argument*⟩}, interfacing with the Lua module `spintent.lua` to validate, parse, and normalize chronological values.

This command work in dual-mode context awareness, operating uniformly across both *text mode* and *math mode*. When executed within *math mode*, it constructs an accessible `MathML` `intent` expression tree. Conversely, when invoked in standard *text mode*, it encapsulates the rendered output within a native `PDF` tagging structure (such as `/Alt (#1)`) containing an `alt` key.

`\l__spintent_spsiglo_luaset_error_str`
`\l__spintent_spsiglo_luaset_arabic_str`
`\l__spintent_spsiglo_luaset_roman_min_str`
`\l__spintent_spsiglo_print_era_str`

These internal string variables handle data transfer and synchronization with the Lua module `spintent.lua`. They isolate the exception evaluation states, the Arabic numeral required for speech synthesis, the lowercase Roman numeral designated for small caps typesetting via `\textsc`, and the localized chronological era modifier.

```
\str_new:N \l__spintent_spsiglo_luaset_error_str
\str_new:N \l__spintent_spsiglo_luaset_arabic_str
\str_new:N \l__spintent_spsiglo_luaset_roman_min_str
\str_new:N \l__spintent_spsiglo_print_era_str
```

(*End of definition for* `\l__spintent_spsiglo_luaset_error_str` *and others.*)

11.19.1 Definition of error messages for `\spsiglo`

`invalid-century-format`

This internal error message is issued when the provided century identifier fails both Arabic and Roman numeral validation. It is triggered when the Lua module `spintent.lua` cannot map the {⟨*argument*⟩} to an integer within the valid range of `1` to `4000`, preventing incorrect speech synthesis.

```
\msg_new:nnnn { spintent } { invalid-century-format }
  {
    Invalid~century~format~in~'#1'.
  }
  {
    The~argument~'#2'~is~not~a~valid~Arabic~or~Roman~numeral.~
    Ensure~the~value~is~between~1~and~4000.
  }
```

(*End of definition for* `invalid-century-format`.)

11.19.2 Definition of keys for `\spsiglo`

`print-era`

Now we add the key `print-era`. User configuration determining if an era designation is appended, available values are `none`, `ac` (antes de cristo) and `dc` (después de cristo).

```
\keys_define:nn { spintent / spsiglo }
  {
    print-era             .choices:nn =
      { none , ac , dc }
      {
        \int_case:nn { \l_keys_choice_int }
          {
```

```
              {1} { \str_set:Nn \l__spintent_spsiglo_print_era_str { none } }
              {2} { \str_set:Nn \l__spintent_spsiglo_print_era_str { ac } }
              {3} { \str_set:Nn \l__spintent_spsiglo_print_era_str { dc } }
            }
        },
      print-era            .initial:n  = none,
      print-era            .value_required:n = true,
      print-era / unknown .code:n =
        \msg_error:nneee { spintent } { unknown-choice }
          { print-era } { none, ~ac, ~dc } { \exp_not:n {#1} },
      unknown              .code:n =
        {
          \str_set:Nn \l__spintent_command_name_str { spsiglo }
          \__spintent_unknown_keys_cmd:n {#1}
        },
    }
```

(*End of definition for* `print-era`.)

11.19.3 Internal functions for \spsiglo

__spintent_spsiglo_render_simple:nn
__spintent_spsiglo_render_simple:VV
__spintent_spsiglo_render_with_era:nnn
__spintent_spsiglo_render_with_era:VVV

The functions `\__spintent_spsiglo_render_simple:nn` and `\__spintent_spsiglo_render_with_era:nnn` generate `MathML` structures when century designations are evaluated within *math mode*. They construct `MathML intent` structures, mapping the Arabic numerals to their phonetic equivalents. Furthermore, `\__spintent_spsiglo_render_with_era:nnn` integrates the era modifiers `ac` or `dc`. It adds invisible spacing and postfix intents to ensure correct `NVDA/MathCAT` pronunciation.

```
\cs_new_protected:Nn \__spintent_spsiglo_render_simple:nn
  {
    \MathMLintent { #1:number } { \text { \textsc {#2} } }
  }
\cs_generate_variant:Nn \__spintent_spsiglo_render_simple:nn { VV }
\cs_new_protected:Nn \__spintent_spsiglo_render_with_era:nnn
  {
    \str_if_eq:nnTF { #1 } { ac }
      {
        \MathMLintent { antes-de-cristo:postfix($z) }
          {
            \__spintent_invisible_sep:
            \MathMLarg { z }
              {
                \MathMLintent { #2:number }
                  {
                    \text { \textsc {#3}~a.~C. }
                  }
              }
          }
      }
      {
        \MathMLintent { después-de-cristo:postfix($z) }
          {
            \__spintent_invisible_sep:
            \MathMLarg { z }
              {
                \MathMLintent { #2:number }
                  {
                    \text { \textsc {#3}~d.~C. }
                  }
              }
          }
      }
  }
\cs_generate_variant:Nn \__spintent_spsiglo_render_with_era:nnn { VVV }
```

(*End of definition for* `\__spintent_spsiglo_render_simple:nn` *and* `\__spintent_spsiglo_render_with_era:nnn`.)

__spintent_spsiglo_tag_span:nn
__spintent_spsiglo_tag_span:en

The function `\__spintent_spsiglo_tag_span:nn` handles century formatting within *text mode*. It utilizes the `tagpdf` package to enclose the content inside a PDF `/Alt (#1)` structure. By passing the {⟨*argument*⟩} to the `alt` key, it embeds the chronological value directly into the PDF, ensuring correct `NVDA/MathCAT` pronunciation without relying on `MathML` intents.

```
\cs_new_protected:Nn \__spintent_spsiglo_tag_span:nn
  {
    \tag_mc_end_push:
```



```
      \tag_struct_begin:n { tag=Span, alt={ #1 } }
        \tag_mc_begin:n { tag=Span }
          \tag_suspend:n { \spsiglo }
            #2
          \tag_resume:n { \spsiglo }
        \tag_mc_end:
      \tag_struct_end:
    \tag_mc_begin_pop:n {}
  }
\cs_generate_variant:Nn \__spintent_spsiglo_tag_span:nn { en }
```

(*End of definition for* `\__spintent_spsiglo_tag_span:nn`.)

`\__spintent_spsiglo_print_math:n`
`\__spintent_spsiglo_print_text:n`

The functions `\__spintent_spsiglo_print_math:n` and `\__spintent_spsiglo_print_text:n` format century designations based on the active mode. Both first check the "error" status in the variable `\l__spintent_spsiglo_luaset_error_str` returned by the Lua module `spintent.lua` and issue an "error" if validation fails.

If the {⟨*argument*⟩} is valid, the function `\__spintent_spsiglo_print_math:n` operates in *math mode*, applying `MathML` structures depending on whether an era modifier is present. Conversely, the function `\__spintent_spsiglo_print_text:n` operates in *text mode*, utilizing the `tag` and `alt` keys to expand the value of the `print-era` option into full text (e.g., "antes de cristo") for `NVDA/MathCAT`.

```
\cs_new_protected:Nn \__spintent_spsiglo_print_math:n
  {
    \str_if_eq:VnTF \l__spintent_spsiglo_luaset_error_str { true }
      {
        \msg_error:nnnn { spintent } { invalid-century-format }
          { \spsiglo } { #1 }
      }
      {
        \str_if_eq:VnTF \l__spintent_spsiglo_print_era_str { none }
          {
            \__spintent_spsiglo_render_simple:VV
              \l__spintent_spsiglo_luaset_arabic_str
              \l__spintent_spsiglo_luaset_roman_min_str
          }
          {
            \__spintent_spsiglo_render_with_era:VVV
              \l__spintent_spsiglo_print_era_str
              \l__spintent_spsiglo_luaset_arabic_str
              \l__spintent_spsiglo_luaset_roman_min_str
          }
      }
  }
\cs_new_protected:Nn \__spintent_spsiglo_print_text:n
  {
    \str_if_eq:VnTF \l__spintent_spsiglo_luaset_error_str { true }
      {
        \msg_error:nnnn { spintent } { invalid-century-format }
          { \spsiglo } { #1 }
      }
      {
         \str_case:Vn \l__spintent_spsiglo_print_era_str
          {
            { none }
            {
              \__spintent_spsiglo_tag_span:en
                { \l__spintent_spsiglo_luaset_arabic_str }
                { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } }
            }
            { ac }
            {
              \__spintent_spsiglo_tag_span:en
                { \l__spintent_spsiglo_luaset_arabic_str ~ antes ~ de ~ cristo }
                { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } ~ a.~C. }
            }
            { dc }
            {
              \__spintent_spsiglo_tag_span:en
                { \l__spintent_spsiglo_luaset_arabic_str ~ después ~ de ~ cristo }
                { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } ~ d.~C. }
            }
```

```
          }
      }
  }
```

(*End of definition for* __spintent_spsiglo_print_math:n *and* __spintent_spsiglo_print_text:n*.*)

\spsiglo

The public command \spsiglo is the main interface for century typesetting. It opens a local group and evaluates any optional ⟨*keys*⟩ passed in '#1', then executes the function __spintent_luafun_spsiglo_parse:n defined in the Lua module spintent.lua passed in '#2' and formats the output for *math mode* or *text mode*.

```
\NewDocumentCommand \spsiglo { O{} m }
  {
    \group_begin:
      \keys_set:nn { spintent / spsiglo } { #1 }
      \__spintent_luafun_spsiglo_parse:n { #2 }
      \mode_if_math:TF
        { \__spintent_spsiglo_print_math:n { #2 } }
        {
          \mode_leave_vertical:
          \__spintent_spsiglo_print_text:n { #2 }
        }
    \group_end:
  }
```

(*End of definition for* \spsiglo*. This function is documented on page* 11*.*)

11.20 The command \sptime

The command \sptime is the main public interface for typesetting and tagging time expressions. It parses time string formats (such as HH:MM or HH:MM:SS) using the Lua function __spintent_luafun_sptime_parse:n. This separates the hours, minutes, seconds, and fractions to construct a MathML intent tree for NVDA/MathCAT.

\l__spintent_sptime_luaset_seconds_str
\l__spintent_sptime_luaset_has_seconds_str
\l__spintent_sptime_luaset_error_str
\l__spintent_sptime_luaset_is_fraction_str
\l__spintent_sptime_luaset_base_str

These internal string variables store the values returned from the Lua module spintent.lua. They hold the extracted time units, error flags, and boolean indicators (such as the presence of seconds or decimal fractions) used during formatting.

```
\str_new:N \l__spintent_sptime_luaset_seconds_str
\str_new:N \l__spintent_sptime_luaset_has_seconds_str
\str_new:N \l__spintent_sptime_luaset_error_str
\str_new:N \l__spintent_sptime_luaset_is_fraction_str
\str_new:N \l__spintent_sptime_luaset_base_str
```

(*End of definition for* \l__spintent_sptime_luaset_seconds_str *and others.*)

11.20.1 Definition of error messages

invalid-time-format

The error message invalid-time-format defines the text displayed when an invalid time input is detected. It is triggered when an argument fails the format checks or contains values outside normal ranges, such as hours exceeding 23 or minutes and seconds exceeding 59.

```
\msg_new:nnnn { spintent } { invalid-time-format }
  {
    Invalid~time~format~in~#1. \\
    The~argument~'#2'~does~not~match~any~supported~time~format.
  }
  {
    Allowed~formats:~HH:MM,~HH:MM:SS,~or~HH:MM:SS.ss. \\
    Please~verify~that~hours~are~0-23~and~minutes/seconds~are~0-59.
  }
```

(*End of definition for* invalid-time-format*.*)

11.20.2 Internal functions for \sptime

__spintent_sptime_build_seconds:nn

The internal function __spintent_sptime_build_seconds:nn builds the MathML structure for the seconds component of a time expression. It wraps the numeric value '#2' in speech intent tags, applying a connector prefix (passed in '#1') and the "segundos" postfix so it is read correctly by NVDA/MathCAT.

```
\cs_new_protected:Nn \__spintent_sptime_build_seconds:nn
  {
    \MathMLintent { #1:prefix($z) }
      {
        \text { : }
        \MathMLarg { z }
          {
            \MathMLintent { segundos:postfix($m) }
              {
```

```
                    \__spintent_invisible_sep:
                    \MathMLarg { m }
                      {
                        \text { #2 }
                      }
                  }
              }
          }
  }
```

(*End of definition for* `\__spintent_sptime_build_seconds:nn`.)

`\__spintent_sptime_build_tree:`

The internal function `\__spintent_sptime_build_tree:` builds the complete `MathML` structure for the time expression. It checks the string variables `\l__spintent_sptime_luaset_has_seconds_str` and `\l__spintent_sptime_luaset_is_fraction_str` to determine if the time includes seconds or decimal fractions. Based on these values, it inserts the base time string and either calls `\__spintent_sptime_build_seconds:` with the appropriate prefix ('con' or 'minutos-con'), or simply appends the 'minutos' intent tag if no seconds are present.

```
\cs_new_protected:Nn \__spintent_sptime_build_tree:
  {
    \str_if_eq:VnTF \l__spintent_sptime_luaset_has_seconds_str { true }
      {
        \MathMLintent { :time }
          {
            \text { \l__spintent_sptime_luaset_base_str }
          }
        \str_if_eq:VnTF \l__spintent_sptime_luaset_is_fraction_str { true }
          {
            \__spintent_sptime_build_seconds:nn { con }
              { \l__spintent_sptime_luaset_seconds_str }
          }
          {
            \__spintent_sptime_build_seconds:nn { minutos-con }
              { \l__spintent_sptime_luaset_seconds_str }
          }
      }
      {
        \MathMLintent { :time }
          {
            \text { \l__spintent_sptime_luaset_base_str }
          }
        \str_if_eq:VnT \l__spintent_sptime_luaset_is_fraction_str { false }
          {
            \MathMLintent { minutos }
              {
                \__spintent_invisible_sep:
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_sptime_build_tree:`.)

`\sptime`

The command `\sptime` is the public interface for formatting time expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it parses the input using `\__spintent_luafun_sptime_parse:n` and checks the error status in `\l__spintent_sptime_luaset_error_str`. If an invalid format is detected, it issues the `invalid-time-format` error; otherwise, it calls `\__spintent_sptime_build_tree:` to assemble the `MathML` output.

```
\NewDocumentCommand \sptime { m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_luafun_sptime_parse:n { #1 }
          \str_if_eq:VnTF \l__spintent_sptime_luaset_error_str { true }
            {
               \msg_error:nnnn { spintent } { invalid-time-format }
                 { \sptime } { #1 }
            }
            {
               \__spintent_sptime_build_tree:
```

```
          }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \sptime } }
  }
```

(*End of definition for* `\sptime`. *This function is documented on page 11.*)

11.21 The command \spdate

The command `\spdate` is the main public command for typesetting and tagging date expressions. It parses date string formats (such as DD/MM/YYYY or YYYY-MM-DD) using the Lua function `\__spintent_luafun_spdate_parse:n`. This verifies the input and constructs a MathML intent tree for NVDA/MathCAT.

`\l__spintent_spdate_luaset_output_str`
`\l__spintent_spdate_luaset_error_str`

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the formatted date string and the error status used during formatting.

```
\str_new:N \l__spintent_spdate_luaset_output_str
\str_new:N \l__spintent_spdate_luaset_error_str
```

(*End of definition for* `\l__spintent_spdate_luaset_output_str` *and* `\l__spintent_spdate_luaset_error_str`.)

11.21.1 Definition of error messages

`invalid-date-format`

The error message `invalid-date-format` defines the text displayed when an invalid date input is detected. It is triggered when an argument fails the format checks or represents an invalid calendar date, such as 31/02/2024 or using an unsupported pattern like YYYY/MM/DD.

```
\msg_new:nnnn { spintent } { invalid-date-format }
  {
    Invalid~date~or~unsupported~format~in~#1. \\
    The~argument~'#2'~does~not~match~a~supported~format.
  }
  {
    Allowed~formats:~DD\slash MM\slash YYYY,~DD-MM-YYYY,~or~YYYY-MM-DD. \\
    Ensure~the~day~and~month~are~valid~(e.g.,~avoid~31\slash 02\slash 2024).
  }
```

(*End of definition for* `invalid-date-format`.)

11.21.2 Internal functions for \spdate

`\__spintent_spdate_process:n`

The internal function `\__spintent_spdate_process:n` parses the input using `\__spintent_luafun_spdate_parse:n` and checks the error status in `\l__spintent_spdate_luaset_error_str`. If an invalid format is detected, it issues the `invalid-date-format` error. Otherwise, it builds the MathML intent structure using the formatted date stored in `\l__spintent_spdate_luaset_output_str`.

```
\cs_new_protected:Nn \__spintent_spdate_process:n
  {
    \__spintent_luafun_spdate_parse:n { #1 }
    \str_if_eq:VnTF \l__spintent_spdate_luaset_error_str { true }
      {
        \msg_error:nnnn { spintent } { invalid-date-format }
          { \spdate } { #1 }
      }
      {
        \MathMLintent { :date }
          {
            \text { \l__spintent_spdate_luaset_output_str }
          }
      }
  }
```

(*End of definition for* `\__spintent_spdate_process:n`.)

`\spdate`

The command `\spdate` is the public interface for formatting date expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it calls `\__spintent_spdate_process:n` to process the input.

```
\NewDocumentCommand \spdate { m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spdate_process:n { #1 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spdate } }
  }
```

(*End of definition for* `\spdate`. *This function is documented on page* 11.)

11.22 Semantic Operators

This section provides purely semantic operators. Unlike their formatting counterparts (which parse and format numbers), these commands do not process mandatory arguments. Instead, they solely inject a mathematical glyph wrapped in its corresponding regional `MathCAT` intent. This grants teachers absolute control when building complex algebraic expressions, mixed numbers, or fraction divisions.

11.23 The command \spusep

For the use of parentheses, we will provide the command `\spusep` with only two ⟨*keys*⟩ `size` and `silent`. Most of the time we can use `\left` and `\right` or `\Uleft` and `\Uright`, but both forms create a group, and this hinders the code for *tagged* `PDF` when we want to interrupt something between them.

`spusep-empty-arg`
`spusep-invalid-arg`

The first "error" message `spusep-empty-arg` is raised when `\spusep` receives an empty or blank {⟨*argument*⟩}, the second this is when an input is invalid.

```
\msg_new:nnnn { spintent } { spusep-empty-arg }
  { Empty~argument~in~\token_to_str:N \spusep. }
  { \token_to_str:N \spusep~requires~a~non-empty~argument. }
\msg_new:nnn { spintent } { spusep-invalid-arg }
  {
    Invalid~argument~'#1'~for~\token_to_str:N \spusep.
  }
```

(*End of definition for* `spusep-empty-arg` *and* `spusep-invalid-arg`.)

`\l__spintent_spusep_arg_tl`
`\l__spintent_spusep_left_tl`
`\l__spintent_spusep_right_tl`

The variable `\l__spintent_spusep_arg_tl` stores the {⟨*argument*⟩}, the variable `\l__spintent_spusep_left_tl` and `\l__spintent_spusep_right_tl` stores the parentheses.

```
\tl_new:N \l__spintent_spusep_arg_tl
\tl_new:N \l__spintent_spusep_left_tl
\tl_new:N \l__spintent_spusep_right_tl
```

(*End of definition for* `\l__spintent_spusep_arg_tl`, `\l__spintent_spusep_left_tl`, *and* `\l__spintent_spusep_right_tl`.)

`size`
`silent`
`unknown`

We define the ⟨*keys*⟩ `size` and `silent`.

```
\keys_define:nn { spintent / spusep }
  {
    size           .choices:nn =
      { none , big , Big, bigg, Bigg }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            { 1 }
              {
                \tl_clear:N \l__spintent_spusep_left_tl
                \tl_clear:N \l__spintent_spusep_right_tl
              }
            { 2 }
              {
                \tl_set:Nn \l__spintent_spusep_left_tl  { \bigl }
                \tl_set:Nn \l__spintent_spusep_right_tl { \bigr }
              }
            { 3 }
              {
                \tl_set:Nn \l__spintent_spusep_left_tl  { \Bigl }
                \tl_set:Nn \l__spintent_spusep_right_tl { \Bigr }
              }
            { 4 }
              {
                \tl_set:Nn \l__spintent_spusep_left_tl  { \biggl }
                \tl_set:Nn \l__spintent_spusep_right_tl { \biggr }
              }
            { 5 }
              {
                \tl_set:Nn \l__spintent_spusep_left_tl  { \Biggl }
                \tl_set:Nn \l__spintent_spusep_right_tl { \Biggr }
              }
          }
      },
    size           .initial:n  = none,
    size           .value_required:n = true,
    size / unknown .code:n =
```

```
      \msg_error:nneee { spintent } { unknown-choice }
        { size } { none,~big,~Big,~bigg,~Bigg } { \exp_not:n {#1} },
    silent          .bool_set:N = \l__spintent_spusep_silent_bool,
    silent          .initial:n = false,
    silent          .value_required:n = true,
    unknown         .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spusep }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `size` , `silent` , *and* `unknown`.)

`\__spintent_spusep_set_parens:nn`

The function `\__spintent_spusep_set_parens:nn` will place the argument passed in '`#2`' to the right of the variables `\l__spintent_spusep_left_tl` and `\l__spintent_spusep_right_tl` and then check the state of the variable `\l__spintent_spusep_silent_bool` handled by the key `silent` to print.

```
\cs_new_protected:Nn \__spintent_spusep_set_parens:nn
  {
    \tl_put_right:cn { l__spintent_spusep_ #1 _tl } { #2 }
    \bool_if:NTF \l__spintent_spusep_silent_bool
      {
         \MathMLintent { :silent } { \tl_use:c { l__spintent_spusep_ #1 _tl } }
      }
      {
        \tl_use:c { l__spintent_spusep_ #1 _tl }
      }
  }
```

(*End of definition for* `\__spintent_spusep_set_parens:nn`.)

`\__spintent_spusep_convert:n`

The function `\__spintent_spusep_convert:n` captures and stores the {⟨*argument*⟩} passed in '`#1`' in the variable `\l__spintent_spusep_arg_tl`, validates the cases by converting them to the macros provided by `unicode-math`, and passes them to the function `\__spintent_spusep_set_parens:nn` for printing. If the input is not supported or is empty, we return an "error".

```
\cs_new_protected:Nn \__spintent_spusep_convert:n
  {
    \tl_set:Ne \l__spintent_spusep_arg_tl { \tl_trim_spaces:n {#1} }
    \tl_if_empty:NT \l__spintent_spusep_arg_tl
      {
        \msg_error:nnn { spintent } { spusep-invalid-arg } {#1}
      }
    \str_case:VnF \l__spintent_spusep_arg_tl
      {
        { (  }
          {
            \__spintent_spusep_set_parens:nn { left } { \lparen }
          }
        { )  }
          {
            \__spintent_spusep_set_parens:nn { right } { \rparen }
          }
        { [  }
          {
            \__spintent_spusep_set_parens:nn { left } { \lbrack }
          }
        { ]  }
          {
            \__spintent_spusep_set_parens:nn { right } { \rbrack }
          }
        { \{ }
          {
            \__spintent_spusep_set_parens:nn { left } { \lbrace }
          }
        { \} }
          {
            \__spintent_spusep_set_parens:nn { right } { \rbrace }
          }
      }
      {
        \msg_error:nnn { spintent } { spusep-invalid-arg } {#1}
```

```
        }
    }
```

(*End of definition for* `\__spintent_spusep_convert:n`.)

`\spusep` The command `\spusep` checks that it is running in math mode, opens a local group, sets the keys passed in '`#1`' and processes the {⟨*argument*⟩} passed in '`#2`' by means of the function `\__spintent_spusep_convert:n`.

```
\NewDocumentCommand \spusep { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \keys_set:nn { spintent / spusep } {#1}
          \__spintent_spusep_convert:n {#2}
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spusep } }
  }
```

(*End of definition for* `\spusep`. *This function is documented on page 12.*)

11.23.1 The command `\spread`

The command `\spread` provides a semantic read for `MathCAT` of mathematical symbols for situations in which it is necessary to specify gender or plurality in Spanish.

`spread-empty-args` The error message `spread-empty-args` is raised when `\spread` receives an empty or blank {⟨*argument*⟩}.

```
\msg_new:nnnn { spintent } { spread-empty-args }
  { Empty~argument~in~\token_to_str:N \spread. }
  { \token_to_str:N \spread~requires~two~non-empty~arguments. }
```

(*End of definition for* `spread-empty-args`.)

`\l__spintent_spread_intent_tl` The variable `\l__spintent_spread_intent_tl` stores the sanitized semantic reading {⟨*argument*⟩} for symbol to be injected into the `\MathMLintent`.

```
\tl_new:N \l__spintent_spread_intent_tl
```

(*End of definition for* `\l__spintent_spread_intent_tl`.)

`\__spintent_spread_exec:nn` The function `\__spintent_spread_exec:nn` implements the core logic for `\spread`. It verifies that neither {⟨*argument*⟩} is empty or blank. If valid, it evaluates the *reading* {⟨*argument*⟩} '`#1`' using `\__spintent_sanitize_affix:Nn` and embeds it via `\MathMLintent` into the *math symbol* '`#2`'.

```
\cs_new_protected:Nn \__spintent_spread_exec:nn
  {
    \bool_lazy_or:nnTF { \tl_if_blank_p:n { #1 } } { \tl_if_blank_p:n { #2 } }
      { \msg_error:nn { spintent } { spread-empty-args } }
      {
        \__spintent_sanitize_affix:Nn \l__spintent_spread_intent_tl { #1 }
        \MathMLintent { \l__spintent_spread_intent_tl } { #2 }
      }
  }
```

(*End of definition for* `\__spintent_spread_exec:nn`.)

`\spread` The command `\spread` assigns a custom semantic *reading* '`#1`' to any mathematical *symbol* '`#2`'. It enforces math mode and passes execution to `\__spintent_spread_exec:nn`.

```
\NewDocumentCommand \spread { m m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spread_exec:nn { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spread } }
  }
```

(*End of definition for* `\spread`. *This function is documented on page 11.*)

11.23.2 The command \spdsym

\l__spintent_spdsym_read_sym_tl
\l__spintent_spdsym_symbol_tl
\l__spintent_spdsym_prefix_tl
\l__spintent_spdsym_suffix_tl

Variables to store the regional reading string, the visual operator, and the optional semantic affixes specifically for \spdsym.

```
\tl_new:N \l__spintent_spdsym_read_sym_tl
\tl_new:N \l__spintent_spdsym_symbol_tl
\tl_new:N \l__spintent_spdsym_prefix_tl
\tl_new:N \l__spintent_spdsym_suffix_tl
```

(*End of definition for* \l__spintent_spdsym_read_sym_tl *and others.*)

prefix
suffix
set-sym
read-sym
unknown

We define the ⟨*keys*⟩ prefix, suffix, set-sym and read-sym for \spdsym.

```
\keys_define:nn { spintent / spdsym }
  {
    prefix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spdsym_prefix_tl {#1} },
    prefix              .initial:n = ,
    prefix              .value_required:n = true,
    suffix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spdsym_suffix_tl {#1} },
    suffix              .initial:n = ,
    suffix              .value_required:n = true,
    set-sym             .choices:nn =
      { old-style , two-dots , slash }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \tl_set:Nn \l__spintent_spdsym_symbol_tl { \div } }
            {2} { \tl_set:Nn \l__spintent_spdsym_symbol_tl {   :  } }
            {3} { \tl_set:Nn \l__spintent_spdsym_symbol_tl {   /  } }
          }
      },
    set-sym             .initial:n  = two-dots,
    set-sym             .value_required:n = true,
    set-sym / unknown   .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { set-sym } { old-style, ~two-dots, ~slash } { \exp_not:n {#1} },
    read-sym            .choices:nn =
      { dividido , dividido-por , dividido-entre }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido       } }
            {2} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido-por   } }
            {3} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido-entre } }
          }
      },
    read-sym            .initial:n  = dividido,
    read-sym            .value_required:n = true,
    read-sym / unknown  .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-sym } { dividido, ~dividido-por, ~dividido-entre }
        { \exp_not:n {#1} },
    unknown             .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spdsym }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* prefix *and others.*)

\spdsym

Defines the \spdsym operator. It takes a single optional argument for local setup and outputs the configured intent structure, safely attaching prefixes and suffixes if present.

```
\NewDocumentCommand \spdsym { O{} }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \keys_set:nn { spintent / spdsym } { #1 }
          \MathMLintent
            {
```

```
              \tl_if_empty:NF \l__spintent_spdsym_prefix_tl
                { \l__spintent_spdsym_prefix_tl - }
              \l__spintent_spdsym_read_sym_tl
              \tl_if_empty:NF \l__spintent_spdsym_suffix_tl
                { - \l__spintent_spdsym_suffix_tl }
            }
            { \l__spintent_spdsym_symbol_tl }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spdsym } }
  }
```

(*End of definition for* \spdsym. *This function is documented on page* 12.)

11.23.3 The command \spmsym

\l__spintent_spmsym_read_sym_tl
\l__spintent_spmsym_symbol_tl
\l__spintent_spmsym_prefix_tl
\l__spintent_spmsym_suffix_tl

Variables to store the regional reading string, the visual operator, and the optional semantic affixes specifically for \spmsym.

```
\tl_new:N \l__spintent_spmsym_read_sym_tl
\tl_new:N \l__spintent_spmsym_symbol_tl
\tl_new:N \l__spintent_spmsym_prefix_tl
\tl_new:N \l__spintent_spmsym_suffix_tl
```

(*End of definition for* \l__spintent_spmsym_read_sym_tl *and others.*)

prefix
suffix
set-sym
read-sym
not-print
unknown

We define the ⟨*keys*⟩ prefix, suffix, set-sym, read-sym and not-print for \spmsym.

```
\keys_define:nn { spintent / spmsym }
  {
    prefix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spmsym_prefix_tl {#1} },
    prefix              .initial:n = ,
    prefix              .value_required:n = true,
    suffix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spmsym_suffix_tl {#1} },
    suffix              .initial:n = ,
    suffix              .value_required:n = true,
    set-sym             .choices:nn =
      { cross , dot , asterisk }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \times } }
            {2} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \cdot  } }
            {3} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \ast   } }
          }
      },
    set-sym             .initial:n  = dot,
    set-sym             .value_required:n = true,
    set-sym / unknown   .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { set-sym } { cross, ~dot, ~asterisk } { \exp_not:n {#1} },
    read-sym            .choices:nn =
      { por , multiplicado-por , veces }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { por } }
            {2} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { multiplicado-por } }
            {3} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { veces } }
          }
      },
    read-sym            .initial:n  = por,
    read-sym            .value_required:n = true,
    read-sym / unknown  .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-sym } { por, ~multiplicado-por, ~veces } { \exp_not:n {#1} },
    not-print           .bool_set:N = \l__spintent_spmsym_not_print_bool,
    not-print           .initial:n  = false,
    not-print           .value_required:n = true,
    unknown             .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spmsym }
        \__spintent_unknown_keys_cmd:n {#1}
```

```
    },
  }
```

(*End of definition for* `prefix` *and others.*)

`\spmsym` Defines the `\spmsym` operator. Similar to `\spdsym`, it outputs the correct mathematical glyph based on the local ⟨*keys*⟩, safely attaching prefixes and suffixes if present.

```
\NewDocumentCommand \spmsym { O{} }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \keys_set:nn { spintent / spmsym } { #1 }
          \MathMLintent
            {
              \tl_if_empty:NF \l__spintent_spmsym_prefix_tl
                { \l__spintent_spmsym_prefix_tl - }
              \l__spintent_spmsym_read_sym_tl
              \tl_if_empty:NF \l__spintent_spmsym_suffix_tl
                { - \l__spintent_spmsym_suffix_tl }
            }
            {
              \bool_if:NTF \l__spintent_spmsym_not_print_bool
                {
                  \invisibletimes
                }
                { \l__spintent_spmsym_symbol_tl }
            }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmsym } }
  }
```

(*End of definition for* `\spmsym`*. This function is documented on page* 12*.*)

11.24 The command `\spdiv`

The user-level command `\spdiv` formats division representations. It takes two mandatory arguments: the {⟨*dividend*⟩} and the {⟨*divisor*⟩}. Both {⟨*arguments*⟩} are internally processed by the `\spnum` command to ensure proper digit separation. This command enforces strict validation, ensuring that neither the {⟨*dividend*⟩} nor the {⟨*divisor*⟩} is *blank*.

11.24.1 Definition of keys for `\spdiv`

`cifras-sep` `read-sign` `read-period-sep` `notation-sym` `set-sym` `read-sym` `wrap-parens` `read-parens`

We define the ⟨*keys*⟩ `wrap-parens` and `read-parens` for `\spdiv` command. Number-related ⟨*keys*⟩ `cifras-sep`, `read-sign`, `read-period-sep` and `notation-sym` are forwarded to `\spnum` command using `.meta:nn`, while semantic operator ⟨*keys*⟩ `set-sym` and `read-sym` are forwarded to the `\spdsym` command using `.meta:nn`.

```
\keys_define:nn { spintent / spdiv }
  {
    cifras-sep      .meta:nn = { spintent / spnum   } { cifras-sep      = {#1} },
    read-sign       .meta:nn = { spintent / spnum   } { read-sign       = {#1} },
    read-period-sep .meta:nn = { spintent / spnum   } { read-period-sep = {#1} },
    notation-sym    .meta:nn = { spintent / spnum   } { notation-sym    = {#1} },
    set-sym         .meta:nn = { spintent / spdsym } { set-sym         = {#1} },
    read-sym        .meta:nn = { spintent / spdsym } { read-sym        = {#1} },
    wrap-parens     .bool_set:N = \l__spintent_spdiv_wrap_parens_bool,
    wrap-parens     .initial:n  = false,
    wrap-parens     .value_required:n = true,
    read-parens     .bool_set:N = \l__spintent_spdiv_read_parens_bool,
    read-parens     .initial:n  = true,
    read-parens     .value_required:n = true,
    unknown         .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spdiv }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `cifras-sep` *and others.*)

11.24.2 Definition of error messages

spdiv-empty-argument

Raised when either the {⟨*dividend*⟩} or the {⟨*divisor*⟩} is missing or *blank*.

```
\msg_new:nnnn { spintent } { spdiv-empty-argument }
  { The~arguments~for~\token_to_str:N \spdiv~cannot~be~empty. }
  {
    You~must~provide~both~a~dividend~and~a~divisor.\\
  }
```

(*End of definition for* spdiv-empty-argument*.*)

11.24.3 Internal functions for \spdiv

__spintent_spdiv_render_op:nn

The function __spintent_spdiv_render_op:nn parses both {⟨*arguments*⟩} through the \spnum command and embeds the semantic operator from \spdsym.

```
\cs_new_protected:Nn \__spintent_spdiv_render_op:nn
  {
    \spnum {#1}
    \MathMLintent { \l__spintent_spdsym_read_sym_tl }
      {
        \l__spintent_spdsym_symbol_tl
      }
    \spnum {#2}
  }
```

(*End of definition for* __spintent_spdiv_render_op:nn*.*)

__spintent_spdiv_render_parens:nn

The function __spintent_spdiv_render_parens:nn handles the structural parentheses around the division.

```
\cs_new_protected:Nn \__spintent_spdiv_render_parens:nn
  {
    \bool_if:NTF \l__spintent_spdiv_read_parens_bool
      {
        ( \__spintent_spdiv_render_op:nn {#1} {#2} )
      }
      {
        \MathMLintent { :silent } { ( }
        \__spintent_spdiv_render_op:nn {#1} {#2}
        \MathMLintent { :silent } { ) }
      }
  }
```

(*End of definition for* __spintent_spdiv_render_parens:nn*.*)

__spintent_spdiv_process:nnn

The function __spintent_spdiv_process:nnn sets ⟨*keys*⟩, validates {⟨*arguments*⟩}, and decides on parenthesis wrapping.

```
\cs_new_protected:Nn \__spintent_spdiv_process:nnn
  {
    \keys_set:nn { spintent / spdiv } { #1 }
    \tl_if_blank:nTF { #2 }
      { \msg_error:nnn { spintent } { spdiv-empty-argument } }
      {
        \tl_if_blank:nTF { #3 }
          { \msg_error:nnn { spintent } { spdiv-empty-argument } }
          {
            \bool_if:NTF \l__spintent_spdiv_wrap_parens_bool
              { \__spintent_spdiv_render_parens:nn {#2} {#3} }
              { \__spintent_spdiv_render_op:nn {#2} {#3} }
          }
      }
  }
```

(*End of definition for* __spintent_spdiv_process:nnn*.*)

\spdiv

Defines the user command \spdiv. It strictly relies on *math mode*.

```
\NewDocumentCommand \spdiv { O{} m m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spdiv_process:nnn { #1 } { #2 } { #3 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spdiv } }
  }
```

(*End of definition for* `\spdiv`. *This function is documented on page* 13.)

11.25 The command `\spmul`

The user-level command `\spmul` formats multiplication representations. It takes two mandatory arguments: the {⟨*first factor*⟩} and the {⟨*second factor*⟩}. Both {⟨*arguments*⟩} are internally processed by the `\spnum` command to ensure proper digit separation. This command enforces strict validation, ensuring that neither the {⟨*first factor*⟩} nor the {⟨*second factor*⟩} is *blank*.

11.25.1 Definition of keys for `\spmul`

cifras-sep
read-sign
read-period-sep
notation-sym
set-sym
read-sym
wrap-parens
read-parens

We define the ⟨*keys*⟩ `wrap-parens` and `read-parens` for `\spmul` command. Number-related ⟨*keys*⟩ `cifras-sep`, `read-sign`, `read-period-sep` and `notation-sym` are forwarded to `\spnum` command using `.meta:nn`, while semantic operator ⟨*keys*⟩ `set-sym` and `read-sym` are forwarded to the `\spmsym` command using `.meta:nn`.

```
\keys_define:nn { spintent / spmul }
  {
    cifras-sep      .meta:nn = { spintent / spnum  } { cifras-sep      = {#1} },
    read-sign       .meta:nn = { spintent / spnum  } { read-sign       = {#1} },
    read-period-sep .meta:nn = { spintent / spnum  } { read-period-sep = {#1} },
    notation-sym    .meta:nn = { spintent / spnum  } { notation-sym    = {#1} },
    set-sym         .meta:nn = { spintent / spmsym } { set-sym         = {#1} },
    read-sym        .meta:nn = { spintent / spmsym } { read-sym        = {#1} },
    wrap-parens     .bool_set:N = \l__spintent_spmul_wrap_parens_bool,
    wrap-parens     .initial:n  = false,
    wrap-parens     .value_required:n = true,
    read-parens     .bool_set:N = \l__spintent_spmul_read_parens_bool,
    read-parens     .initial:n  = true,
    read-parens     .value_required:n = true,
    unknown         .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spmul }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `cifras-sep` *and others.*)

11.25.2 Definition of error messages

spmul-empty-argument

Raised when either the {⟨*first factor*⟩} or the {⟨*second factor*⟩} is missing or *blank*.

```
\msg_new:nnnn { spintent } { spmul-empty-argument }
  { The~arguments~for~\token_to_str:N \spmul~cannot~be~empty. }
  {
    You~must~provide~both~first~factor~and~second~factor.\\
  }
```

(*End of definition for* `spmul-empty-argument`.)

11.25.3 Internal functions for `\spmul`

__spintent_spmul_render_op:nn

The function `\__spintent_spmul_render_op:nn` parses both {⟨*arguments*⟩} through the `\spnum` command and embeds the semantic operator from `\spmsym`.

```
\cs_new_protected:Nn \__spintent_spmul_render_op:nn
  {
    \spnum {#1}
    \MathMLintent { \l__spintent_spmsym_read_sym_tl }
      {
        \l__spintent_spmsym_symbol_tl
      }
    \spnum {#2}
  }
```

(*End of definition for* `\__spintent_spmul_render_op:nn`.)

__spintent_spmul_render_parens:nn

The function `\__spintent_spmul_render_parens:nn` handles the structural parentheses around the multiplication.

```
\cs_new_protected:Nn \__spintent_spmul_render_parens:nn
  {
    \bool_if:NTF \l__spintent_spmul_read_parens_bool
      {
        ( \__spintent_spmul_render_op:nn {#1} {#2} )
      }
      {
        \MathMLintent { :silent } { ( }
```

```
        \__spintent_spmul_render_op:nn {#1} {#2}
        \MathMLintent { :silent } { ) }
      }
  }
```

(*End of definition for* `\__spintent_spmul_render_parens:nn`.)

`\__spintent_spmul_process:nnn` The function `\__spintent_spmul_process:nnn` sets ⟨*keys*⟩, validates {⟨*arguments*⟩}, and decides on parenthesis wrapping.

```
\cs_new_protected:Nn \__spintent_spmul_process:nnn
  {
    \keys_set:nn { spintent / spmul } { #1 }
    \tl_if_blank:nTF { #2 }
      { \msg_error:nnn { spintent } { spmul-empty-argument } }
      {
        \tl_if_blank:nTF { #3 }
          { \msg_error:nnn { spintent } { spmul-empty-argument } }
          {
            \bool_if:NTF \l__spintent_spmul_wrap_parens_bool
              { \__spintent_spmul_render_parens:nn {#2} {#3} }
              { \__spintent_spmul_render_op:nn {#2} {#3} }
          }
      }
  }
```

(*End of definition for* `\__spintent_spmul_process:nnn`.)

`\spmul` Defines the user command `\spmul`. It strictly relies on *math mode*.

```
\NewDocumentCommand \spmul { O{} m m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spmul_process:nnn { #1 } { #2 } { #3 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmul } }
  }
```

(*End of definition for* `\spmul`. *This function is documented on page* 13.)

11.26 The commands \spmcd and \spmcm

The user-level commands `\spmcd` and `\spmcm` provide a semantic interface for the Greatest Common Divisor (MCD in Spanish) and the Least Common Multiple (MCM in Spanish). They delegate argument validation and calculation to the Lua module `spintent.lua`, constructing a valid `MathCAT` intent.

11.26.1 Variables for \spmcd and \spmcm

`\l__spintent_spmcm_spmcd_luaset_error_str` `\l__spintent_mc_args_seq` `\l__spintent_spmcd_print_result_bool` `\l__spintent_spmcm_print_result_bool` `\l__spintent_spmcd_read_terse_bool` `\l__spintent_spmcm_read_terse_bool` `\l__spintent_spmcd_print_upper_bool` `\l__spintent_spmcm_print_upper_bool` `\l__spintent_spmcd_prefix_tl` `\l__spintent_spmcm_prefix_tl` `\l__spintent_spmcd_suffix_tl` `\l__spintent_spmcm_suffix_tl`

Internal variables generated dynamically via a comma-separated list to avoid code duplication for both symmetric commands.

```
\str_new:N \l__spintent_spmcm_spmcd_luaset_error_str
\seq_new:N \l__spintent_mc_args_seq
\clist_map_inline:nn { mcd , mcm }
  {
    \tl_new:c   { l__spintent_sp #1 _luaset_ #1 _value_tl }
    \bool_new:c { l__spintent_sp #1 _print_result_bool }
    \bool_new:c { l__spintent_sp #1 _read_terse_bool }
    \bool_new:c { l__spintent_sp #1 _print_upper_bool }
    \tl_new:c   { l__spintent_sp #1 _prefix_tl }
    \tl_new:c   { l__spintent_sp #1 _suffix_tl }
  }
```

(*End of definition for* `\l__spintent_spmcm_spmcd_luaset_error_str` *and others.*)

11.26.2 Definition of error messages

`mc-invalid-arguments` Exception triggered when the comma-separated sequence contains negative integers, floats, or alphabetical characters.

```
\msg_new:nnnn { spintent } { mc-invalid-arguments }
  {
    Invalid~arguments~provided~to~#1.
  }
  {
    The~arguments~must~be~a~comma-separated~list. \\
  }
```

(*End of definition for* `mc-invalid-arguments`.)

11.26.3 Keys for \spmcd and \spmcm

result prefix read-cmd upper-case sample-arg silent-cmd

We define the shared ⟨*keys*⟩ `result`, `prefix`, `read-cmd`, `upper-case`, `sample-arg` and `silent-cmd`.

```
\clist_map_inline:nn { mcd , mcm }
  {
    \keys_define:nn { spintent / sp #1 }
      {
        result              .bool_set:c = { l__spintent_sp #1 _print_result_bool },
        result              .initial:n  = false,
        result              .value_required:n = true,
        prefix              .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix              .initial:n = ,
        prefix              .value_required:n = true,
        read-cmd            .choices:nn =
          { terse , clear }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \bool_set_true:c  { l__spintent_sp #1 _read_terse_bool } }
                {2} { \bool_set_false:c { l__spintent_sp #1 _read_terse_bool } }
              }
          },
        read-cmd            .initial:n = clear,
        read-cmd            .value_required:n = true,
        read-cmd / unknown .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-cmd } { terse , ~clear } { \exp_not:n {##1} },
        upper-case          .bool_set:c = { l__spintent_sp #1 _print_upper_bool },
        upper-case          .initial:n  = false,
        upper-case          .value_required:n = true,
        sample-arg          .bool_set:c = { l__spintent_sp #1 _sample_arg_bool },
        sample-arg          .initial:n  = false,
        sample-arg          .value_required:n = true,
        silent-cmd          .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd          .initial:n  = false,
        silent-cmd          .value_required:n = true,
        unknown             .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
            \__spintent_unknown_keys_cmd:n {##1}
          },
      }
  }
```

(*End of definition for* `result` *and others.*)

11.26.4 Internal functions for \spmcd and \spmcm

__spintent_mc_process_args:nn

The function `\__spintent_mc_process_args:nn` parses the operation '`#1`' and its arguments '`#2`', injecting MathML intents.

```
\cs_new_protected:Nn \__spintent_mc_process_args:nn
  {
    \seq_set_from_clist:Nn \l__spintent_mc_args_seq {#2}
    \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
      {
        \seq_set_map:NNn \l__spintent_mc_args_seq \l__spintent_mc_args_seq
          { \MathMLintent { :silent } { ##1 } }
        \MathMLintent { entre:silent } { ( }
        \seq_use:Nnnn \l__spintent_mc_args_seq
          { \MathMLintent { _y:silent } { , } }
          { \MathMLintent { :silent }   { , } }
          { \MathMLintent { _y:silent } { , } }
      }
      {
        \MathMLintent { entre } { ( }
        \seq_use:Nnnn \l__spintent_mc_args_seq
          { \MathMLintent { _y }      { , } }
          { \MathMLintent { :silent } { , } }
          { \MathMLintent { _y }      { , } }
      }
```

```
    \MathMLintent { :silent } { ) }
  }
```

(*End of definition for* `\__spintent_mc_process_args:nn`.)

`\__spintent_spmc_process:nnnn` The internal function `\__spintent_spmc_process:nnnn` serves as a shared central handler for both the `\spmcd` and `\spmcm` commands. It takes the command identifier '#1', its full reading string '#2', the local ⟨*keys*⟩ '#3', and the optional values '#4'. It dynamically resolves variables using the identifier to build the `MathML intent` string (including any prefixes or suffixes) and prints the math operator. If values are provided in '#4', it calculates the result using the Lua backend. If an error occurs, it throws the `mc-invalid-arguments` error; otherwise, it processes the arguments and conditionally prints the calculated result.

```
\cs_new_protected:Nn \__spintent_spmc_process:nnnn
  {
    \keys_set:nn { spintent / sp #1 } { #3 }
    \MathMLintent
      {
        \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
          { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
        \bool_if:cTF { l__spintent_sp #1 _read_terse_bool }
          { #1 } { #2 }
        \bool_if:cT { l__spintent_sp #1 _silent_cmd_bool }
          {
            \tl_if_novalue:nTF {#4}
              { :silent }
              { \bool_if:cT { l__spintent_sp #1 _sample_arg_bool } { :silent } }
          }
      }
      {
        \operatorname
          {
            \bool_if:cTF { l__spintent_sp #1 _print_upper_bool }
              { \text_uppercase:n {#1} } { #1 }
          }
      }
    \tl_if_novalue:nF { #4 }
      {
        \bool_if:cTF { l__spintent_sp #1 _sample_arg_bool }
          {
            \__spintent_mc_process_args:nn {#1} {#4}
          }
          {
            \use:c { __spintent_luafun_calculate_ #1 :n } { #4 }
            \str_if_eq:VnTF \l__spintent_spmcm_spmcd_luaset_error_str { true }
              {
                \msg_error:nne { spintent } { mc-invalid-arguments } { \exp_not:c { sp #1 } }
              }
              {
                \__spintent_mc_process_args:nn {#1} {#4}
                \bool_if:cT { l__spintent_sp #1 _print_result_bool }
                  {
                    \MathMLintent { es-igual-a } { = }
                    \tl_use:c { l__spintent_sp #1 _luaset_ #1 _value_tl }
                  }
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_spmc_process:nnnn`.)

`\spmcd` `\spmcm` The commands `\spmcd` and `\spmcm` act as the public interface wrappers for typesetting the greatest common divisor and least common multiple. They enforce math mode, open a local group, and defer execution to the main processing macro `\__spintent_spmc_process:nnnn`, passing along the appropriate reading strings and arguments.

```
\NewDocumentCommand \spmcd { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spmc_process:nnnn { mcd } { máximo-común-divisor } {#1} {#2}
        \group_end:
```

```
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmcd } }
  }
\NewDocumentCommand \spmcm { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spmc_process:nnnn { mcm } { mínimo-común-múltiplo } {#1} {#2}
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmcm } }
  }
```

(*End of definition for* `\spmcd` *and* `\spmcm`*. These functions are documented on page 15.*)

11.27 The commands `\spnM` and `\spnD`

The user-level commands `\spnM` and `\spnD` provide a semantic interface for the Multiples of a number (*"múltiplos"* in Spanish) and the Divisors of a number (*"divisores"* in Spanish). They delegate argument validation and calculation to the Lua module `spintent.lua`, constructing a valid `MathCAT` intent.

11.27.1 Variables for `\spnM` and `\spnD`

`\l__spintent_md_out_seq`
`\l__spintent_spnM_luaset_result_tl`
`\l__spintent_spnD_luaset_result_tl`
`\l__spintent_spnM_luaset_is_truncated_str`
`\l__spintent_spnD_luaset_is_truncated_str`
`\l__spintent_spnM_print_result_bool`
`\l__spintent_spnD_print_result_bool`
`\l__spintent_spnM_silent_cmd_bool`
`\l__spintent_spnD_silent_cmd_bool`
`\l__spintent_spnM_prefix_tl`
`\l__spintent_spnD_prefix_tl`
`\l__spintent_spnM_print_result_num_tl`
`\l__spintent_spnD_print_result_num_tl`

Internal variables generated dynamically for multiples and divisors.

```
\seq_new:N \l__spintent_md_out_seq
\clist_map_inline:nn { nM , nD }
  {
    \tl_new:c   { l__spintent_sp #1 _luaset_result_tl        }
    \str_new:c  { l__spintent_sp #1 _luaset_is_truncated_str }
    \bool_new:c { l__spintent_sp #1 _print_result_bool       }
    \bool_new:c { l__spintent_sp #1 _silent_cmd_bool         }
    \tl_new:c   { l__spintent_sp #1 _prefix_tl               }
    \tl_new:c   { l__spintent_sp #1 _print_result_num_tl     }
  }
```

(*End of definition for* `\l__spintent_md_out_seq` *and others.*)

11.27.2 Keys for `\spnM` and `\spnD`

`result`
`silent-cmd`
`result-num`
`prefix`
`sample-arg`

We define the shared ⟨*keys*⟩ `result`, `silent-cmd`, `result-num` and `prefix`.

```
\clist_map_inline:nn { nM , nD }
  {
    \keys_define:nn { spintent / sp #1 }
      {
        result     .bool_set:c = { l__spintent_sp #1 _print_result_bool },
        result     .initial:n  = false,
        result     .value_required:n = true,
        silent-cmd .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd .initial:n  = false,
        silent-cmd .value_required:n = true,
        result-num .tl_set:c   = { l__spintent_sp #1 _print_result_num_tl },
        result-num .initial:n  = ,
        prefix     .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix     .initial:n = ,
        prefix     .value_required:n = true,
        sample-arg .bool_set:c = { l__spintent_sp #1 _sample_arg_bool },
        sample-arg .initial:n  = false,
        sample-arg .value_required:n = true,
        unknown    .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
            \__spintent_unknown_keys_cmd:n {##1}
          },
      }
  }
```

(*End of definition for* `result` *and others.*)

11.27.3 Definition of error messages for \spnM and \spnD

md-invalid-argument

Exception triggered when the argument for multiples or divisors is invalid.

```
\msg_new:nnnn { spintent } { md-invalid-argument }
  { Invalid~argument~provided~to~#1. }
  {
    The~argument~must~be~a~single~positive~integer. \\
  }
```

(*End of definition for* md-invalid-argument.)

11.27.4 Internal functions for \spnM and \spnD

__spintent_spnM_spnD_process:nnnnn

The internal function __spintent_spnM_spnD_process:nnnnn acts as the primary entry point for the multiples and divisors commands. It takes the command identifier '#1', its printed mathematical symbol '#2', its base reading string '#3', the local key-value options '#4', and the optional argument '#5'. Its sole purpose is to evaluate and set the provided keys within the current scope before delegating the core branching and typesetting logic to __spintent_spn_M_and_D_exec:nnnn.

```
\cs_new_protected:Nn \__spintent_spnM_spnD_process:nnnnn
  {
    \keys_set:nn { spintent / sp #1 } { #4 }
    \__spintent_spnM_spnD_exec:nnnn { #1 } { #2 } { #3 } { #5 }
  }
```

(*End of definition for* __spintent_spnM_spnD_process:nnnnn.)

__spintent_spnM_spnD_intent:nn

The internal function __spintent_spnM_spnD_intent:nn resolves the main reading intent string. It takes the command identifier '#1' (such as "nM" or "nD") and the base reading string '#2'. It automatically prepends the localized prefix if defined and handles specific string normalizations, such as ensuring the correct accentuation for "múltiplos".

```
\cs_new:Nn \__spintent_spnM_spnD_intent:nn
  {
    \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
      { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
    \str_case:nnF { #2 }
      { { multiplos } { múltiplos } }
      { #2 }
  }
```

(*End of definition for* __spintent_spnM_spnD_intent:nn.)

__spintent_spnM_spnD_silent:nn

The internal function __spintent_spn_M_and_D_silent:nn prints the atomized notation structure in full silent mode. It takes the mathematical symbol '#1' (such as "D" or "M") and the variable or explicit argument '#2'. Every component is wrapped in a silent intent, and an invisible separator is injected between the parentheses to prevent assistive technologies from applying default contextual reading rules like "de".

```
\cs_new_protected:Nn \__spintent_spnM_spnD_silent:nn
  {
    \MathMLintent { :silent } { \operatorname { #1 } }
    \MathMLintent { :silent } { ( }
    \MathMLintent { :silent } { \__spintent_invisible_sep: }
    \MathMLintent { :silent } { #2 }
    \MathMLintent { :silent } { ) }
  }
```

(*End of definition for* __spintent_spnM_spnD_silent:nn.)

__spintent_spnM_spnD_result:n

The internal function __spintent_spn_M_and_D_result:n handles the calculation and accessible formatting of the computed mathematical set. It takes the command identifier '#1'. It invokes the corresponding Lua backend calculation, maps the resulting clist into a sequence, and formats the final printed output utilizing custom accessibility intents for both separators and conjunctions. It also safely manages truncation indicators.

```
\cs_new_protected:Nn \__spintent_spnM_spnD_result:n
  {
    \use:c { __spintent_luafun_calcular_ #1 :nn }
      { \l__spintent_luaset_part_int_tl }
      { \tl_use:c { l__spintent_sp #1 _print_result_num_tl } }
    \MathMLintent { son } { = \{ }
    \seq_set_from_clist:Ne
      \l__spintent_md_out_seq { \tl_use:c { l__spintent_sp #1 _luaset_result_tl } }
    \seq_use:Nnnn \l__spintent_md_out_seq
      { \MathMLintent { _y }       { , } }
      { \MathMLintent { :silent } { , } }
      { \MathMLintent { _y }       { , } }
```

```
    \str_if_eq:vnT
      { l__spintent_sp #1 _luaset_is_truncated_str } { true }
        {
          \MathMLintent { :silent } { , } ~ \dots
        }
    \MathMLintent { :silent } { \} }
  }
```

(*End of definition for* `\__spintent_spnM_spnD_result:n`.)

`\__spintent_spnM_spnD_exec:nnnn`

The function `\__spintent_spn_M_and_D_exec:nnnn` implements the multiples and divisors commands. It takes the {⟨*argument*⟩} '`#1`' (the command name), the {⟨*argument*⟩} '`#2`' (the mathematical symbol), the base reading {⟨*argument*⟩} '`#3`', and the optional {⟨*argument*⟩} '`#4`'. If no {⟨*argument*⟩} is provided, it evaluates the key `silent-cmd` using the default token 'n'. If an {⟨*argument*⟩} is given, it evaluates the key `sample-arg`. If true, it processes the {⟨*argument*⟩} directly and evaluates the key `silent-cmd`. If false, it bypasses the *silent mode*, validates the {⟨*argument*⟩} as a natural number, and evaluates the key `result` optionally calculate and append the resulting set.

```
\cs_new_protected:Nn \__spintent_spnM_spnD_exec:nnnn
  {
    \tl_if_novalue:nTF { #4 }
      {
        \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
          { \__spintent_spnM_spnD_silent:nn { #2 } { n } }
          {
            \MathMLintent
              { \__spintent_spnM_spnD_intent:nn { #1 } { #3 } }
              { \operatorname { #2 } ( n ) }
          }
      }
      {
        \bool_if:cTF { l__spintent_sp #1 _sample_arg_bool }
          {
            \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
              { \__spintent_spnM_spnD_silent:nn { #2 } { #4 } }
              {
                \MathMLintent
                  { \__spintent_spnM_spnD_intent:nn { #1 } { #3 } }
                  { \operatorname { #2 } } ( #4 )
              }
          }
          {
            \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #4 } }
            \str_if_eq:VnTF \l__spintent_luaset_has_natural_str { true }
              {
                \MathMLintent
                  { \__spintent_spnM_spnD_intent:nn { #1 } { #3 } }
                  { \operatorname { #2 } } ( #4 )
                \bool_if:cT { l__spintent_sp #1 _print_result_bool }
                  { \__spintent_spnM_spnD_result:n { #1 } }
              }
              {
                \msg_error:nne { spintent } { md-invalid-argument } { \exp_not:c { sp #1 } }
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_spnM_spnD_exec:nnnn`.)

`\spnM`
`\spnD`

The commands `\spnM` and `\spnD` act as the public interface wrappers for typesetting multiples and divisors, respectively. They enforce math mode, open a local group, and defer execution to the function `\__spintent_spnM_spnD_process:nnnnn`, passing along the optional ⟨*keys*⟩ '`#1`' and the numeric {⟨*argument*⟩} '`#2`'.

```
\NewDocumentCommand \spnM { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spnM_spnD_process:nnnnn { nM } { M } { multiplos } {#1} {#2}
        \group_end:
      }
```

```
    { \msg_error:nnn { spintent } { need-math-mode } { \spnM } }
  }
\NewDocumentCommand \spnD { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spnM_spnD_process:nnnnn { nD } { D } { divisores } {#1} {#2}
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spnD } }
  }
```

(*End of definition for* `\spnM` *and* `\spnD`*. These functions are documented on page* 14*.*)

11.28 The command \spang

The command `\spang` is the main public command for typesetting and tagging sexagesimal angle expressions (degrees, minutes, and seconds). It parses the input using the Lua function `\__spintent_luafun_spang_parse:n`. This separates the degrees, minutes, and seconds to construct a `MathML intent` tree with the correct postfixes for `NVDA/MathCAT`.

`\l__spintent_spang_luaset_grado_str`
`\l__spintent_spang_luaset_minuto_str`
`\l__spintent_spang_luaset_segundo_str`
`\l__spintent_spang_luaset_error_str`

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the extracted numerical values for degrees, minutes, and seconds, as well as the error status used during formatting.

```
\str_new:N  \l__spintent_spang_luaset_grado_str
\str_new:N  \l__spintent_spang_luaset_minuto_str
\str_new:N  \l__spintent_spang_luaset_segundo_str
\str_new:N  \l__spintent_spang_luaset_error_str
```

(*End of definition for* `\l__spintent_spang_luaset_grado_str` *and others.*)

11.28.1 Definition of error messages for \spang

`invalid-angle-format`

The error message `invalid-angle-format` defines the text displayed when an invalid angle input is detected. It is triggered when an argument fails the format checks, does not match the expected sexagesimal pattern, or contains invalid measurement symbols.

```
\msg_new:nnnn { spintent } { invalid-angle-format }
  { Invalid~angle~format~in~#1. }
  {
    The~argument~'#2'~could~not~be~parsed. \\
  }
```

(*End of definition for* `invalid-angle-format`*.*)

11.28.2 Internal functions for \spang

`\__spintent_spang_process:n`

The internal function `\__spintent_spang_process:n` parses the angle input using `\__spintent_luafun_spang_parse:n` after resetting the error flag in `\l__spintent_spang_luaset_error_str`. If an invalid format is detected, it issues the `invalid-angle-format` error. Otherwise, it checks the extracted string variables `\l__spintent_spang_luaset_grado_str`, `\l__spintent_spang_luaset_minuto_str`, and `\l__spintent_spang_luaset_segundo_str`. For each non-empty variable, it constructs the corresponding `MathML intent` structure with the correct measurement symbols and postfixes for `NVDA/MathCAT`.

```
\cs_new_protected:Nn \__spintent_spang_process:n
  {
    \str_set:Nn \l__spintent_spang_luaset_error_str { false }
    \__spintent_luafun_spang_parse:n { #1 }
    \str_if_eq:VnTF \l__spintent_spang_luaset_error_str { true }
      {
        \msg_error:nnnn { spintent } { invalid-angle-format } { \spang } { #1 }
      }
      {
        \str_if_empty:NF \l__spintent_spang_luaset_grado_str
          {
            \spunit{ \l__spintent_spang_luaset_grado_str °}
          }
        \str_if_empty:NF \l__spintent_spang_luaset_minuto_str
          {
            \MathMLintent { minutos : postfix($m) }
              {
                \MathMLarg { m }
                  {
                    \spunit{ \l__spintent_spang_luaset_minuto_str '}
```

```
                    }
                }
            }
        \str_if_empty:NF \l__spintent_spang_luaset_segundo_str
            {
                \MathMLintent { segundos : postfix($s) }
                    {
                        \MathMLarg { s }
                            {
                                \spunit { \l__spintent_spang_luaset_segundo_str "}
                            }
                    }
            }
        }
    }
```

(*End of definition for* `\__spintent_spang_process:n`.)

`\spang` The command `\spang` is the public interface for formatting sexagesimal angle expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it calls `\__spintent_spang_process:n` to process the {⟨*argument*⟩}.

```
\NewDocumentCommand \spang { m }
    {
        \mode_if_math:TF
            {
                \group_begin:
                    \__spintent_spang_process:n { #1 }
                \group_end:
            }
            { \msg_error:nnn { spintent } { need-math-mode } { \spang } }
    }
```

(*End of definition for* `\spang`. *This function is documented on page* 21.)

11.29 The command \spnZ

The user-level command `\spnZ` formats integers (the set ℤ). It includes specific options to automatically enclose numbers (particularly negative ones) in visual parentheses and allows you to control whether these parentheses are read by `MathCAT`.

This command verifies that the {⟨*argument*⟩} is indeed an *integer*, checking for the complete absence of decimal separators or attached units of measurement.

11.29.1 Definition of keys for \spnZ

`cifras-sep` `read-sign` `wrap-parens` `read-parens` Now we define the ⟨*keys*⟩ `cifras-sep` and `read-sign`. These are `.meta:nn` ⟨*keys*⟩ passed directly to the `\spnum` command. We also introduce `wrap-parens`, which determines whether the number is enclosed in parentheses, and `read-parens`, which controls if these parentheses are read by `MathCAT`.

```
\keys_define:nn { spintent / spnZ }
    {
        cifras-sep  .meta:nn = { spintent / spnum } { cifras-sep = {#1} },
        read-sign   .meta:nn = { spintent / spnum } { read-sign  = {#1} },
        wrap-parens .bool_set:N = \l__spintent_spnz_wrap_parens_bool,
        wrap-parens .initial:n  = false,
        wrap-parens .value_required:n = true,
        read-parens .bool_set:N = \l__spintent_spnz_read_parens_bool,
        read-parens .initial:n  = true,
        read-parens .value_required:n = true,
        unknown     .code:n =
            {
                \str_set:Nn \l__spintent_command_name_str { spnZ }
                \__spintent_unknown_keys_cmd:n {#1}
            },
    }
```

(*End of definition for* `cifras-sep` *and others.*)

11.29.2 Definition of error messages

`spnz-not-integer` Raised when the {⟨*argument*⟩} contains decimal separators or attached units, which conflict with the constraints of a pure integer representation.

```
\msg_new:nnnn { spintent } { spnz-not-integer }
    { The~value~'#1'~is~not~a~valid~integer. }
    {
```

```
    Command~\token_to_str:N \spnZ~only~accepts~integers.\\
  }
```

(*End of definition for* spnz-not-integer.)

11.29.3 Internal functions for \spnZ

__spintent_spnz_render_parens:

The function __spintent_spnz_render_parens: first checks the state of the boolean variable \l__spintent_spnz (controlled by the read-parens key). If it is true, it prints the number surrounded by parentheses. Otherwise, it applies the :silent *function intent* to the parentheses while printing the number.

```
\cs_new_protected:Nn \__spintent_spnz_render_parens:
  {
    \bool_if:NTF \l__spintent_spnz_read_parens_bool
      {
        ( \__spintent_spnum_print_num_start: )
      }
      {
        \MathMLintent { :silent } { ( }
        \__spintent_spnum_print_num_start:
        \MathMLintent { :silent } { ) }
      }
  }
```

(*End of definition for* __spintent_spnz_render_parens:.)

__spintent_spnz_process:nn

The function __spintent_spnz_process:nn first processes the ⟨*keys*⟩ passed in '#1' and then calls __spintent_luafun_clean_split_and_set:n on '#2'. It verifies that the input is a valid integer by checking if the variable \l__spintent_luaset_has_integer_str is true. If not true, an "error" is raised.

If it is true, it checks whether \l__spintent_luaset_part_int_tl is *empty* and raises an "error" if so. Otherwise, it checks \l__spintent_spnz_wrap_parens_bool set by wrap-parens and calls either the function __spintent_spnz_render_parens: or __spintent_spnum_print_num_start:.

```
\cs_new_protected:Nn \__spintent_spnz_process:nn
  {
    \keys_set:nn { spintent / spnZ } {#1}
    \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#2} }
    \str_if_eq:VnTF \l__spintent_luaset_has_integer_str { true }
      {
        \tl_if_empty:NTF \l__spintent_luaset_part_int_tl
          { \msg_error:nnn { spintent } { spnum-no-digits } {#2} }
          {
            \bool_if:NTF \l__spintent_spnz_wrap_parens_bool
              { \__spintent_spnz_render_parens: }
              { \__spintent_spnum_print_num_start: }
          }
      }
      { \msg_error:nnn { spintent } { spnz-not-integer } {#2} }
  }
```

(*End of definition for* __spintent_spnz_process:nn.)

\spnZ

Defines the user command \spnZ, which is executed within a *group* and strictly enforces *math mode*.

```
\NewDocumentCommand \spnZ { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spnz_process:nn { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spnZ } }
  }
```

(*End of definition for* \spnZ. *This function is documented on page* 16.)

11.30 The command \spmQ

The command \spmQ typesets *"mixed numbers"* within Spanish mathematical contexts (e.g., $3\frac{1}{2}$). It delegates the scaling of the integer part to the Lua module spintent.lua via __spintent_luafun_spmQ_scale_int:Vn, which measures both the integer glyph and the fraction box at the node level and computes a scaling factor to align their heights visually. Finally, it constructs the correct MathML intent for MathCAT.

11.30.1 Definition of variables for \spmQ

\l__spintent_spmQ_join_word_str
\l__spintent_spmQ_math_style_tl
\l__spintent_spnQ_intent_read_sign_str
\l__spintent_spnQ_wrap_parens_l_tl
\l__spintent_spnQ_wrap_parens_r_tl
\l__spintent_saved_luamml_flag_int

The variable \l__spintent_spmQ_join_word_str stores the value set by the key join-word which will be verbalized by MathCAT as a grammatical connector between the integer and fractional parts.

The variable \l__spintent_spmQ_math_style_tl stores the current mathematical style used by the function __spintent_spmQ_set_wrap_parens_size:.

The variable \l__spintent_spnQ_intent_read_sign_str stores the :intent passed to \MathMLintent when the signs '+' or '−' are present in the integer part, used by the function __spintent_spmQ_print_scale_int:n

The variables \l__spintent_spnQ_wrap_parens_l_tl and \l__spintent_spnQ_wrap_parens_r_tl store the sized parentheses set by the function __spintent_spmQ_set_wrap_parens_size: and used by the key wrap-parens.

The integer variable \l__spintent_saved_luamml_flag_int temporarily stores the current value of \l__luamml_flag_int before disabling it during the injection of the scaled integer node, and restores it afterwards. This prevents luamml from generating a spurious <math> element around the sub_box processed in isolation, before \frac has expanded.

```
\str_new:N \l__spintent_spmQ_join_word_str
\tl_new:N  \l__spintent_spmQ_math_style_tl
\str_new:N \l__spintent_spnQ_intent_read_sign_str
\tl_new:N  \l__spintent_spnQ_wrap_parens_l_tl
\tl_new:N  \l__spintent_spnQ_wrap_parens_r_tl
\int_new:N \l__spintent_saved_luamml_flag_int
```

(*End of definition for* \l__spintent_spmQ_join_word_str *and others.*)

11.30.2 Definition of error messages for \spmQ

spmQ-zero-denominator
spmQ-not-integer

The error message spmQ-zero-denominator is raised when a zero denominator is passed to \spmQ. The second error spmQ-not-integer is returned when the integer part passed in {⟨*argument*⟩} is not actually an integer.

```
\msg_new:nnnn { spintent } { spmQ-zero-denominator }
  { Zero~denominator~in~\token_to_str:N \spmQ. }
  {
    The~denominator~of~a~fraction~cannot~be~zero.\\
  }
\msg_new:nnnn { spintent } { spmQ-not-integer }
  { The~value~'#1'~is~not~a~valid~integer. }
  {
    Command~\token_to_str:N \spmQ \c_space_tl only~accepts~integers.\\
  }
```

(*End of definition for* spmQ-zero-denominator *and* spmQ-not-integer*.*)

11.30.3 Definition of keys for \spmQ

join-word
use-spnZ
wrap-parens
read-parens
print-dfrac
unknown

Now we add the keys join-word, use-spnZ, wrap-parens, read-parens and print-dfrac.

```
\keys_define:nn { spintent / spmQ }
  {
    join-word         .choices:nn =
      { entero , enteros , y }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spmQ_join_word_str { entero } }
            {2} { \str_set:Nn \l__spintent_spmQ_join_word_str { enteros } }
            {3} { \str_set:Nn \l__spintent_spmQ_join_word_str { y } }
          }
      },
    join-word         .initial:n  = y,
    join-word         .value_required:n = true,
    join-word/unknown .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { join-word } { entero, ~enteros, ~y } { \exp_not:n {#1} },
    use-spnZ          .bool_set:N = \l__spintent_spmQ_use_spnZ_bool,
    use-spnZ          .initial:n = true,
    use-spnZ          .value_required:n = true,
    print-dfrac       .bool_set:N = \l__spintent_spmQ_print_dfrac_bool,
    print-dfrac       .initial:n = false,
    print-dfrac       .value_required:n = true,
    wrap-parens       .bool_set:N = \l__spintent_spnQ_wrap_parens_bool,
    wrap-parens       .initial:n  = false,
    wrap-parens       .value_required:n = true,
```

```
    read-parens       .bool_set:N = \l__spintent_spnQ_read_parens_bool,
    read-parens       .initial:n  = true,
    read-parens       .value_required:n = true,
    unknown           .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spnQ }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `join-word` *and others.*)

11.30.4 Internal functions for \spnQ

`\__spintent_spnQ_set_wrap_parens_size:`

The function `\__spintent_spnQ_set_wrap_parens_size:` captures the current mathematical style using the primitive `\mathstyle` from LuaTeX and sets the variables `\l__spintent_spnQ_wrap_parens_l_tl` and `\l__spintent_spnQ_wrap_parens_r_tl` with the appropriate size for the parentheses. Display styles (`0` and `1`) receive `\Bigl\lparen` and `\Bigr\rparen`; text styles (`2` and `3`) receive `\bigl\lparen` and `\bigr\rparen`; any other style falls back to unsized `\lparen` and `\rparen`.

If the key `print-dfrac` is active, the size is always forced to `\Bigl`/`\Bigr` regardless of the current mathematical style, since `\dfrac` always produces a display-height fraction.

```
\cs_new_protected:Nn \__spintent_spnQ_set_wrap_parens_size:
  {
    \int_case:nnF { \mathstyle }
      {
        { 0 }
          {
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_l_tl { \Bigl \lparen }
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_r_tl { \Bigr \rparen }
          }
        { 1 }
          {
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_l_tl { \Bigl \lparen }
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_r_tl { \Bigr \rparen }
          }
        { 2 }
          {
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_l_tl { \bigl \lparen }
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_r_tl { \bigr \rparen }
          }
        { 3 }
          {
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_l_tl { \bigl \lparen }
            \tl_set:Nn \l__spintent_spnQ_wrap_parens_r_tl { \bigr \rparen }
          }
      }
      {
        \tl_set:Nn \l__spintent_spnQ_wrap_parens_l_tl { \lparen }
        \tl_set:Nn \l__spintent_spnQ_wrap_parens_r_tl { \rparen }
      }
```

If the key `print-dfrac` is active, we must reset the variables `\l__spintent_spnQ_wrap_parens_l_tl` and `\l__spintent_spnQ_wrap_parens_r_tl` to `\Bigl`/`\Bigr` unconditionally.

```
    \bool_if:NT \l__spintent_spnQ_print_dfrac_bool
      {
        \tl_set:Nn \l__spintent_spnQ_wrap_parens_l_tl { \Bigl \lparen }
        \tl_set:Nn \l__spintent_spnQ_wrap_parens_r_tl { \Bigr \rparen }
      }
  }
```

(*End of definition for* `\__spintent_spnQ_set_wrap_parens_size:`.)

`\__spintent_spnQ_start_wrap_parens:`
`\__spintent_spnQ_stop_wrap_parens:`

The functions `\__spintent_spnQ_start_wrap_parens:` and `\__spintent_spnQ_stop_wrap_parens:` open and close the parentheses around the *“mixed number”* when the key `wrap-parens` is active. They first call `\__spintent_spnQ_set_wrap_parens_size:` to determine the correct size for the current mathematical style, then print the appropriate parenthesis stored in `\l__spintent_spnQ_wrap_parens_l_tl` or `\l__spintent_spnQ_wrap_parens_r_tl`.

If the variable `\l__spintent_spnQ_read_parens_bool`, controlled by the key `read-parens`, is *false*, each parenthesis is wrapped inside `\MathMLintent{:silent}` so that NVDA/MathCAT does not verbalize them.

```
\cs_new_protected:Nn \__spintent_spnQ_start_wrap_parens:
  {
```

```
    \bool_if:NT \l__spintent_spnQ_wrap_parens_bool
      {
        \__spintent_spmQ_set_wrap_parens_size:
        \bool_if:NTF \l__spintent_spnQ_read_parens_bool
          { \l__spintent_spnQ_wrap_parens_l_tl }
          { \MathMLintent { :silent } { \l__spintent_spnQ_wrap_parens_l_tl } }
      }
  }
\cs_new_protected:Nn \__spintent_spmQ_stop_wrap_parens:
  {
    \bool_if:NT \l__spintent_spnQ_wrap_parens_bool
      {
        \__spintent_spmQ_set_wrap_parens_size:
        \bool_if:NTF \l__spintent_spnQ_read_parens_bool
          { \l__spintent_spnQ_wrap_parens_r_tl }
          { \MathMLintent { :silent } { \l__spintent_spnQ_wrap_parens_r_tl } }
      }
  }
```

(*End of definition for* `\__spintent_spmQ_start_wrap_parens:` *and* `\__spintent_spmQ_stop_wrap_parens:`.)

`\__spintent_spmQ_set_join_word:`

The function `\__spintent_spmQ_set_join_word:` injects the grammatical connector between the integer part and the fraction by dispatching on the value stored in `\l__spintent_spmQ_join_word_str`, set by the key `join-word`. In all three cases the visual output is the Unicode invisible separator `U+2063` via `\c__spintent_invisible_sep_tl`, which produces a `<mi>` node in the `MathML` structure; what differs is the `:intent` text attached to it via `\MathMLintent`, which `NVDA`/`MathCAT` will verbalize as *«entero»*, *«enteros»* or *«y»* respectively.

```
\cs_new_protected:Nn \__spintent_spmQ_set_join_word:
  {
    \str_case:Vn \l__spintent_spmQ_join_word_str
      {
        { entero  } { \MathMLintent { _entero  } { \c__spintent_invisible_sep_tl } }
        { enteros } { \MathMLintent { _enteros } { \c__spintent_invisible_sep_tl } }
        { y       } { \MathMLintent { _y-      } { \c__spintent_invisible_sep_tl } }
      }
  }
```

(*End of definition for* `\__spintent_spmQ_set_join_word:`.)

`\__spintent_spmQ_print_scale_int:`

The function `\__spintent_spmQ_print_scale_int:` prints the integer part of the *"mixed number"* scaled to match the visual height of the accompanying fraction. Before calling the Lua function `\__spintent_luafun_spmQ_sca` it temporarily disables `luamml` by saving the current value of `\l__luamml_flag_int` into `\l__spintent_saved_luamml_flag_int` and setting it to `0` (skip completely). This prevents `luamml` from generating a spurious `<math>` wrapper around the `sub_box` node that holds the scaled integer, which is processed in isolation before `\frac` has expanded. Once the Lua function returns, the original flag value is restored via `\l__spintent_saved_luamml_flag_int`.

The Lua function receives the integer digits from `\l__spintent_luaset_part_int_tl` and the fraction command name (`\frac` or `\dfrac`, determined by the key `print-dfrac`) as its two arguments, and inserts the appropriately scaled `sub_box` node directly into the current horizontal list.

```
\cs_new_protected:Nn \__spintent_spmQ_print_scale_int:
  {
    \int_set_eq:Nc \l__spintent_saved_luamml_flag_int { l__luamml_flag_int }
    \int_set:Nn \l__luamml_flag_int { 0 }
    \bool_if:NTF \l__spintent_spmQ_print_dfrac_bool
      { \__spintent_luafun_spmQ_scale_int:Vn \l__spintent_luaset_part_int_tl { dfrac } }
      { \__spintent_luafun_spmQ_scale_int:Vn \l__spintent_luaset_part_int_tl { frac  } }
    \int_set_eq:cN { l__luamml_flag_int } \l__spintent_saved_luamml_flag_int
  }
```

(*End of definition for* `\__spintent_spmQ_print_scale_int:`.)

`\__spintent_spmQ_print_scale_int:n`

The function `\__spintent_spmQ_print_scale_int:n` takes the integer part of the *"mixed number"* passed in '`#1`'. It first calls `\__spintent_luafun_clean_split_and_set:n` on '`#1`' to parse and populate the shared Lua variables, then checks the status of `\l__spintent_luaset_has_integer_str`.

If the status is `true`, it inspects `\l__spintent_luaset_sign_tl`. When the sign is *empty*, it delegates directly to `\__spintent_spmQ_print_scale_int:` to render the scaled integer. When a sign is present, it sets `\l__spintent_spnQ_intent_read_sign_str` to either `_más:prefix($e)` or `_menos:prefix($e)` and wraps the sign glyph together with a `MathML` argument `($e)` containing the scaled integer inside a `\MathMLintent`, so that `NVDA`/`MathCAT` verbalizes the sign as a prefix before the integer value. An

`\__spintent_invisible_sep:` call between the sign and the `MathMLarg` forces the opening of a new `<mrow>` in the `MathML` structure, which is required for the intent tree to be well-formed.

If the status is `false`, the `spmQ-not-integer` "error" is raised immediately.

```
\cs_new_protected:Nn \__spintent_spmQ_print_scale_int:n
  {
    \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnTF \l__spintent_luaset_has_integer_str { true }
      {
        \tl_if_empty:NTF \l__spintent_luaset_sign_tl
          {
            \__spintent_spmQ_print_scale_int:
          }
          {
            \str_if_eq:VnTF \l__spintent_luaset_sign_tl { + }
              {
                \str_set:Nn
                  \l__spintent_spnQ_intent_read_sign_str { _más:prefix($e) }
              }
              {
                \str_set:Nn
                  \l__spintent_spnQ_intent_read_sign_str { _menos:prefix($e) }
              }
            \MathMLintent { \l__spintent_spnQ_intent_read_sign_str }
              {
                \l__spintent_luaset_sign_tl \__spintent_invisible_sep:
                \MathMLarg { e } { \__spintent_spmQ_print_scale_int: }
              }
          }
      }
      { \msg_error:nn { spintent } { spmQ-not-integer } }
  }
```

(*End of definition for* `\__spintent_spmQ_print_scale_int:n`.)

`\__spintent_spmQ_print_spnZ_dfrac:nn`

The function `\__spintent_spmQ_print_spnZ_dfrac:nn` typesets the fractional part of the *"mixed number"*. It takes the numerator in '`#1`' and the denominator in '`#2`'. It first calls `\__spintent_luafun_clean_split_and_set:` on '`#1`' to validate that the numerator is an integer; if `\l__spintent_luaset_has_integer_str` is `false`, the `spmQ-not-integer` "error" is raised immediately.

If the validation passes, the function dispatches on the two boolean variables `\l__spintent_spmQ_use_spnZ_bool` and `\l__spintent_spmQ_print_dfrac_bool`, controlled respectively by the keys `use-spnZ` and `print-dfrac`. When `use-spnZ` is `true`, both the numerator and denominator are passed through `\spnZ` to gain `MathML` integer semantics; otherwise, the raw token lists '`#1`' and '`#2`' are used directly. In both branches, the fraction command is selected dynamically via `\use:c`: if `print-dfrac` is `true`, `dfrac` is assembled; otherwise, `frac` is used.

```
\cs_new_protected:Nn \__spintent_spmQ_print_spnZ_dfrac:nn
  {
    \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnT \l__spintent_luaset_has_integer_str { false }
      { \msg_error:nn { spintent } { spmQ-not-integer } }
    \bool_if:NTF \l__spintent_spmQ_use_spnZ_bool
      {
        \use:c { \bool_if:NT \l__spintent_spmQ_print_dfrac_bool { d } frac }
          { \spnZ { #1 } } { \spnZ { #2 } }
      }
      {
        \use:c { \bool_if:NT \l__spintent_spmQ_print_dfrac_bool { d } frac }
          { #1 } { #2 }
      }
  }
```

(*End of definition for* `\__spintent_spmQ_print_spnZ_dfrac:nn`.)

`\__spintent_spmQ_print:nnn`

The function `\__spintent_spmQ_print:nnn` is the main rendering orchestrator for `\spmQ`. It takes three arguments: the integer part '`#1`', the numerator '`#2`', and the denominator '`#3`'. It opens the optional parentheses via `\__spintent_spmQ_start_wrap_parens:`, then calls `\__spintent_spmQ_print_scale_int:n` to scale and render the integer part. Next, it injects the grammatical connector between the integer and the fraction via `\__spintent_spmQ_set_join_word:`. Then it calls `\__spintent_spmQ_print_spnZ_dfrac:nn` to typeset the fraction using the keys `print-dfrac` and `use-spnZ`. Finally, it closes the optional parentheses via `\__spintent_spmQ_stop_wrap_parens:`.

```
\cs_new_protected:Nn \__spintent_spmQ_print:nnn
```

```
  {
    \__spintent_spmQ_start_wrap_parens:
    \__spintent_spmQ_print_scale_int:n { #1 }
    \__spintent_spmQ_set_join_word:
    \__spintent_spmQ_print_spnZ_dfrac:nn { #2 } { #3 }
    \__spintent_spmQ_stop_wrap_parens:
  }
```

(*End of definition for* `\__spintent_spmQ_print:nnn`.)

`\__spintent_spmQ_exec:nnnn`

The internal function `\__spintent_spmQ_exec:nnnn` takes four arguments: the optional ⟨*keys*⟩ '`#1`', the integer '`#2`', the numerator '`#3`', and the denominator '`#4`'. First sets the optional ⟨*keys*⟩ from '`#1`', then processes the denominator input in '`#4`' using `\__spintent_luafun_clean_split_and_set:n` to validate it is a non-zero integer.

It checks the status of `\l__spintent_luaset_has_integer_str`. If `true` and the integer part is exactly zero, it issues the `spmQ-zero-denominator` "error". Otherwise, whether the status is `false` or the integer part is *non-zero*, it calls `\__spintent_spmQ_print:nnn` to format the *"mixed number"*.

```
\cs_new_protected:Nn \__spintent_spmQ_exec:nnnn
  {
    \keys_set:nn { spintent / spmQ } { #1 }
    \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #4 } }
    \str_if_eq:VnTF \l__spintent_luaset_has_integer_str { true }
      {
        \int_compare:nNnTF { \l__spintent_luaset_part_int_tl } = { 0 }
          { \msg_error:nn { spintent } { spmQ-zero-denominator } }
          { \__spintent_spmQ_print:nnn { #2 } { #3 } { #4 } }
      }
      { \__spintent_spmQ_print:nnn { #2 } { #3 } { #4 } }
  }
```

(*End of definition for* `\__spintent_spmQ_exec:nnnn`.)

`\spmQ`

The command `\spmQ` is the public interface for formatting *"mixed numbers"*. It first checks if it is being used in *math mode*, issuing an "error" if not. Inside a local group, it calls `\__spintent_spmQ_exec:nnnn` to process the optional ⟨*keys*⟩ and the {⟨*arguments*⟩}.

```
\NewDocumentCommand \spmQ { O{} m m m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spmQ_exec:nnnn { #1 } { #2 } { #3 } { #4 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmQ } }
  }
```

(*End of definition for* `\spmQ`. *This function is documented on page* 16.)

11.31 The commands of the geo family

The commands described in this section provide a semantic interface for geometric objects in Spanish mathematical contexts. They all delegate parsing to `spintent.lua`, sharing the variables `\l__spintent_geo_luaset_error_str`, `\l__spintent_geo_luaset_intent_str` and `\l__spintent_geo_luaset_print_tl`, constructing a valid `MathCAT` intent.

11.31.1 Definition of shared variables

`\l__spintent_geo_luaset_error_str`
`\l__spintent_geo_luaset_intent_str`
`\l__spintent_geo_luaset_print_tl`
`\l__spintent_geo_luaset_points_str`
`\l__spintent_geo_luaset_is_name_str`
`\l__spintent_geo_luaset_poly_sym_str`
`\l__spintent_geo_luaset_intent_pts_str`
`\l__spintent_geo_luaset_angulo_case_str`
`\l__spintent_geo_intent_tl`
`\l__spintent_geo_fig_op_tl`
`\l__spintent_geo_luaset_concept_str`
`\l__spintent_geo_luaset_letra_word_str`
`\l__spintent_geo_luaset_sub_spoken_str`

Internal variables shared across all commands of the family to avoid repeated calls into Lua.

```
\str_new:N \l__spintent_geo_luaset_error_str
\str_new:N \l__spintent_geo_luaset_intent_str
\tl_new:N  \l__spintent_geo_luaset_print_tl
\str_new:N \l__spintent_geo_luaset_points_str
\str_new:N \l__spintent_geo_luaset_is_name_str
\str_new:N \l__spintent_geo_luaset_poly_sym_str
\str_new:N \l__spintent_geo_luaset_intent_pts_str
\str_new:N \l__spintent_geo_luaset_angulo_case_str
\tl_new:N  \l__spintent_geo_intent_tl
\tl_new:N  \l__spintent_geo_fig_op_tl
\str_new:N \l__spintent_geo_luaset_concept_str
\str_new:N \l__spintent_geo_luaset_letra_word_str
\str_new:N \l__spintent_geo_luaset_sub_spoken_str
```

(*End of definition for* `\l__spintent_geo_luaset_error_str` *and others*.)

11.32 The command \spPoint

The user-level command \spPoint provides a semantic interface for a single geometric point. It delegates argument validation to the Lua module spintent.lua, constructing a valid MathCAT intent.

11.32.1 Definition of variables for \spPoint

\l__spintent_spPoint_mode_str
\l__spintent_spPoint_prefix_tl
\l__spintent_spPoint_intent_arg_tl
\l__spintent_spPoint_kv_tl

Internal variables holding the values set by read-cmd and prefix.

```
\str_new:N \l__spintent_spPoint_mode_str
\tl_new:N  \l__spintent_spPoint_prefix_tl
\tl_new:N  \l__spintent_spPoint_intent_arg_tl
\tl_new:N  \l__spintent_spPoint_kv_tl
```

(*End of definition for* \l__spintent_spPoint_mode_str *and others.*)

11.32.2 Definition of error messages for \spPoint

spPoint-invalid-point

The "error" spPoint-invalid-point is raised when the argument passed in '#1' is not a single uppercase letter with an optional subscript or prime modifier.

```
\msg_new:nnnn { spintent } { spPoint-invalid-point }
  {
    Invalid~point~format~in~\token_to_str:N \spPoint:~'#1'.
  }
  {
    The~argument~must~be~a~single~uppercase~letter~with~an~optional~
    subscript~(\_{...})~or~prime~(').\\
  }
```

(*End of definition for* spPoint-invalid-point.)

11.32.3 Definition of keys for \spPoint

read-cmd
prefix
silent-cmd
unknown

We define the ⟨*keys*⟩ read-cmd, prefix, silent-cmd and concept.

```
\keys_define:nn { spintent / spPoint }
  {
    read-cmd            .choices:nn = { school , mathcat }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spPoint_mode_str { school  } }
            {2} { \str_set:Nn \l__spintent_spPoint_mode_str { mathcat } }
          }
      },
    read-cmd            .initial:n  = school,
    read-cmd            .value_required:n = true,
    read-cmd / unknown  .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
    prefix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spPoint_prefix_tl {#1} },
    prefix              .initial:n  = ,
    prefix              .value_required:n = true,
    silent-cmd          .bool_set:N = \l__spintent_spPoint_silent_cmd_bool,
    silent-cmd          .initial:n  = false,
    silent-cmd          .value_required:n = true,
    concept             .bool_set:N = \l__spintent_spPoint_concept_bool,
    concept             .initial:n  = true,
    concept             .value_required:n = true,
    unknown             .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spPoint }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* read-cmd *and others.*)

11.32.4 Internal functions for \spPoint

__spintent_spPoint_print:n

The internal function __spintent_spPoint_print:n process the {⟨*argument*⟩} passed in '#1'. First, call the function __spintent_luafun_geo_parse_and_set:n to process the {⟨*argument*⟩}, returning an "error" if the input is invalid. Then, if silent-cmd is true or concept is false, set \l__spintent_spPont_intent_arg to :silent($pt); otherwise set it to punto:prefix($pt), prefixed with prefix when given. Finally, call __spintent_spPoint_mathml:.

```
\cs_new_protected:Nn \__spintent_spPoint_print:n
```

```
  {
    \__spintent_luafun_geo_point_parse_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nne { spintent } { spPoint-invalid-point } { \exp_not:n {#1} }
      }
    \bool_if:nTF
      {
        \bool_lazy_or_p:nn
          { \l__spintent_spPoint_silent_cmd_bool }
          { ! \l__spintent_spPoint_concept_bool }
      }
      {
        \tl_set:Nn \l__spintent_spPoint_intent_arg_tl { :silent($pt) }
      }
      {
        \tl_set:Ne \l__spintent_spPoint_intent_arg_tl
          {
            \tl_if_empty:NF \l__spintent_spPoint_prefix_tl
              {
                \tl_use:N \l__spintent_spPoint_prefix_tl -
              }
            punto:prefix($pt)
          }
      }
    \__spintent_spPoint_mathml:
  }
```

(*End of definition for* `\__spintent_spPoint_print:n`.)

`\__spintent_spPoint_mathml:`

The function `\__spintent_spPoint_mathml:` generates the <mrow> and prints `\l__spintent_geo_luaset_print` `\__spintent_invisible_sep:` is called first (`luamml` issue #26).
The kv string is `arg="pt", intent="_-"` when `silent-cmd` is `true`; otherwise, for `read-cmd=school` it is `arg="pt", intent="_body-"` (literal spelling); for `read-cmd=mathcat` it is `arg="pt"` alone. A single call to `\__spintent_luafun_wrap_mrow_intent_arg:nn` applies it, with `\l__spintent_spPont_intent_arg_tl` as the concept `intent`.

```
\cs_new_protected:Nn \__spintent_spPoint_mathml:
  {
    \__spintent_invisible_sep:
    \bool_if:NTF \l__spintent_spPoint_silent_cmd_bool
      {
        \tl_set:Nn \l__spintent_spPoint_kv_tl { arg = "pt", intent = "_-" }
      }
      {
        \str_if_eq:VnTF \l__spintent_spPoint_mode_str { school }
          {
            \tl_set:Ne \l__spintent_spPoint_kv_tl
              { arg = "pt", intent = "_ \l__spintent_geo_luaset_intent_str -" }
          }
          {
            \tl_set:Nn \l__spintent_spPoint_kv_tl { arg = "pt" }
          }
      }
    \__spintent_luafun_wrap_mrow_intent_arg:nn
      { \l__spintent_spPoint_intent_arg_tl }
      { \l__spintent_spPoint_kv_tl }
    { \l__spintent_geo_luaset_print_tl }
  }
```

(*End of definition for* `\__spintent_spPoint_mathml:`.)

`\__spintent_spPoint_exec:nn`

The function `\__spintent_spPoint_exec:nn` receives the optional ⟨*keys*⟩ passed in '#1' and the {⟨*argument*⟩} passed in '#2' delegating the rendering of the latter to `\__spintent_spPoint_print:n`.

```
\cs_new_protected:Nn \__spintent_spPoint_exec:nn
  {
    \keys_set:nn { spintent / spPoint } { #1 }
    \__spintent_spPoint_print:n { #2 }
  }
```

(*End of definition for* `\__spintent_spPoint_exec:nn`.)

\spPoint

The public command `\spPoint` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spPoint_exec:nn`.

```
\NewDocumentCommand \spPoint { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spPoint_exec:nn { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spPoint } }
  }
```

(*End of definition for* `\spPoint`. *This function is documented on page 17.*)

__spintent_mathmlarg_point:nnn

Annotates one point (`#3`) with `arg` and, when applicable, a literal `intent`, without wrapping it in its own `<mrow>`. `#1` is the reference name (`arg="#1"`); `#2` is the calling command's name, used to look up its own `silent-cmd`/`read-cmd` variables. Same kv logic as `\__spintent_spPoint_mathml:`. Concept `intent` is passed empty on purpose: the caller applies it once, externally, via a native `\MathMLintent`.

```
\tl_new:N   \l__spintent_mathmlarg_point_ref_tl
\str_new:N  \l__spintent_mathmlarg_point_mode_str
\tl_new:N   \l__spintent_mathmlarg_point_kv_tl
\bool_new:N \l__spintent_mathmlarg_point_silent_bool
\cs_new_protected:Nn \__spintent_mathmlarg_point:nnn
  {
    % referencia del intent ($ref)
    \tl_set:Nn \l__spintent_mathmlarg_point_ref_tl {#1}
    \__spintent_luafun_geo_point_parse_and_set:n { \exp_not:n {#3} }
    % Ajustamos la lectura acorde al comando
    \bool_set_eq:Nc \l__spintent_mathmlarg_point_silent_bool { l__spintent_ #2 _silent_cmd_bool }
    \str_set_eq:Nc  \l__spintent_mathmlarg_point_mode_str  { l__spintent_ #2 _mode_str }
    \__spintent_invisible_sep:
    \bool_if:NTF \l__spintent_mathmlarg_point_silent_bool
      {
        \tl_set:Ne \l__spintent_mathmlarg_point_kv_tl
          { arg = "\l__spintent_mathmlarg_point_ref_tl", intent = "_-" }
      }
      {
        \str_if_eq:VnTF \l__spintent_mathmlarg_point_mode_str { school }
          {
            \tl_set:Ne \l__spintent_mathmlarg_point_kv_tl
              { arg = "\l__spintent_mathmlarg_point_ref_tl", intent = "_ \l__spintent_geo_luaset_
          }
          {
            \tl_set:Ne \l__spintent_mathmlarg_point_kv_tl { arg = "\l__spintent_mathmlarg_point_r
          }
      }
    \__spintent_luafun_wrap_mrow_intent_arg:nn
      {  } % debe ir vacío
      { \l__spintent_mathmlarg_point_kv_tl }
    { \l__spintent_geo_luaset_print_tl }
  }
```

(*End of definition for* `\__spintent_mathmlarg_point:nnn`.)

11.33 The command \sprecta

The user command `\sprecta` supports two types of notation for the input {⟨*argument*⟩}: when the argument contains two points (separated by an optional comma), `\overleftrightarrow` is applied; when the argument is a single point (without comma) or a lowercase letter, it is formatted without any visual decorator. Both notations are automatically detected by the Lua module.

11.33.1 Definition of variables for \sprecta

\l__spintent_sprecta_mode_str
\l__spintent_sprecta_prefix_tl
\l__spintent_sprecta_pt_seq
\l__spintent_sprecta_refs_tl
\l__spintent_sprecta_intent_arg_tl
\l__spintent_sprecta_pt_idx_int

The variable `\l__spintent_sprecta_mode_str` store the values set by the key `read-cmd` and the variable `\l__spintent_sprayo_prefix_tl` store the value set by the key `prefix`.

```
\str_new:N \l__spintent_sprecta_mode_str
\tl_new:N  \l__spintent_sprecta_prefix_tl
\seq_new:N \l__spintent_sprecta_pt_seq
\tl_new:N  \l__spintent_sprecta_refs_tl
\tl_new:N  \l__spintent_sprecta_intent_arg_tl
\int_new:N \l__spintent_sprecta_pt_idx_int
```

(*End of definition for* `\l__spintent_sprecta_mode_str` *and others.*)

11.33.2 Definition of error messages for \sprecta

sprecta-invalid-points

The "error" `sprecta-invalid-points` is raised when the argument matches neither the two-point notation nor the name notation.

```
\msg_new:nnnn { spintent } { sprecta-invalid-points }
  {
    Invalid~format~in~\token_to_str:N \sprecta:~'#1'~\msg_line_context:.
  }
  {
    See~the~documentation~of~\token_to_str:N \sprecta~for~the~accepted~argument~forms.
  }
```

(*End of definition for* `sprecta-invalid-points`.)

11.33.3 Definition of keys for \sprecta

read-cmd
prefix
silent-cmd
unknown

We define the ⟨*keys*⟩ `read-cmd`, `prefix` and `silent-cmd`.

```
\keys_define:nn { spintent / sprecta }
  {
    read-cmd            .choices:nn = { school , mathcat }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_sprecta_mode_str { school  } }
            {2} { \str_set:Nn \l__spintent_sprecta_mode_str { mathcat } }
          }
      },
    read-cmd            .initial:n  = school,
    read-cmd            .value_required:n = true,
    read-cmd / unknown  .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
    prefix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_sprecta_prefix_tl {#1} },
    prefix              .initial:n  = ,
    prefix              .value_required:n = true,
    silent-cmd          .bool_set:N = \l__spintent_sprecta_silent_cmd_bool,
    silent-cmd          .initial:n  = false,
    silent-cmd          .value_required:n = true,
    concept             .bool_set:N = \l__spintent_sprecta_concept_bool,
    concept             .initial:n  = true,
    concept             .value_required:n = true,
    unknown             .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { sprecta }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `read-cmd` *and others.*)

11.33.4 Internal functions for \sprecta

__spintent_sprecta_print:n

The internal function `\__spintent_sprecta_print:n` process the {⟨*argument*⟩} passed in '`#1`'. It calls `\__spintent_luafun_geo_parse_and_set:n` to process the {⟨*argument*⟩}, returning an "error" if the input is invalid, then calls `\__spintent_sprecta_mathml:`. Unlike `\spPoint`, no `intent` is built here: both branches of `\__spintent_sprecta_mathml:` build their own, since a proper name (`m`, `l`, ...) and a list of points need different amounts of it (a single `$pt` versus `$pt1,$pt2,\ldots`).

```
\cs_new_protected:Nn \__spintent_sprecta_print:n
  {
    \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nne { spintent } { sprecta-invalid-points } { \exp_not:n {#1} }
      }
    \__spintent_sprecta_mathml:
  }
```

(*End of definition for* `\__spintent_sprecta_print:n`.)

__spintent_sprecta_mathml:

The function `\__spintent_sprecta_mathml:` branches on `\l__spintent_geo_luaset_is_name_str`. For a proper name (`m`, `l`, ...), the logic mirrors `\__spintent_spPoint_mathml:` exactly: the concept `intent` is `:silent($pt)` when `silent-cmd` is `true` *or* `concept` is `false`; otherwise it is `recta:prefix($pt)`,

prefixed with prefix when given. The kv string, however, depends *only* on silent-cmd — concept=false alone must not silence the letter itself: arg="pt", intent="_-" when silent; otherwise arg="pt" plus a literal intent for read-cmd=school, or arg="pt" alone for read-cmd=mathcat. A single call to __spintent_luafun_wrap_mrow_intent_arg:nn applies both.
For a list of points, \l__spintent_geo_luaset_points_str (set by Lua, one raw point per element, unmerged) is split into \l__spintent_sprecta_pt_seq and walked once to build \l__spintent_sprecta_refs_tl, the comma-separated $pt1,$pt2,... reference list, naming each point by position. The concept intent is _-:nofix(refs) when silent-cmd is true *or* concept is false; otherwise recta:prefix(refs), prefixed as above. It is applied with a native \MathMLintent wrapping \overleftrightarrow, preceded by __spintent_luafun_inner_sep: (a second sibling so the sole <mrow> does not collapse into <math>, luamml issue #26). The sequence is walked a second time, calling __spintent_mathmlarg_point:nnn once per point with its pt*n* reference name and silent-cmd/read-cmd looked up under sprecta: it alone decides arg/literal intent per point, so concept plays no role at that level.

```
\cs_new_protected:Nn \__spintent_sprecta_mathml:
  {
    \str_if_eq:VnTF \l__spintent_geo_luaset_is_name_str { true }
      {
        \__spintent_invisible_sep:
        \bool_if:nTF
          {
            \bool_lazy_or_p:nn
              { \l__spintent_sprecta_silent_cmd_bool }
              { ! \l__spintent_sprecta_concept_bool }
          }
          {
            \tl_set:Nn \l__spintent_sprecta_intent_arg_tl { :silent($pt) }
          }
          {
            \tl_set:Ne \l__spintent_sprecta_intent_arg_tl
              {
                \tl_if_empty:NF \l__spintent_sprecta_prefix_tl
                  { \tl_use:N \l__spintent_sprecta_prefix_tl - }
                recta:prefix($pt)
              }
          }
        \bool_if:NTF \l__spintent_sprecta_silent_cmd_bool
          {
            \tl_set:Nn \l__spintent_sprecta_kv_tl { arg = "pt", intent = "_-" }
          }
          {
            \str_if_eq:VnTF \l__spintent_sprecta_mode_str { school }
              {
                \tl_set:Ne \l__spintent_sprecta_kv_tl
                  { arg = "pt", intent = "_ \l__spintent_geo_luaset_intent_str -" }
              }
              {
                \tl_set:Nn \l__spintent_sprecta_kv_tl { arg = "pt" }
              }
          }
        \__spintent_luafun_wrap_mrow_intent_arg:nn
          { \l__spintent_sprecta_intent_arg_tl }
          { \l__spintent_sprecta_kv_tl }
        { \l__spintent_geo_luaset_print_tl }
      }
      {
        \seq_set_from_clist:NN \l__spintent_sprecta_pt_seq \l__spintent_geo_luaset_points_str
        \tl_clear:N \l__spintent_sprecta_refs_tl
        \int_zero:N \l__spintent_sprecta_pt_idx_int
        \seq_map_inline:Nn \l__spintent_sprecta_pt_seq
          {
            \int_incr:N \l__spintent_sprecta_pt_idx_int
            \tl_if_empty:NF \l__spintent_sprecta_refs_tl
              { \tl_put_right:Nn \l__spintent_sprecta_refs_tl { , } }
            \tl_put_right:Ne \l__spintent_sprecta_refs_tl { $ pt \int_use:N \l__spintent_sprecta_
          }
        \bool_if:nTF
          {
            \bool_lazy_or_p:nn
              { \l__spintent_sprecta_silent_cmd_bool }
              { ! \l__spintent_sprecta_concept_bool }
```

```
            }
            {
              \tl_set:Ne \l__spintent_sprecta_intent_arg_tl { _-:nofix( \l__spintent_sprecta_refs_t
            }
            {
              \tl_set:Ne \l__spintent_sprecta_intent_arg_tl
                {
                  \tl_if_empty:NF \l__spintent_sprecta_prefix_tl
                    { \tl_use:N \l__spintent_sprecta_prefix_tl - }
                  recta:prefix( \l__spintent_sprecta_refs_tl )
                }
            }
          \__spintent_invisible_sep:
          \MathMLintent { \l__spintent_sprecta_intent_arg_tl }
            {
              \__spintent_luafun_inner_sep:
              \overleftrightarrow
                {
                  \int_zero:N \l__spintent_sprecta_pt_idx_int
                  \seq_map_inline:Nn \l__spintent_sprecta_pt_seq
                    {
                      \int_incr:N \l__spintent_sprecta_pt_idx_int
                      \__spintent_mathmlarg_point:nnn
                        { pt \int_use:N \l__spintent_sprecta_pt_idx_int }
                        { sprecta }
                        { ##1 }
                    }
                }
            }
        }
    }
```

(*End of definition for* `\__spintent_sprecta_mathml:`.)

`\__spintent_sprecta_exec:nn`

The function `\__spintent_sprecta_exec:nn` receives the optional ⟨*keys*⟩ passed in '`#1`' and the {⟨*argument*⟩} passed in '`#2`' delegating the rendering of the latter to `\__spintent_sprecta_print:n`.

```
\cs_new_protected:Nn \__spintent_sprecta_exec:nn
  {
    \keys_set:nn { spintent / sprecta } { #1 }
    \__spintent_sprecta_print:n { #2 }
  }
```

(*End of definition for* `\__spintent_sprecta_exec:nn`.)

`\sprecta`

The public command `\sprecta` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sprecta_exec:nn`.

```
\NewDocumentCommand \sprecta { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_sprecta_exec:nn { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \sprecta } }
  }
```

(*End of definition for* `\sprecta`. *This function is documented on page* *18*.)

11.34 The command `\sprayo`

The user command `\sprayo` provides a semantic interface for a ray. The key `read-sty` selects the verbalized term, `rayo` or `semirrecta`, both rendered identically with `\overrightarrow`.

11.34.1 Definition of variables for `\sprayo`

`\l__spintent_sprayo_mode_str`
`\l__spintent_sprayo_prefix_tl`
`\l__spintent_sprayo_read_sty_str`

The variable `\l__spintent_sprayo_mode_str` store the values set by the key `read-cmd`; the variable `\l__spintent_sprayo_prefix_tl` store the value set by the key `prefix` and the variable `\l__spintent_sprayo_r` stores the value set by the key `read-sty`.

The variables defined here are assigned using `.choices:nn` or `.code:n` within `\keys_define:nn`.

```
\str_new:N \l__spintent_sprayo_mode_str
\tl_new:N  \l__spintent_sprayo_prefix_tl
\str_new:N \l__spintent_sprayo_read_sty_str
```

(*End of definition for* \l__spintent_sprayo_mode_str *,* \l__spintent_sprayo_prefix_tl *, and* \l__spintent_sprayo_read_-sty_str*.*)

11.34.2 Definition of error messages for \sprayo

sprayo-invalid-points

Raised when the argument is not a comma-separated pair of uppercase letters.

```
\msg_new:nnnn { spintent } { sprayo-invalid-points }
  {
    Invalid~point~format~in~\token_to_str:N \sprayo:~'#1'~\msg_line_context:.
  }
  {
    See~the~documentation~of~\token_to_str:N \sprayo~for~the~accepted~point~formats.
  }
```

(*End of definition for* sprayo-invalid-points*.*)

11.34.3 Definition of keys for \sprayo

read-cmd
prefix
silent-cmd
read-sty
unknown

We define the ⟨*keys*⟩ read-cmd, prefix, silent-cmd, concept and read-sty.

```
\keys_define:nn { spintent / sprayo }
  {
    read-cmd            .choices:nn = { school , mathcat }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_sprayo_mode_str { school  } }
            {2} { \str_set:Nn \l__spintent_sprayo_mode_str { mathcat } }
          }
      },
    read-cmd            .initial:n  = school,
    read-cmd            .value_required:n = true,
    read-cmd / unknown  .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
    prefix              .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_sprayo_prefix_tl {#1} },
    prefix              .initial:n  = ,
    prefix              .value_required:n = true,
    silent-cmd          .bool_set:N = \l__spintent_sprayo_silent_cmd_bool,
    silent-cmd          .initial:n  = false,
    silent-cmd          .value_required:n = true,
    concept             .bool_set:N = \l__spintent_sprayo_concept_bool,
    concept             .initial:n  = true,
    concept             .value_required:n = true,
    read-sty            .choices:nn = { rayo , semirrecta }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_sprayo_read_sty_str { rayo       } }
            {2} { \str_set:Nn \l__spintent_sprayo_read_sty_str { semirrecta } }
          }
      },
    read-sty            .initial:n  = rayo,
    read-sty            .value_required:n = true,
    read-sty / unknown  .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-sty } { rayo,~semirrecta } { \exp_not:n {#1} },
    unknown             .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { sprayo }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* read-cmd *and others.*)

11.34.4 Internal functions for \sprayo

\l__spintent_sprayo_pt_seq
\l__spintent_sprayo_refs_tl
\l__spintent_sprayo_intent_arg_tl
\l__spintent_sprayo_pt_idx_int

Auxiliary variables used by __spintent_sprayo_mathml: to split \l__spintent_geo_luaset_points_str and build the $pts1,$pts2,... reference list.

```
\seq_new:N \l__spintent_sprayo_pt_seq
\tl_new:N  \l__spintent_sprayo_refs_tl
\tl_new:N  \l__spintent_sprayo_intent_arg_tl
\int_new:N \l__spintent_sprayo_pt_idx_int
```

(*End of definition for* `\l__spintent_sprayo_pt_seq` *and others.*)

`\__spintent_sprayo_print:n`

The internal function `\__spintent_sprayo_print:n` process the {⟨*argument*⟩} passed in '`#1`'. It calls `\__spintent_luafun_geo_parse_and_set:n` to process the {⟨*argument*⟩}, returning an "error" if the input is invalid, then calls `\__spintent_sprayo_mathml:`. Unlike `\sprecta`, `\sprayo` has no proper-name argument: it always takes a list of points, so no `intent` is built here.

```
\cs_new_protected:Nn \__spintent_sprayo_print:n
  {
    \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nne { spintent } { sprayo-invalid-points } { \exp_not:n {#1} }
      }
    \__spintent_sprayo_mathml:
  }
```

(*End of definition for* `\__spintent_sprayo_print:n`*.*)

`\__spintent_sprayo_mathml:`

The function `\__spintent_sprayo_mathml:` splits `\l__spintent_geo_luaset_points_str` (set by Lua, one raw point per element) into `\l__spintent_sprayo_pt_seq` and walks it once to build `\l__spintent_sprayo_re` the comma-separated `$pts1,$pts2,...` reference list, naming each point by position.
The concept `intent` is `_-:nofix(refs)` when `silent-cmd` is `true` *or* `concept` is `false`; otherwise it is `\l__spintent_sprayo_read_sty_str:prefix(refs)` — the concept word itself (`rayo` or `semirrecta`) comes from the `read-sty` key, not a literal, since `\sprayo` verbalizes under two names — prefixed with `prefix` when given. It is applied with a native `\MathMLintent` wrapping `\overrightarrow`, preceded by `\__spintent_luafun_inner_sep:` (a second sibling so the sole <mrow> does not collapse into <math>, `luamml` issue #26). The sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `pts`*n* reference name and `silent-cmd`/`read-cmd` looked up under `sprayo`: it alone decides `arg`/literal `intent` per point, so `concept` plays no role at that level.

```
\cs_new_protected:Nn \__spintent_sprayo_mathml:
  {
    \seq_set_from_clist:NN \l__spintent_sprayo_pt_seq \l__spintent_geo_luaset_points_str
    \tl_clear:N \l__spintent_sprayo_refs_tl
    \int_zero:N \l__spintent_sprayo_pt_idx_int
    \seq_map_inline:Nn \l__spintent_sprayo_pt_seq
      {
        \int_incr:N \l__spintent_sprayo_pt_idx_int
        \tl_if_empty:NF \l__spintent_sprayo_refs_tl
          { \tl_put_right:Nn \l__spintent_sprayo_refs_tl { , } }
        \tl_put_right:Ne \l__spintent_sprayo_refs_tl { $ pts \int_use:N \l__spintent_sprayo_pt_id
      }
    \bool_if:nTF
      {
        \bool_lazy_or_p:nn
          { \l__spintent_sprayo_silent_cmd_bool }
          { ! \l__spintent_sprayo_concept_bool }
      }
      {
        \tl_set:Ne \l__spintent_sprayo_intent_arg_tl { _-:nofix( \l__spintent_sprayo_refs_tl ) }
      }
      {
        \tl_set:Ne \l__spintent_sprayo_intent_arg_tl
          {
            \tl_if_empty:NF \l__spintent_sprayo_prefix_tl
              { \tl_use:N \l__spintent_sprayo_prefix_tl - }
            \l__spintent_sprayo_read_sty_str :prefix( \l__spintent_sprayo_refs_tl )
          }
      }
    \__spintent_invisible_sep:
    \MathMLintent { \l__spintent_sprayo_intent_arg_tl }
      {
        \__spintent_luafun_inner_sep:
        \overrightarrow
          {
            \int_zero:N \l__spintent_sprayo_pt_idx_int
            \seq_map_inline:Nn \l__spintent_sprayo_pt_seq
              {
                \int_incr:N \l__spintent_sprayo_pt_idx_int
                \__spintent_mathmlarg_point:nnn
                  { pts \int_use:N \l__spintent_sprayo_pt_idx_int }
```

```
                    { sprayo }
                    { ##1 }
                }
            }
        }
    }
```

(*End of definition for* `\__spintent_sprayo_mathml:`.)

`\__spintent_sprayo_exec:nn`

The function `\__spintent_sprayo_exec:nn` receives the optional ⟨*keys*⟩ passed in '`#1`' and the {⟨*argument*⟩} passed in '`#2`' delegating the rendering of the latter to `\__spintent_sprayo_print:n`.

```
\cs_new_protected:Nn \__spintent_sprayo_exec:nn
  {
    \keys_set:nn { spintent / sprayo } { #1 }
    \__spintent_sprayo_print:n { #2 }
  }
```

(*End of definition for* `\__spintent_sprayo_exec:nn`.)

`\sprayo`

The public command `\sprayo` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sprayo_exec:nn`.

```
\NewDocumentCommand \sprayo { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_sprayo_exec:nn { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \sprayo } }
  }
```

(*End of definition for* `\sprayo`. *This function is documented on page* 18.)

11.35 The commands `\spside` and `\spmside`

The user commands `\spside` and `\spmside` provide a semantic interface for a side, segment or trace determined by two points. The key `read-sty` selects the verbalized term — `lado`, `segmento`, or `trazo`, the latter used in straightedge-and-compass constructions — and the key `read-arg` overrides that term entirely with a free-form string. Additionally, the command `\spmside` provides the key `print-m`, controlling whether `\operatorname{m}` is typeset before the argument.

11.35.1 Definition of variables for `\spside` and `\spmside`

`\l__spintent_spside_mode_str`
`\l__spintent_spmside_mode_str`
`\l__spintent_spside_prefix_tl`
`\l__spintent_spmside_prefix_tl`
`\l__spintent_spside_read_sty_str`
`\l__spintent_spmside_read_sty_str`
`\l__spintent_spside_read_arg_tl`
`\l__spintent_spmside_read_arg_tl`

Each pair of variables is declared once for both `\spside` and `\spmside` via `\clist_map_inline:nn`. The variables `\l__spintent_spside_mode_str` and `\l__spintent_spside_mode_str` store the values set by the key `read-cmd`; the variables `\l__spintent_spside_prefix_tl` and `\l__spintent_spmside_prefix_tl` store the values set by the key `prefix`.

The variables `\l__spintent_spside_read_sty_str` and `\l__spintent_spmside_read_sty_str` stores the value set by the key `read-sty`.

The variables `\l__spintent_spside_read_arg_tl` and `\l__spintent_spmside_read_arg_tl` stores the value set by the key `read-arg`.

The variables are defined here as they are assigned using `.choices:nn` or `.code:n` rather than a direct property within `\keys_define:nn`.

```
\clist_map_inline:nn { side , mside }
  {
    \str_new:c { l__spintent_sp #1 _mode_str     }
    \tl_new:c  { l__spintent_sp #1 _prefix_tl    }
    \str_new:c { l__spintent_sp #1 _read_sty_str }
    \tl_new:c  { l__spintent_sp #1 _read_arg_tl  }
  }
```

(*End of definition for* `\l__spintent_spside_mode_str` *and others*.)

11.35.2 Definition of error messages for \spside and \spmside

spside-invalid-points
spmside-invalid-points

We define an error message for the commands \spside and \spmside. The "error" is triggered if the user includes a invalid *"point"* in the {⟨*argument*⟩} passed in '#1' to the command.

```
\clist_map_inline:nn { side , mside }
  {
    \msg_new:nnnn { spintent } { sp #1 -invalid-points }
      {
        Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~the~accepted~point~formats.
      }
  }
```

(*End of definition for* spside-invalid-points *and* spmside-invalid-points*.*)

11.35.3 Definition of keys for \spside and \spmside

read-cmd
prefix
silent-cmd
concept
read-sty
read-arg
print-m
unknown

We define the shared ⟨*keys*⟩ read-cmd, prefix, silent-cmd, concept, read-sty, read-arg and print-m.

```
\clist_map_inline:nn { side , mside }
  {
    \keys_define:nn { spintent / sp #1 }
      {
        read-cmd            .choices:nn = { school , mathcat }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school  } }
                {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
              }
          },
        read-cmd            .initial:n  = school,
        read-cmd            .value_required:n = true,
        read-cmd / unknown  .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
        prefix              .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix              .initial:n  = ,
        prefix              .value_required:n = true,
        silent-cmd          .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd          .initial:n  = false,
        silent-cmd          .value_required:n = true,
        concept             .bool_set:c = { l__spintent_sp #1 _concept_bool },
        concept             .initial:n  = true,
        concept             .value_required:n = true,
        read-sty            .choices:nn = { lado , segmento , trazo }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { lado      } }
                {2} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { segmento  } }
                {3} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { trazo     } }
              }
          },
        read-sty            .initial:n  = lado,
        read-sty            .value_required:n = true,
        read-sty / unknown  .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-sty } { lado,~segmento,~trazo } { \exp_not:n {##1} },
        read-arg            .code:n =
          {
            \__spintent_sanitize_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1}
          },
        read-arg            .initial:n  = ,
        read-arg            .value_required:n = true,
        print-m             .bool_set:c = { l__spintent_sp #1 _print_m_bool },
        print-m             .initial:n  = false,
        print-m             .value_required:n = true,
        unknown             .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
```

```
              \__spintent_unknown_keys_cmd:n {##1}
            },
        }
    }
```

The keys `print-m` and `concept` are defined for both commands for simplicity, but has no effect on `\spside`.

(*End of definition for* `read-cmd` *and others.*)

`\l__spintent_spside_intent_tl`
`\l__spintent_spside_pt_seq`
`\l__spintent_spside_refs_tl`
`\l__spintent_spside_intent_arg_tl`
`\l__spintent_spside_pt_idx_int`

`\l__spintent_spside_intent_tl` holds the concept word (`read-arg` or `read-sty`). The rest are scratch variables shared by both commands (the function itself is shared) to split `\l__spintent_geo_luaset_points_str` and build the `$pts1,$pts2,...` reference list.

```
\tl_new:N  \l__spintent_spside_intent_tl
\seq_new:N \l__spintent_spside_pt_seq
\tl_new:N  \l__spintent_spside_refs_tl
\tl_new:N  \l__spintent_spside_intent_arg_tl
\int_new:N \l__spintent_spside_pt_idx_int
```

(*End of definition for* `\l__spintent_spside_intent_tl` *and others.*)

`\__spintent_spside_process:nn`

The internal function `\__spintent_spside_process:nn` receives the short name of the command (`side` or `mside`) in '`#1`' and the {⟨*argument*⟩} in '`#2`'. It calls `\__spintent_luafun_geo_parse_and_set:n` to process '`#2`', returning an "error" if invalid, then builds `\l__spintent_spside_intent_tl`.
When `read-arg` is empty, it is set from `read-sty`, prefixed with '`medida-del-`' for `\spmside`. When `read-arg` is given, it is used as is, and '`medida-del-`' is prepended only if `concept` is `true`: with `concept=false`, `read-arg` is the verbatim escape hatch for a full user-written phrase (e.g. `read-arg={medida de la hipotenusa}`), gender and article left to the user. No `intent` is built here: it needs the reference list, which `\__spintent_spside_mathml` builds after splitting the points.

```
\cs_new_protected:Nn \__spintent_spside_process:nn
  {
    \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#2} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nnee { spintent } { sp #1 -invalid-points }
          { \c_backslash_str sp #1 } { \exp_not:n {#2} }
      }
    \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
      {
        \tl_set:Ne \l__spintent_spside_intent_tl { \str_use:c { l__spintent_sp #1 _read_sty_str }
        \str_if_eq:eeT {#1} { mside }
          {
            \tl_put_left:Nn \l__spintent_spside_intent_tl { medida-del- }
          }
      }
      {
        \tl_set:Ne \l__spintent_spside_intent_tl { \tl_use:c { l__spintent_sp #1 _read_arg_tl } }
        \bool_if:cT { l__spintent_sp #1 _concept_bool }
          {
            \str_if_eq:eeT {#1} { mside }
              {
                \tl_put_left:Nn \l__spintent_spside_intent_tl { medida-del- }
              }
          }
      }
    \__spintent_spside_mathml:n { #1 }
  }
```

(*End of definition for* `\__spintent_spside_process:nn`*.*)

`\__spintent_spside_mathml:n`

The function `\__spintent_spside_mathml:n` receives the short command name in '`#1`'. It splits `\l__spintent_geo_luaset_points_str` into `\l__spintent_spside_pt_seq` and walks it once to build `\l__spintent_spside_refs_tl`, the comma-separated `$pts1,$pts2,...` reference list, naming each point by position.
The concept `intent` is `_(refs)` when `silent-cmd` is `true`; otherwise, if `read-arg` is empty, it is `_(refs)` when `concept` is `false` or `\l__spintent_spside_intent_tl:prefix(refs)` when `true`; if `read-arg` is given, it is always `\l__spintent_spside_intent_tl:prefix(refs)` — concept never silences a user-given `read-arg` (see `\__spintent_spside_process:nn`). The `:prefix(...)` form is prefixed with `prefix` when given. It is applied with a native `\MathMLintent`, preceded by `\__spintent_luafun_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, `luamml` issue #26).
Inside, for `#1=side` the points are always wrapped in `\overline`; for `#1=mside`, when `print-m` is `true` a silent `\operatorname{m}` precedes the `\overline`-wrapped points; when `false`, the points are typeset

bare, with no `\overline` at all — the recursive lookup in the shared `mathml_filter` callback finds each point either way, whether nested inside `\overline` or sitting directly in the `\MathMLintent` group. In every case the sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `pts`*n* reference name and `silent-cmd/read-cmd` looked up under `sp#1`: it alone decides `arg`/literal `intent` per point, so `concept` plays no role at that level.

```
\cs_new_protected:Nn \__spintent_spside_mathml:n
  {
    \seq_set_from_clist:NN \l__spintent_spside_pt_seq \l__spintent_geo_luaset_points_str
    \tl_clear:N \l__spintent_spside_refs_tl
    \int_zero:N \l__spintent_spside_pt_idx_int
    \seq_map_inline:Nn \l__spintent_spside_pt_seq
      {
        \int_incr:N \l__spintent_spside_pt_idx_int
        \tl_if_empty:NF \l__spintent_spside_refs_tl
          { \tl_put_right:Nn \l__spintent_spside_refs_tl { , } }
        \tl_put_right:Ne \l__spintent_spside_refs_tl { $ pts \int_use:N \l__spintent_spside_pt_id
      }
    \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
      {
        \tl_set:Ne \l__spintent_spside_intent_arg_tl { _( \l__spintent_spside_refs_tl ) }
      }
      {
        \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
          {
            \bool_if:cTF { l__spintent_sp #1 _concept_bool }
              {
                \tl_set:Ne \l__spintent_spside_intent_arg_tl
                  {
                    \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                      { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                    \l__spintent_spside_intent_tl :prefix( \l__spintent_spside_refs_tl )
                  }
              }
              {
                \tl_set:Ne \l__spintent_spside_intent_arg_tl { _( \l__spintent_spside_refs_tl ) }
              }
          }
          {
            \tl_set:Ne \l__spintent_spside_intent_arg_tl
              {
                \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                  { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                \l__spintent_spside_intent_tl :prefix( \l__spintent_spside_refs_tl )
              }
          }
      }
    \__spintent_invisible_sep: % (luamml issues #26)
    \MathMLintent { \l__spintent_spside_intent_arg_tl }
      {
        \__spintent_luafun_inner_sep:
        \str_if_eq:eeTF {#1} { side }
          {
            \overline { \__spintent_spside_points_loop:n {#1} }
          }
          {
            \bool_if:cTF { l__spintent_sp #1 _print_m_bool }
              {
                \__spintent_luafun_wrap_mrow_intent_arg:nn
                  { } { intent = ":silent" }
                { \operatorname { m } }
                \overline { \__spintent_spside_points_loop:n {#1} }
              }
              {
                \__spintent_spside_points_loop:n {#1}
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_spside_mathml:n`.)

`\__spintent_spside_points_loop:n`

Walks `\l__spintent_spside_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once per point. Factored out of `\__spintent_spside_mathml:n` since it is needed identically in three places there (`side`, `mside` with and without `print-m`); it takes the short command name in '`#1`' again since a separate macro does not inherit its caller's own `#1`.

```
\cs_new_protected:Nn \__spintent_spside_points_loop:n
  {
    \int_zero:N \l__spintent_spside_pt_idx_int
    \seq_map_inline:Nn \l__spintent_spside_pt_seq
      {
        \int_incr:N \l__spintent_spside_pt_idx_int
        \__spintent_mathmlarg_point:nnn
          { pts \int_use:N \l__spintent_spside_pt_idx_int }
          { sp #1 }
          { ##1 }
      }
  }
```

(*End of definition for* `\__spintent_spside_points_loop:n`.)

`\__spintent_spside_exec:nnn`

The function `\__spintent_spside_exec:nnn` receives the *short name* of the command passed in '`#1`', the optional ⟨*keys*⟩ passed in '`#2`' and the {⟨*argument*⟩} passed in '`#3`', delegating the rendering to `\__spintent_spside_process:nn`.

```
\cs_new_protected:Nn \__spintent_spside_exec:nnn
  {
    \keys_set:nn { spintent / sp #1 } { #2 }
    \__spintent_spside_process:nn { #1 } { #3 }
  }
```

(*End of definition for* `\__spintent_spside_exec:nnn`.)

`\spside`
`\spmside`

Both public commands `\spside` and `\spmside` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spside_exec:nnn`.

```
\NewDocumentCommand \spside { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spside_exec:nnn { side } { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spside } }
  }
\NewDocumentCommand \spmside { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spside_exec:nnn { mside } { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmside } }
  }
```

(*End of definition for* `\spside` *and* `\spmside`. *These functions are documented on page* 18.)

11.36 The commands `\spangle` and `\spmangle`

The user commands `\spangle` and `\spmangle`, provide a semantic interface for angles, following the same generic pattern as `\spside`/`\spmside`. The key `print-m` chooses between the `\angle`/`\rightanglesqr` pair, typeset together with an explicit `\operatorname{m}`, and the `\measuredangle`/`\measuredrightangle` pair, which already conveys the notion of measure on its own; the key `recto` independently chooses the right-angle variant of whichever pair is active.

11.36.1 Definition of variables for `\spangle` and `\spmangle`

`\l__spintent_spangle_mode_str`
`\l__spintent_spmangle_mode_str`
`\l__spintent_spangle_prefix_tl`
`\l__spintent_spmangle_prefix_tl`

Each pair of variables is declared once for both `\spangle` and `\spmangle` via `\clist_map_inline:nn`. The variables `\l__spintent_spangle_mode_str` and `\l__spintent_spmangle_mode_str` store the values set by the key `read-cmd`; the variables `\l__spintent_spangle_prefix_tl` and `\l__spintent_spmangle_prefix` store the values set by the key `prefix`.

The variables defined here are assigned using `.choices:nn` or `.code:n` within `\keys_define:nn`.

```
\clist_map_inline:nn { angle , mangle }
  {
    \str_new:c { l__spintent_sp #1 _mode_str  }
```

```
    \tl_new:c { l__spintent_sp #1 _prefix_tl }
  }
```

(*End of definition for* `\l__spintent_spangle_mode_str` *and others.*)

11.36.2 Definition of error messages for \spangle and \spmangle

spangle-invalid-arg
spmangle-invalid-arg

We define an "error" message for the commands `\spangle` and `\spmangle`. The "error" is triggered if the user includes a invalid {⟨*argument*⟩} passed in '#1' to the command.

```
\clist_map_inline:nn { angle , mangle }
  {
    \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
      {
        Invalid~format~in~##1:~'##2'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~the~accepted~argument~forms.
      }
  }
```

(*End of definition for* `spangle-invalid-arg` *and* `spmangle-invalid-arg`*.*)

11.36.3 Definition of keys for \spangle and \spmangle

read-cmd
prefix
recto
print-m
silent-cmd
concept
unknown

We define the shared ⟨*keys*⟩ `read-cmd`, `prefix`, `recto`, `print-m`, `silent-cmd` and `concept`,

```
\clist_map_inline:nn { angle , mangle }
  {
    \keys_define:nn { spintent / sp #1 }
      {
        read-cmd             .choices:nn = { school , mathcat }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school  } }
                {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
              }
          },
        read-cmd             .initial:n  = school,
        read-cmd             .value_required:n = true,
        read-cmd / unknown   .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
        prefix               .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix               .initial:n  = ,
        prefix               .value_required:n = true,
        recto                .bool_set:c = { l__spintent_sp #1 _recto_bool },
        recto                .initial:n  = false,
        recto                .value_required:n = true,
        print-m              .bool_set:c = { l__spintent_sp #1 _print_m_bool },
        print-m              .initial:n  = false,
        print-m              .value_required:n = true,
        silent-cmd           .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd           .initial:n  = false,
        silent-cmd           .value_required:n = true,
        concept              .bool_set:c = { l__spintent_sp #1 _concept_bool },
        concept              .initial:n  = true,
        concept              .value_required:n = true,
        unknown              .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
            \__spintent_unknown_keys_cmd:n {##1}
          },
      }
  }
```

The keys `print-m` and `concept` are defined for both commands for simplicity, but has no effect on `\spangle`.

(*End of definition for* `read-cmd` *and others.*)

11.36.4 Internal functions for \spangle and \spmangle

__spintent_spangle_sym:n

The internal function __spintent_spangle_sym:n dispatches on #1, print-m and recto to select one of four symbols: \angle/\rightanglesqr when print-m is true; \measuredangle/\measuredrightangle otherwise. When #1 is angle, only recto applies, choosing between \angle and \rightangle.

```
\cs_new_protected:Nn \__spintent_spangle_sym:n
  {
    \str_if_eq:eeTF {#1} { mangle }
      {
        \bool_if:cTF { l__spintent_sp #1 _print_m_bool }
          {
            \bool_if:cTF { l__spintent_sp #1 _recto_bool }
              { \__spintent_right_angle_sqr: } { \angle }
          }
          {
            \bool_if:cTF { l__spintent_sp #1 _recto_bool }
              { \measuredrightangle } { \measuredangle }
          }
      }
      {
        \bool_if:cTF { l__spintent_sp #1 _recto_bool }
          { \__spintent_right_angle_sqr: } { \angle }
      }
  }
```

(*End of definition for __spintent_spangle_sym:n.*)

\l__spintent_spangle_intent_tl
\l__spintent_spangle_pt_seq
\l__spintent_spangle_refs_tl
\l__spintent_spangle_intent_arg_tl
\l__spintent_spangle_pt_idx_int
\l__spintent_spangle_print_m_now_bool

\l__spintent_spangle_intent_tl holds the concept word (ángulo or medida-del-ángulo, '-recto'-suffixed when recto is true). The rest are scratch variables shared by both commands to split \l__spintent_geo_luas and build the reference list — used only in the tres-puntos and un-punto cases (see __spintent_spangle_mathml:

```
\tl_new:N   \l__spintent_spangle_intent_tl
\seq_new:N  \l__spintent_spangle_pt_seq
\tl_new:N   \l__spintent_spangle_refs_tl
\tl_new:N   \l__spintent_spangle_intent_arg_tl
\int_new:N  \l__spintent_spangle_pt_idx_int
\str_new:N  \l__spintent_spangle_kv_tl
\bool_new:N \l__spintent_spangle_print_m_now_bool
```

(*End of definition for \l__spintent_spangle_intent_tl and others.*)

__spintent_spangle_process:nn

The internal function __spintent_spangle_process:nn receives the short name of the command (angle or mangle) in '#1' and the {⟨*argument*⟩} in '#2'. It calls __spintent_luafun_geo_angulo_parse_and_set:n to process '#2', returning an "error" if invalid, then sets \l__spintent_spangle_intent_tl to ángulo (or medida-del-ángulo for \spmangle), suffixed with '-recto' when recto is true. No intent is built here: it depends on \l__spintent_geo_luaset_angulo_case_str (tres-puntos, un-punto or nombre), which __spintent_spangle_mathml:n branches on.

```
\cs_new_protected:Nn \__spintent_spangle_process:nn
  {
    \__spintent_luafun_geo_angulo_parse_and_set:n { \exp_not:n {#2} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nnee { spintent } { sp #1 -invalid-arg }
          { \c_backslash_str sp #1 } { \exp_not:n {#2} }
      }
    \str_if_eq:eeTF {#1} { mangle }
      {
        \tl_set:Nn \l__spintent_spangle_intent_tl { medida-del-ángulo }
      }
      {
        \tl_set:Nn \l__spintent_spangle_intent_tl { ángulo }
      }
    \bool_if:cT { l__spintent_sp #1 _recto_bool }
      {
        \tl_put_right:Nn \l__spintent_spangle_intent_tl { -recto }
      }
    \__spintent_spangle_mathml:n {#1}
  }
```

(*End of definition for __spintent_spangle_process:nn.*)

__spintent_spangle_mathml:n

For tres-puntos/un-punto, a single native \MathMLintent wraps the symbol and the points together, with \l__spintent_spangle_intent_tl:prefix(refs) — the same shape as __spintent_spside_mathml:n. The earlier split into two sibling \MathMLintent calls worked around a MathCAT bug with several explicit $-references in :prefix(...) that only affects the demo build; the stable NVDA/updated Acrobat Reader combination reads this shape correctly, so the split is no longer needed and both commands now share the same structure.
For \spmangle with print-m true, a silent \operatorname{m} precedes __spintent_spangle_sym:n inside the same \MathMLintent, exactly as __spintent_spside_mathml:n does for its own m.
nombre is unchanged (see below): it never references more than one thing, so it was never affected by the bug this simplifies away from.

```
\cs_new_protected:Nn \__spintent_spangle_mathml:n
  {
    \__spintent_invisible_sep: % (luamml issues #26)
    \str_if_eq:VnTF \l__spintent_geo_luaset_angulo_case_str { nombre }
      {
        \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
          {
            \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
          }
          {
            \bool_if:cTF { l__spintent_sp #1 _concept_bool }
              {
                \tl_set:Ne \l__spintent_spangle_intent_arg_tl
                  {
                    \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                      { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                    \l__spintent_spangle_intent_tl :prefix
                  }
              }
              {
                \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
              }
          }
        \tl_set:Ne \l__spintent_spangle_kv_tl
          { intent = "\l__spintent_spangle_intent_arg_tl" }
        \__spintent_luafun_wrap_mrow_intent_arg:nn
          { } { \l__spintent_spangle_kv_tl }
        { \__spintent_spangle_sym:n {#1} }
        \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
          {
            \__spintent_luafun_wrap_mrow_intent_arg:nn { } { intent = "_-" }
            { \l__spintent_geo_luaset_print_tl }
          }
          {
            \str_if_eq:VnTF \l__spintent_geo_luaset_nombre_es_num_str { true }
              {
                \tl_set:Ne \l__spintent_spangle_kv_tl
                  { intent = "_ \l__spintent_geo_luaset_intent_str -" }
                \__spintent_luafun_wrap_mrow_intent_arg:nn
                  { } { \l__spintent_spangle_kv_tl }
                { \l__spintent_geo_luaset_print_tl }
              }
              { \l__spintent_geo_luaset_print_tl }
          }
      }
      {
        \seq_set_from_clist:NN \l__spintent_spangle_pt_seq \l__spintent_geo_luaset_points_str
        \tl_clear:N \l__spintent_spangle_refs_tl
        \int_zero:N \l__spintent_spangle_pt_idx_int
        \seq_map_inline:Nn \l__spintent_spangle_pt_seq
          {
            \int_incr:N \l__spintent_spangle_pt_idx_int
            \tl_if_empty:NF \l__spintent_spangle_refs_tl
              { \tl_put_right:Nn \l__spintent_spangle_refs_tl { , } }
            \tl_put_right:Ne \l__spintent_spangle_refs_tl { $ pt \int_use:N \l__spintent_spangle_
          }
        \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
          {
            \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
          }
```

```
          {
            \bool_if:cTF { l__spintent_sp #1 _concept_bool }
              {
                \tl_set:Ne \l__spintent_spangle_intent_arg_tl
                  {
                    \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                      { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                    \l__spintent_spangle_intent_tl :prefix( \l__spintent_spangle_refs_tl )
                  }
              }
              {
                \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
              }
          }
        \bool_set_false:N \l__spintent_spangle_print_m_now_bool
        \str_if_eq:eeT {#1} { mangle }
          {
            \bool_if:cT { l__spintent_sp #1 _print_m_bool }
              { \bool_set_true:N \l__spintent_spangle_print_m_now_bool }
          }
        \MathMLintent { \l__spintent_spangle_intent_arg_tl }
          {
            \__spintent_luafun_inner_sep:
            \bool_if:NT \l__spintent_spangle_print_m_now_bool
              {
                \__spintent_luafun_wrap_mrow_intent_arg:nn
                  { } { intent = ":silent" }
                { \operatorname { m } }
              }
            \__spintent_luafun_wrap_mrow_intent_arg:nn
              { } { intent = ":silent" }
            { \__spintent_spangle_sym:n {#1} }
            \int_zero:N \l__spintent_spangle_pt_idx_int
            \seq_map_inline:Nn \l__spintent_spangle_pt_seq
              {
                \int_incr:N \l__spintent_spangle_pt_idx_int
                \__spintent_mathmlarg_point:nnn
                  { pt \int_use:N \l__spintent_spangle_pt_idx_int }
                  { sp#1 }
                  { ##1 }
              }
          }
      }
  }
```

(End of definition for `\__spintent_spangle_mathml:n`*.)*

`\__spintent_spangle_exec:nnn`

The function `\__spintent_spangle_exec:nnn` receives the *short name* of the command passed in '#1', the optional ⟨*keys*⟩ passed in '#2' and the {⟨*argument*⟩} passed in '#3', delegating the rendering to `\__spintent_spangle_process:nn`.

```
\cs_new_protected:Nn \__spintent_spangle_exec:nnn
  {
    \keys_set:nn { spintent / sp #1 } { #2 }
    \__spintent_spangle_process:nn { #1 } { #3 }
  }
```

(End of definition for `\__spintent_spangle_exec:nnn`*.)*

`\spangle`
`\spmangle`

Both public commands `\spangle` and `\spmangle` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spangle_exec:nnn`.

```
\NewDocumentCommand \spangle { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spangle_exec:nnn { angle } { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spangle } }
  }
\NewDocumentCommand \spmangle { O{} m }
```

```
    {
      \mode_if_math:TF
        {
          \group_begin:
            \__spintent_spangle_exec:nnn { mangle } { #1 } { #2 }
          \group_end:
        }
        { \msg_error:nnn { spintent } { need-math-mode } { \spmangle } }
    }
```

(*End of definition for* `\spangle` *and* `\spmangle`. *These functions are documented on page* 20.)

11.37 The commands `\sparc` and `\spmarc`

The user commands `\sparc` and `\spmarc` provide a semantic interface for arcs. Unlike `\spmside` and `\spmangle`, `\spmarc` has no `print-m` key: `\operatorname{m}` is always present, since $\mathrm{m}\,\widehat{AB}$ is the only standard notation for the measure of an arc.

11.37.1 Definition of variables for `\sparc` and `\spmarc`

`\l__spintent_sparc_mode_str`
`\l__spintent_spmarc_mode_str`
`\l__spintent_sparc_prefix_tl`
`\l__spintent_smparc_prefix_tl`

Each pair of variables is declared once for both `\sparc` and `\spmarc` via `\clist_map_inline:nn`. The variables `\l__spintent_sparc_mode_str` and `\l__spintent_spmarc_mode_str` store the values set by the key `read-cmd`; the variables `\l__spintent_sparc_prefix_tl` and `\l__spintent_smparc_prefix_tl` store the values set by the key `prefix`.

```
\clist_map_inline:nn { arc , marc }
  {
    \str_new:c { l__spintent_sp #1 _mode_str  }
    \tl_new:c  { l__spintent_sp #1 _prefix_tl }
  }
```

(*End of definition for* `\l__spintent_sparc_mode_str` *and others.*)

11.37.2 Definition of error messages for `\sparc` and `\spmarc`

`sparc-invalid-points`
`spmarc-invalid-points`

We define an "error" message for the commands `\sparc` and `\spmarc`. The "error" is triggered if the user includes a invalid {⟨*argument*⟩} passed in '`#1`' to the command.

```
\clist_map_inline:nn { arc , marc }
  {
    \msg_new:nnnn { spintent } { sp #1 -invalid-points }
      {
        Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~the~accepted~point~formats.
      }
  }
```

(*End of definition for* `sparc-invalid-points` *and* `spmarc-invalid-points`.)

11.37.3 Definition of keys for `\sparc` and `\spmarc`

`read-cmd`
`prefix`
`silent-cmd`
`concept`
`unknown`

We define the shared ⟨*keys*⟩ `read-cmd`, `prefix`, `silent-cmd` and `concept`.

```
\clist_map_inline:nn { arc , marc }
  {
    \keys_define:nn { spintent / sp #1 }
      {
        read-cmd            .choices:nn = { school , mathcat }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school  } }
                {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
              }
          },
        read-cmd            .initial:n  = school,
        read-cmd            .value_required:n = true,
        read-cmd / unknown  .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
        prefix              .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix              .initial:n  = ,
        prefix              .value_required:n = true,
        silent-cmd          .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd          .initial:n  = false,
```

```
        silent-cmd          .value_required:n = true,
        concept             .bool_set:c = { l__spintent_sp #1 _concept_bool },
        concept             .initial:n  = true,
        concept             .value_required:n = true,
        unknown             .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
            \__spintent_unknown_keys_cmd:n {##1}
          },
      }
  }
```

The key print-m is defined for both commands for simplicity, but has no effect on \sparc.

(*End of definition for* read-cmd *and others.*)

11.37.4 Internal functions for \sparc and \spmarc

__spintent_sparc_wide_arc:n

The function __spintent_sparc_wide_arc:n acts as a fallback when the mathematical font used in the document does not provide the command \widearc. The definition of this function is a direct adaptation of a response given by Clea F. Rees (@cfr) to a query in the chat room of TeX-SX.

```
\cs_new_protected_nopar:Nn \__spintent_sparc_wide_arc:n
  {
    \Umathaccent 7 \symoperators "023DC \scan_stop: {#1}
  }
```

(*End of definition for* __spintent_sparc_wide_arc:n.)

__spintent_sparc_print_arc:n

The function __spintent_sparc_print_arc:n applies the command \widearc to the argument passed to \sparc or \spmarc. If the mathematical font set in the document does not provide the command \widearc, we will use the built-in function __spintent_sparc_wide_arc:n.

```
\cs_new_protected:Nn \__spintent_sparc_print_arc:n
  {
    \cs_if_exist:NTF \widearc
      { \widearc {#1} }
      { \__spintent_sparc_wide_arc:n {#1} }
  }
```

(*End of definition for* __spintent_sparc_print_arc:n.)

\l__spintent_sparc_intent_tl
\l__spintent_sparc_pt_seq
\l__spintent_sparc_refs_tl
\l__spintent_sparc_intent_arg_tl
\l__spintent_sparc_pt_idx_int

\l__spintent_sparc_intent_tl holds the concept word (arco or medida-del-arco). The rest are scratch variables to split \l__spintent_geo_luaset_points_str and build the $pt1,$pt2,... reference list used in the concept intent.

```
\tl_new:N  \l__spintent_sparc_intent_tl
\seq_new:N \l__spintent_sparc_pt_seq
\tl_new:N  \l__spintent_sparc_refs_tl
\tl_new:N  \l__spintent_sparc_intent_arg_tl
\int_new:N \l__spintent_sparc_pt_idx_int
```

(*End of definition for* \l__spintent_sparc_intent_tl *and others.*)

__spintent_sparc_process:nn

The internal function __spintent_sparc_process:nn receives the short name of the command (arc or marc) in '#1' and the {⟨*argument*⟩} in '#2'. It calls __spintent_luafun_geo_parse_and_set:n to process '#2', returning an "error" if invalid, then sets \l__spintent_sparc_intent_tl to arco (or medida-del-arco for \spmarc). No intent is built here: it needs the reference list, which __spintent_sparc_math builds after splitting the points.

```
\cs_new_protected:Nn \__spintent_sparc_process:nn
  {
    \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#2} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nnee { spintent } { sp #1 -invalid-points }
          { \c_backslash_str sp #1 } { \exp_not:n {#2} }
      }
    \str_if_eq:eeTF {#1} { marc }
      { \tl_set:Nn \l__spintent_sparc_intent_tl { medida-del-arco } }
      { \tl_set:Nn \l__spintent_sparc_intent_tl { arco } }
    \__spintent_sparc_mathml:n {#1}
  }
```

(*End of definition for* __spintent_sparc_process:nn.)

__spintent_sparc_mathml:n

The function __spintent_sparc_mathml:n receives the short command name in '#1'. It splits \l__spintent_geo_ into \l__spintent_sparc_pt_seq and walks it once to build \l__spintent_sparc_refs_tl, the comma-separated $pt1,$pt2,... reference list, naming each point by position.

The concept intent is _- when silent-cmd is true or concept is false; otherwise \l__spintent_sparc_intent_ prefixed with prefix when given. It is applied with a single native \MathMLintent wrapping __spintent_sparc_pr directly (same shape as __spintent_spside_mathml:n), preceded by __spintent_luafun_inner_sep: (a second sibling so the sole <mrow> does not collapse into <math>, luamml issue #26). A silent \operatorname{m} precedes it for \spmarc (always, no print-m key here). The sequence is walked a second time, calling __spintent_mathmlarg_point:nnn once per point with its pt*n* reference name, looked up under sp#1.

```
\cs_new_protected:Nn \__spintent_sparc_mathml:n
  {
    \seq_set_from_clist:NN \l__spintent_sparc_pt_seq \l__spintent_geo_luaset_points_str

    \tl_clear:N \l__spintent_sparc_refs_tl
    \int_zero:N \l__spintent_sparc_pt_idx_int
    \seq_map_inline:Nn \l__spintent_sparc_pt_seq
      {
        \int_incr:N \l__spintent_sparc_pt_idx_int
        \tl_if_empty:NF \l__spintent_sparc_refs_tl
          { \tl_put_right:Nn \l__spintent_sparc_refs_tl { , } }
        \tl_put_right:Ne \l__spintent_sparc_refs_tl { $ pt \int_use:N \l__spintent_sparc_pt_idx_i
      }

    \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
      {
        \tl_set:Nn \l__spintent_sparc_intent_arg_tl { _- }
      }
      {
        \bool_if:cTF { l__spintent_sp #1 _concept_bool }
          {
            \tl_set:Ne \l__spintent_sparc_intent_arg_tl
              {
                \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                  { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                \l__spintent_sparc_intent_tl :prefix( \l__spintent_sparc_refs_tl )
              }
          }
          {
            \tl_set:Nn \l__spintent_sparc_intent_arg_tl { _- }
          }
      }

    \__spintent_invisible_sep: % need by MathCAT (bug in luamml issues #26)
    \MathMLintent { \l__spintent_sparc_intent_arg_tl }
      {
        \__spintent_luafun_inner_sep:
        \str_if_eq:eeT {#1} { marc }
          {
            \__spintent_luafun_wrap_mrow_intent_arg:nn
              { } { intent = ":silent" }
            { \operatorname { m } }
          }
        \__spintent_sparc_print_arc:n
          {
            \int_zero:N \l__spintent_sparc_pt_idx_int
            \seq_map_inline:Nn \l__spintent_sparc_pt_seq
              {
                \int_incr:N \l__spintent_sparc_pt_idx_int
                \__spintent_mathmlarg_point:nnn
                  { pt \int_use:N \l__spintent_sparc_pt_idx_int }
                  { sp #1 }
                  { ##1 }
              }
          }
      }
  }
```

(*End of definition for* __spintent_sparc_mathml:n*.*)

__spintent_sparc_exec:nnn

The function __spintent_sparc_exec:nnn receives the *short name* of the command passed in '#1', the optional ⟨*keys*⟩ passed in '#2' and the {⟨*argument*⟩} passed in '#3', delegating the rendering to __spintent_sparc_process:nn.

```
\cs_new_protected:Nn \__spintent_sparc_exec:nnn
  {
    \keys_set:nn { spintent / sp #1 } { #2 }
    \__spintent_sparc_process:nn { #1 } { #3 }
  }
```

(*End of definition for* __spintent_sparc_exec:nnn.)

\sparc
\spmarc

Both public commands \sparc and \spmarc enforces *math mode*, opens a local group, and delegates execution to __spintent_sparc_exec:nnn.

```
\NewDocumentCommand \sparc { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_sparc_exec:nnn { arc } { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \sparc } }
  }
\NewDocumentCommand \spmarc { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_sparc_exec:nnn { marc } { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spmarc } }
  }
```

(*End of definition for* \sparc *and* \spmarc*. These functions are documented on page 21.*)

11.38 The command \spPoly

The user command \spPoly typesets a polygon defined by its vertices. The figure type and its visual symbol are inferred from the vertex count: three points produce a triangle, four a square, five or more a generic polygon with no symbol. It never sends a boolean to Lua: \l__spintent_geo_luaset_intent_str (the full intent, with the figure name) and \l__spintent_geo_luaset_intent_pts_str (vertices only) are both returned unconditionally, and expl3 picks between them according to print-sym.

11.38.1 Definition of variables for \spPoly

\l__spintent_spPoly_mode_str
\l__spintent_spPoly_prefix_tl

The variable \l__spintent_spPoly_mode_str stores the value set by the key read-cmd; the variable \l__spintent_spPoly_prefix_tl stores the value set by the key prefix.

```
\str_new:N \l__spintent_spPoly_mode_str
\tl_new:N  \l__spintent_spPoly_prefix_tl
```

(*End of definition for* \l__spintent_spPoly_mode_str *and* \l__spintent_spPoly_prefix_tl.)

11.38.2 Definition of error messages for \spPoly

spPoly-invalid-points

Raised when the argument is not a comma-separated list of at least three uppercase letters.

```
\msg_new:nnnn { spintent } { spPoly-invalid-points }
  {
    Invalid~point~format~in~\token_to_str:N \spPoly:~'#1'~\msg_line_context:.
  }
  {
    See~the~documentation~of~\token_to_str:N \spPoly~for~the~accepted~point~formats.
  }
```

(*End of definition for* spPoly-invalid-points.)

11.38.3 Definition of keys for \spPoly

read-cmd
prefix
print-sym
silent-cmd
concept
unknown

We define the ⟨*keys*⟩ read-cmd, prefix, print-sym, silent-cmd and concept.

```
\keys_define:nn { spintent / spPoly }
  {
    read-cmd            .choices:nn = { school , mathcat }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spPoly_mode_str { school  } }
```

```
            {2} { \str_set:Nn \l__spintent_spPoly_mode_str { mathcat } }
          }
      },
    read-cmd              .initial:n  = school,
    read-cmd              .value_required:n = true,
    read-cmd / unknown    .code:n =
      \msg_error:nneee { spintent } { unknown-choice }
        { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
    prefix                .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spPoly_prefix_tl {#1} },
    prefix                .initial:n  = ,
    prefix                .value_required:n = true,
    print-sym             .bool_set:N = \l__spintent_spPoly_print_sym_bool,
    print-sym             .initial:n  = true,
    print-sym             .value_required:n = true,
    silent-cmd            .bool_set:N = \l__spintent_spPoly_silent_cmd_bool,
    silent-cmd            .initial:n  = false,
    silent-cmd            .value_required:n = true,
    concept               .bool_set:N = \l__spintent_spPoly_concept_bool,
    concept               .initial:n  = true,
    concept               .value_required:n = true,
    unknown               .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spPoly }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `read-cmd` *and others.*)

11.38.4 Internal functions for `\spPoly`

`\l__spintent_spPoly_pt_seq`
`\l__spintent_spPoly_refs_tl`
`\l__spintent_spPoly_intent_arg_tl`
`\l__spintent_spPoly_pt_idx_int`

Scratch variables to split `\l__spintent_geo_luaset_points_str` and build the `$pt1,$pt2,...` reference list, and to hold the final concept `intent` once built.

```
\seq_new:N \l__spintent_spPoly_pt_seq
\tl_new:N  \l__spintent_spPoly_refs_tl
\tl_new:N  \l__spintent_spPoly_intent_arg_tl
\int_new:N \l__spintent_spPoly_pt_idx_int
```

(*End of definition for* `\l__spintent_spPoly_pt_seq` *and others.*)

`\__spintent_spPoly_sym:`

Unchanged: reads `\l__spintent_geo_luaset_poly_sym_str` and typesets the corresponding symbol.

```
\cs_new_protected:Nn \__spintent_spPoly_sym:
  {
    \str_case:Vn \l__spintent_geo_luaset_poly_sym_str
      {
        { triangle } { \triangle      }
        { square   } { \mdlgwhtsquare }
      }
  }
```

(*End of definition for* `\__spintent_spPoly_sym:`.)

`\__spintent_spPoly_build_refs:`

Splits `\l__spintent_geo_luaset_points_str` into `\l__spintent_spPoly_pt_seq` and walks it once to build `\l__spintent_spPoly_refs_tl`, the comma-separated `$pt1,$pt2,...` reference list, naming each vertex by position.

```
\cs_new_protected:Nn \__spintent_spPoly_build_refs:
  {
    \seq_set_from_clist:NN \l__spintent_spPoly_pt_seq \l__spintent_geo_luaset_points_str
    \tl_clear:N \l__spintent_spPoly_refs_tl
    \int_zero:N \l__spintent_spPoly_pt_idx_int
    \seq_map_inline:Nn \l__spintent_spPoly_pt_seq
      {
        \int_incr:N \l__spintent_spPoly_pt_idx_int
        \tl_if_empty:NF \l__spintent_spPoly_refs_tl
          { \tl_put_right:Nn \l__spintent_spPoly_refs_tl { , } }
        \tl_put_right:Ne \l__spintent_spPoly_refs_tl { $ pt \int_use:N \l__spintent_spPoly_pt_idx_
      }
  }
```

(*End of definition for* `\__spintent_spPoly_build_refs:`.)

__spintent_spPoly_build_concept: Builds \l__spintent_spPoly_intent_arg_tl, independently of print-sym: _(refs) when silent-cmd is true or concept is false; otherwise \l__spintent_geo_luaset_concept_str:prefix(refs), prefixed with prefix when given. Must be called after __spintent_spPoly_build_refs:.

```
\cs_new_protected:Nn \__spintent_spPoly_build_concept:
  {
    \bool_if:NTF \l__spintent_spPoly_silent_cmd_bool
      {
        \tl_set:Ne \l__spintent_spPoly_intent_arg_tl { _( \l__spintent_spPoly_refs_tl ) }
      }
      {
        \bool_if:NTF \l__spintent_spPoly_concept_bool
          {
            \tl_set:Ne \l__spintent_spPoly_intent_arg_tl
              {
                \tl_if_empty:NF \l__spintent_spPoly_prefix_tl
                  { \tl_use:N \l__spintent_spPoly_prefix_tl - }
                \l__spintent_geo_luaset_concept_str :prefix( \l__spintent_spPoly_refs_tl )
              }
          }
          {
            \tl_set:Ne \l__spintent_spPoly_intent_arg_tl { _( \l__spintent_spPoly_refs_tl ) }
          }
      }
  }
```

(*End of definition for* __spintent_spPoly_build_concept:.)

__spintent_spPoly_points_loop: Walks \l__spintent_spPoly_pt_seq a second time, calling __spintent_mathmlarg_point:nnn once per vertex with its pt*n* reference name, looked up under spPoly: it alone decides arg/literal intent per vertex — replacing the single combined \MathMLarg{pts}{...} that lost arg on any vertex with a subscript or prime (the same annotate_attribute/sub-sup limitation already found and worked around for \spPoint).

```
\cs_new_protected:Nn \__spintent_spPoly_points_loop:
  {
    \int_zero:N \l__spintent_spPoly_pt_idx_int
    \seq_map_inline:Nn \l__spintent_spPoly_pt_seq
      {
        \int_incr:N \l__spintent_spPoly_pt_idx_int
        \__spintent_mathmlarg_point:nnn
          { pt \int_use:N \l__spintent_spPoly_pt_idx_int }
          { spPoly }
          { ##1 }
      }
  }
```

(*End of definition for* __spintent_spPoly_points_loop:.)

__spintent_spPoly_process:n Calls __spintent_luafun_geo_poly_parse_and_set:n, reports spPoly-invalid-points on failure, then delegates to __spintent_spPoly_mathml:. No intent is built here: it needs the reference list, which __spintent_spPoly_mathml: builds after splitting the vertices.

```
\cs_new_protected:Nn \__spintent_spPoly_process:n
  {
    \__spintent_luafun_geo_poly_parse_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nne { spintent } { spPoly-invalid-points }
          { \exp_not:n {#1} }
      }
    \__spintent_spPoly_mathml:
  }
```

(*End of definition for* __spintent_spPoly_process:n.)

__spintent_spPoly_mathml: Builds the reference list and the concept intent (__spintent_spPoly_build_refs:, __spintent_spPoly_buil
then wraps everything in a single native \MathMLintent, preceded by __spintent_luafun_inner_sep: (a second sibling so the sole <mrow> does not collapse into <math>, luamml issue #26).
print-sym and concept/silent-cmd are independent: the former only decides whether the glyph (__spintent_spPoly_sym:, silenced with __spintent_luafun_wrap_mrow_intent_arg:nn) is attempted — it is skipped anyway when \l__spintent_geo_luaset_poly_sym_str is empty (no symbol defined for that vertex count, e.g. pentagon and above) — and never affects whether the concept word itself is spoken.

```
\cs_new_protected:Nn \__spintent_spPoly_mathml:
```

```
  {
    \__spintent_invisible_sep: % (luamml issues #26)
    \__spintent_spPoly_build_refs:
    \__spintent_spPoly_build_concept:
    \MathMLintent { \l__spintent_spPoly_intent_arg_tl }
      {
        \__spintent_luafun_inner_sep:
        \bool_if:NT \l__spintent_spPoly_print_sym_bool
          {
            \str_if_empty:NF \l__spintent_geo_luaset_poly_sym_str
              {
                \__spintent_luafun_wrap_mrow_intent_arg:nn
                  { } { intent = ":silent" }
                { \__spintent_spPoly_sym: }
              }
          }
        \__spintent_spPoly_points_loop:
      }
  }
```

(*End of definition for* `\__spintent_spPoly_mathml:`.)

`\__spintent_spPoly_exec:nn`

The function `\__spintent_spPoly_exec:nn` receives the optional keys in `#1` and the argument in `#2`, delegating the rendering to `\__spintent_spPoly_process:n`.

```
\cs_new_protected:Nn \__spintent_spPoly_exec:nn
  {
    \keys_set:nn { spintent / spPoly } { #1 }
    \__spintent_spPoly_process:n { #2 }
  }
```

(*End of definition for* `\__spintent_spPoly_exec:nn`.)

`\spPoly`

The public command `\spPoly` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spPoly_exec:nn`.

```
\NewDocumentCommand \spPoly { O{} m }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \__spintent_spPoly_exec:nn { #1 } { #2 }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spPoly } }
  }
```

(*End of definition for* `\spPoly`. *This function is documented on page 22.*)

11.39 The commands `\spAfig` and `\spPfig`

The user commands `\spAfig` and `\spPfig` typeset the area and the perimeter of a polygonal figure, using the delimited-optional argument syntax `d()` already established by `\spmcd`/`\spmcm`. The key `print-sym` is never sent to Lua: the symbol type is returned unconditionally in `\l__spintent_geo_luaset_poly_sym_str`, and expl3 decides whether to wrap it in `\MathMLintent{⟨:silent⟩}` — the same mechanism already used by `\spPoly`.

11.39.1 Definition of variables for `\spAfig` and `\spPfig`

`\l__spintent_geo_sym_word_tl`
`\l__spintent_geo_final_print_tl`
`\l__spintent_geo_final_intent_str`
`\l__spintent_geo_mathcat_concept_str`

Scratch variables used by `\__spintent_spfig_process:nn` to assemble the visual and the `intent`. `\l__spintent_geo_sym_word_tl` holds the spoken symbol word (`triángulo`/`cuadrado`, or empty). `\l__spintent_geo_final_print_tl` holds the final visual placed under `\mathcal`'s subscript. `\l__spintent_geo` and `\l__spintent_geo_mathcat_concept_str` hold, respectively, the final literal `intent` and the concept forwarded when `read-cmd` is `mathcat`.

```
\tl_new:N  \l__spintent_geo_sym_word_tl
\tl_new:N  \l__spintent_geo_final_print_tl
\str_new:N \l__spintent_geo_final_intent_str
\str_new:N \l__spintent_geo_mathcat_concept_str
```

(*End of definition for* `\l__spintent_geo_sym_word_tl` *and others.*)

`\l__spintent_spAfig_mode_str`
`\l__spintent_spPfig_mode_str`
`\l__spintent_spAfig_print_sym_str`
`\l__spintent_spPfig_print_sym_str`
`\l__spintent_spAfig_prefix_tl`
`\l__spintent_spPfig_prefix_tl`

These variables store the value set by each corresponding key: `read-cmd`, `print-sym`, `prefix` and `read-arg`, respectively. The value set by `read-arg` overrides the figure name inferred from the vertex count, independently of `print-sym`.

```
\clist_map_inline:nn { Afig , Pfig }
```

```
  {
    \str_new:c { l__spintent_sp #1 _mode_str      }
    \str_new:c { l__spintent_sp #1 _print_sym_str }
    \tl_new:c  { l__spintent_sp #1 _prefix_tl     }
    \tl_new:c  { l__spintent_sp #1 _read_arg_tl   }
  }
```

(*End of definition for* `\l__spintent_spAfig_mode_str` *and others.*)

11.39.2 Definition of error messages for `\spAfig` and `\spPfig`

spAfig-invalid-points
spPfig-invalid-points
spAfig-invalid-arg
spPfig-invalid-arg
spAfig-sym-letra-conflict
spPfig-sym-letra-conflict
spAfig-sym-cardinality-conflict
spPfig-sym-cardinality-conflict

All eight messages are generated by the same map, using the same `##1`/`##2` doubling convention as `\spside`/`\spmside`. sp `#1-invalid-points` is currently unused by `\__spintent_spfig_process:nn`, which raises only sp `#1-invalid-arg` for any parse failure, regardless of whether a comma was present. sp `#1-sym-letra-conflict` and sp `#1-sym-cardinality-conflict` are raised when `print-sym` is forced to a value incompatible with the actual argument.

```
\clist_map_inline:nn { Afig , Pfig }
  {
    \msg_new:nnnn { spintent } { sp #1 -invalid-points }
      {
        Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~the~accepted~point~formats.
      }
    \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
      {
        Invalid~argument~format~in~##1:~'##2'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~the~accepted~argument~forms.
      }
    \msg_new:nnnn { spintent } { sp #1 -sym-letra-conflict }
      {
        print-sym~cannot~be~combined~with~a~lettered~argument~in~'##1'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~details~on~print-sym.
      }
    \msg_new:nnnn { spintent } { sp #1 -sym-cardinality-conflict }
      {
        print-sym~does~not~match~the~vertex~count~in~'##1'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~details~on~print-sym.
      }
  }
```

(*End of definition for* `spAfig-invalid-points` *and others.*)

11.39.3 Definition of keys for `\spAfig` and `\spPfig`

read-cmd
prefix
print-sym
read-arg
read-sub
silent-cmd
concept
unknown

The key `print-sym` chooses the visual symbol: `auto` (default) infers it from the vertex count (three points give `\triangle`, four give `\mdlgwhtsquare`, five or more give none); `triangle`/`square` force one, requiring the vertex count to match; `none` omits it, even with vertices. With a bare number/letter argument, `auto` and `none` behave the same (no symbol); `triangle`/`square` nest the argument under the forced symbol instead, and cannot be combined with a table letter (`F`/`P`/`B`/`L`). The key `read-arg` overrides the spoken figure name for real vertices, independently of `print-sym`. The key `read-sub` controls whether the bare number/letter argument is read with the word «sub» or as a plain number.

```
\clist_map_inline:nn { Afig , Pfig }
  {
    \keys_define:nn { spintent / sp #1 }
      {
        read-cmd             .choices:nn = { school , mathcat }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school  } }
                {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
              }
          },
        read-cmd             .initial:n  = school,
        read-cmd             .value_required:n = true,
```

```
        read-cmd / unknown  .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
        prefix              .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix              .initial:n  = ,
        prefix              .value_required:n = true,
        print-sym           .choices:nn = { auto , triangle , square , none }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { auto     } }
                {2} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { triangle } }
                {3} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { square   } }
                {4} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { none     } }
              }
          },
        print-sym           .initial:n  = auto,
        print-sym           .value_required:n = true,
        print-sym / unknown  .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { print-sym } { auto,~triangle,~square,~none } { \exp_not:n {##1} },
        read-arg            .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1} },
        read-arg            .initial:n  = ,
        read-arg            .value_required:n = true,
        read-sub            .bool_set:c = { l__spintent_sp #1 _read_sub_bool },
        read-sub            .initial:n  = false,
        read-sub            .value_required:n = true,
        silent-cmd          .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd          .initial:n  = false,
        silent-cmd          .value_required:n = true,
        concept             .bool_set:c = { l__spintent_sp #1 _concept_bool },
        concept             .initial:n  = true,
        concept             .value_required:n = true,
        unknown             .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
            \__spintent_unknown_keys_cmd:n {##1}
          },
      }
  }
```

(*End of definition for* `read-cmd` *and others.*)

11.39.4 Internal functions for \spAfig and \spPfig

`\__spintent_spfig_sym:`

The function `\__spintent_spfig_sym:` reads `\l__spintent_geo_print_sym_str` and typesets the corresponding symbol: `\triangle` for the value `triangle`, `\mdlgwhtsquare` for the value `square`.

```
\cs_new_protected:Nn \__spintent_spfig_sym:
  {
    \str_case:Vn \l__spintent_geo_print_sym_str
      {
        { triangle } { \triangle      }
        { square   } { \mdlgwhtsquare }
      }
  }
```

(*End of definition for* `\__spintent_spfig_sym:`.)

`\l__spintent_spfig_op_tl`
`\l__spintent_spfig_intent_tl`
`\l__spintent_spfig_symbol_tl`
`\l__spintent_spfig_pt_seq`
`\l__spintent_spfig_refs_tl`
`\l__spintent_spfig_intent_arg_tl`
`\l__spintent_spfig_pt_idx_int`

`\l__spintent_spfig_op_tl` holds `área` or `perímetro`. `\l__spintent_spfig_intent_tl` holds the concept word for the vertices case (`read-arg` or the auto-detected figure name). The rest are scratch variables to split `\l__spintent_geo_luaset_points_str` and build the `$pt1,$pt2,...` reference list.

```
\tl_new:N  \l__spintent_spfig_op_tl
\tl_new:N  \l__spintent_spfig_intent_tl
\tl_new:N  \l__spintent_spfig_desc_tl
\tl_new:N  \l__spintent_spfig_symbol_tl
\seq_new:N \l__spintent_spfig_pt_seq
\tl_new:N  \l__spintent_spfig_refs_tl
\tl_new:N  \l__spintent_spfig_intent_arg_tl
\int_new:N \l__spintent_spfig_pt_idx_int
```

(*End of definition for* `\l__spintent_spfig_op_tl` *and others.*)

__spintent_spfig_process:nn

Receives the short command name (Afig or Pfig) in '#1' and the {⟨argument⟩} in '#2'. Calls __spintent_luafun_ge or __spintent_luafun_geo_pfig_parse_and_set:n, reports sp#1-invalid-arg on failure, sets \l__spintent_spfig_op_tl to área/perímetro, then delegates to __spintent_spfig_mathml:n.

```
\cs_new_protected:Nn \__spintent_spfig_process:nn
  {
    \str_if_eq:eeTF {#1} { Afig }
      { \__spintent_luafun_geo_afig_parse_and_set:n { \exp_not:n {#2} } }
      { \__spintent_luafun_geo_pfig_parse_and_set:n { \exp_not:n {#2} } }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nnee { spintent } { sp #1 -invalid-arg }
          { \c_backslash_str sp #1 } { \exp_not:n {#2} }
      }
    \str_if_eq:eeTF {#1} { Afig }
      { \tl_set:Nn \l__spintent_spfig_op_tl { área } }
      { \tl_set:Nn \l__spintent_spfig_op_tl { perímetro } }
    \str_set:Ne \l__spintent_geo_print_sym_str { \str_use:c { l__spintent_sp #1 _print_sym_str } }
    \__spintent_spfig_mathml:n {#1}
  }
```

(*End of definition for* __spintent_spfig_process:nn.)

__spintent_spfig_build_simple:n

Handles the argument-less, table-letter and bare number/letter cases (no vertices). Builds \l__spintent_geo_final_pr and \l__spintent_spfig_desc_tl (the descriptive part, without área/perímetro). \l__spintent_spfig_inten is \l__spintent_spfig_op_tl-\l__spintent_spfig_desc_tl. silent-cmd overrides \l__spintent_spfig_ with _-; with silent-cmd=false and concept false, it becomes _\l__spintent_spfig_desc_tl- (the descriptive part alone, literally, without área/perímetro); with concept true it is \l__spintent_spfig_intent_tl Preserves the sym-letra-conflict check (a table letter cannot combine with a forced print-sym). Receives the short command name in '#1'.

```
\cs_new_protected:Nn \__spintent_spfig_build_simple:n
  {
    \tl_if_empty:nTF { \l__spintent_geo_luaset_print_tl }
      {
        \str_case:VnF \l__spintent_geo_print_sym_str
          {
            { auto }
              {
                \tl_clear:N \l__spintent_geo_final_print_tl
                \tl_clear:N \l__spintent_spfig_desc_tl
              }
            { none }
              {
                \tl_clear:N \l__spintent_geo_final_print_tl
                \tl_clear:N \l__spintent_spfig_desc_tl
              }
          }
          {
            \tl_set:Nn \l__spintent_geo_final_print_tl { \__spintent_spfig_sym: }
            \tl_set:Ne \l__spintent_spfig_desc_tl
              {
                \str_if_eq:VnTF \l__spintent_geo_print_sym_str { triangle }
                  { triángulo } { cuadrado }
              }
          }
      }
      {
        \str_if_empty:NTF \l__spintent_geo_luaset_letra_word_str
          {
            \str_case:VnF \l__spintent_geo_print_sym_str
              {
                { auto } { \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_pr
                { none } { \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_p
              }
              {
                \tl_set:Nn \l__spintent_geo_final_print_tl
                  {
                    \__spintent_spfig_sym:
                    \c_math_subscript_token { \l__spintent_geo_luaset_print_tl }
                  }
              }
            \str_case:VnF \l__spintent_geo_print_sym_str
```

```
              {
                { auto }
                  {
                    \tl_set:Ne \l__spintent_spfig_desc_tl
                      {
                        \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub - }
                        \l__spintent_geo_luaset_sub_spoken_str
                      }
                  }
                { none }
                  {
                    \tl_set:Ne \l__spintent_spfig_desc_tl
                      {
                        \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub - }
                        \l__spintent_geo_luaset_sub_spoken_str
                      }
                  }
              }
              {
                \tl_set:Ne \l__spintent_spfig_desc_tl
                  {
                    \str_if_eq:VnTF \l__spintent_geo_print_sym_str { triangle }
                      { triángulo } { cuadrado }
                    -
                    \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub - }
                    \l__spintent_geo_luaset_sub_spoken_str
                  }
              }
          }
          {
            \str_case:VnF \l__spintent_geo_print_sym_str
              {
                { auto } { }
                { none } { }
              }
              {
                \msg_error:nne { spintent } { sp #1 -sym-letra-conflict }
                  { \c_backslash_str sp #1 }
              }
            \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_print_tl }
            \tl_set:Ne \l__spintent_spfig_desc_tl
              {
                \str_if_empty:NF \l__spintent_geo_luaset_letra_conector_str
                  { \l__spintent_geo_luaset_letra_conector_str - }
                \l__spintent_geo_luaset_letra_word_str
                \str_if_empty:NF \l__spintent_geo_luaset_sub_spoken_str
                  {
                    -
                    \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub - }
                    \l__spintent_geo_luaset_sub_spoken_str
                  }
              }
          }
      }
    \tl_set:Ne \l__spintent_spfig_intent_tl
      {
        \tl_if_empty:NTF \l__spintent_spfig_desc_tl
          { \l__spintent_spfig_op_tl }
          { \l__spintent_spfig_op_tl - \l__spintent_spfig_desc_tl }
      }
    \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
      { \tl_set:Nn \l__spintent_spfig_intent_arg_tl { _- } }
      {
        \bool_if:cTF { l__spintent_sp #1 _concept_bool }
          { \tl_set:Ne \l__spintent_spfig_intent_arg_tl { \l__spintent_spfig_intent_tl } }
          { \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _ \l__spintent_spfig_desc_tl - } }
      }
  }
```

(End of definition for `\__spintent_spfig_build_simple:n`*.)*

`\__spintent_spfig_build_refs:` Splits `\l__spintent_geo_luaset_points_str` into `\l__spintent_spfig_pt_seq` and walks it once to build \l__spintent_spfig_refs_tl, the comma-separated $pt1,$pt2,... reference list, naming each vertex by position.

```
\cs_new_protected:Nn \__spintent_spfig_build_refs:
  {
    \seq_set_from_clist:NN \l__spintent_spfig_pt_seq \l__spintent_geo_luaset_points_str
    \tl_clear:N \l__spintent_spfig_refs_tl
    \int_zero:N \l__spintent_spfig_pt_idx_int
    \seq_map_inline:Nn \l__spintent_spfig_pt_seq
      {
        \int_incr:N \l__spintent_spfig_pt_idx_int
        \tl_if_empty:NF \l__spintent_spfig_refs_tl
          { \tl_put_right:Nn \l__spintent_spfig_refs_tl { , } }
        \tl_put_right:Ne \l__spintent_spfig_refs_tl { $ pt \int_use:N \l__spintent_spfig_pt_idx_i
      }
  }
```

(*End of definition for* __spintent_spfig_build_refs:.)

__spintent_spfig_build_concept_vertices:n

Handles the vertices case. read-arg, when given, overrides the auto-detected figure name; with concept true it is still prefixed with área-del-/perímetro-del-, with concept false it is used verbatim, and never silenced (only silent-cmd silences when read-arg is given — same convention as __spintent_spside_process:nn). With read-arg empty, concept controls silencing as usual. Also determines \l__spintent_spfig_symbol_tl (empty when none applies), preserving the sym-cardinality-conflict check (a forced print-sym must match the actual vertex count). Receives the short command name in '#1'. Must be called after __spintent_spfig_build_refs:.

```
\cs_new_protected:Nn \__spintent_spfig_build_concept_vertices:n
  {
    \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
      { \tl_set:Ne \l__spintent_spfig_intent_tl { \l__spintent_geo_luaset_concept_str } }
      {
        \bool_if:cTF { l__spintent_sp #1 _concept_bool }
          {
            \tl_set:Ne \l__spintent_spfig_intent_tl
              {
                \l__spintent_spfig_op_tl -del-
                \tl_use:c { l__spintent_sp #1 _read_arg_tl }
              }
          }
          {
            \tl_set:Ne \l__spintent_spfig_intent_tl { \tl_use:c { l__spintent_sp #1 _read_arg_tl
          }
      }

    \str_case:VnF \l__spintent_geo_print_sym_str
      {
        { auto }
          {
            \str_case:Vn \l__spintent_geo_luaset_poly_sym_str
              {
                { triangle } { \tl_set:Nn \l__spintent_spfig_symbol_tl { \triangle      } }
                { square   } { \tl_set:Nn \l__spintent_spfig_symbol_tl { \mdlgwhtsquare } }
              }
            \str_if_empty:NT \l__spintent_geo_luaset_poly_sym_str
              { \tl_clear:N \l__spintent_spfig_symbol_tl }
          }
        { none }
          { \tl_clear:N \l__spintent_spfig_symbol_tl }
      }
      {
        \str_if_eq:VVTF \l__spintent_geo_print_sym_str \l__spintent_geo_luaset_poly_sym_str
          { \tl_set:Nn \l__spintent_spfig_symbol_tl { \__spintent_spfig_sym: } }
          {
            \msg_error:nne { spintent } { sp #1 -sym-cardinality-conflict }
              { \c_backslash_str sp #1 }
          }
      }

    \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
      {
        \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _( \l__spintent_spfig_refs_tl ) }
      }
```

```
      {
        \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
          {
            \bool_if:cTF { l__spintent_sp #1 _concept_bool }
              {
                \tl_set:Ne \l__spintent_spfig_intent_arg_tl
                  {
                    \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                      { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                    \l__spintent_spfig_intent_tl :prefix( \l__spintent_spfig_refs_tl )
                  }
              }
              {
                \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _( \l__spintent_spfig_refs_tl ) }
              }
          }
          {
            \tl_set:Ne \l__spintent_spfig_intent_arg_tl
              {
                \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                  { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                \l__spintent_spfig_intent_tl :prefix( \l__spintent_spfig_refs_tl )
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_spfig_build_concept_vertices:n`.)

`\__spintent_spfig_points_loop:n`

Walks `\l__spintent_spfig_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once per vertex with its `pt`n reference name, looked up under `sp#1`. Receives the short command name in '`#1`'.

```
\cs_new_protected:Nn \__spintent_spfig_points_loop:n
  {
    \int_zero:N \l__spintent_spfig_pt_idx_int
    \seq_map_inline:Nn \l__spintent_spfig_pt_seq
      {
        \int_incr:N \l__spintent_spfig_pt_idx_int
        \__spintent_mathmlarg_point:nnn
          { pt \int_use:N \l__spintent_spfig_pt_idx_int }
          { sp #1 }
          { ##1 }
      }
  }
```

(*End of definition for* `\__spintent_spfig_points_loop:n`.)

`\__spintent_spfig_mathml:n`

Receives the short command name in '`#1`'. Branches on whether `\l__spintent_geo_luaset_points_str` is empty.
Without vertices, `\__spintent_spfig_build_simple:n` builds the content; a `\MathMLintent` wraps it only when `concept` is `true` — with `concept=false` the content is typeset directly, with no `intent` at all.
With vertices, `\__spintent_spfig_build_refs:` then `\__spintent_spfig_build_concept_vertices:n` build the reference list and the concept `intent`, applied with a single native `\MathMLintent`. Inside, `\__spintent_mathcal:n`{⟨A | P⟩} is silenced with `\__spintent_luafun_wrap_mrow_intent_arg:nn` (it would otherwise be claimed by the queue meant for the symbol/vertices, the same ordering issue already found and fixed for `\spangle`'s own symbol); `\l__spintent_spfig_symbol_tl` is silenced the same way when non-empty; the sequence is then walked by `\__spintent_spfig_points_loop:n`. All of this sits nested inside a subscript (`\c_math_subscript_token`), reached by the shared `mathml_filter` callback's recursive lookup through a noad's `sub` field, not just its `nucleus`.
`\__spintent_invisible_sep:` precedes both branches (a second sibling so the sole <`mrow`> does not collapse into <`math`>, `luamml` issue #26).

```
\cs_new_protected:Nn \__spintent_spfig_mathml:n
  {
    \__spintent_invisible_sep: % need by MathCAT (bug in luamml issues #26)
    \str_if_empty:NTF \l__spintent_geo_luaset_points_str
      {
        \__spintent_spfig_build_simple:n {#1}
        \MathMLintent { \l__spintent_spfig_intent_arg_tl }
          {
            \__spintent_luafun_inner_sep:
            \str_if_eq:eeTF {#1} { Afig } { \__spintent_mathcal:n { A } } { \__spintent_mathcal:n
```

```
            \tl_if_empty:NF \l__spintent_geo_final_print_tl
              {
                \c_math_subscript_token { \l__spintent_geo_final_print_tl }
              }
          }
      }
      {
        \__spintent_spfig_build_refs:
        \__spintent_spfig_build_concept_vertices:n {#1}
        \MathMLintent { \l__spintent_spfig_intent_arg_tl }
          {
            \__spintent_luafun_inner_sep:
            \__spintent_luafun_wrap_mrow_intent_arg:nn
              { } { intent = ":silent" }
            {
              \str_if_eq:eeTF {#1} { Afig } { \__spintent_mathcal:n { A } } { \__spintent_mathcal
            }
            \c_math_subscript_token
              {
                \tl_if_empty:NF \l__spintent_spfig_symbol_tl
                  {
                    \__spintent_luafun_wrap_mrow_intent_arg:nn
                      { } { intent = ":silent" }
                    { \tl_use:N \l__spintent_spfig_symbol_tl }
                  }
                \__spintent_spfig_points_loop:n {#1}
              }
          }
      }
  }
```

(*End of definition for* `\__spintent_spfig_mathml:n`.)

`\__spintent_spfig_exec:nnn`

The function `\__spintent_spfig_exec:nnn` receives the short name of the command in `#1`, the optional keys in `#2`, and the argument in `#3`, delegating the rendering to `\__spintent_spfig_process:nn`.

```
\cs_new_protected:Nn \__spintent_spfig_exec:nnn
  {
    \keys_set:nn { spintent / sp #1 } { #2 }
    \__spintent_spfig_process:nn { #1 } { #3 }
  }
```

(*End of definition for* `\__spintent_spfig_exec:nnn`.)

`\spAfig`
`\spPfig`

Both public commands `\spAfig` and `\spPfig` enforce math mode, open a local group, and delegate execution to `\__spintent_spfig_exec:nnn`.

```
\NewDocumentCommand \spAfig { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \tl_if_novalue:nTF {#2}
            { \__spintent_spfig_exec:nnn { Afig } { #1 } { } }
            { \__spintent_spfig_exec:nnn { Afig } { #1 } { #2 } }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spAfig } }
  }
\NewDocumentCommand \spPfig { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \tl_if_novalue:nTF {#2}
            { \__spintent_spfig_exec:nnn { Pfig } { #1 } { } }
            { \__spintent_spfig_exec:nnn { Pfig } { #1 } { #2 } }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spPfig } }
  }
```

(*End of definition for* `\spAfig` *and* `\spPfig`. *These functions are documented on page* 22.)

11.40 The command \spcirc

The user command \spcirc provides a semantic interface for a circle or a circumference, using the delimited-optional argument syntax d() already established by \spmcd/\spmcm and \spAfig/\spPfig. The argument holds a required center point and an optional radius, separated by a comma. The radius is never required to declare its own kind via a key: its shape alone decides whether it is a pure number, a measure (number + unit), a bare letter, or a two-letter segment — reusing, respectively, \spnum, \spunit, and \spside/\spmside directly as sub-commands, rather than reimplementing any of their parsing.

11.40.1 Definition of variables for \spcirc

_spintent_geo_luaset_circ_radio_type_str
_spintent_geo_luaset_circ_radio_raw_str

Shared outputs of __spintent_luafun_geo_circ_parse_and_set:n. \l__spintent_geo_luaset_circ_radi holds one of number, measure, label, segment, or an empty string when no radius was given. \l__spintent_geo_lu holds the corresponding raw text, ready to be forwarded to the sub-command that renders it.

```
\str_new:N \l__spintent_geo_luaset_circ_radio_type_str
\str_new:N \l__spintent_geo_luaset_circ_radio_raw_str
```

(*End of definition for* \l__spintent_geo_luaset_circ_radio_type_str *and* \l__spintent_geo_luaset_circ_radio_raw_str.)

\l__spintent_spcirc_mode_str
\l__spintent_spcirc_prefix_tl
\l__spintent_spcirc_print_sym_str
\l__spintent_spcirc_read_sty_str
\l__spintent_spcirc_radio_sty_str
\l__spintent_spcirc_sub_index_str
\l__spintent_spcirc_articulo_str

The variable \l__spintent_spcirc_mode_str stores the value set by the key read-cmd; the variable \l__spintent_spcirc_prefix_tl stores the value set by the key prefix. The variable \l__spintent_spcirc_pr stores the value set by the key print-sym (usual or odot). The variable \l__spintent_spcirc_read_sty_str stores the value set by the key read-sty (circun or circle). The variable \l__spintent_spcirc_radio_sty_str stores the value set by the key radio-sty, only meaningful when the radius is a segment. The variable \l__spintent_spcirc_sub_index_str stores the value set by the key sub-index. The variable \l__spintent_spcirc_articulo_str is set internally by __spintent_spcirc_word_and_article: to la or el, matching the grammatical gender of the chosen word.

```
\str_new:N \l__spintent_spcirc_mode_str
\tl_new:N  \l__spintent_spcirc_prefix_tl
\str_new:N \l__spintent_spcirc_print_sym_str
\str_new:N \l__spintent_spcirc_read_sty_str
\str_new:N \l__spintent_spcirc_radio_sty_str
\str_new:N \l__spintent_spcirc_sub_index_str
\str_new:N \l__spintent_spcirc_articulo_str
\tl_new:N  \l__spintent_spcirc_read_arg_tl
```

(*End of definition for* \l__spintent_spcirc_mode_str *and others.*)

\l__spintent_geo_circ_word_tl
\l__spintent_geo_circ_intent_str

Scratch variables used by __spintent_spcirc_process:n to assemble the chosen word (circunferencia/círculo or the read-arg override) and the final literal intent string.

```
\tl_new:N  \l__spintent_geo_circ_word_tl
\str_new:N \l__spintent_geo_circ_intent_str
```

(*End of definition for* \l__spintent_geo_circ_word_tl *and* \l__spintent_geo_circ_intent_str.)

11.40.2 Definition of error messages for \spcirc

spcirc-invalid-arg

Raised when the argument does not match a single center point, optionally followed by a comma and a valid radius shape.

```
\msg_new:nnnn { spintent } { spcirc-invalid-arg }
  {
    Invalid~argument~format~in~\token_to_str:N \spcirc:~'#1'~\msg_line_context:.
  }
  {
    See~the~documentation~of~\token_to_str:N \spcirc~for~the~accepted~argument~forms.
  }
```

(*End of definition for* spcirc-invalid-arg.)

11.40.3 Definition of keys for \spcirc

read-cmd
prefix
print-sym
read-sty
radio-sty
sub-index
read-sub
read-arg
silent-cmd
spside
spmside
unknown

The key print-sym chooses the visual symbol: usual for a plain C, odot (default) for \odot. The key read-sty chooses the verbalized word and its article: circun (default) for *«la circunferencia»*, circle for *«el círculo»*. The key radio-sty chooses, only when the radius is a two-letter segment, whether it is rendered as the *object* (side, via \spside) or its *measure* (mside, default, via \spmside). The key sub-index sets a subscript on the symbol; the key read-sub controls whether it is read with the word «sub» or as a plain number, matching the same convention as \spAfig.

```
\keys_define:nn { spintent / spcirc }
  {
    read-cmd     .choices:nn = { school , mathcat }
      {
        \int_case:nn { \l_keys_choice_int }
```

```
          {
            {1} { \str_set:Nn \l__spintent_spcirc_mode_str { school  } }
            {2} { \str_set:Nn \l__spintent_spcirc_mode_str { mathcat } }
          }
      },
    read-cmd     .initial:n = school,
    read-cmd     .value_required:n = true,
    prefix       .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spcirc_prefix_tl {#1} },
    prefix       .initial:n = ,
    prefix       .value_required:n = true,
    print-sym    .choices:nn = { usual , odot }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spcirc_print_sym_str { usual } }
            {2} { \str_set:Nn \l__spintent_spcirc_print_sym_str { odot  } }
          }
      },
    print-sym    .initial:n = odot,
    print-sym    .value_required:n = true,
    read-sty     .choices:nn = { circun , circle }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spcirc_read_sty_str { circun } }
            {2} { \str_set:Nn \l__spintent_spcirc_read_sty_str { circle } }
          }
      },
    read-sty     .initial:n = circun,
    read-sty     .value_required:n = true,
    radio-sty    .choices:nn = { side , mside }
      {
        \int_case:nn { \l_keys_choice_int }
          {
            {1} { \str_set:Nn \l__spintent_spcirc_radio_sty_str { side  } }
            {2} { \str_set:Nn \l__spintent_spcirc_radio_sty_str { mside } }
          }
      },
    radio-sty    .initial:n = mside,
    radio-sty    .value_required:n = true,
    sub-index    .code:n =
      { \str_set:Nn \l__spintent_spcirc_sub_index_str {#1} },
    sub-index    .initial:n = ,
    sub-index    .value_required:n = true,
    read-sub     .bool_set:N = \l__spintent_spcirc_read_sub_bool,
    read-sub     .initial:n = false,
    read-sub     .value_required:n = true,
    read-arg     .code:n =
      { \__spintent_sanitize_affix:Nn \l__spintent_spcirc_read_arg_tl {#1} },
    read-arg     .initial:n = ,
    read-arg     .value_required:n = true,
    silent-cmd   .bool_set:N = \l__spintent_spcirc_silent_cmd_bool,
    silent-cmd   .initial:n = false,
    silent-cmd   .value_required:n = true,
    spside       .meta:nn = { spintent / spside  } { #1 },
    spside       .value_required:n = true,
    spmside      .meta:nn = { spintent / spmside } { #1 },
    spmside      .value_required:n = true,
    unknown      .code:n =
      {
        \str_set:Nn \l__spintent_command_name_str { spcirc }
        \__spintent_unknown_keys_cmd:n {#1}
      },
  }
```

(*End of definition for* `read-cmd` *and others.*)

11.40.4 Internal functions for \spcirc

`\__spintent_spcirc_sym:`

The function `\__spintent_spcirc_sym:` typesets `\bigodot` or the plain letter `C`, according to the value set by the key `print-sym`.

```
\cs_new_protected:Nn \__spintent_spcirc_sym:
```

```
    {
      \str_if_eq:VnTF \l__spintent_spcirc_print_sym_str { odot }
        {
          \group_begin:
            \textstyle\bigodot
          \group_end:
        }
        { C }
    }
```

(End of definition for `\__spintent_spcirc_sym:`*.)*

`\__spintent_spcirc_word_and_article:`

The function `\__spintent_spcirc_word_and_article:` sets `\l__spintent_geo_circ_word_tl` to the chosen word — either the value set by the key `read-arg`, or `circunferencia`/`círculo` according to the key `read-sty` — and `\l__spintent_spcirc_articulo_str` to the matching article (`la`/`el`). The article always follows `read-sty`, even when `read-arg` overrides the word itself.

```
\cs_new_protected:Nn \__spintent_spcirc_word_and_article:
  {
    \tl_if_empty:NTF \l__spintent_spcirc_read_arg_tl
      {
        \str_if_eq:VnTF \l__spintent_spcirc_read_sty_str { circun }
          {
            \tl_set:Nn \l__spintent_geo_circ_word_tl { circunferencia }
            \str_set:Nn \l__spintent_spcirc_articulo_str { la }
          }
          {
            \tl_set:Nn \l__spintent_geo_circ_word_tl { círculo }
            \str_set:Nn \l__spintent_spcirc_articulo_str { el }
          }
      }
      {
        \tl_set:Nn \l__spintent_geo_circ_word_tl { \l__spintent_spcirc_read_arg_tl }
        \str_if_eq:VnTF \l__spintent_spcirc_read_sty_str { circun }
          { \str_set:Nn \l__spintent_spcirc_articulo_str { la } }
          { \str_set:Nn \l__spintent_spcirc_articulo_str { el } }
      }
  }
```

(End of definition for `\__spintent_spcirc_word_and_article:`*.)*

`\__spintent_spcirc_radio_print:`

The function `\__spintent_spcirc_radio_print:` dispatches on `\l__spintent_geo_luaset_circ_radio_type` reusing the actual public sub-commands rather than reimplementing any parsing: `number` and `measure` delegate to `\spnum`/`\spunit`; `segment` delegates to `\spside`/`\spmside` according to the key `radio-sty` — both already accept the concatenated two-letter form that Lua hands over here. `label` is typeset as a bare letter, with no special intent wrapping. In every case, `\exp_args:No` is required to expand `\l__spintent_geo_luaset_circ_radio_raw_str` into the sub-command's argument, since a plain brace group does not expand a `str` variable on its own.

```
\cs_new_protected:Nn \__spintent_spcirc_radio_print:
  {
    \str_case:Vn \l__spintent_geo_luaset_circ_radio_type_str
      {
        { number  } { \exp_args:No \spnum { \l__spintent_geo_luaset_circ_radio_raw_str } }
        { measure } { \exp_args:No \spunit { \l__spintent_geo_luaset_circ_radio_raw_str } }
        { label   } { \l__spintent_geo_luaset_circ_radio_raw_str }
        { segment }
          {
            \str_if_eq:VnTF \l__spintent_spcirc_radio_sty_str { side }
              { \exp_args:No \spside  { \l__spintent_geo_luaset_circ_radio_raw_str } }
              { \exp_args:No \spmside { \l__spintent_geo_luaset_circ_radio_raw_str } }
          }
      }
  }
```

(End of definition for `\__spintent_spcirc_radio_print:`*.)*

`\__spintent_spcirc_process:n`

The function `\__spintent_spcirc_process:n` calls `\__spintent_luafun_geo_circ_parse_and_set:n` to resolve the center and classify the radius, then assembles `\l__spintent_geo_circ_intent_str` — the word, an optional `sub-index` suffix, `de-centro-en-`, the center, and `-y-radio` when a radius is present — and delegates the rendering to `\__spintent_spcirc_mathml:`.

```
\cs_new_protected:Nn \__spintent_spcirc_process:n
  {
```

```
    \__spintent_luafun_geo_circ_parse_and_set:n { \exp_not:n {#1} }
    \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
      {
        \msg_error:nne { spintent } { spcirc-invalid-arg } { \exp_not:n {#1} }
      }
    \__spintent_spcirc_word_and_article:
    \str_set:Ne \l__spintent_geo_circ_intent_str { \l__spintent_geo_circ_word_tl }
    \str_if_empty:NF \l__spintent_spcirc_sub_index_str
      {
        \str_put_right:Ne \l__spintent_geo_circ_intent_str
          {
            -
            \bool_if:NT \l__spintent_spcirc_read_sub_bool { sub - }
            \l__spintent_spcirc_sub_index_str
          }
      }
    \str_put_right:Ne \l__spintent_geo_circ_intent_str
      {
        -de-centro-en-\l__spintent_geo_luaset_print_tl
      }
    \str_if_empty:NF \l__spintent_geo_luaset_circ_radio_type_str
      {
        \str_put_right:Nn \l__spintent_geo_circ_intent_str { -y-radio }
      }
    \__spintent_spcirc_mathml:
  }
```

(*End of definition for* `\__spintent_spcirc_process:n`.)

`\__spintent_spcirc_mathml:`

The function `\__spintent_spcirc_mathml:` reads the state already set by `\__spintent_spcirc_process:n` and renders it. When a radius is present, the visual is split across *two* separate `\MathMLintent` calls with `\__spintent_spcirc_radio_print:` — which typesets a sub-command carrying its own independent `intent` — sitting as a *sibling* in between, rather than nested inside either one. Embedding a full sub-command's own `\MathMLintent` inside another command's content argument makes MathCAT read only the outer literal string, the same problem already seen with `mover`/`msub` structures; as *siblings*, both are read in sequence as one continuous phrase.

```
\cs_new_protected:Nn \__spintent_spcirc_mathml:
  {
    \__spintent_invisible_sep: % need by MathCAT (bug in luamml https://github.com/latex3/luamml/issues/26)
    \str_if_empty:NTF \l__spintent_geo_luaset_circ_radio_type_str
      {
        \MathMLintent
          {
            \tl_if_empty:NF \l__spintent_spcirc_prefix_tl
              {
                \tl_use:N \l__spintent_spcirc_prefix_tl -
              }
            \l__spintent_geo_circ_intent_str
          }
          {
            \c__spintent_invisible_sep_tl
            \__spintent_spcirc_sym:
            \str_if_empty:NF \l__spintent_spcirc_sub_index_str
              {
                \c_math_subscript_token { \l__spintent_spcirc_sub_index_str }
              }
            ( \l__spintent_geo_luaset_print_tl )
          }
      }
      {
        \MathMLintent
          {
            \tl_if_empty:NF \l__spintent_spcirc_prefix_tl
              {
                \tl_use:N \l__spintent_spcirc_prefix_tl -
              }
            \l__spintent_geo_circ_intent_str
          }
          {
            \c__spintent_invisible_sep_tl
            \__spintent_spcirc_sym:
```

```
                \str_if_empty:NF \l__spintent_spcirc_sub_index_str
                  {
                    \c_math_subscript_token { \l__spintent_spcirc_sub_index_str }
                  }
                ( \l__spintent_geo_luaset_print_tl ,
              }
          \__spintent_invisible_sep:
          \__spintent_spcirc_radio_print:
          \MathMLintent { :silent } { ) }
        }
    }
```

(*End of definition for* `\__spintent_spcirc_mathml:`.)

`\__spintent_spcirc_exec:nn`

The function `\__spintent_spcirc_exec:nn` receives the optional keys in `#1` and the argument in `#2`, delegating the rendering to `\__spintent_spcirc_print:n`.

```
\cs_new_protected:Nn \__spintent_spcirc_exec:nn
  {
    \keys_set:nn { spintent / spcirc } { #1 }
    \__spintent_spcirc_process:n { #2 }
  }
```

(*End of definition for* `\__spintent_spcirc_exec:nn`.)

`\spcirc`

The public command `\spcirc` enforces math mode, opens a local group, and delegates execution to `\__spintent_spcirc_exec:nn`. It follows the same `d()`-argument pattern as `\spmcd`/`\spmcm` and `\spAfig`/`\spPfig`; an omitted radius forwards an explicit empty group rather than the `-NoValue-` marker.

```
\NewDocumentCommand \spcirc { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \tl_if_novalue:nTF {#2}
            { \__spintent_spcirc_exec:nn { #1 } { } }
            { \__spintent_spcirc_exec:nn { #1 } { #2 } }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spcirc } }
  }
```

(*End of definition for* `\spcirc`. *This function is documented on page* ??.)

11.41 The commands `\spAcirc` and `\spPcirc`

The user commands `\spAcirc` and `\spPcirc` typeset the area and the perimeter of a circle or a circumference. The argument follows `\spcirc`'s own point-and-radius grammar for a real center (with an optional radius); when no comma is present and no valid center is found, it falls back to a bare number/letter shortcut, reusing the same Lua entry point as `\spAfig`/`\spPfig`. A table letter matched by that fallback is rejected, since it has no meaning for a circle.

11.41.1 Definition of variables for `\spAcirc` and `\spPcirc`

`\l__spintent_spAcirc_mode_str`
`\l__spintent_spPcirc_mode_str`
`\l__spintent_spAcirc_print_sym_str`
`\l__spintent_spPcirc_print_sym_str`
`\l__spintent_spAcirc_read_sty_str`
`\l__spintent_spPcirc_read_sty_str`
`\l__spintent_spAcirc_radio_sty_str`
`\l__spintent_spPcirc_radio_sty_str`
`\l__spintent_spAcirc_sub_index_str`
`\l__spintent_spPcirc_sub_index_str`
`\l__spintent_spAcirc_prefix_tl`
`\l__spintent_spPcirc_prefix_tl`
`\l__spintent_spAcirc_read_arg_tl`
`\l__spintent_spPcirc_read_arg_tl`

These variables store the value set by each corresponding key: `read-cmd`, `print-sym`, `read-sty`, `radio-sty`, `sub-index`, `prefix` and `read-arg`, respectively.

```
\clist_map_inline:nn { Acirc , Pcirc }
  {
    \str_new:c { l__spintent_sp #1 _mode_str      }
    \str_new:c { l__spintent_sp #1 _print_sym_str }
    \str_new:c { l__spintent_sp #1 _read_sty_str  }
    \str_new:c { l__spintent_sp #1 _radio_sty_str }
    \str_new:c { l__spintent_sp #1 _sub_index_str }
    \tl_new:c  { l__spintent_sp #1 _prefix_tl     }
    \tl_new:c  { l__spintent_sp #1 _read_arg_tl   }
  }
```

(*End of definition for* `\l__spintent_spAcirc_mode_str` *and others*.)

`\l__spintent_geo_circfig_mode_str`

The variable `\l__spintent_geo_circfig_mode_str` stores whether the argument was resolved as a real circle (`circle`) or as the bare number/letter shortcut (`bare`).

```
\str_new:N \l__spintent_geo_circfig_mode_str
```

(*End of definition for* `\l__spintent_geo_circfig_mode_str`.)

11.41.2 Definition of error messages for \spAcirc and \spPcirc

spAcirc-invalid-arg
spPcirc-invalid-arg

The error message shown when the argument does not match any of the accepted forms: empty, a center point (optionally followed by a comma and a radius), or a bare number/letter.

```
\clist_map_inline:nn { Acirc , Pcirc }
  {
    \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
      {
        Invalid~argument~in~##1:~'##2'~\msg_line_context:.
      }
      {
        See~the~documentation~of~##1~for~the~accepted~argument.
      }
```

(End of definition for spAcirc-invalid-arg *and* spPcirc-invalid-arg*.)*

11.41.3 Definition of keys for \spAcirc and \spPcirc

read-cmd
prefix
print-sym
read-sty
radio-sty
sub-index
read-sub
read-arg
silent-cmd
spside
spmside
unknown

We define the shared ⟨*keys*⟩ read-cmd, prefix, print-m, read-sty, sub-index, read-sub, read-arg, and silent-cmd.

The keys spside and spmside are of the type .meta:nn passed to the commands \spside and \spmside

```
    \keys_define:nn { spintent / sp #1 }
      {
        read-cmd             .choices:nn = { school , mathcat }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school  } }
                {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
              }
          },
        read-cmd             .initial:n  = school,
        read-cmd             .value_required:n = true,
        read-cmd / unknown   .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
        prefix               .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
        prefix               .initial:n  = ,
        prefix               .value_required:n = true,
        print-sym            .choices:nn = { usual , odot }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { usual } }
                {2} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { odot  } }
              }
          },
        print-sym            .initial:n  = odot,
        print-sym            .value_required:n = true,
        print-sym / unknown .code:n =
          \msg_error:nneee { spintent } { unknown-choice }
            { print-sym } { usual,~odot } { \exp_not:n {##1} },
        read-sty             .choices:nn = { circun , circle }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { circun } }
                {2} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { circle } }
              }
          },
        read-sty             .initial:n  = circun,
        read-sty             .value_required:n = true,
        radio-sty            .choices:nn = { side , mside }
          {
            \int_case:nn { \l_keys_choice_int }
              {
                {1} { \str_set:cn { l__spintent_sp #1 _radio_sty_str } { side  } }
                {2} { \str_set:cn { l__spintent_sp #1 _radio_sty_str } { mside } }
              }
          },
        radio-sty            .initial:n  = mside,
```

```
        radio-sty           .value_required:n = true,
        sub-index           .code:n =
          { \str_set:cn { l__spintent_sp #1 _sub_index_str } {##1} },
        sub-index           .initial:n  = ,
        sub-index           .value_required:n = true,
        read-sub            .bool_set:c = { l__spintent_sp #1 _read_sub_bool },
        read-sub            .initial:n  = false,
        read-sub            .value_required:n = true,
        read-arg            .code:n =
          { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1} },
        read-arg            .initial:n  = ,
        read-arg            .value_required:n = true,
        silent-cmd          .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
        silent-cmd          .initial:n  = false,
        silent-cmd          .value_required:n = true,
        spside              .meta:nn = { spintent / spside  } { ##1 },
        spside              .value_required:n = true,
        spmside             .meta:nn = { spintent / spmside } { ##1 },
        spmside             .value_required:n = true,
        unknown             .code:n =
          {
            \str_set:Nn \l__spintent_command_name_str { sp #1 }
            \__spintent_unknown_keys_cmd:n {##1}
          },
      }
  }
```

(*End of definition for* `read-cmd` *and others.*)

11.41.4 Internal functions for `\spAcirc` and `\spPcirc`

`\__spintent_spcircfig_sym:n`

The function `\__spintent_spcircfig_sym:n` typesets `\bigodot` or the plain letter `C`, according to the value set by the key `print-sym`.

```
\cs_new_protected:Nn \__spintent_spcircfig_sym:n
  {
    \str_if_eq:vnTF { l__spintent_sp #1 _print_sym_str } { odot }
      {
        \bigodot
      }
      { C }
  }
```

(*End of definition for* `\__spintent_spcircfig_sym:n`.)

`\__spintent_spcircfig_radio_print:n`

The function `\__spintent_spcircfig_radio_print:n` typesets the radius according to the type set by `\__spintent_luafun_geo_circ_parse_and_set:n`: `\spnum` for a plain number, `\spunit` for a measure, the raw text for a label, or `\spside`/`\spmside` for a segment, according to the value set by the key `radio-sty`.

```
\cs_new_protected:Nn \__spintent_spcircfig_radio_print:n
  {
    \str_case:Vn \l__spintent_geo_luaset_circ_radio_type_str
      {
        { number  } { \exp_args:No \spnum  { \l__spintent_geo_luaset_circ_radio_raw_str } }
        { measure } { \exp_args:No \spunit { \l__spintent_geo_luaset_circ_radio_raw_str } }
        { label   } { \l__spintent_geo_luaset_circ_radio_raw_str }
        { segment }
          {
            \str_if_eq:vnTF { l__spintent_sp #1 _radio_sty_str } { side }
              { \exp_args:No \spside  { \l__spintent_geo_luaset_circ_radio_raw_str } }
              { \exp_args:No \spmside { \l__spintent_geo_luaset_circ_radio_raw_str } }
          }
      }
  }
```

(*End of definition for* `\__spintent_spcircfig_radio_print:n`.)

`\__spintent_spcircfig_word_and_article:n`

The function `\__spintent_spcircfig_word_and_article:n` sets `\l__spintent_geo_circ_word_tl` to the chosen word — either the value set by the key `read-arg`, or `circunferencia`/`círculo` according to the key `read-sty` — and `\l__spintent_spcirc_articulo_str` to `la` or `el`, matching the grammatical gender of the chosen word.

```
\cs_new_protected:Nn \__spintent_spcircfig_word_and_article:n
  {
    \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
```

```
      {
        \str_if_eq:vnTF { l__spintent_sp #1 _read_sty_str } { circun }
          {
            \tl_set:Nn \l__spintent_geo_circ_word_tl { circunferencia }
            \str_set:Nn \l__spintent_spcirc_articulo_str { la }
          }
          {
            \tl_set:Nn \l__spintent_geo_circ_word_tl { círculo }
            \str_set:Nn \l__spintent_spcirc_articulo_str { el }
          }
      }
      {
        \tl_set:Ne \l__spintent_geo_circ_word_tl { \tl_use:c { l__spintent_sp #1 _read_arg_tl } }
        \str_if_eq:vnTF { l__spintent_sp #1 _read_sty_str } { circun }
          { \str_set:Nn \l__spintent_spcirc_articulo_str { la } }
          { \str_set:Nn \l__spintent_spcirc_articulo_str { el } }
      }
  }
```

(*End of definition for* `\__spintent_spcircfig_word_and_article:n`.)

`\__spintent_spcircfig_process:nn`

The function `\__spintent_spcircfig_process:nn` sets `\l__spintent_geo_fig_op_tl` to `área` or `perímetro`. When `#2` is not empty, it calls `\__spintent_luafun_geo_circ_parse_and_set:n`; if that call reports an error and `#2` contains no comma, it falls back to `\__spintent_luafun_geo_afig_parse_and_set:n` / `\__spintent_luafun_geo_pfig_parse_and_set:n` (the same Lua entry point used by `\spAfig`/`\spPfig`) to try the bare number/letter shortcut, setting `\l__spintent_geo_circfig_mode_str` to `bare` on success. A table letter match from that fallback is rejected (forced back to an error), since table letters are not meaningful for a circle.

`\__spintent_spcircfig_word_and_article:n` sets the word/article used to build `\l__spintent_geo_circ_int` for the `bare`, empty, and real-center cases, delegating the rendering to `\__spintent_spcircfig_mathml:nn`.

```
\cs_new_protected:Nn \__spintent_spcircfig_process:nn
  {
    \str_if_eq:eeTF {#1} { Acirc }
      { \tl_set:Nn \l__spintent_geo_fig_op_tl { área } }
      { \tl_set:Nn \l__spintent_geo_fig_op_tl { perímetro } }
    \str_set:Nn \l__spintent_geo_circfig_mode_str { circle }
    \tl_if_empty:nF {#2}
      {
        \__spintent_luafun_geo_circ_parse_and_set:n { \exp_not:n {#2} }
        \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
          {
            \tl_if_in:nnF {#2} { , }
              {
                \str_if_eq:eeTF {#1} { Acirc }
                  { \__spintent_luafun_geo_afig_parse_and_set:n { \exp_not:n {#2} } }
                  { \__spintent_luafun_geo_pfig_parse_and_set:n { \exp_not:n {#2} } }
                \str_if_eq:VnF \l__spintent_geo_luaset_error_str { true }
                  {
                    \str_if_empty:NTF \l__spintent_geo_luaset_letra_word_str
                      { \str_set:Nn \l__spintent_geo_circfig_mode_str { bare } }
                      { \str_set:Nn \l__spintent_geo_luaset_error_str { true } }
                  }
              }
          }
        \str_if_eq:VnF \l__spintent_geo_circfig_mode_str { bare }
          {
            \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
              {
                \msg_error:nnee { spintent } { sp #1 -invalid-arg }
                  { \c_backslash_str sp #1 } { \exp_not:n {#2} }
              }
          }
      }
    \__spintent_spcircfig_word_and_article:n {#1}
    \str_if_eq:VnTF \l__spintent_geo_circfig_mode_str { bare }
      {
        \str_set:Ne \l__spintent_geo_circ_intent_str
          {
            \l__spintent_geo_fig_op_tl -
            \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
            \l__spintent_geo_luaset_sub_spoken_str
```

```
          }
        \str_if_eq:eeT { \str_use:c { l__spintent_sp #1 _mode_str } } { mathcat }
          {
            \str_set:Ne \l__spintent_geo_circ_intent_str
              {
                _ \l__spintent_geo_fig_op_tl -
                \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
                \l__spintent_geo_luaset_sub_spoken_str :prefix($pts)
              }
          }
      }
      {
        \tl_if_empty:nTF {#2}
          {
            \str_set:Ne \l__spintent_geo_circ_intent_str
              {
                \l__spintent_geo_fig_op_tl -de- \l__spintent_spcirc_articulo_str -
                \l__spintent_geo_circ_word_tl
              }
            \str_if_eq:eeT { \str_use:c { l__spintent_sp #1 _mode_str } } { mathcat }
              {
                \str_set:Ne \l__spintent_geo_circ_intent_str
                  {
                    _ \l__spintent_geo_fig_op_tl -de-
                    \l__spintent_spcirc_articulo_str -
                    \l__spintent_geo_circ_word_tl :prefix($pts)
                  }
              }
          }
          {
            \str_set:Ne \l__spintent_geo_circ_intent_str
              {
                \l__spintent_geo_fig_op_tl -de- \l__spintent_spcirc_articulo_str -
                \l__spintent_geo_circ_word_tl
              }
            \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
              {
                \str_put_right:Ne \l__spintent_geo_circ_intent_str
                  {
                    -
                    \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
                    \str_use:c { l__spintent_sp #1 _sub_index_str }
                  }
              }
            \str_put_right:Ne \l__spintent_geo_circ_intent_str
              { -de-centro-en- \l__spintent_geo_luaset_print_tl }
            \str_if_empty:NF \l__spintent_geo_luaset_circ_radio_type_str
              { \str_put_right:Nn \l__spintent_geo_circ_intent_str { -y-radio } }
            \str_if_eq:eeT { \str_use:c { l__spintent_sp #1 _mode_str } } { mathcat }
              {
                \str_set:Ne \l__spintent_geo_circ_intent_str
                  {
                    _ \l__spintent_geo_fig_op_tl -de-
                    \l__spintent_spcirc_articulo_str -
                    \l__spintent_geo_circ_word_tl :prefix($pts)
                  }
              }
          }
      }
    \__spintent_spcircfig_mathml:nn { #1 } { #2 }
  }
```

(*End of definition for* `\__spintent_spcircfig_process:nn`.)

`\__spintent_spcircfig_mathml:nn` The function `\__spintent_spcircfig_mathml:nn` reads the state already set by `\__spintent_spcircfig_proces` and renders it. Three cases follow: the `bare` shortcut prints the symbol with the shortcut value as its own subscript; an empty `#2` prints the symbol alone; and a real center (with an optional radius) prints the symbol, the optional `sub-index` subscript, and the center between parentheses. When a radius is present, `\mathcal` (`\__spintent_mathcal:n`) gets its own `\MathMLintent` holding the full spoken intent, and the circle's own visual pieces — symbol/center/comma, the radius (via `\__spintent_spcircfig_radio_print:n`, kept audible), and the closing parenthesis — become three sibling `\MathMLintent` calls sharing a single

\c_math_subscript_token group, so the whole notation is typeset at subscript size.

```
\cs_new_protected:Nn \__spintent_spcircfig_mathml:nn
  {
    \__spintent_invisible_sep:
    \str_if_eq:VnTF \l__spintent_geo_circfig_mode_str { bare }
      {
        %%\__spintent_invisible_sep:
        \MathMLintent
          {
            \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
              { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
            \l__spintent_geo_circ_intent_str
          }
          {
            \c__spintent_invisible_sep_tl
            \str_if_eq:eeTF {#1} { Acirc }
              { \__spintent_mathcal:n { A } }
              { \__spintent_mathcal:n { P } }
            \c_math_subscript_token
              {
                \__spintent_spcircfig_sym:n {#1}
                \c_math_subscript_token { \l__spintent_geo_luaset_print_tl }
              }
          }
      }
      {
        \tl_if_empty:nTF {#2}
          {
            %%\__spintent_invisible_sep:
            \MathMLintent
              {
                \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                  { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                \l__spintent_geo_circ_intent_str
              }
              {
                \c__spintent_invisible_sep_tl
                \str_if_eq:eeTF {#1} { Acirc }
                  { \__spintent_mathcal:n { A } }
                  { \__spintent_mathcal:n { P } }
                \c_math_subscript_token { \__spintent_spcircfig_sym:n {#1} }
              }
          }
          {
            %%\__spintent_invisible_sep:
            \str_if_empty:NTF \l__spintent_geo_luaset_circ_radio_type_str
              {
                \MathMLintent
                  {
                    \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                      { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                    \l__spintent_geo_circ_intent_str
                  }
                  {
                    \c__spintent_invisible_sep_tl
                    \str_if_eq:eeTF {#1} { Acirc }
                      { \__spintent_mathcal:n { A } }
                      { \__spintent_mathcal:n { P } }
                    \c_math_subscript_token
                      {
                        \__spintent_spcircfig_sym:n {#1}
                        \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
                          {
                            \c_math_subscript_token
                              { \str_use:c { l__spintent_sp #1 _sub_index_str } }
                          }
                        ( \l__spintent_geo_luaset_print_tl )
                      }
                  }
              }
              {
```

```
                    \MathMLintent
                      {
                        \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
                          { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
                        \l__spintent_geo_circ_intent_str
                      }
                      {
                        \c__spintent_invisible_sep_tl
                        \str_if_eq:eeTF {#1} { Acirc }
                          { \__spintent_mathcal:n { A } }
                          { \__spintent_mathcal:n { P } }
                      }
                    \c_math_subscript_token
                      {
                        \MathMLintent
                          { :silent }
                          {
                            \MathMLintent{ :silent } { \__spintent_spcircfig_sym:n {#1} }
                            \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
                              {
                                \c_math_subscript_token
                                  { \str_use:c { l__spintent_sp #1 _sub_index_str } }
                              }
                            ( \l__spintent_geo_luaset_print_tl , \,
                          }
                        %%\__spintent_invisible_sep:
                        \__spintent_spcircfig_radio_print:n {#1}
                        \MathMLintent { :silent } { ) }
                      }
                  }
              }
          }
      }
```

(*End of definition for* `\__spintent_spcircfig_mathml:nn`.)

`\__spintent_spcircfig_exec:nnn`

The function `\__spintent_spcircfig_exec:nnn` receives the short name of the command in `#1`, the optional keys in `#2`, and the argument in `#3`, delegating the rendering to `\__spintent_spcircfig_process:nn`.

```
\cs_new_protected:Nn \__spintent_spcircfig_exec:nnn
  {
    \keys_set:nn { spintent / sp #1 } { #2 }
    \__spintent_spcircfig_process:nn { #1 } { #3 }
  }
```

(*End of definition for* `\__spintent_spcircfig_exec:nnn`.)

`\spAcirc`
`\spPcirc`

Both public commands `\spAcirc` and `\spPcirc` enforce math mode, open a local group, and delegate execution to `\__spintent_spcircfig_exec:nnn`.

```
\NewDocumentCommand \spAcirc { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \tl_if_novalue:nTF {#2}
            { \__spintent_spcircfig_exec:nnn { Acirc } { #1 } { } }
            { \__spintent_spcircfig_exec:nnn { Acirc } { #1 } { #2 } }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spAcirc } }
  }
\NewDocumentCommand \spPcirc { O{} d() }
  {
    \mode_if_math:TF
      {
        \group_begin:
          \tl_if_novalue:nTF {#2}
            { \__spintent_spcircfig_exec:nnn { Pcirc } { #1 } { } }
            { \__spintent_spcircfig_exec:nnn { Pcirc } { #1 } { #2 } }
        \group_end:
      }
      { \msg_error:nnn { spintent } { need-math-mode } { \spPcirc } }
  }
```

(*End of definition for* `\spAcirc` *and* `\spPcirc`*. These functions are documented on page* **??**.)

11.42 Finish package

Finish package implementation.

```
\file_input_stop:
⟨/package⟩
```

12 Índice de la Implementación

The italic numbers denote the pages where the corresponding entry is described, the numbers underlined and all others indicate the line on which they are implemented in the package code.